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Research

Q2 2026 Global Markets Survey

June 2026

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With special thanks to Anthony Chaimowitz from the dbDataInsights team

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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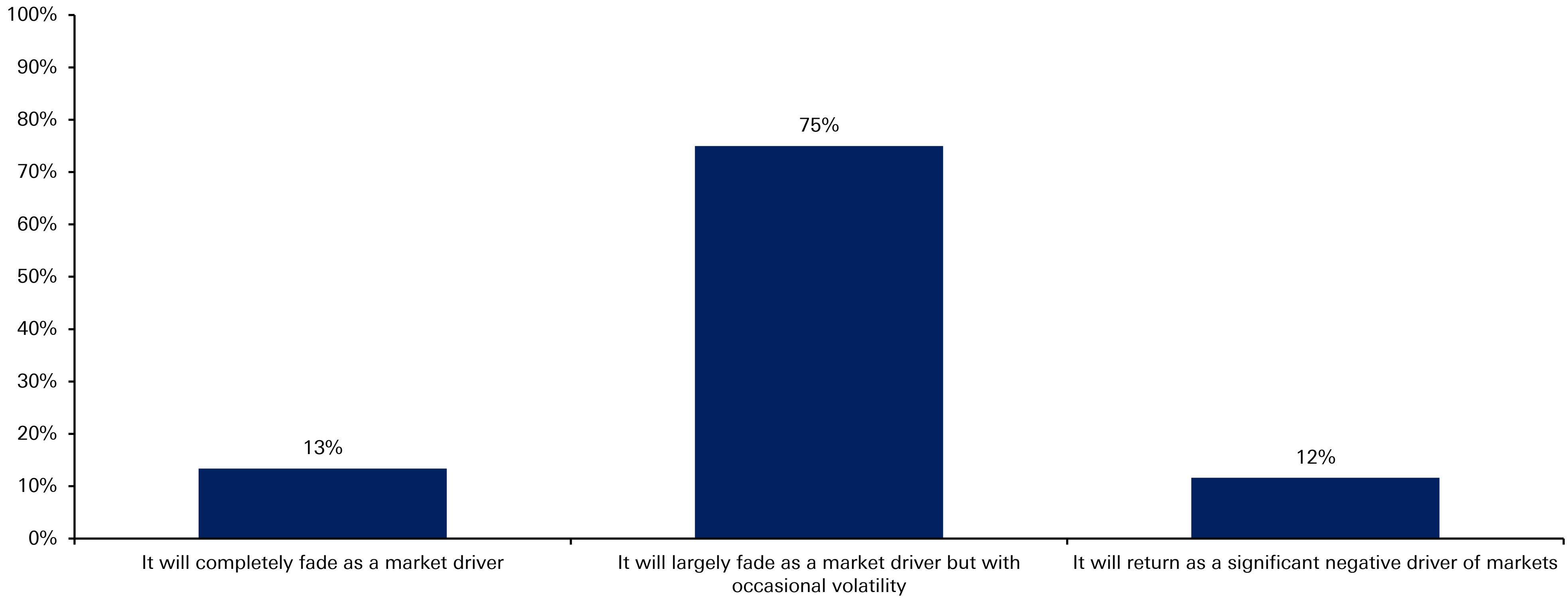
Our Q2 2026 global financial market survey had 275 responses from around the world, conducted between June 22nd and June 26th.

An overwhelming 75% believe the Iran conflict will now largely fade as a market driver into year-end, albeit with occasional volatility. With the 13% who think it will completely fade as a factor that makes 88% who are reasonably sanguine about events in the Middle East. Just 12% think it will return as a big negative driver.

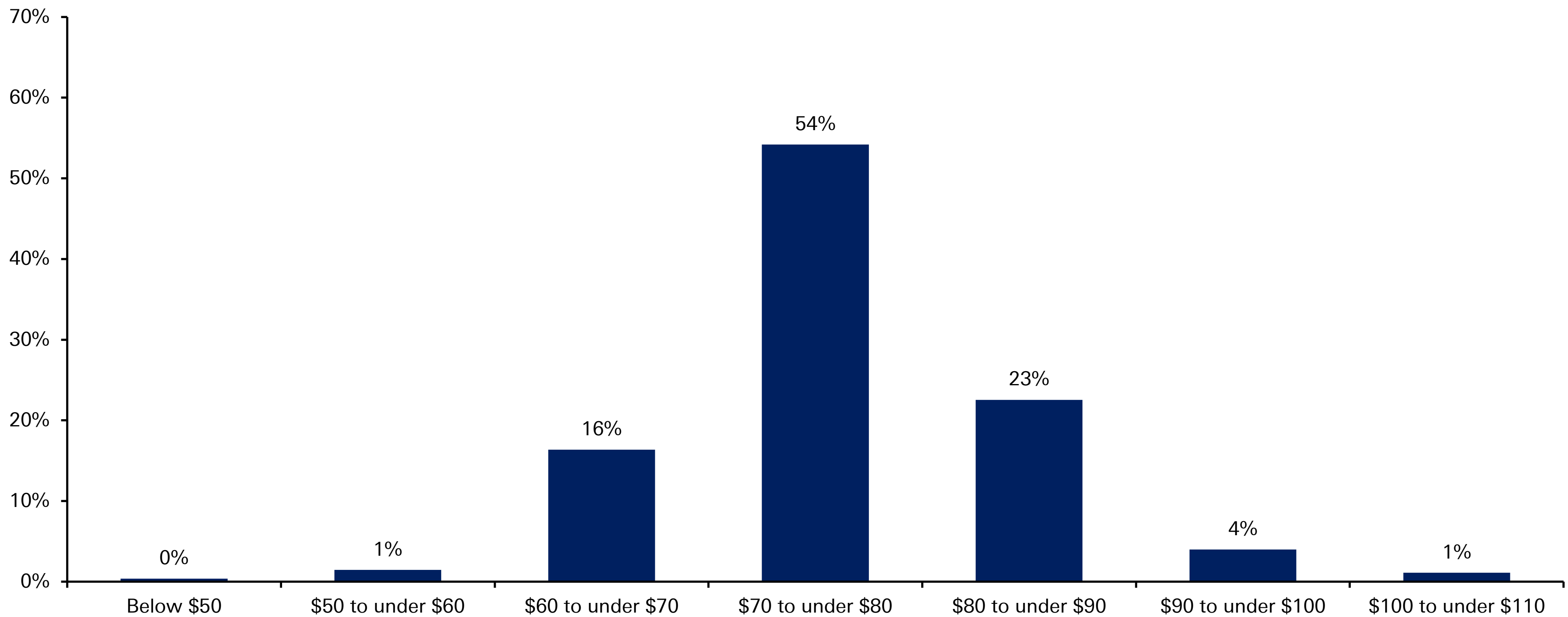
When thinking about the Iran conflict, which of the following best reflects your view of its impact on financial markets for the rest of 2026?



Source: dbDataInsights Survey, Deutsche Bank Research

The more sanguine view on Iran has also led investors to dial down expectations of higher oil prices. A majority of respondents now think Brent will stay between $70-80/bbl for the remainder of the year. Compare that to April, when Brent peaked at ~$118/bbl.

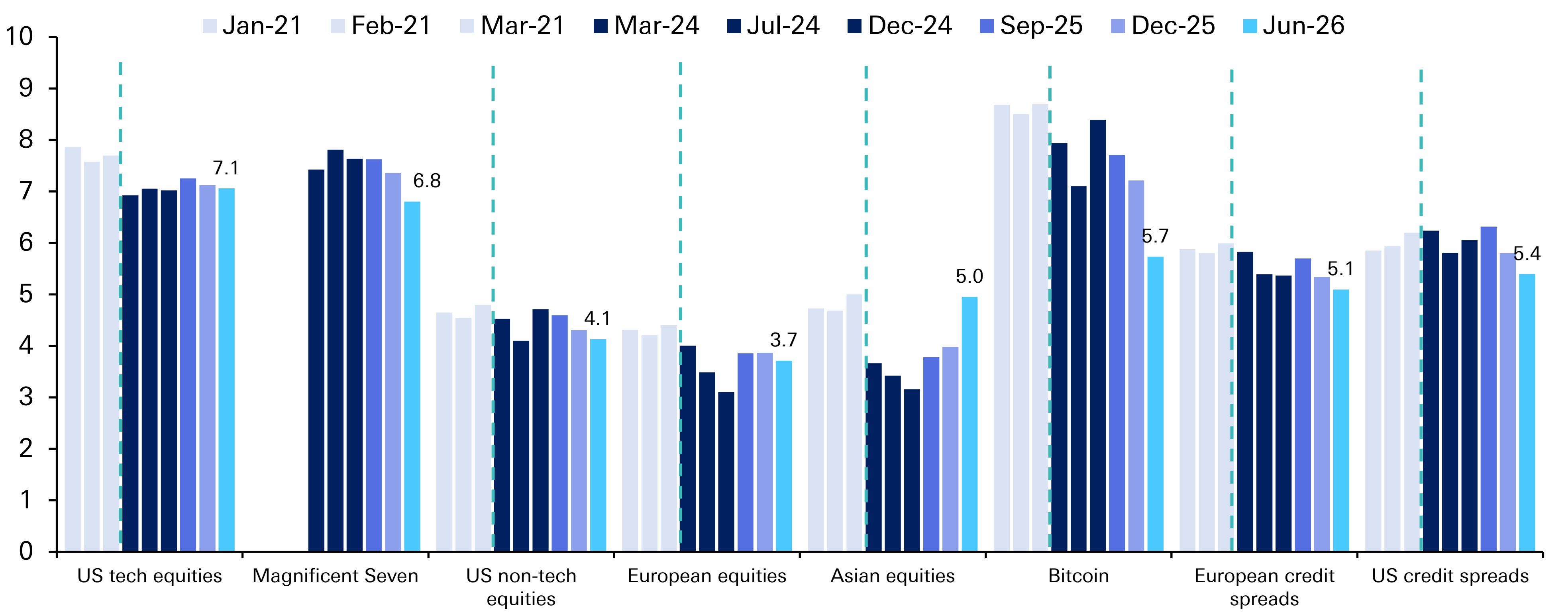
Brent crude oil prices started the year at around $60, rose to approximately $70 prior to the Iran conflict, peaked near $120, and have since declined to around $80. With this in mind, where do you expect Brent crude prices to average for the remainder of the year?



Source: dbDataInsights Survey, Deutsche Bank Research

In our regular bubble question that we first asked in 2021, and then after a gap, in most quarters since 2024, we've seen the lowest perceived bubble risk for the Mag-7. Both US tech and Mag-7 still score around 7 out of 10 on our bubble risk scale, though. Note that Bitcoin bubble risk has collapsed to 5.7 after the near halving since last October.

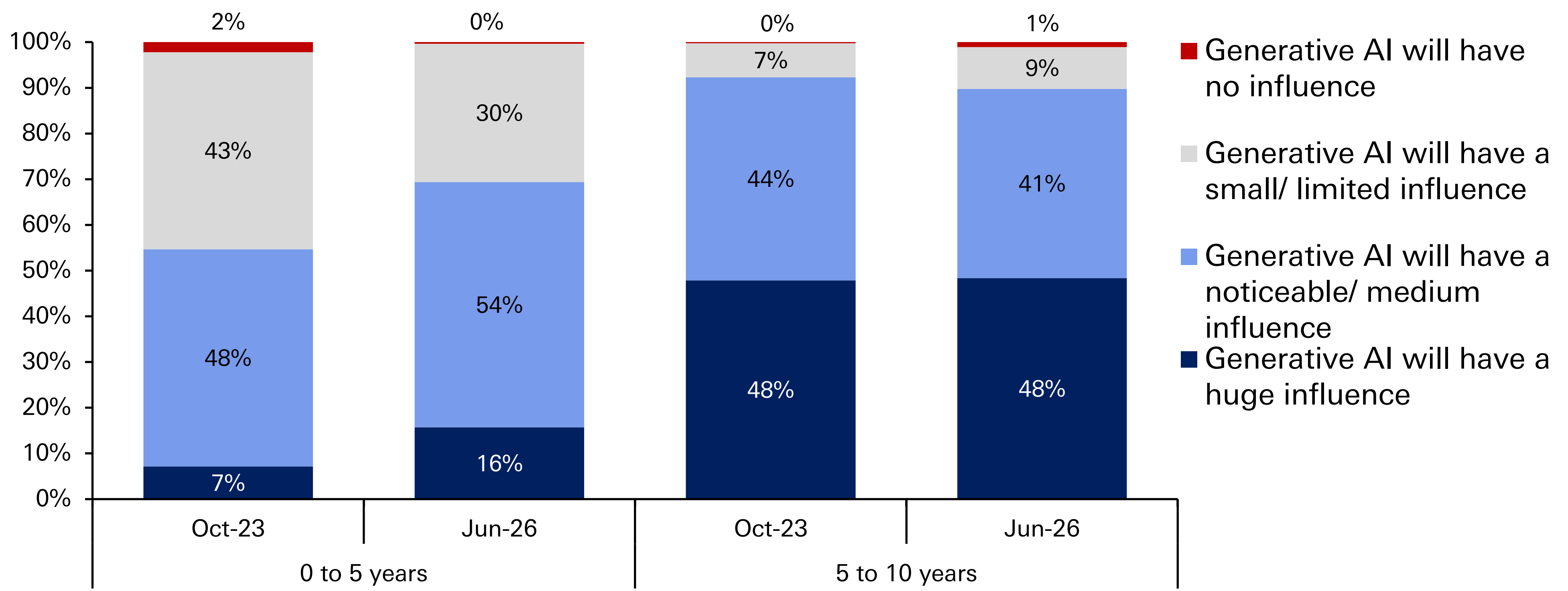
Looking at the following assets and using a scale of 0 to 10, where 0 means 'No Bubble' and 10 means 'Extreme Bubble', please tell us where you think these assets currently are.



Source: dbDataInsights Survey, Deutsche Bank Research

We last asked about Gen AI's potential for productivity growth back in October 2023. Its expected near-term positive impact has increased since then – although some of that likely reflects the nearly 3 years of extra time since we last asked. Surprisingly, the anticipated 5- to 10-year impact has hardly changed, even if just under half still think it will have a huge impact on productivity.

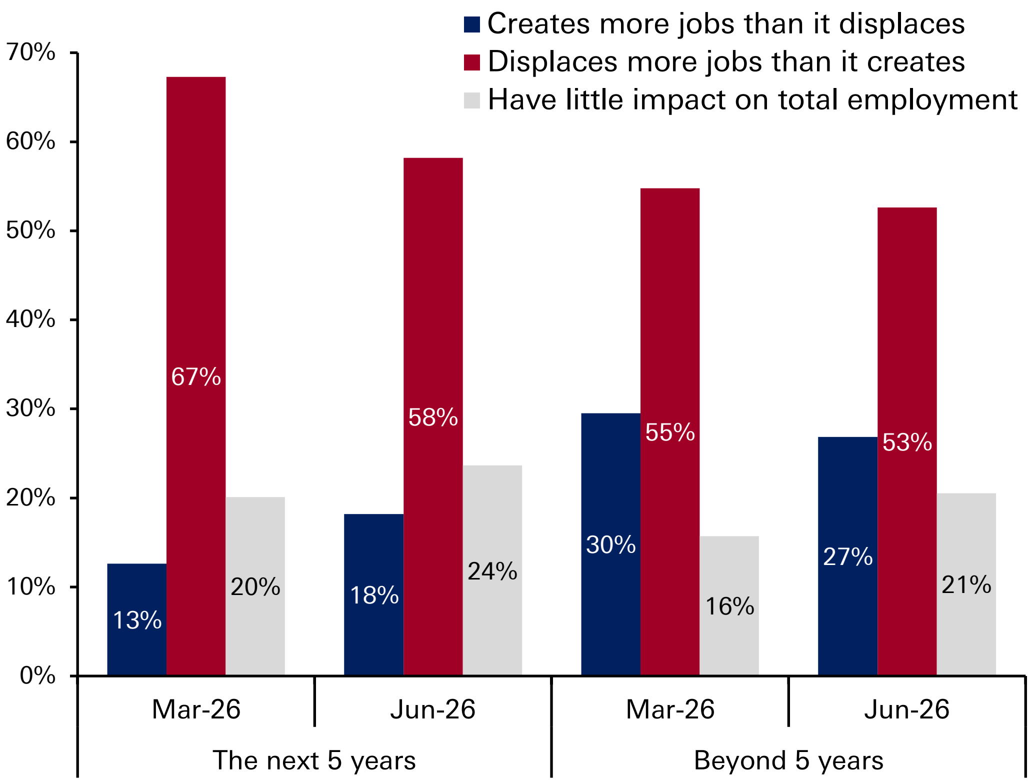
What influence, if any, do you think future iterations of Generative AI will have on productivity growth within the following time frames?

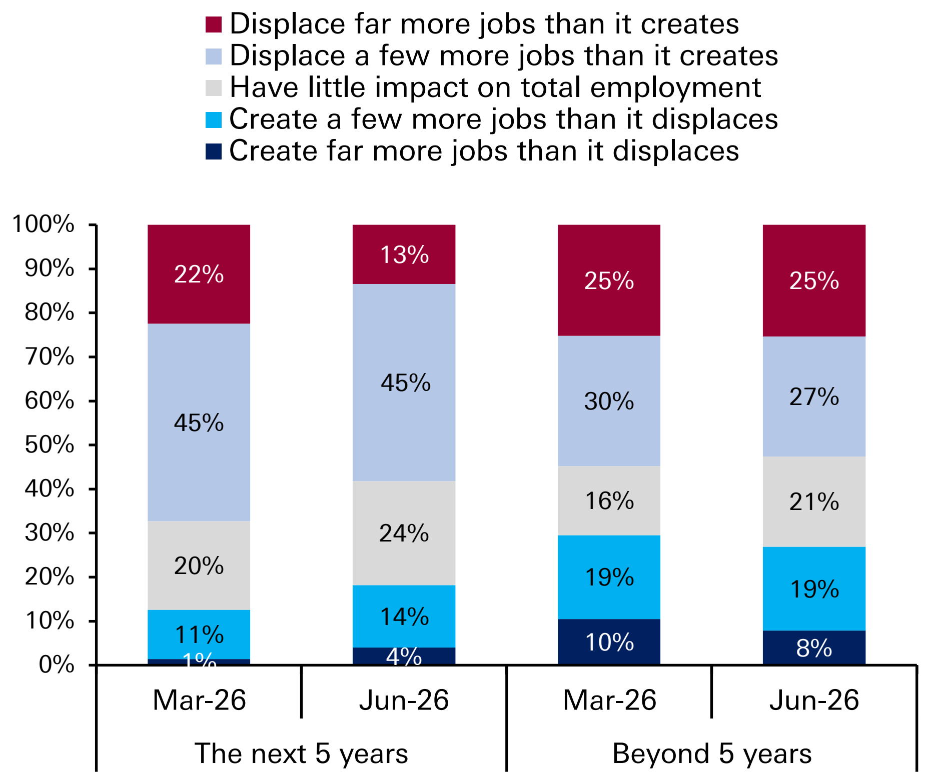


Source: dbDataInsights Survey, Deutsche Bank Research

Market sentiment on AI's impact on jobs remains pessimistic, but we've seen a small improvement since we asked in March. The first chart aggregates the data into binary positive or negative, and the second breaks it down as to how positive or negative.

For the following time horizons, which best describes your expectation for AI's impact on employment? Assume a scenario where AI adoption stays on its current trajectory.





Source: dbDataInsights Survey, Deutsche Bank Research

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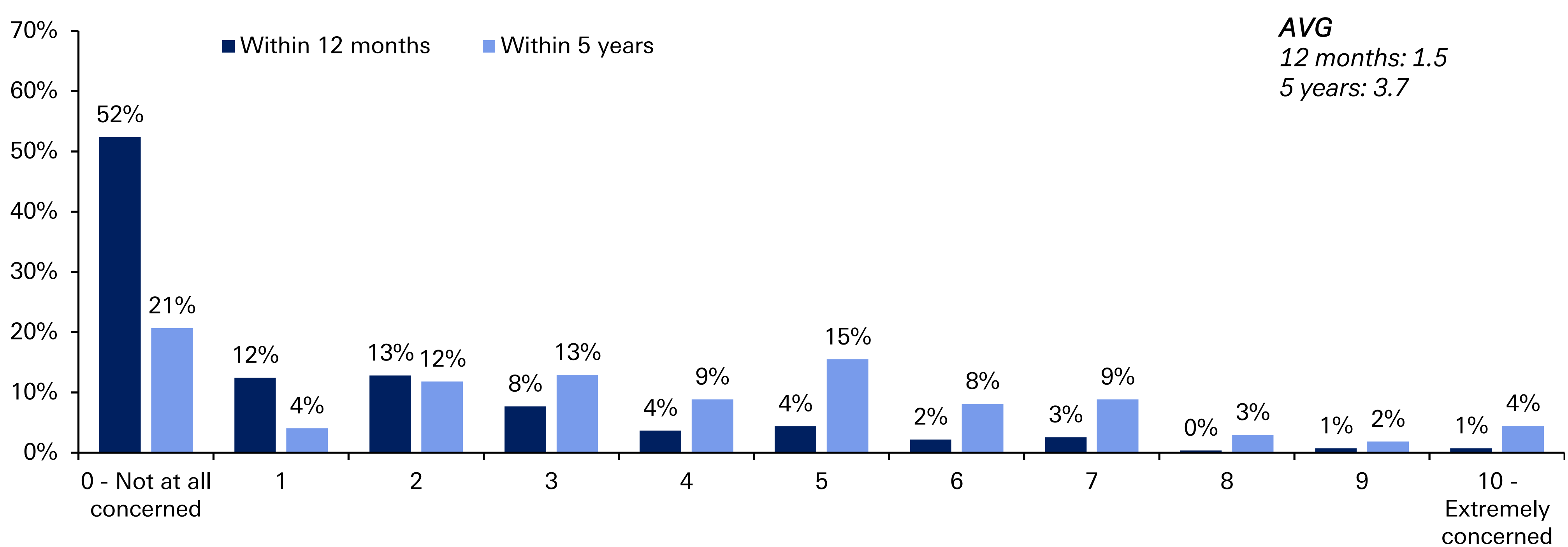
52% say they are not concerned at all about their own jobs because of AI within the next 12 months. This drops to 21% within the next 5 years. The average scores (scale of 1-10) for these periods are 1.5 and 3.7, respectively. So, combining the last two charts suggests that people are more concerned about other people's jobs than their own due to AI.

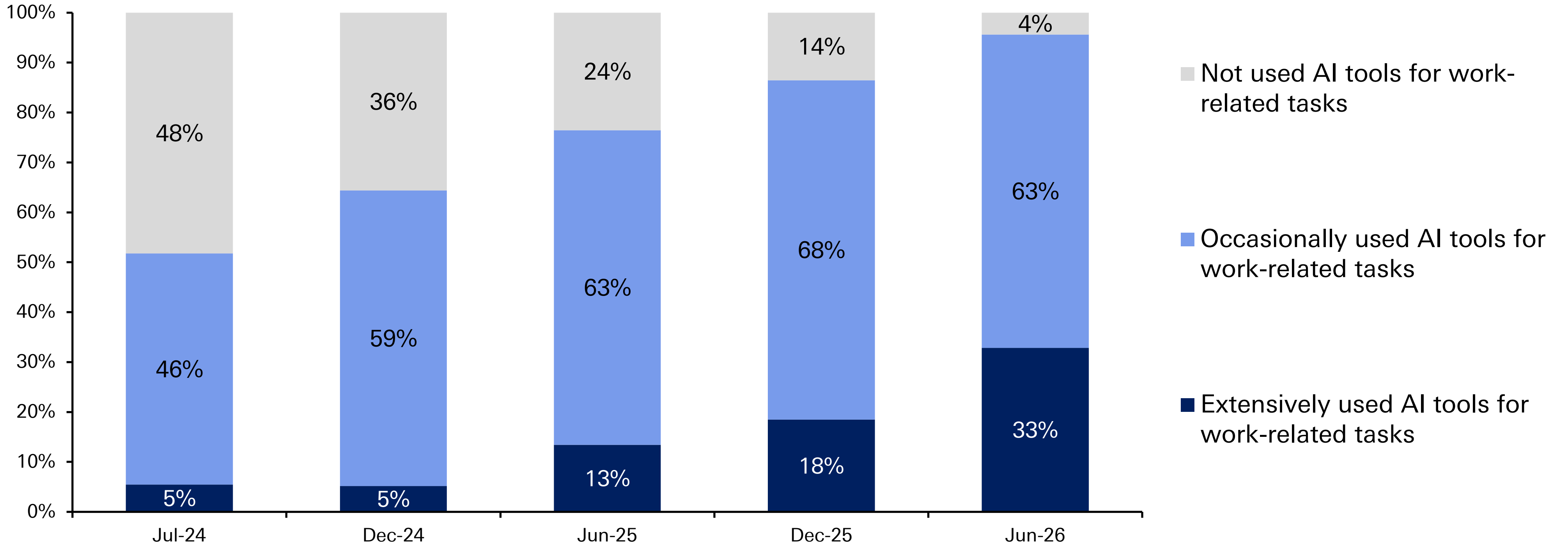
Using a scale of 0 to 10, where 0 is 'Not at all concerned' and 10 is 'Extremely concerned', how concerned are you that you will need to look for a new job within the following time frames because of AI?



Source: dbDataInsights Survey, Deutsche Bank Research

Since March, AI adoption at work continues to accelerate as it has done steadily since we first asked 2 years ago. Now only 4% have not used AI for work, compared to 14% last December and 48% 2 years ago. "Extensively used" has also risen from 18% to 33% now since December – a pretty big increase.

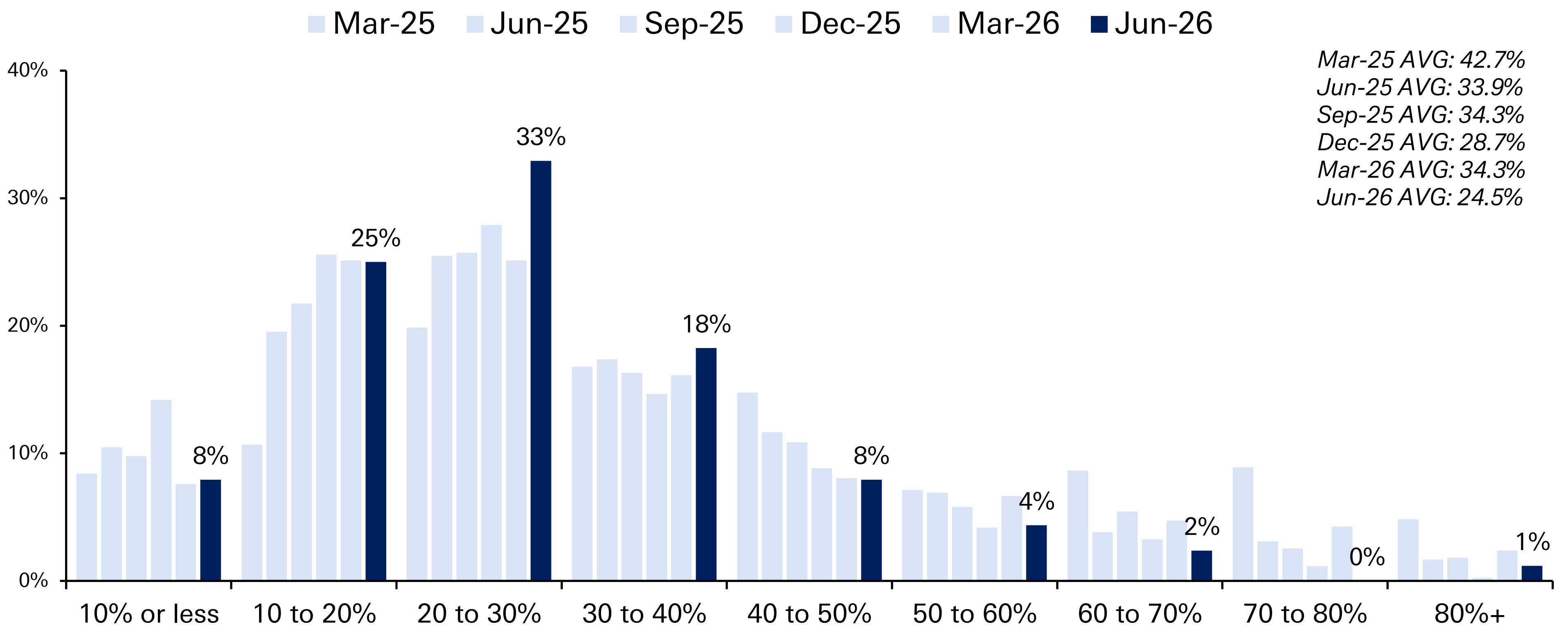
Which of the following best describes your use of AI tools for work-related tasks within the last three months?



Source: dbDataInsights Survey, Deutsche Bank Research

Average perception of 12-month US recession risk is back down to 24.5%, its lowest score since we asked the question in this format last March. Back then it was 42.7% – and even this March, when the Iran War was in its early stages, it was 34.3%.

What do you think the probability of a US recession in the next 12 months is?



Source: dbDataInsights Survey, Deutsche Bank Research

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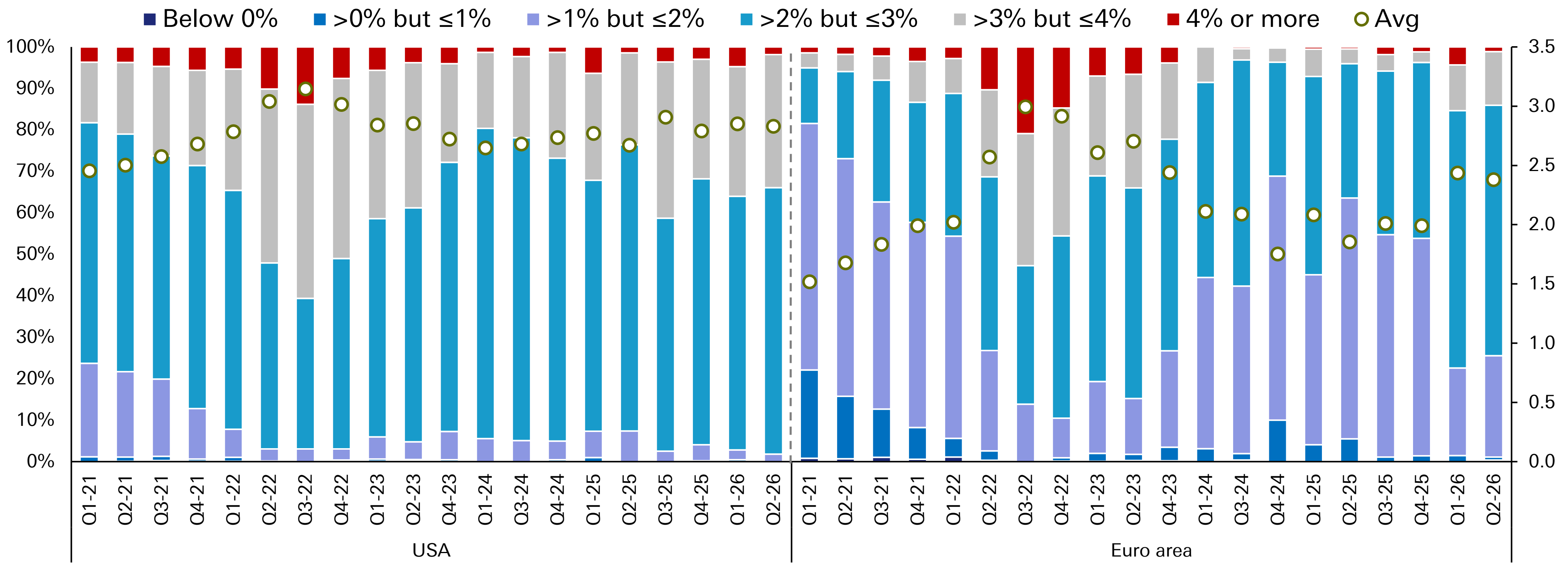
Europe inflation expectations over the next 5 years are unchanged from March, at 2.4%. 5-year US inflation expectations have edged back down to 2.8% from 2.9% in March.

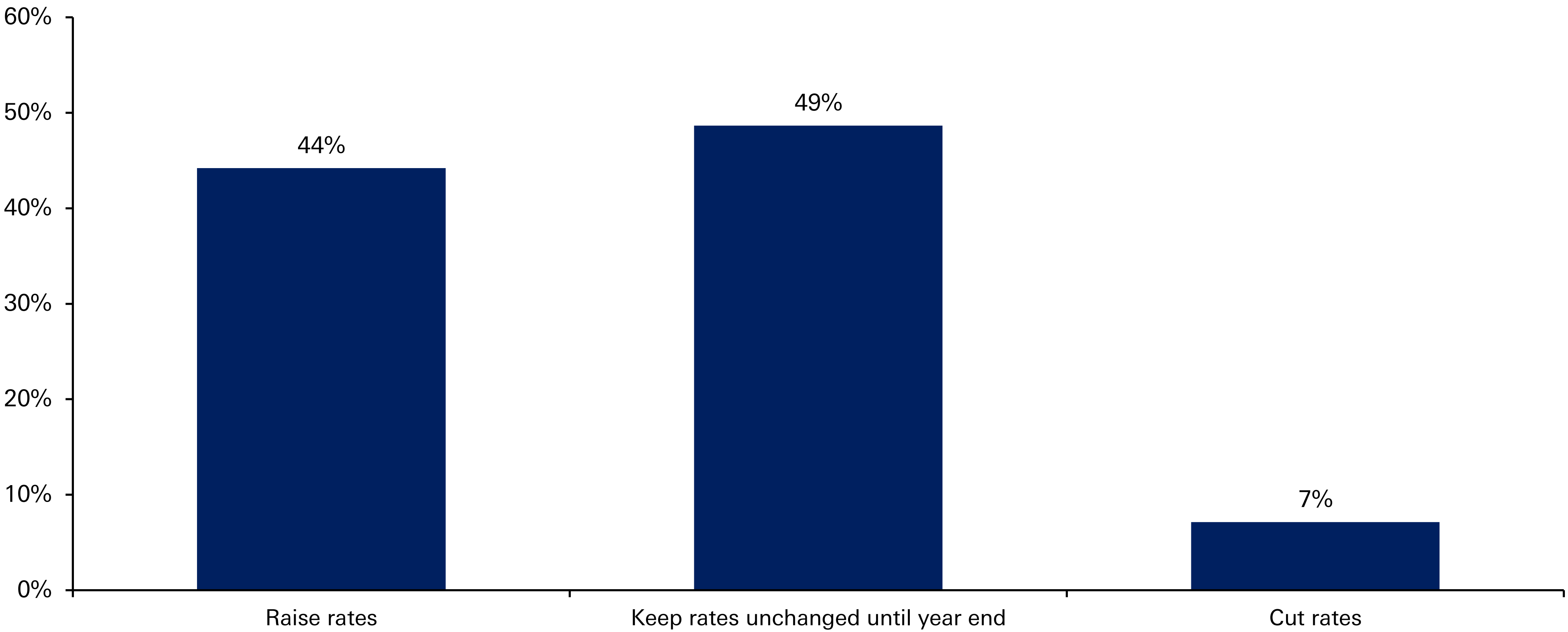
Looking at the below, what do you think inflation will average in the following regions over the next 5 years?



Source: dbDataInsights Survey, Deutsche Bank Research

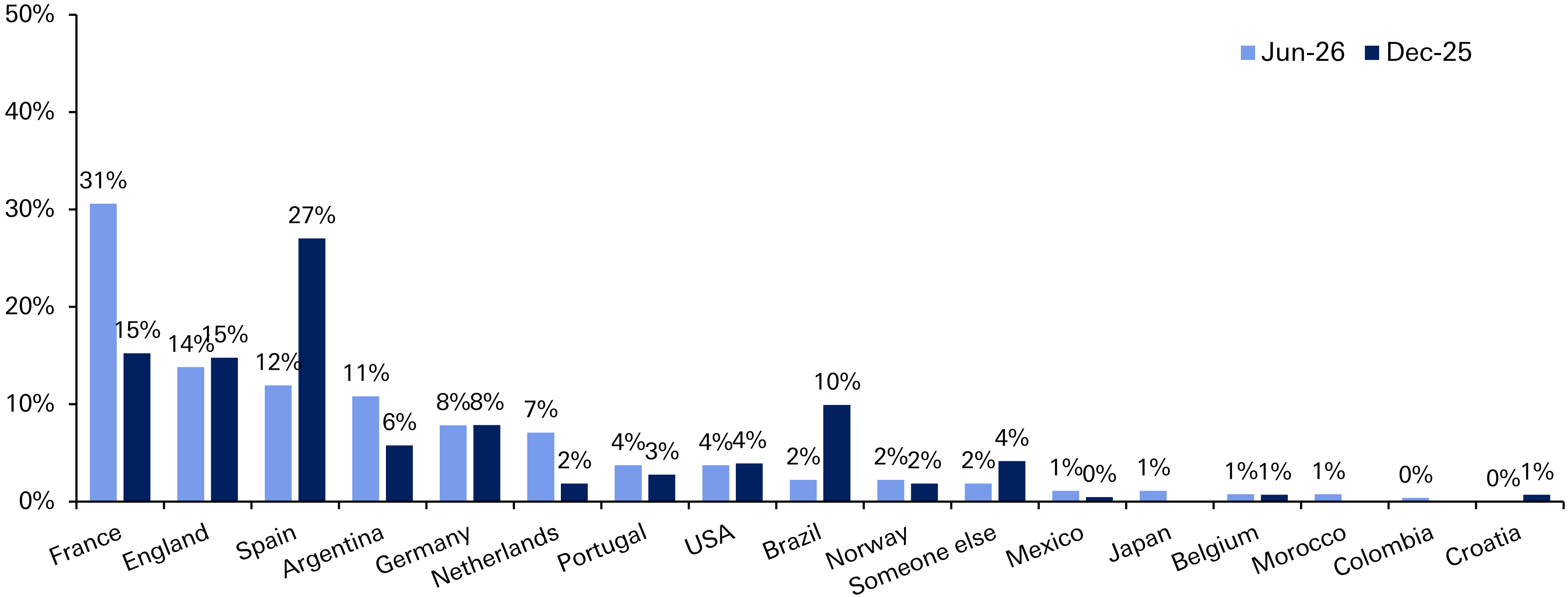
Markets are pricing in almost one-and-a-half hikes by December. Yet perhaps the receding threat of an oil-driven shock has made our respondents lean slightly more dovish, with 49% thinking the Fed will stay on hold this year and 7% thinking cuts are still on the agenda. 44% think a hike is coming.

What do you think the Fed's next move will be in 2026?



And finally, a majority (31%) think France will win the World Cup this year, followed by England (14%). Spain and Brazil have seen the biggest shifts in opinion since December.

Who do you think will win the 2026 football World Cup?



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Appendix 1

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US at 250 – Why has the US been so successful and can it continue?

June 24, 2026

US at 250

To mark the 250 year anniversary of the US Declaration of Independence, this paper looks at how the US emerged as a global superpower, and why it's likely to remain one despite new challenges.

First, we consider how the US went from being a comparatively small country to the world's pre-eminent global power. These reasons range from the US' natural advantages, like favourable geography, to factors like its institutional stability and risk-tolerant capital markets.

We then consider the challenges that threaten US outperformance. China's rapid growth means the US faces its biggest rival since its own emergence as the world's dominant power. China's rise also coincides with several other issues: the rules-based international system that the US helped create is under immense strain; the reserve currency status of the US Dollar is under pressure; and the country's public debt-to-GDP ratio is set to hit new records in the coming years.

Finally, we look at why the US is likely to sustain its outperformance, thanks to a collection of reinforcing advantages. In fact, the US has consistently emerged from challenging periods successfully, be that after the Great Depression, the malaise of the 1970s, and again around the GFC and its aftermath. We conclude with a few lessons for investors and for the rest of the world.

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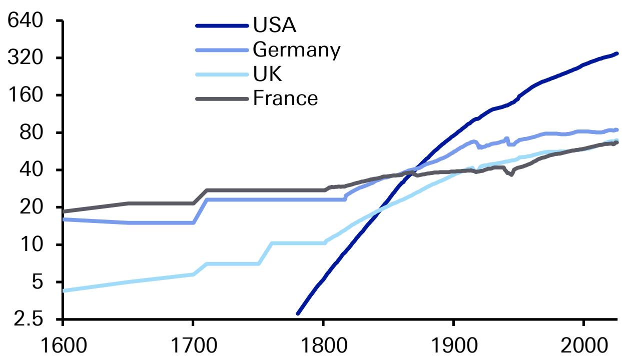
1. The Lessons from History: What underpins America's Success?

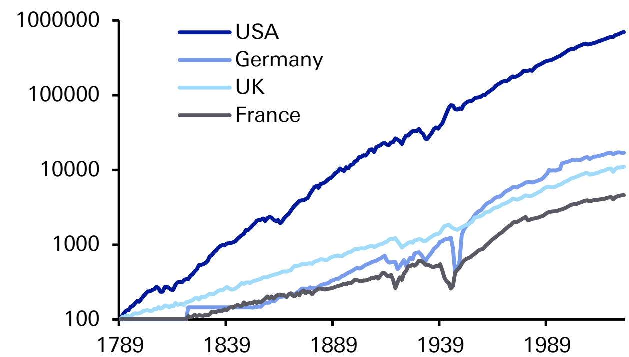
Before delving into the drivers of US success, let us bring with a few highlights of its historical outperformance. Since the United States' founding, the country has achieved remarkable economic success on a whole range of metrics. For one, the US saw rapid growth in population. It started as a comparatively small entity at its founding, but it quickly surged to overtake the large European countries in the mid-19th century and has continued to outpace them since. And with more people, it was little surprise that US GDP rapidly outpaced other countries too.

Figure 1: Population on a log scale (millions)



Source : Finaeon, Deutsche Bank

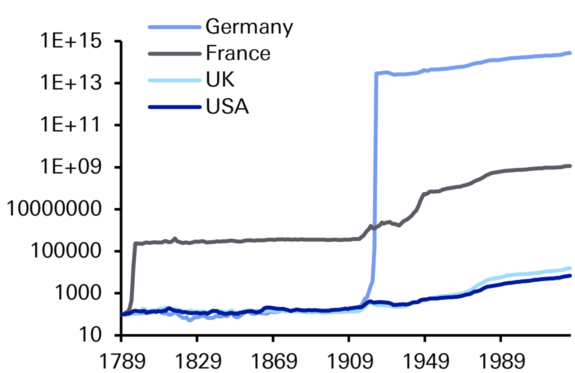
Figure 2: Real GDP on a log scale (1789 = 100)



Source : Finaeon, Deutsche Bank

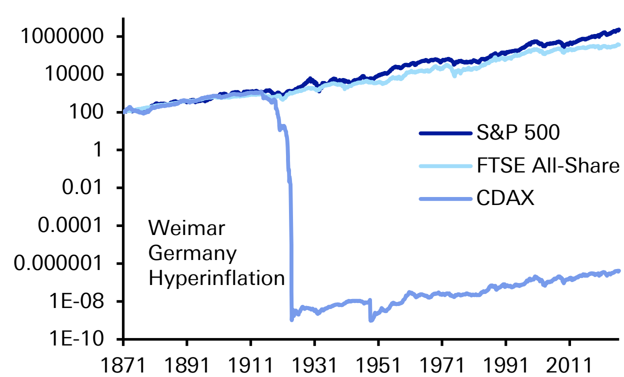
This success extends beyond GDP growth. Unlike France or Germany, the US has never had a bout of hyperinflation in its history, with its currency maintaining its value comparatively well. Meanwhile, the US' long-term equity performance has also been very strong. In the time since equity data is available from the late-19th century, real returns have outpaced the UK and Germany by substantial margins.

Figure 3: Consumer Price Index on a log scale (1789 = 100)



Source : Finaeon, Deutsche Bank

Figure 4: Real equity returns on a log scale (Jan 1871 = 100)



Source : Finaeon, Deutsche Bank

So what has caused this incredible success?

The first reason, and one of the most critical historically, has been the US' political and institutional stability. Clearly, there have been moments of intense turmoil, most notably with the Civil War in the 1860s. Its historical path also should not be over-romanticised, with US territorial expansion during the 19th century having many similarities to that of the Old World colonial empires. Nonetheless, the US is incredibly rare in that its political system is recognisably the same over the last 200 years. This relative stability and early development of property rights provided a fertile environment for long-term investments, which in turn has aided economic growth over the centuries.

The second reason is its geographic advantages. The US possesses vast arable land, navigable rivers, large coastlines, and access to two oceans. It therefore found itself more insulated from the destructive effects of the world wars, with productive capacity not impacted in the same way. Moreover, the country borders Mexico and Canada, who in both population and GDP terms are much smaller than the United States, meaning it didn't face the security risks that many European powers faced over the 19th and early-20th centuries.

The third reason has been its abundance of energy resources, particularly relative to Europe. In large part a corollary of its geographic strengths, this energy abundance has given the US several advantages. First of all, lowering costs for households and industry, which has helped the economy be more resilient against geopolitical shocks. Moreover, with the US becoming a net energy exporter in recent years, it also strengthens the external position.

The fourth reason for the US' relative success was that its main competitors in Europe were deeply affected by the world wars and wider political turmoil. Their productive and financial capacity was severely degraded, with the destruction causing huge loss of life as well. Even among those who survived, many scientists, engineers and entrepreneurs left for the US in the first half of the 20th century. Whilst not a US "success" as such, this meant that on a relative basis, the US' divergence widened considerably. Indeed, in the first half of the 20th century, the US economy grew by 5.7 times, Germany by 3.4 times, the UK by 2.0 times, and France by 1.8 times.

The fifth reason for the US' outperformance is scale. It has a large domestic market of over 300m people, with high average incomes, a common language, and low internal barriers to trade. This has given firms the ability to reach a large-scale domestically before expanding abroad. Indeed, it is notable that 8 of the world's 10 largest firms are based in the US, whereas none are in Europe.

The sixth reason is the structural advantage of the US Dollar, which remains dominant in global trade and FX reserves. This dollar demand matters because it lowers borrowing costs and raises demand for US Treasuries. In turn, it leaves the US with an exceptional capacity to run bigger fiscal and external deficits without facing a funding crisis, and expands the geopolitical leverage that the US possesses. This has been dubbed the "exorbitant privilege", and has been a major advantage in recent decades.

The seventh reason is financial depth. The US has a large banking system, but also a wide range of non-bank financing, which means startups have access to other sources of capital. In fact, in the decade from 2013 to 2023, annual venture capital financing was 0.7% of GDP in the US, compared to just 0.2% in the EU. This financial depth is important because innovative firms can often be loss-making for lengthy periods, so countries with bigger pools of patient risk capital are better placed to commercialise new technologies.

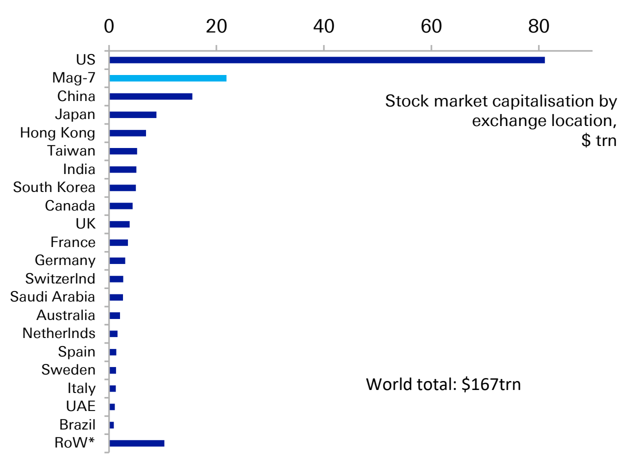
The eighth reason is its self-compounding advantages in education and research. The US has many of the world's strongest research universities, including 7 of the top 10 in the Times Higher Education World University Rankings for 2026. Moreover, this research also has an economic angle, as scientific discoveries feed into startups, workforce training and industrial scale-ups. In turn, this becomes self-compounding, because top global talent is attracted to the US, and ensures it remains at the forefront of new sectors such as artificial intelligence.

The ninth reason is its pro-business architecture, including a greater tolerance of business failure than many European systems. For instance, Chapter 11 reorganisation is designed to preserve and restructure viable firms rather than liquidate them. This more positive approach to business failure ties into the empirical literature, which suggests that more debtor-friendly and efficient insolvency frameworks are associated with greater entrepreneurship and innovation.

The tenth reason is adaptability. The cultural acceptance of failure and capacity to reinvent itself have boosted US ability to adapt to the changing world. This has allowed it to navigate repeat boom-and-bust cycles, as well swings of the policy pendulum – between openness and isolationism, between protectionism and free trade – without threatening overall the institutional stability we highlighted above. And while capitalism has been a key underlying driving force, it is pragmatism rather than ideology that have driven continued success over time.

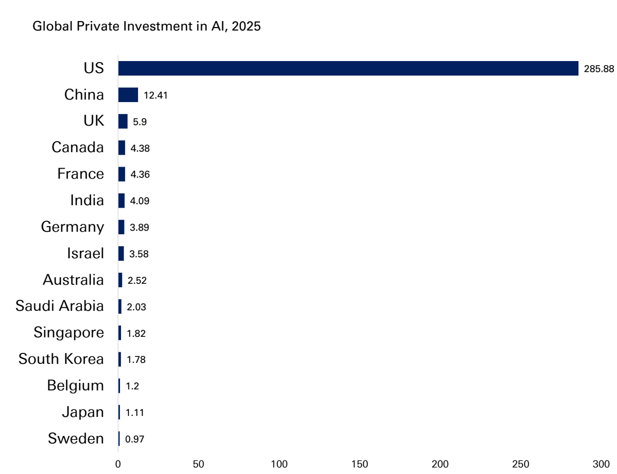
It is also important to note that these advantages do not play out individually, but are mutually reinforcing, with the interaction between them allowing the US to benefit from network and externality effects that few if any countries can match.

Figure 5: The US accounts for nearly half of the world's public stock market capitalisation



Source : Bloomberg Finance LP, Deutsche Bank Research; Data as of 22 June 2026
* Rest of World comprises of a further 62 country exchanges.

Figure 6: US private AI investment reached $285bn in 2025, more 20x as much as in China



Source : Stanford University 2026 Artificial Intelligence Index Report, Deutsche Bank Research

US history has been marked by challenges and recoveries

That said, this success story was far from a straight line. **US outperformance has frequently been questioned, yet it has repeatedly defied the sceptics, including at numerous points in the last century.**

Right from the country's founding, questions were raised about its long-term potential, and in the Civil War of the 1860s, the nation's survival was at stake after less than a century.

Even over the past 100 years, which has ostensibly been a period of US pre-eminence, there have frequently been doubts. For example, after the Wall Street Crash of 1929, the **Great Depression** saw unemployment peak above 25% and remain above 10% for an entire decade. The associated stock market collapse saw the S&P 500 fall by -86% from peak-to-trough, not reaching its 1929 peak again until 1954. This was a global shock, but it undermined faith in the US system of free-market capitalism.

However, the country eventually pulled out of the slump. That started with Franklin Roosevelt's New Deal program, which helped to restore confidence and economic activity. Then the start of WWII saw economic activity surge even further, particularly as there wasn't fighting on US soil. By the end of WWII, the US had never been in a stronger position relative to its peers, not least with Europe having to rebuild after the war.

Similar doubts were clear as the US grappled with the various **crises of the 1960s and 70s**. This period saw major political turmoil and multiple assassinations, including President Kennedy in 1963, his brother and presidential candidate Robert Kennedy in 1968, and civil rights leader Martin Luther King Jr. in 1968. A few years later, Richard Nixon became the first president to resign from office amidst the Watergate scandal. US military power was also facing questions at this time, as the Vietnam War turned into a protracted conflict that also faced substantial domestic opposition.

Then on the economic front, inflation started to gather pace from the late-1960s, surging further after the first oil shock of 1973, leading to a 16-month recession. Measures were imposed to conserve energy, including the National Maximum Speed Limit, which wasn't repealed until 1995. Then in 1979, a second oil shock drove inflation up again.

This turmoil raised questions about whether policymakers could effectively tackle the crises of the day. In 1979, President Jimmy Carter delivered what was widely referred to as the "malaise" speech, although it was actually called "A crisis of confidence". He said there was "growing doubt about the meaning of our own lives and in the loss of a unity of purpose for our nation." He spoke of a "growing disrespect" for institutions, and said how the public often "see paralysis and stagnation and drift." Even the US president was acknowledging that the country was facing huge issues.

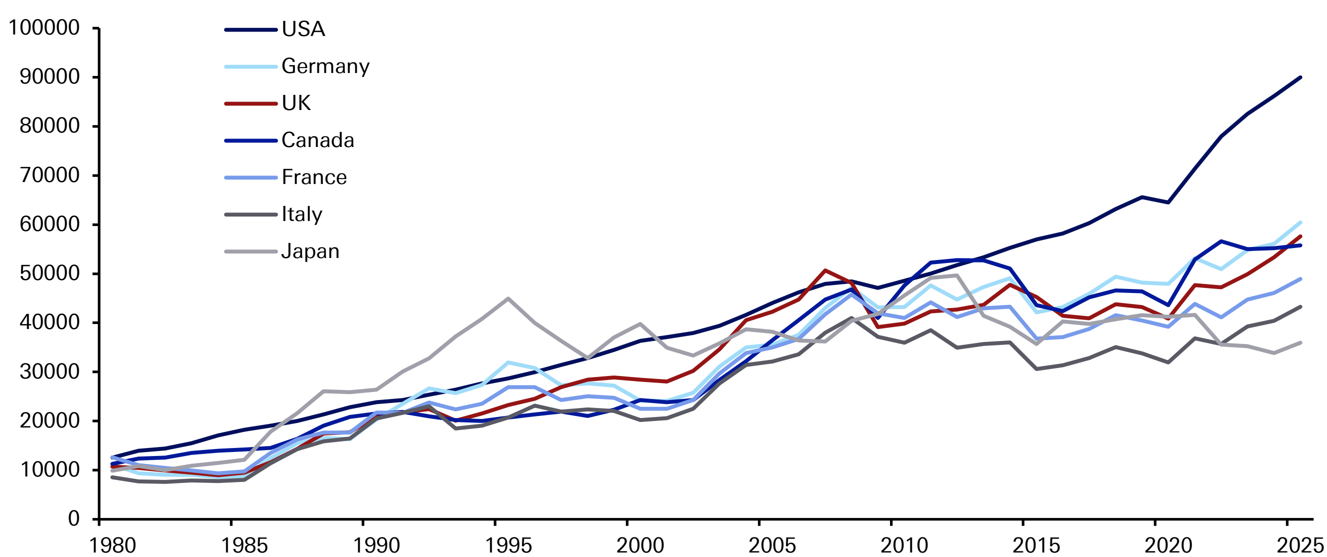
Nevertheless, the US then recovered strongly into the 1980s and 90s. Inflation was tamed, albeit with drastic monetary tightening, and the economy saw rapid growth once it recovered from the early 1980s recession. In 1991, the dissolution of the Soviet Union marked an end to the Cold War, and the 1990s were widely considered a unipolar moment where the US was left as the unmatched global superpower. So once again, the US came through a period of doubt over its success.

But the unipolar moment proved brief, as the **global financial crisis** cast fresh doubt over US prestige. Unemployment surged above 10%, and the economy entered an 18-month recession, the longest since the Great Depression. Moreover, the subsequent recovery was very sluggish, with growth rates well below those of previous decades. Economists resurrected the idea of “secular stagnation”, originally put forward by Alvin Hansen in the 1930s, suggesting the US faced an era of permanently slower growth.

These questions about US pre-eminence came as China was growing rapidly at the same time. In 2005, China’s GDP (in USD terms) was still only 18% of the US total, but that surged to 62% by 2015. That was an incredible shift in relative economic weight in the space of a decade.

But for all the doubts of the 2010s, the secular stagnation narrative has since disappeared in the 2020s. The US economy bounced back remarkably quickly from the pandemic, aided by huge quantities of stimulus that were far larger than 2008. Moreover, recent years have seen the US at the forefront of AI developments, raising hopes of a new productivity surge, with its lead extending over the other advanced economies. So yet again today, the US has emerged from a rockier period around the financial crisis and its aftermath, into a relatively strong position that includes the highest GDP per capita of any G7 country.

Figure 7: GDP per capita (USD terms)



Source : Haver Analytics, Deutsche Bank

2. The Challenges Ahead: Why the US now faces a crucial decade

The above discussion highlights how the US has repeatedly overcome periods of doubt, with predictions of American decline proving premature each time. However, in several respects today’s challenges feel more acute. The US is not as dominant a geopolitical power as it used to be, given China’s emergence as a serious rival. Moreover, its debt-to-GDP ratio is on the verge of new records, meaning its fiscal space is far more constrained to deal with new challenges. As such, the next decade will offer a further test of US adaptability, and whether the political system can respond to these interacting pressures.

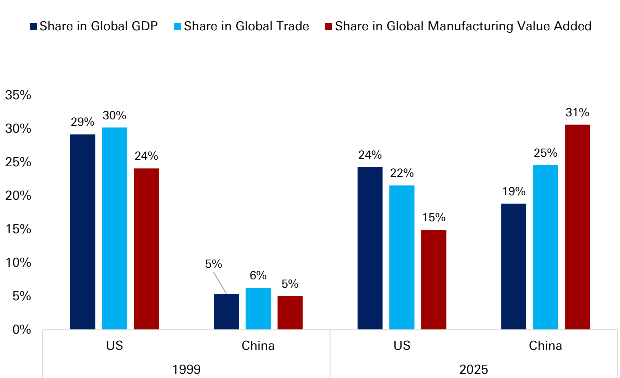
What are the biggest near-term challenges for the United States?

Since the end of the Cold War, the most significant adjustment has been the end of the unipolar moment and China's emergence as a genuine peer competitor, which is different in scale and capability from any rival the US has faced since the late-19th century. China has already overtaken the US on manufacturing output, merchandise trade, and GDP (on a PPP basis), and is rapidly closing the gap in advanced technologies and semiconductors. The US retains decisive leads in nominal GDP, capital markets depth, the dollar system, and frontier innovation, but the gap has narrowed substantially.

The military balance tells a similar story. US defence spending remains roughly three times China's at market exchange rates, but the PPP-adjusted gap is far smaller. More broadly, the gap between US military spending and other major economies has shrunk over the past 15 years. That comes as the proliferation of low-cost precision and autonomous systems is eroding the historic advantages of expeditionary power projection, raising the cost of underwriting the global security order the US has provided since 1945.

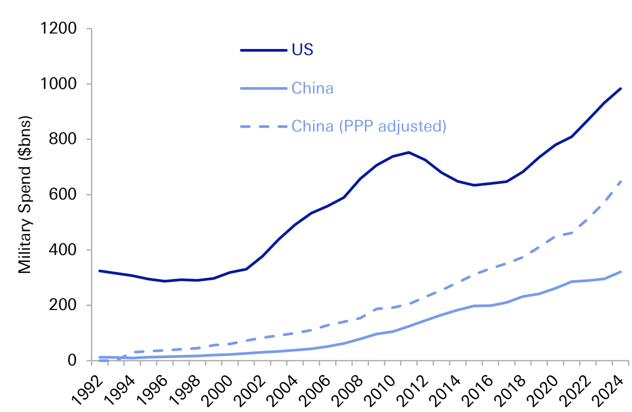
We can also see the emergence of Chinese leadership in a whole range of areas. Among others, China accounts for a third of all global manufacturing activity, with a sector as large as the US, Japan, Germany and South Korea combined. The cost of electricity there is a fraction of what people pay in Europe or the US. And just as the US dollar's role has come under pressure, China is strengthening the global influence of the RMB, in part because of its vast domestic savings surplus.

Figure 8: China has overtaken the US as the world's largest trade and manufacturing hub, though the US is still ahead on nominal GDP



Source : Deutsche Bank Research, UN Industrial Development Organization, Haver. All in Constant 2020 USD, Data for 2025

Figure 9: US lead on military spending has narrowed over the past 15 years



Source : World Development Indicators, World Bank Data, Deutsche Bank Research

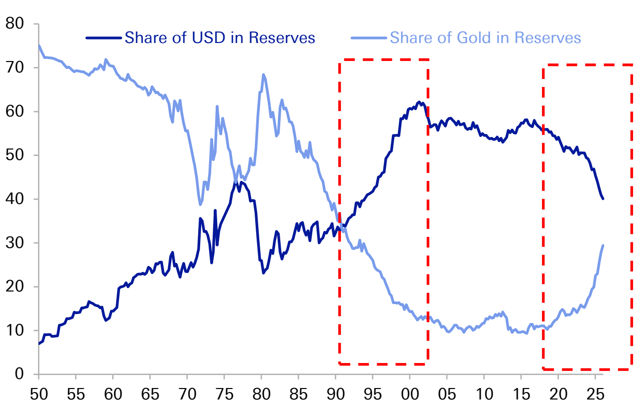
The emergence of China comes as the rules-based international system designed and led by the US for eight decades is under strain from multiple directions. Admittedly, these strains have not just emerged. Even before the Trump administration, emerging markets had grown in relative size and become less willing to accept Western-designed institutions. Meanwhile, the political support for globalisation has frayed across advanced economies.

But recent US policy choices have raised profound questions about the future global order. In 2025, the US effective tariff rate rose to its highest since the 1930s, representing the largest peacetime trade barrier increase in a century. Strategically, the US alliance network has been an enormous economic asset, underwriting dollar dominance, securing trade routes, and providing the soft-power foundation for global rule-setting. Yet US membership of NATO has become an openly debated question, and tariffs have been raised on US allies as well as rivals.

This uncertainty has contributed to growing pressure on the US' status as the global reserve currency. The dollar's share of global reserves has fallen from roughly 72% to 58% over two decades — a gradual decline rather than a collapse, but a clear trend nevertheless. The reduction has accrued to a basket of "non-traditional" reserves and to gold, which has seen the largest central bank buying programme in over half a century. In addition to the overall transition away from a 1990s unipolar world, specific factors have added to the challenges that dollar dominance faces. This includes the weaponisation of the dollar through sanctions — notably the freezing of Russian central bank reserves in 2022, which gave non-Western-aligned countries a strategic incentive to develop alternatives even at material cost — as well as the structural decline of the petrodollar arrangement as energy trade has fragmented and the US has become a net oil exporter.

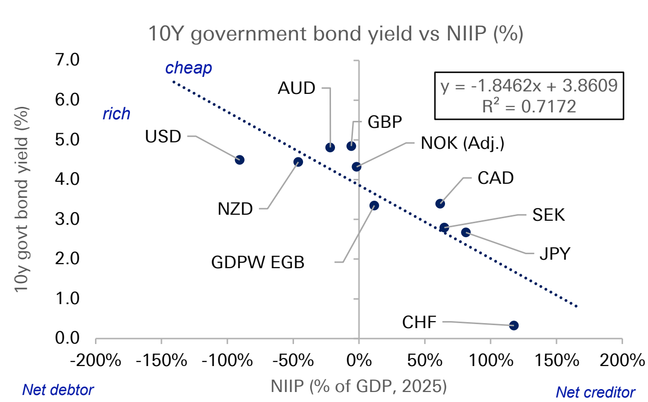
Irrespective of the cause, were the US "exorbitant privilege" to erode further, this would bring major costs to the US and allow other currencies to benefit (see also the DBRI note last year on the potential benefits for the euro). As mentioned earlier, the structural subsidy that the US receives from its reserve currency status includes lower borrowing costs, the ability to run persistent external deficits without the discipline imposed on other economies and the advantages of the dollar being the unit of account for global trade, commodities and cross-border lending.

Figure 10: The "end of history" in the 1990s saw a sharp rise in the USD and decline in gold's share of reserves



Source : Deutsche Bank, Bloomberg Finance LP, Eichengreen, Chitu and Mehl (2014), IMF International Liquidity.

Figure 11: 10yr yields vs Net international investment Position – An illustration of US "exorbitant privilege"



Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP, IMF
Updated 22 June 2026

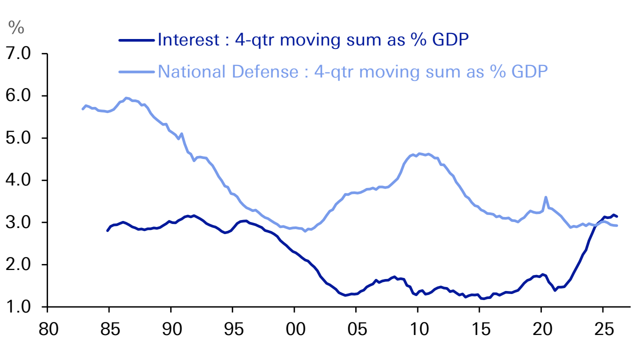
None of this implies imminent displacement. No alternative is remotely ready to assume the dollar's role. Network effects and inertia in trade invoicing and central bank balance sheets all favour continued dollar dominance. Indeed, the pound sterling held a prominent role for many decades after the US had emerged as the world's largest economy. So the realistic risk for the next decade is not collapse but gradual erosion, with a slow loss of the privilege margin that has boosted US economic performance.

The US fiscal trajectory is the most plausible catalyst to accelerate that erosion, and for institutional investors is the single most concrete macroeconomic risk facing the United States. Over the last four years since 2022, the federal deficit has been consistently running at around 5-6% of GDP. Those are the highest peacetime deficits in US history outside of a major recession, and this is happening in a full-employment economy. Debt held by the public is set to surpass 100% of GDP this year, and the CBO projections point to an unsustainable trajectory. Interest payments on the federal debt now exceed defence spending and are the fastest-growing line item in the budget.

Most significantly, the binding constraints are now set to arrive within the next decade. The latest estimates project that the Social Security trust fund will be depleted by late 2032, which would trigger an automatic benefit cut absent legislative action. The Medicare hospital insurance trust fund faces a similar reckoning shortly after. So while the unsustainability of the US public debt trajectory has been in the headlines for many years, these events are now more imminent – ones that the next US administration that takes office after the 2028 election will have to face.

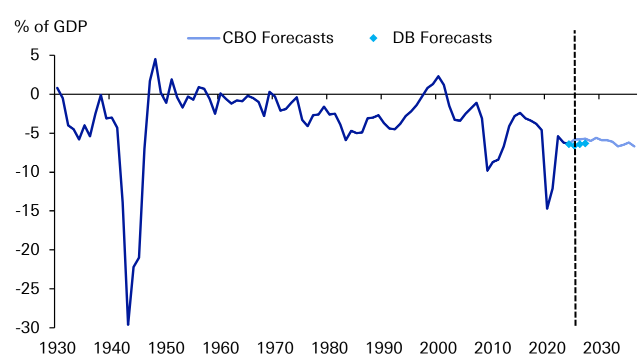
To be fair, the fiscal challenge is far from arithmetically catastrophic – a combination of AI-driven productivity growth and moderate fiscal consolidation could lead to significant improvements in the long-term debt profile. The difficulty is whether the political system can deliver such an outcome. Past fiscal consolidations required bipartisan compromise. Yet the current political environment offers little prospect of similar cooperation, with polarisation a mounting issue. Attempts in recent decades (like the Simpson-Bowles Commission in 2010) have failed to gain sufficient support.

Figure 12: The Crossover: Rising Federal Interest Payments have Overtaken Defense Spending



Source :CBO, OMB, Haver Analytics, Deutsche Bank Research

Figure 13: A Century of Deficits: US Federal Budget Balance from the Great Depression to the 2030s

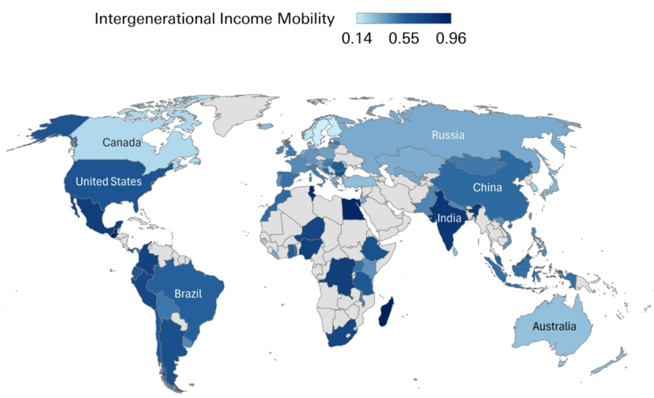


Source : CBO, OMB, Haver Analytics, Deutsche Bank Research

In the meantime, demographic trends are compounding both the fiscal and innovation challenges. The US fertility rate has fallen to 1.6 and the dependency ratio is set to deteriorate sharply as the baby boomer cohort fully enters retirement. Even though the population is comparatively younger than many other developed countries, it is still ageing. For example, the UN projects that the share of the US population aged 65 and over will rise from 19% in 2026 to 22% by 2036, rising further to 28% by the end of the century.

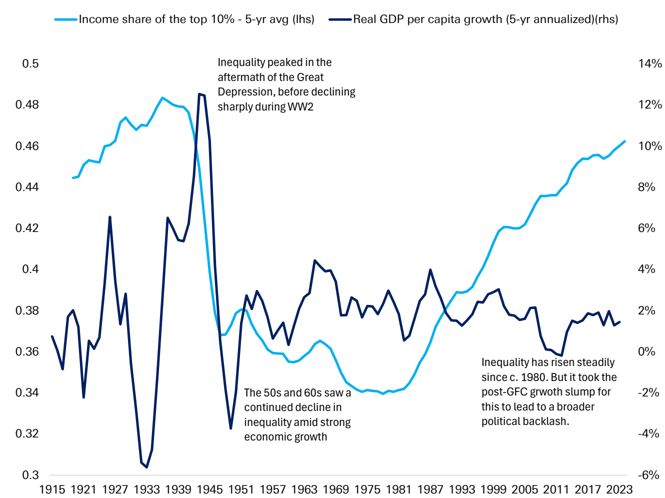
An important connecting driver across fiscal, political and demographic challenges has been a rise in inequality. Its steady rise since the 1980s has gone hand-in-hand with a rise in political polarisation, especially once combined with the economic shock that followed the GFC and the sluggish recovery in its aftermath. A corollary of this is that the traditional "American dream" of social mobility has found itself under mounting pressure. The US now has lower intergenerational income mobility than almost all DM economies, which risks corroding the social contract on which open markets and creative destruction depend. It has also led to negative social effects, notably a widening of the life expectancy gap between Europe and the US since the 1980s.

Figure 14: The US has lower intergenerational mobility than most DM economies
(lower score = higher mobility)



Source : World Bank Data, Deutsche Bank Research

Figure 15: The rise in inequality since 1980 added to the political challenges facing the US, especially after growth slowed post-GFC



Source : Finaeon, World Inequality Report, Deutsche Bank Research

3. Why the US Should Remain a Long-Term Success Story

The challenges facing the US are real, but the weight of evidence still suggests it will remain the world's leading economy for the foreseeable future. Its collective structural advantages remain difficult to replicate, and it retains the same adaptability that's helped it recover from previous difficulties.

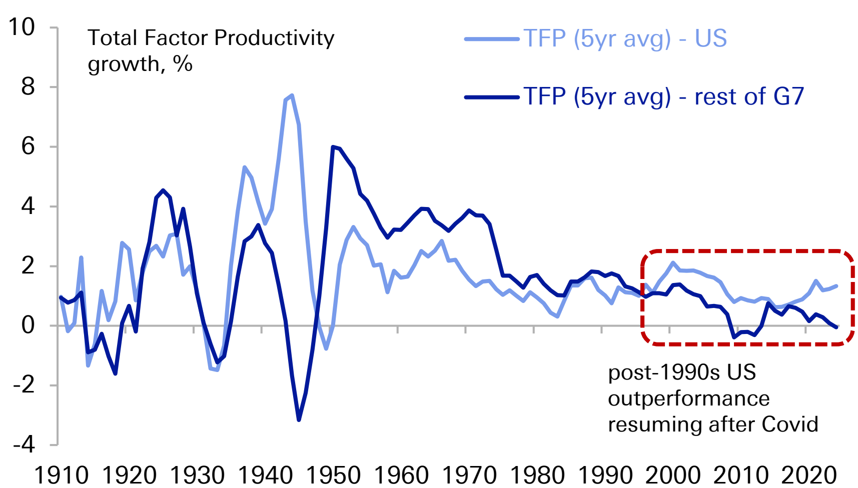
The starting point is that the US should remain the world's largest economy well beyond the critical decade coming up. A global demographic slowdown means that previous assumptions of rapid population and economic growth in emerging markets no longer look inevitable. Conversely, US economic growth has generally surprised on the upside since the pandemic, with the "secular stagnation" narrative of the 2010s quietly fading. Indeed, the US economy has successfully weathered multiple shocks in recent years, ranging from rapid Fed rate hikes, to tariffs, to the Iran conflict. Moreover, there is further upside potential to productivity growth (and hence economic growth) from the US' significant lead on AI.

This AI boom is the latest expression of the US capacity for reinvention that stretches back through its history: a willingness to embrace disruptive, capital-intensive technology and absorb the dislocation it brings. AI is one of many new technologies that have often led to economic volatility (and boom-bust cycles on occasion), but whose benefits have delivered broad, economy-wide productivity gains over time. That was clear historically with canals, railroads, and electrification, and also with more recent innovations like the internet.

Thus, the US is well-positioned to lead the AI cycle for several mutually reinforcing reasons: an innovation-friendly ecosystem of research universities and venture capital; a cultural and institutional acceptance of the boom-bust investment cycles that frontier technology requires; the deepest capital markets in the world to fund it; and abundant domestic energy. The shale revolution has transformed the US into a net energy exporter, and cheap, plentiful power is now a decisive input advantage in the race to build and run data centres. Few competitors can match this combination.

The AI boost comes at a time when the US has already re-established its productivity outperformane in recent years. Since the pandemic, US productivity growth has risen above its post-1970 average, which in turn is boosting GDP and the revenue base. The potential fiscal arithmetic is powerful. According to CBO estimates, if productivity growth were just half a point above their baseline in the coming decade, then debt held by the public would only rise to 110% rather than 120% by 2036 – see our colleagues' note on the US debt outlook last autumn for more. However, productivity gains alone are unlikely to resolve the fiscal issue, in part because the political incentive is to spend the fiscal dividend rather than save it and in part given the uncertainty of how the gains would be distributed — a theme we return to below.

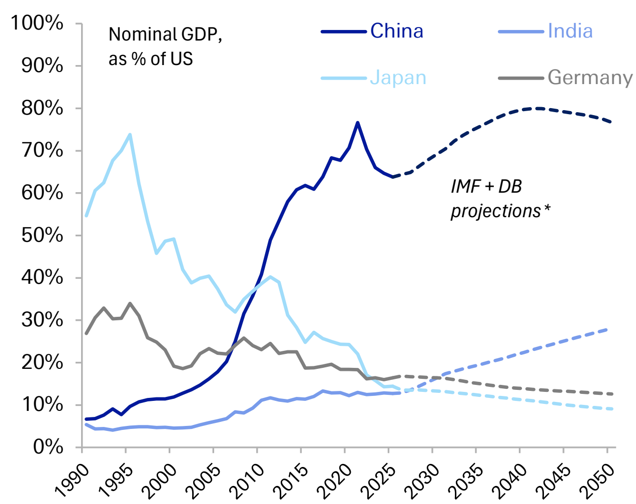
Figure 16: US has resumed its productivity outperformance in recent years



Source : Deutsche Bank Research, www.longtermproductivity.com
Rest of G7 is the simple average of Canada, France, Germany, Italy, Japan and the UK.

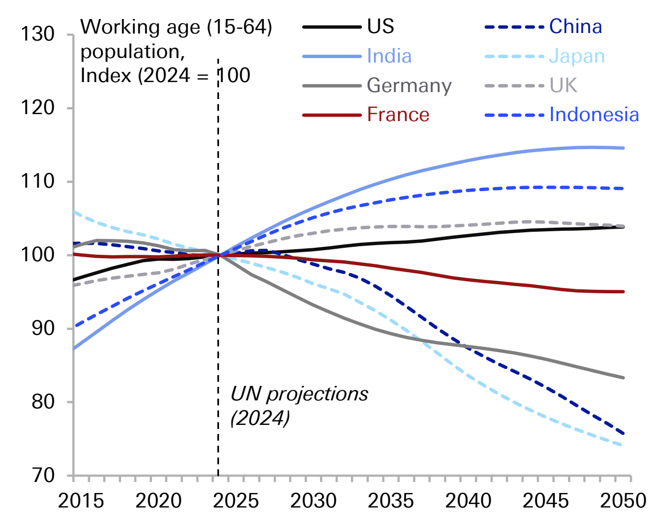
On demographics, the US also enjoys a clear advantage over its principal rivals. China's working-age population will begin a pronounced contraction in the 2030s, with Europe in aggregate also seeing a decline, whereas the US will see moderate growth under a normalised migration trend[1]. With a growing working-age population, the US would avoid this drag from a shrinking labour force. Moreover, the US has had a consistent capacity to absorb immigration at levels few other developed economies have managed. And with the demographic slowdown now reaching emerging markets, few major economies now have clearly better long-term demographic potential than the US. For instance, while India will maintain positive demographic momentum over the next couple of decades, with its fertility rate now below replacement rate, its advantage will shrink by the middle of the century.

Figure 17: US likely to remain the world's largest economy for the foreseeable future



Source : IMF World Economic Outlook, Deutsche Bank Research
** IMF projections to 2031; DB estimate to 2050 based on (i) UN population projections (2024), (ii) stable productivity growth among DM economies, (iii) gradually slowing labour productivity growth for China (to 2.0% by 2050) and India (3.5% by 2050), (iv) stable REER exchange rates.*

Figure 18: The US has better demographics than most DM countries and China, while India's advantage is set to peter out over time



Source : UN Population Database (2024), Deutsche Bank Research
Recent research suggests the 2024 UN demographic projections are likely too optimistic (e.g. www.mercatus.org/macro-musings/jesus-fernandez-villaverde-quandary-global-demographic-decline). However, this downside is largely global and unlikely to see relative US underperformance under normalised migrations trends.

Even on the financial side, whilst the dollar's role has come under pressure, it still retains a dominant and unrivalled position. The network effects that sustain it are exceptionally strong, and there is no obvious successor in the wings, unlike when the dollar overtook sterling in the interwar period. The depth and liquidity of US capital markets, inseparable from the dollar's role, are themselves a structural draw that even US rivals rely upon. Indeed, it is notable that the asset seeing a major increase in official reserves is gold, rather than another currency. And while new payment technologies could bring new risks to dollar dominance, US dominance in the stablecoin space will help limit these risks for the foreseeable future.

1 The UN population projections assume net migration into the US of c. 1.2m annually in the coming years. This would be roughly in line with the average migration rate since 1960 (0.3pp of population annually), while being well below the c.2.5m peak reached in 2023 but well above the c.300k latest Census projection for 2026.

The Conditions for Continued Success: Pitfalls to Avoid

Yet much as there are credible reasons for US outperformance to continue, this cannot be taken for granted.

The first key challenge is to ensure that the gains from future technological growth are broad-based, as excessive concentration risks adding to political and institutional dysfunction. We know from history that technological progress does not automatically deliver broadly shared prosperity. For instance, during the early Industrial Revolution, aggregate output rose for decades as wages for ordinary workers stagnated, a period that has been dubbed "Engels' pause". This offers a cautionary lesson on AI, and with the labour share of income in the US already near historic lows today, a further deterioration risks intensifying the social and political backlash building against the technology. It's important to avoid that, as a backlash against technological advances would in turn negate its gains. A related risk is that if AI gains accrue to capital through profit margins while labour is shed, this would negate the fiscal benefits from higher productivity given the focus of the US tax system on labour taxation.

This brings us to the second challenge – managing the fiscal adjustment and the related timing risk. The US is currently running budget deficits at levels previously unprecedented outside of a major recession or war. The ideal scenario would be some sort of bipartisan consensus, as in the early 1990s, to deal with the fiscal challenges ahead in a pre-emptive manner. However, political incentives to reduce deficits appear limited, raising the prospect that it will be up to markets to ultimately force consolidation. The risk is that this could come at a moment of heightened vulnerability, such as a wider economic crisis or recession, resulting in a harsher correction. A degree of financial repression, especially if combined with genuine consolidation, could end up playing a role in managing the debt burden. But a full embrace of that route — in an extreme case, via debt monetisation by the Fed — would merely accelerate the erosion of the dollar's special status.

The third challenge is protecting the US' institutional foundations. Institutional stability (and the consistent rule of law and secure property rights underpinning it) have been the bedrock on which the capital-markets and dollar advantages ultimately rest. If that eroded over the coming decades, it would create a more uncertain climate for long-term investments and dampen the international appeal that has attracted top talent and financial capital to the United States.

The fourth challenge is a global one, in that the US has benefited immensely from the post-war system of US leadership. However, that settlement has come under increasing strain in recent years, particularly as emerging markets that have grown in size seek greater influence in global affairs. After all, the US only makes up around 26% of global GDP in USD terms. But with its allies, the influence of the US increases substantially. For instance, the G7 as a whole makes up 44% of global GDP, and if you include further US allies such as Australia and South Korea, its effective alliance network still comprises more than 50% of global GDP in dollar terms. With the US standing as the clear and unsurpassed leader of this group, maintaining this network has enormous benefits.

What will the US' ongoing success imply, and what are the key takeaways?

To wrap up, let us briefly consider two concluding questions. First, what does the above analysis mean for investors looking at the US over the long term. And second, what lessons can the rest of the world learn from the success of the US model over time.

For those allocating capital, the case for the US as the world's pre-eminent investment destination remains intact. An investor into the US buys a bundle that is hard to replicate – productivity leadership, the deepest and most liquid capital markets in the world, the rule of law that underpins them, and the flexibility conferred by the dollar's reserve role. That combination should persist. However, the risk premium around the US has risen, as fiscal strains and a wider geopolitical reshuffling point to a wider distribution of outcomes and bigger tail risks.

In practice, this could play out in several ways. First, the dollar's "exorbitant privilege" is unlikely to disappear within the next decade, but gradual diversification of both reserves and payments is real and persistent. Already, the dollar's global importance has ebbed in recent years. These growing question marks around the dollar's status are the first risk. Second, US equity leadership is now heavily concentrated in a handful of AI-exposed names, so the bull case increasingly rests on a small group of companies, again widening the tail risk outcomes. And third, there is a tail risk of a correlated scenario where a fiscal reckoning coincides with a correction in the AI investment cycle, pressuring equities, duration and the dollar simultaneously, and undermining the diversification that normally cushions portfolios. This is a scenario worth planning for, even if it is not the central case.

What can the rest of the world learn from the US' success? In part, it is simply down to luck, including geographic advantages and an abundance of resources, such as energy, that cannot be replicated elsewhere. However, there are several features that could be reproduced elsewhere.

The first is its resilient institutional architecture, with a legal system and property rights that have provided a favourable environment for long-term investment. Although the US has faced severe crises in the last 250 years, including a civil war, its system of government has been recognisably the same throughout that time.

Second, it has a well developed financial system, and a tolerance for risk that's enabled the US to embrace new technologies and emerge quicker from downturns, with resources shifting quickly to new and productive sectors. US history demonstrates how suppressing volatility and preventing resource re-allocation can be counterproductive over the long term.

Third, a key component of US success has been its adaptability to different contexts, and its ability to leverage existing strengths when under pressure. These range across its energy resources, capital markets, universities, military power, reserve currency status, and technological leadership. Even when one point has appeared under threat, others have been able to compensate.

Appendix 1

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Remembering Alan Greenspan

June 22, 2026

Authors

Peter Hooper
Vice-Chair of Research

First Impressions

Alan Greenspan, who passed away today at the age of 100, was one of the most consequential figures in the history of American economic policy—and someone I had the privilege of working with for more than two decades, first as a colleague at the Federal Reserve and later as a friend and adviser at Deutsche Bank. His passion for economic analysis, his respect for data, and his willingness to challenge conventional wisdom profoundly influenced my own approach to economics and policy.

I joined the Federal Reserve staff in September 1973 as a newly minted PhD in Economics from the University of Michigan—four years after my new Fed colleague Don Kohn—and served through the chairmanships of Arthur Burns, G. William Miller, and Paul Volcker. When Greenspan arrived in August 1987, I was in mid-level management in the Board's International Finance economic research staff. I still remember the day he arrived. He toured the building with my colleague Ted Truman, meeting many members of the staff, and I had the opportunity to shake his hand and welcome him personally. It was a small gesture, but a characteristic one: Greenspan appreciated the value of his staff and moved quickly to engage them.

A Remarkable User of Expertise

That engagement was immediately substantive. Greenspan had ambitious ideas about how to use the Fed's deep reservoir of talent. He was fascinated by data—not only the headline numbers but obscure series and unusual relationships that others overlooked.

Much of my early Fed career predated the Internet. Yet the institution possessed an extraordinary asset: specialists in virtually every corner of the economy. One could simply pick up the phone and speak with an expert on housing, productivity, corporate profits, or almost any topic imaginable. It was, in effect, a pre-Internet web of distributed expertise. Greenspan understood this instinctively and made exceptional use of it.

Don Kohn, in his remembrance published today at Brookings, recalled that Greenspan arrived "like a kid in a candy store," eager to tap the knowledge embedded throughout the organization. That description captured him perfectly.

Models, Markets, and Instinct

Greenspan brought macroeconomic instincts somewhat different from those of most Fed staff. I had been trained in the neo-Keynesian tradition, while Alan placed greater faith in markets and the informational efficiency of prices.

Having spent much of my own career modeling trade flows and global economic interactions, I came to appreciate both the power and limitations of formal econometric models. Greenspan shared my strong support for free trade, but he was less inclined than many Fed economists to rely heavily on model-based forecasts. Instead, he searched for revealing relationships in data and drew heavily on information gathered from his extensive network of business leaders and industry contacts.

Over the years I occasionally worked on speeches for him, mostly in an editorial capacity, since he inevitably rewrote drafts into precisely what he wanted to say.

"Irrational Exuberance"

One memorable episode involved the now-famous phrase "irrational exuberance" in a speech prepared for the American Enterprise Institute in December 1996. Several staff members, myself included, were skeptical. Greenspan insisted. History vindicated him.

Years later I asked him the secret of his remarkable longevity and productivity. His answer surprised me: soaking in a hot tub. The same ritual that, as he later recalled, had inspired the concept of "irrational exuberance" sustained him through his tenth decade.

The Productivity Revolution

The 1990s were, in many ways, a magical time for the Federal Reserve. Greenspan earned his reputation as the "Maestro"—the title Bob Woodward gave him in his 2000 book—in large part because of his success in steering the economy through the long expansion of that decade.

By the mid-1990s, many policymakers worried that growth was running beyond the economy's capacity and that inflation pressures were building. Greenspan disagreed. He became convinced that information technology and the computerization of business processes were producing a genuine acceleration in productivity growth, allowing the economy to sustain faster expansion without inflation.

Greenspan's recognition of the productivity surge before most others demonstrated his ability to identify anomalies that did not fit conventional wisdom and pursue them to their logical conclusion. He was right. The economy grew faster, and inflation remained subdued. My good friend Steve Landefeld, former Director of the Bureau of Economic Analysis, well remembers how Alan, one of their most ardent supporters, frequently challenged him to expand their measures of sustainable GDP growth.

Communication and Transparency

Greenspan believed that some ambiguity served monetary policy well. He thought the central bank retained greater flexibility when markets focused on incoming economic data rather than endlessly parsing every word uttered by policymakers.

Nevertheless, under his leadership the Fed began its gradual move toward greater transparency, introducing post-meeting statements and the policy bias. Ben Bernanke later accelerated that transition through press conferences, forward guidance, and eventually the dot plot.

Bringing Greenspan to Deutsche Bank

Greenspan was more than a mentor to me; he was an inspiration who demonstrated the importance of rigorous analysis and intellectual curiosity.

He was gracious enough to acknowledge our relationship in his own way. When he inscribed a copy of his memoir, The Age of Turbulence, he referred to me as a mentor—a generous inversion that I treasure, even though I would reverse the description.

After leaving the Fed in 1999 and joining Deutsche Bank, I became convinced that Alan would make a valuable adviser to the firm. With David Folkerts-Landau's blessing, I approached him after his retirement in January 2006, but he wanted first to finish his book. Months later, after learning he had accepted a position with PIMCO, I reminded him of our earlier conversation. He agreed to join Deutsche Bank as a senior adviser for two years, beginning in 2007.

Around the World with the Maestro

Alan saw clients globally for Deutsche Bank, and the attraction was exactly what the firm had hoped when he joined as senior adviser: access to his strategic insight and his extraordinary perspective on markets, monetary policy, and risk. Clients everywhere relished hearing the Maestro's latest thinking—not simply because he had once chaired the Federal Reserve, but because he could translate complex economic developments into practical judgments that sophisticated market participants could use.

Intellectual Honesty

Alan inspired me in many ways, though we had our differences. I was uncomfortable with his early confidence that private institutions could regulate themselves more effectively than government authorities—a view that events eventually proved overly optimistic.

Yet those disagreements never diminished my admiration for him. What mattered most was that Alan continued to interrogate his own views when events challenged them. Testifying before Congress in October 2008, Greenspan acknowledged that he had "made a mistake" in presuming that the self-interest of banks and other institutions would be sufficient to protect their shareholders. That willingness to confront error openly, rather than evade it or rationalize it away, was one of the clearest expressions of his intellectual honesty—and one of the qualities I most respected in him.

His Living Legacy

What living legacy does Greenspan leave at the Federal Reserve?

Part of the answer may be found in Kevin Warsh, who was sworn in as Fed chairman just one month ago. At his White House ceremony, Warsh paid explicit tribute: "Chairman Greenspan was the first to tell me and show me what this role demands."

Warsh's instincts resemble Greenspan's in two important respects. First, just as Greenspan resisted premature tightening in the 1990s because he believed information technology was driving a productivity revolution, Warsh has argued that artificial intelligence could prove a significant disinflationary force. In both cases, the judgment rested on perceiving structural change before the data fully confirmed it.

Second, Greenspan's preference for brevity when discussing the outlook for monetary policy appears to have found an echo in Warsh's early approach. In his first press conference, he reduced the Fed's post-meeting statement from 341 words to 132. As my Deutsche Bank colleague Matthew Luzzetti observed, that move reverses a decades-long trend toward ever greater transparency.

Whether Warsh's views on AI prove as well founded as Greenspan's insights about information technology will be closely watched. Inflation today is stickier, geopolitical risks are greater, and the enormous investment required to build AI infrastructure may itself prove inflationary in the near term.

“We Had Fun”

Don Kohn concluded his remembrance by recalling Greenspan summing up their years together with a simple observation: “We had fun, didn’t we, Don?”

Yes, Alan—we had fun.

And that spirit of curiosity, intellectual excitement, and joy in understanding how the world works enriched economic policymaking in the United States and around the globe. It certainly enriched my own life and career, and strengthened my desire to remain active in professional economics. For that, and for much else, I will always be grateful.

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Brexit 10 Years On: what's worked, what hasn't, what's next?

June 22, 2026

Ten years ago, the UK voted to leave the European Union with a small majority. This marked the biggest self-imposed trade disruption in history (at the time). Much ink has been spilt on understanding the effects of Brexit, and much uncertainty still remains.

In this special report as part of the Deutsche Bank Research Institute, we revisit the Brexit impact 10 years on. We look at what's worked, what hasn't, and what's next?

To be sure, the effects of Brexit are still felt today. The UK economy continues to suffer from tight trade restrictions via non-tariff barriers. And we estimate some meaningful costs to GDP and employment as a result of the UK's departure from the EU. The cost of living has also been modestly higher on the back of Brexit.

It's not all doom and gloom, however. While Brexit may not have been the success some claimed it would be, it hasn't unfolded quite as badly as others thought either. While our Brexit Scorecard shows an economic cost to the UK's exit, our Brexit Balance Sheet brings together things that have been more positive for the UK as well as things which have not, and where we remain unsure.

We conclude on where we think the UK should focus its efforts in the near term while asking the big medium-term question: is there a path back to the EU? No matter how ambitious the UK's reset with the EU may be in the coming years, the UK will need to garner full support from both the electorate and the EU to come to terms on a deal fit for the British economy of the future. The bottom line: while the cost of Brexit cannot be ignored, there may be political limitations on how far the UK can go in re-integrating with the EU.

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Executive Summary

On the 23rd of June 2016, the UK economy and political ecosystem changed forever with the public voting to leave the European Union (EU). Fragmented politics gave way to a fragmented economy with the UK the first country to break from the European Union – marking, at the time, the biggest unilateral trade disruption in history.

The effects of Brexit are still felt today. The UK economy continues to suffer from tight trade restrictions via non-tariff barriers. The changing tide of immigration has also impacted businesses, from agriculture to hospitality and retail. To be sure, the costs of Brexit – though perhaps slightly exaggerated – are real. The economy is smaller as a result of its departure from the EU. The labour market is also smaller and more dislocated. And prices are running higher, given increased cost of EU imports.

Still, the path ahead – though uncertain – is filled with opportunity. The UK's Brexit freedoms have opened a path to better trade deals with other nations. Regulatory autonomy has positioned the UK to stay at the frontier of AI – the next biggest technological revolution. The speed of deregulation has also allowed the UK to benefit from stronger investment growth. And outside the EU, the UK has been able to embark on a more aggressive form of industrialisation.

In this report, we look at how the economy has taken shape a decade since the Brexit referendum. First, we present a scorecard of economic and market measures over the last decade, comparing the UK to its G7 peers (where compatible). We also set out a doppelganger analysis to estimate the running costs of Brexit. Second, we set out a balance sheet approach looking at what's worked and what's not worked. And third, we conclude with the path ahead. Low hanging fruit to improve the UK-EU Trade & Cooperation Agreement (TCA) remains ripe for picking. At the same time, however, big questions around the UK's future with the EU remain unanswered.

Our key conclusions:

- The cost of Brexit, we estimate, is a smaller economy (-4%), lower employment (-2%) and higher prices (+0.7%). While meaningful, these are slightly better readings than the overarching consensus.
- From a market perspective, investors have taken a dim view of Brexit and its consequences. While the currency has been broadly stable for much of the last decade, the disruption to the supply side of the economy has seen the cost of borrowing rise more sharply and seen equities underperform. From a current account standpoint, while income and trade balances have improved, the external economy is more unbalanced between goods and services than ever before.
- The good news is that the options to improve the TCA – without taking any big decisions on deep integration - are meaningful. We estimate that such improvements to the TCA can lift UK GDP by 0.4-0.8%.
- It takes two to tango. No matter how ambitious the UK's reset with the EU may be, the UK will need full support from both the electorate (to avoid any political paralysis) and the EU to come to terms for a better deal.

Introduction: Brexit Impact – 10 Years On

Ten years on from the June 2016 EU referendum, the United Kingdom's economic journey since Brexit remains one of the most contested empirical questions in modern economic history.

Isolating the structural Brexit impact (higher trade friction, reduced labour mobility, regulatory divergence) however, remains fraught with complexity. The overlapping shocks of a global pandemic, a historic energy shock, a global trade war, and a pronounced monetary tightening cycle affected virtually every advanced economy.

In this report we take a multi-step approach to disentangle some of the effects of Brexit and look ahead to what's next.

We start by providing a **brief overview** on how Brexit was delivered, highlighting the two phases that dominated markets and the economy: the initial referendum shock and the constitutional crisis arising as a result of delivering Brexit, and secondly the post-pandemic shock in which the final Trade & Cooperation Agreement (TCA) was both negotiated and implemented.

We then present a **UK scorecard** looking at how the economy fared during the Brexit transition – comparing it to other G7 economies. This includes GDP growth, the labour market, inflation, trade and market performance. The UK – despite Brexit – has outperformed many of its peers in some areas, underscoring its importance on the world stage. Strikingly though, multiple areas and asset classes have still seen material underperformance. Domestic-focused stocks have delivered one-third of the return of peers since the UK's departure from the EU.

As part of our scorecard, we run an empirical estimation exploring the cost of Brexit (as things stand) using a synthetic control method (i.e. the doppelganger method). Here, we cover GDP, employment, and prices (CPI). Under this framework, UK GDP sits nearly 4% smaller. Employment is 2% lower. And prices are 0.7% higher.

The next section presents our **Brexit Balance Sheet**: a non-exhaustive list of what we think has and hasn't worked, and where the jury is still out. To be sure, while the negatives of Brexit are largely well-explored, we also set out some of the positives that position the UK into a more fragmented world.

Turning to the **future**, we estimate the impact on the UK economy of potential deals with the EU. There are plenty of low-hanging fruit to improve the TCA. We set these out in detail. In short, we think there may be something like 0.4-0.8% upside to GDP growth if the UK and EU worked together to reduce some of the non-tariff barriers.

We conclude with some bigger picture thoughts on the longer-term future of the UK's relationship with the EU, and our view on the optimal path forward. To be sure, no matter how ambitious the UK's reset with the EU may be, the UK will need full support from both the electorate and the EU to come to terms on a better deal. Bottom line, while the cost of Brexit cannot be ignored, there may be political limitations on how far the UK can go in re-integrating with the EU.

The Long Goodbye - How The UK Delivered Brexit

The UK's exit from the European Union was anything but swift. While a slim majority of the British public voted to leave the European Union on 23 June 2016, the path to the eventual Brexit Day on 31 January 2020 was a near four-year journey. We start by walking down history lane and reviewing how Brexit was delivered. Indeed, this is important in understanding how Brexit talks induced uncertainty and ultimately impacted the economy and UK financial markets.

Following the referendum decision to leave the EU, the UK was plunged into a state of economic uncertainty and political chaos. Indeed, Brexit was only delivered three prime ministers later in 2020 as part of the Trade and Cooperation Agreement. The end state was a far removed trade deal than many would have anticipated in 2016, with the UK unilaterally engineering the biggest trade disruption in its economic history with a "hard Brexit".

The decade since the Brexit referendum can be divided into two distinct phases – each with their own economic and political characteristics.

PHASE I	PHASE II
2016–2019	2020-2025
Referendum Shock/Constitutional Crisis	Pandemic/TAC/Energy Shock

Phase 1 entailed the referendum shock, including a constitutional crisis.

The immediate aftermath of the referendum result was messy.

The UK saw an immediate constitutional convulsion on how actually to deliver Brexit, raising uncertainty (politically and economically). The resignation of PM Cameron ushered in a new government under PM May, where she set out twelve Brexit principles to guide future negotiations – including establishing the infamous 'red-lines' in securing a deal with the EU.

To shore up support and strengthen her hand, PM Theresa May called a snap general election in 2017, which ended up stripping the Conservatives of their working majority and creating dependency on the Northern Irish Democratic Unionist Party (DUP) via a confidence and supply agreement.

Over the next several quarters, the UK was plunged into a constitutional crisis. This in turn saw renewed political uncertainty, with investment and business decisions coming to a standstill. The core political paradox was that Parliament had a majority against every available Brexit outcome simultaneously: against no-deal, against May's deal, against remaining, and against any second referendum.

Eventually, PM Theresa May called time on her premiership, opening the path towards a new Prime Minister. Boris Johnson's premiership from July 2019 brought resolution through confrontation. PM Johnson prorogued Parliament (declared unlawful), threatened a no-deal Brexit, expelled 21 Tory MPs who voted against him, and ultimately secured a renegotiated Withdrawal Agreement by solving the

Irish border through a customs border in the Irish Sea. To then obtain a parliamentary majority for his deal, PM Johnson called a general election for December 2019, in which the Conservatives won a large majority.

The UK formally left the EU under the Withdrawal agreement at the end of January 2020, but was to stay in the single market and customs union during a one-year transition period in which a full trade deal was to be negotiated.

Phase 2 was underscored by a global pandemic – with the final rush to secure a trade deal with the EU.

Despite the pandemic, trade negotiations between the UK and EU continued, but under extraordinary conditions. With the Covid-19 pandemic consuming essentially 100% of legislative bandwidth from March 2020 to mid-2021, the UK-EU Trade and Cooperation Agreement (TCA) — the most important trade negotiation in the country's post-war history — was conducted entirely via video conference.

After years of negotiation and deliberation, the culmination of the Brexit journey – at least as things stand – was the TCA being agreed on Christmas Eve 2020 – 1,645 days after the UK voted to leave the EU. For the avoidance of doubt, the TCA represented a marked, "hard Brexit," departure from the UK's previous relationship with the EU. Figure 1 highlights all the key dates on this multi-year journey.

Figure 1:Key events in the run up to Brexit

Date	Key Political Event
	Phase 1: The Referendum Shock
23-Jun-16	**Referendum: 51.9% Leave, 48.1% Remain. Turnout 72.2%.**
24-Jun-16	David Cameron announces resignation
13-Jul-16	Theresa May becomes Prime Minister.
17-Jan-17	**Lancaster House speech: May sets 12 Brexit principles.**
29-Mar-17	Article 50 formally triggered. Two-year countdown begins.
08-Jun-17	**Snap general election. Conservatives lose majority: 318 seats (–13).**
6–9 Jul 2018	**Chequers Agreement. Cabinet 'white paper' Brexit plan agreed.**
14-Nov-18	Draft Withdrawal Agreement published.
10-Apr-19	Article 50 extended to 31 October 2019. UK must hold European Parliament elections.
24-May-19	May announces resignation. Cites failure to deliver Brexit.
	Conservative leadership contest: 10 candidates. Boris Johnson wins.
24-Jul-19	**Boris Johnson becomes PM.**
17-Oct-19	Revised Withdrawal Agreement agreed with EU. Irish Sea border accepted.
12-Dec-19	**General election. Conservatives win 365 seats: 80-seat majority.**
	Phase 2: The Deal And Pandemic Collision
31-Jan-20	UK formally leaves the European Union at 11:00pm. Transition period begins (to year end).
24-Dec-20	**Trade and Cooperation Agreement (TCA) agreed. Christmas Eve deal.**
01-Jan-21	TCA in force. UK customs border operational for first time.

Source : Deutsche Bank, News Sources

The UK Scorecard: economic performance relative to the G7

How did the UK fare relative to its counterparts in the G7 over the past decade? In this section, we build on the UK's economic performance since its decision to leave the European Union via a UK scorecard. We group this into three time periods.

The first (2010-2015) is the pre-referendum period – which in itself encapsulated its own economic issues as the UK dealt with the aftermath of the global financial crisis (GFC) and the government's austerity.

The second period is the referendum shock and the constitutional crisis (2016-2019) as the UK's Withdrawal Agreement with the EU was negotiated amidst domestic political turbulence.

And the third period looks at the pandemic period and beyond (2020-2025), where the UK and EU negotiated and implemented its TCA during a global pandemic, and shortly after which saw the Russian invasion of Ukraine and energy crisis of 2022.

We briefly set out some headline statistics as part of an economic scorecard looking at GDP growth, business investment, trade, employment and inflation (Figure 2).

Figure 2: The Broad UK Economic Scorecard

UK Economic Scorecard			
	2010-2015	2016-2019	2020-2025
Real GDP Growth	1.9%	2.0%	1.1%
% annual growth	(4)	(3)	(4)
Gross Fixed Capital Formation	3.8%	2.9%	2.3%
% annual growth	(2)	(7)	(2)
Current Account Balance, % of GDP	-3.6%	-3.7%	-2.5%
	(7)	(7)	(6)
Employment Rate	71.6%	75.2%	75.0%
	(3)	(2)	(3)
CPI Inflation	2.4%	1.9%	4.3%
% annual rate	(1)	(2)	(1)
*G7 Rank In Parantheses (1= highest, 7=lowest)			

Source : Deutsche Bank, IMF, OECD

GDP

On the surface, UK real GDP has expanded sizably since the referendum – especially in 2016 and 2017. During the early post-referendum years, UK GDP annual growth averaged 2% – above the G7 average and behind only the USA and Canada. From 2020 onwards, UK GDP growth slipped meaningfully to 1.1% – converging to the G7 average (Figure 3).

What drove some of the upward momentum in GDP growth? Part of the reason for the outperformance could be linked to easier monetary policy. Indeed, the Bank of England did its best to support economic activity by unveiling a new package of liquidity measures immediately following the referendum result, cutting Bank Rate to 0.25% in August 2016, and pushing through a new quantitative easing package comprising GBP 60bn of government bond purchases and GBP 10bn of corporate bond purchases.

Stockpiling also helped support activity. Businesses began stockpiling inventories ahead of the referendum with Q4-15 seeing the largest single quarterly inventories on record. And then between Q4-2018 and Q1-2019, inventories shot up again by a cumulative GBP 11.1bn, supporting activity as the first Brexit deadline drew closer.

Real GDP *per capita*, however, tells a slightly different story. Much of the jump in growth – unsurprisingly – was concentrated rather than distributed across the population. While real GDP per capita annual growth sat slightly above its G7 comparators for much of the decade preceding the pandemic and Brexit deal, the UK then fell well behind its peers from 2020, averaging 0.3% compared to a G7 average of 0.7%. Part of this was a reflection of a deeper pandemic drop, while it may have been the case that Brexit slowed the UK's catchup with its peers.

To be sure, the GDP gap cannot be solely attributed to Brexit and its uncertainty effect. Post 2010, the UK was suffering from a cocktail of cyclical and structural issues, including a chronically low savings rate, the hangover of the GFC, weak manufacturing productivity, and regional economic concentration – all of which may have constrained long-run growth.

Figure 3: UK GDP Scorecard

	GDP, Average Annual Growth Rate		
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
United Kingdom	1.9%	2.0%	1.1%
G7 avg (ex-UK)	***1.6%***	***1.6%***	***1.1%***
United States	2.4%	2.5%	2.4%
France	1.2%	1.7%	0.9%
Canada	2.3%	2.2%	1.9%
Japan	1.7%	0.7%	0.4%
Germany	2.1%	1.8%	0.1%
Italy	-0.3%	1.0%	1.2%

Source : Deutsche Bank, IMF; bars represent relative strength for each country across the three periods.

Business Investment

If there was a single macroeconomic variable that most reflected the Brexit effect — separated from pandemic noise, global inflation, and cyclical forces — it was business investment.

Why was investment impacted even before Brexit happened? Despite no change in trading relations between the UK and EU, political uncertainty was starting to have far reaching implications on economic uncertainty – with businesses uncertain around the future trading outlook. Indeed, it became difficult for firms to plan supply chains, staffing, and/or regulatory compliance.

Investment decisions, in the end, were delayed or shelved entirely with firms grappling with uncertainty around future trade arrangements, regulatory environments, and labour market conditions.

The impact of deferred investment cannot be overstated, as its effects compound over time, impacting growth and living standards. Put simply, foregone investment meant foregone productive capacity, foregone technology adoption, and foregone productivity growth.

The investment paralysis was deep and broad. UK business investment fell in four of the eight quarters between Q1-2018 and the eve of the pandemic, Q4-2019. This was the largest sustained investment shortfall in the post-war UK record outside of the 2008-09 financial crisis. As our scorecard highlights, gross fixed capital formation slowed dramatically since the Brexit referendum – moving from a 3.8% annualised pace to 2.9% per annum between 2016 and 2019 and slowing further in the 2020s (2.3% per annum) (Figure 4). As a share of GDP, UK investment relative to its G7 peers continued to underperform. And on the World Bank's own scoring measure of 'Government Effectiveness', the UK went from 'above average' to 'below average'.

What impacted investment plans? Constant threats of a 'no-deal' Brexit left just-in-time sectors particularly exposed – automotive, pharmaceuticals, food manufacturing, etc. Several auto manufacturers pre-announced production shutdowns timed to avoid a no-deal Brexit scenario. SMMT estimated that the collective cost of these shutdowns amounted to GBP 130m per day of lost production.

In numbers, the OBR estimated that Brexit uncertainty was costing approximately 0.5pp of annual GDP growth, equivalent to c. GBP 10-12bn of foregone output per year. A separate paper by Bloom et al. (2019) estimated that Brexit anticipation itself gradually cut investment by 11% and lowered productivity by 2-5%.

Major corporates deferred plans as a direct consequence of the paralysis: the Honda Swindon closure (February 2019, 3,500 jobs); Dyson HQ moved to Singapore; Goldman Sachs, JP Morgan, HSBC, Barclays, and UBS all moved substantial EU-facing operations to Dublin, Paris, Frankfurt, or Amsterdam. EY's Brexit Tracker (Q4-19) noted that 7k+ financial services positions had formally relocated to EU cities. Productivity growth effectively dropped to 0% with UK total factor productivity moving into negative territory.

It's not all bad news though. As our own scorecard shows, FDI and R&D expenditure has picked up since the referendum. Net FDI inflows jumped from 1.8% of GDP in the pre-Brexit period, rising to 4.1% between 2016 and 2019, before slowing to 2.4% of GDP since 2020. R&D expenditure (as a share of GDP) also improved over the past decade or so, pushing from below the G7 (ex-UK) average to above it since 2016. So while business investment dropped, international investors still sought to maintain exposure to UK markets and its role as a domestic innovator.

Figure 4: UK Investment Scorecard

Investment Scorecard	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Gross Fixed Capital Formation, % Annual Growth	3.8%	2.9%	2.3%
G7 ex UK	*2.5%*	*2.5%*	*1.5%*
Gross Fixed Capital Formation, % of GDP	*16.8%*	*18.8%*	*18.7%*
G7 ex UK	*19.9%*	*20.4%*	*21.3%*
FDI (Net), % of GDP	*1.8%*	*4.1%*	*2.4%*
G7 ex UK	1.5%	1.8%	1.7%
R&D Expenditure, % of GDP	*1.8%*	*2.5%*	*2.8%*
G7 ex UK	2.3%	2.4%	2.6%
Government Effectiveness (std dev from world avg)	*1.53*	*1.54*	*1.26*
G7 ex UK	1.4	1.5	1.5

Source : Deutsche Bank, OECD, IMF; bars represent relative strength for each variable across the three periods

Employment

The UK labour market has witnessed three distinct phases since the GFC. The pre-Brexit period saw the UK jobless rate peak at a multi-decade high of over 8% in late 2011. By the time the EU referendum came around, the jobless rate had receded to around 5%.

Post-referendum, the UK saw a new wave of EU migration – the share of EU-born workers in the labour force jumped from 5.5% to 7.2% (as EU migrants likely sought to bypass any potential future immigration restrictions). But by the time the pandemic rolled around and Brexit was enacted, the share of EU workers then dropped. In fact, since the pandemic began, the average annual rate change in the EU born population in the UK has been -0.2%.

Relative to the G7, the UK maintained a higher employment rate (over the last decade +), with the jobless rate tracking below its peers – a reflection of its more flexible labour market. Unit labour cost growth slowed during the pandemic and widened dramatically during the pandemic/TCA period (Figure 5).

The headline statistics, however, fail to capture the impact of Brexit. The replacement of EU free movement with a points-based immigration system fundamentally altered the supply of workers in sectors that previously relied on low-to-medium skilled EU nationals: hospitality, social care, agriculture, haulage, and food processing. The resulting shortages have had two consequences — wage growth at the bottom of the distribution (a genuine distributional gain for low-paid UK workers) and persistent vacancies in those low-wage sectors.

Empirically, it's likely that in the absence of Brexit, employment levels would have been higher. As investment plans were shelved and/or staff were relocated elsewhere, the case for higher employment should the UK have remained in the EU is strong. Breinlich & Magli in 2024 noted that UK services firms dodged Brexit barriers by changing how they served the EU — increasingly via local affiliate sales (commercial presence) rather than cross-border exports from the UK. This protected firm-level services exports but at the cost of lower domestic employment — jobs and activity grew at the new EU affiliates instead of in the UK.

Figure 5: UK Labour Market Scorecard

Labour Market Scorecard	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Employment Rate (16-64)	71.6%	75.2%	75.0%
G7 ex UK	*67.3%*	*70.0%*	*70.9%*
Unemployment Rate (16+)	*7.2%*	*4.3%*	*4.4%*
G7 ex UK	*7.5%*	*6.1%*	*5.5%*
Vacancy Rate	*1.9%*	*2.5%*	*2.7%*
G7 ex UK	1.7%	2.6%	3.2%
Unit Labour Costs, % Growth	*0.7%*	*2.1%*	*0.5%*
G7 ex UK	0.8%	1.5%	3.1%
EU Nationals, % of UK Labour Force	*5.5%*	*7.2%*	*6.7%*
Non-EU Nationals, % of UK Labour Force	*9.4%*	*10.3%*	*12.7%*

Source : Deutsche Bank, IMF, OECD; bars represent relative strength for each variable across the three periods

Prices

UK inflation since 2016 has been shaped by four overlapping forces: a Brexit-specific component (sterling depreciation pass-through, non-tariff barrier (NTB) cost pass-through, labour market restructuring); a global pandemic component (supply chain disruption, goods demand surge); a global energy component (Russia-Ukraine, gas prices); and a domestic demand component (tight labour market, fiscal support).

Unsurprisingly, UK inflation scored well above its G7 peers through the last several years. The first inflation wave came just after the Brexit referendum, when trade-weighted sterling collapsed by around 7%. The fall in sterling fed directly into import prices with non-energy industrial goods prices jumping over 200bps compared to 2016. Food inflation, over the same period, pushed over 400bps. Breinlich et al. (2017) estimated that the sterling depreciation cost the average household GBP 404 a year by mid-2017.

The gap between the UK and the G7 widened the most after the onset of the pandemic, when the UK officially left the EU (Figure 6). While the aftermath of the pandemic and Russia-Ukraine war led to a big spike in inflationary pressures, the effects of Brexit were hard felt too. Structural gaps in the labour market – owing to a re-jigging of EU labour – added to wage pressures. And most directly, the UK's exposure to goods/food inflation was laid bare by rises in non-tariff costs. A study from the LSE's CEP identified a 6% cumulative increase in food import prices attributable to sanitary and phytosanitary checks and customs friction, which they estimate contributed approximately 0.3-0.5pp to headline CPI on a sustained basis. Reduced import competition may have also shifted prices further. With fewer EU producers competing freely in the UK market, domestic services pricing faced less competitive constraint.

Figure 6:UK Prices Scorecard

Inflation Scorecard			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Headline CPI, YoY	2.4%	1.9%	4.3%
G7 ex UK	*1.4%*	*1.3%*	*3.2%*
Unit Labour Costs, YoY	*2.2%*	*1.9%*	*3.9%*
G7 ex UK	*1.2%*	*1.2%*	*2.6%*
Import Prices, YoY	*0.0%*	*3.5%*	*3.2%*
G7 ex UK	2.6%	1.3%	5.1%

Source : Deutsche Bank, OECD, IMF; bars represent relative strength for each variable across the three periods

Trade and external accounts

Brexit was perhaps first and foremost a trade shock, with the UK breaking from decades of closer integration with its largest trading partner. Ten years on, however, the scorecard on the UK's external *balances* is relatively favourable, which may be surprising to some. Strip out erratics like gold, and the UK's current account balance improved notably. This improvement was seen across both the trade balance and the income balance components. To be sure, the average G7 current account balance also improved over the same time period – but the UK's improvement was particularly welcome given its very weak external position at the time of the referendum (Figure 7).

Figure 7: UK external balance improved in the decade since Brexit

UK trade scorecard (%GDP unless otherwise specified			
	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Current account balance	-3.1%	-3.9%	-2.4%
G7 ex UK	*-0.7%*	*1.6%*	*0.9%*
Trade balance	*-1.0%*	*-1.3%*	*-0.6%*
Of which goods	*-6.2%*	*-6.3%*	*-6.8%*
Of which services	*5.2%*	*5.0%*	*6.2%*
Of which balance with EU	*-2.3%*	*-3.6%*	*-2.7%*
Of which balance with non-EU countries	*1.1%*	*2.1%*	*2.3%*
Income balance	*-2.1%*	*-2.6%*	*-1.9%*
Goods exports growth (cumulative, real terms)	*18%*	*-5%*	*-11%*
G7 ex UK	20%	4%	-4%

Source :Deutsche Bank, Haver Analytics; bars represent relative strength for each variable across the three periods

On the trade balance, improvement came from continued growth in services exports, which more than offset a widening goods balance – something we explore in more detail later on. Geographically, the UK's trade surplus with non-EU countries increased, more than offsetting a slight deterioration in the balance with the EU vs the pre-referendum period. To be sure, the trade scorecard still contains some dark marks. Perhaps most strikingly, growth in UK goods' exports volumes underperformed the rest of the G7 over the past ten years.

Empirically, many studies have set out the effects of Brexit. Freeman et al. in 2022, noted that the UK saw a sharp drop in trade once new barriers set in from January 2021. Many smaller UK firms simply stopped exporting to the EU because they could not absorb the fixed compliance costs (rules of origin, SPS paperwork, etc.). Larger firms coped; the long tail of small exporters did not.

Financial markets

Financial markets are in many ways a real time scorecard of economic performance. The past decade for the UK does not make for pretty reading on this front (Figure 8).

The pound is perhaps the single cleanest, catch-all measure of relative UK financial market performance, memorably falling over 7% (in trade-weighted terms) as the referendum results came in. Since then, over the past decade the pound has gradually recovered in nominal terms. And combined with higher inflation in the UK, in real terms the pound is now actually slightly stronger than right before the vote, albeit still below its average from the few years preceding the referendum.

Perhaps the main takeaway from looking at the currency, however, is its relative stability over the past ten years, with far narrower ranges than have historically been the norm. To be sure, that's consistent with a global pattern of broadly lower currency volatility over the past ten years, but in the UK's case, it is still notable given what at times has felt like a continued stream of headlines and events.

Figure 8: Higher relative borrowing costs, weaker pound, underperforming equities relative to pre-referendum period

Financial market scorecard	Pre-Brexit (2010-2015)	Brexit Ref/Pre-Pandemic (2016-2019)	Pandemic+ (2020-2025)
Govt. borrowing costs: 10y gilt yield average (%)	2.5%	1.0%	2.7%
G7 ex UK	*2.2%*	*1.1%*	*2.0%*
Major equities: FTSE 100 cumulative total return (%)	43%	43%	63%
G7 ex UK	*64%*	*42%*	*116%*
Midcap equities: FTSE 250 cumulative total return (%)	120%	40%	21%
G7 ex UK	102%	37%	67%
Purchasing power of a pound: GBP FX Index average	80.6	74.0	74.2

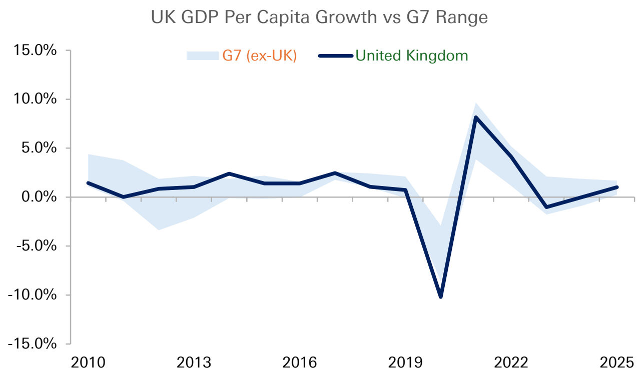
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics; bars represent relative strength for each variable across the three periods

In the fixed income space, the UK government's cost of borrowing has risen sharply in the past few years and more so than in other G7 countries (Figure 18). The decision to leave the EU was far from a primary force at play here, with the dominating factors being the monetary policy responses to the inflation shocks of recent years, looser fiscal policy, and also a more aggressive programme of active quantitative tightening.

Finally, the underperformance of UK equities since the pandemic is noteworthy. The FTSE 100 having a lower weighting on sectors which have outperformed globally, (such as tech), may be one factor at play and is explored in more detail later. Meanwhile, for the more domestically focused FTSE 250, the larger increase in UK borrowing costs vs peers may also be playing a role. In any case, it has provided only one-third of the return vs the G7 average since 2020.

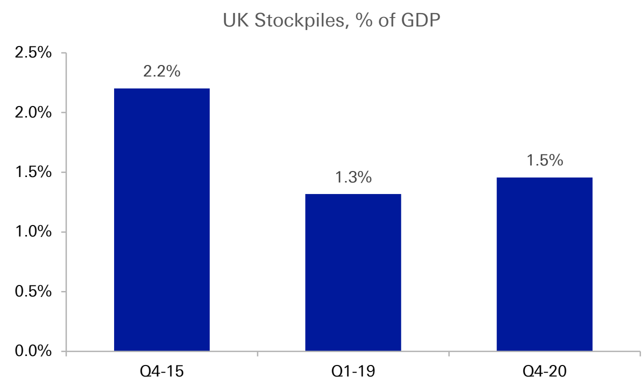
UK Scorecards in Charts

Figure 9: UK GDP per capita falling to bottom of G7 pack since the referendum



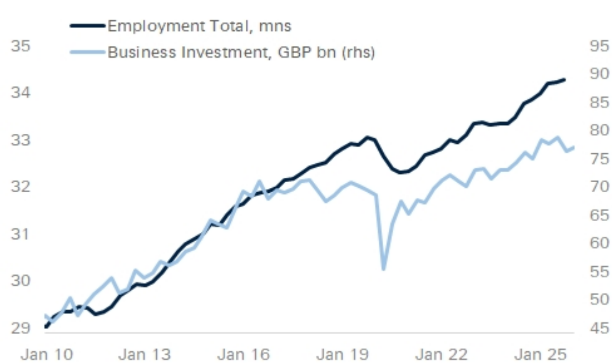
Source : Deutsche Bank, IMF

Figure 10: Stockpiling added materially during period of uncertainty through the pre-Brexit period



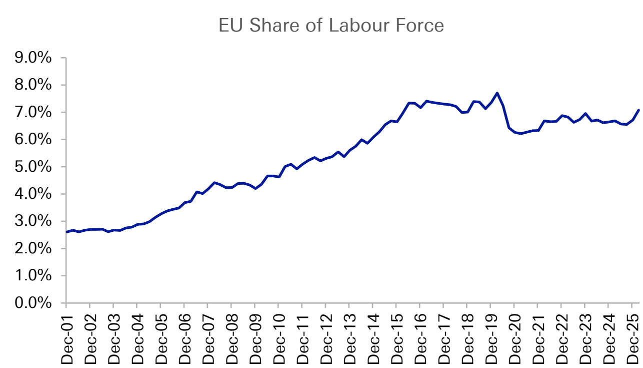
Source : Deutsche Bank, Macrobond

Figure 11: Employment and business investment decoupled as firms substituted labour for capital



Source : Deutsche Bank, Macrobond

Figure 12: Share of EU nationals in the UK saw small spike just after the referendum but has fallen meaningfully since

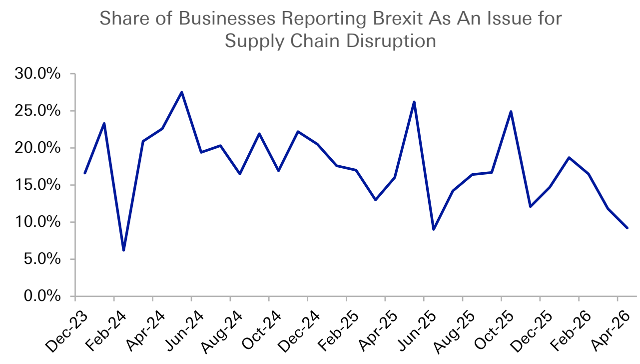


Source : Deutsche Bank, ONS

Figure 13: UK goods exports to the EU have fallen in real terms since the vote

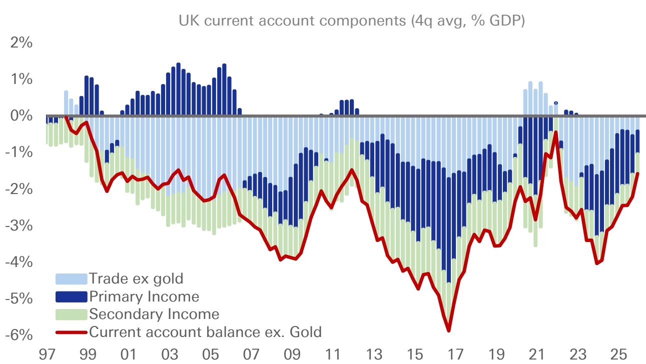
Source : Deutsche Bank, Macrobond

Figure 14: Around 10% of firms still blame Brexit for supply chain disruptions



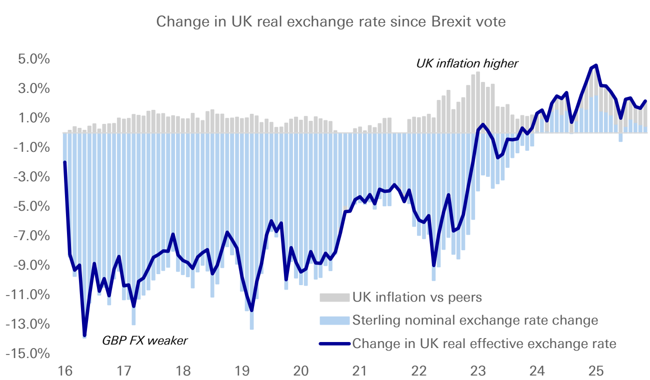
Source : Deutsche Bank, ONS

Figure 15: UK current account deficit improved over the past decade



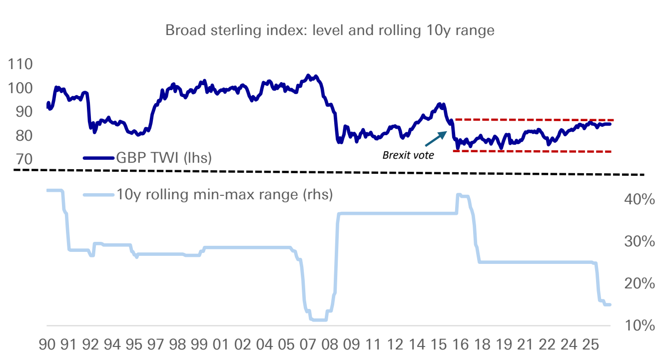
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 16: Pound in real terms is marginally stronger in real terms than before vote….



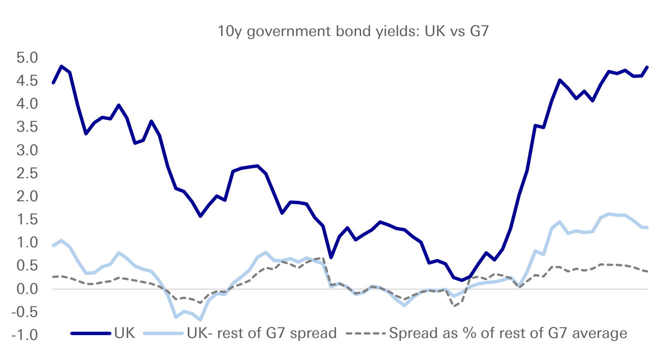
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 17: ..and has been very stable by historical standards since initial Brexit shock



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 18: Cost of UK borrowing firmly higher than peers but hard to isolate Brexit effect



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Marking to market the Brexit impact: a doppelganger analysis

Real GDP	Employment	Prices
-4.0%	-2.0%	0.7%
GBP 1,700 per person	680k jobs	CPI, 4q

Deutsche Bank SCM Analyses

The preceding two sections set out what the UK economy actually produced, invested, traded, employed, and priced over the post-referendum decade. But they cannot, on their own, answer the question that matters most to policy-makers and investors: how much of this was Brexit?

Indeed, several factors have shaped the economic outlook over the past decade. The UK experienced Brexit, a pandemic, an energy crisis, and a sharp monetary tightening cycle essentially simultaneously. Every metric examined above was influenced by all four shocks in different proportions. Disentangling them requires a counterfactual — a credible picture of what the UK economy would have looked like absent Brexit, controlling for the other shocks.

Turning to a Synthetic Control

As mentioned above, the UK – alongside other economies – experienced several shocks at the same time. Resorting to a doppelganger type framework (SCM) has several advantages.

First, it addresses the 'no simple control group' problem. Indeed, we cannot observe a 'UK without Brexit' in principle. Every conventional comparison — UK vs eurozone, UK vs G7 average, UK vs pre-referendum trend — conflates the Brexit effect with differences in economic structure, pandemic policy (furlough generosity, lockdown timing), energy exposure, monetary policy frameworks, and starting conditions. Simple comparisons, therefore, fail to estimate a 'genuine' Brexit impact.

Second, there are overlapping shocks. As noted above, Brexit was not the only problem facing the UK. The pandemic hit in 2020; the TCA came into force in 2021; monetary tightening began in late 2021; and energy prices spiked in 2022. Each of these shocks interacts with the Brexit structural effect — amplifying it in some cases (HGV shortage = Brexit × COVID), and attenuating it in others (pandemic emergency spending temporarily offsetting investment gap).

Third, phase dependency. As highlighted separately, the impact of Brexit can be traced through various phases. What began via consistent political uncertainty, led to economic disruption, which in turn led to the biggest single re-write of trade relations in UK history. Put differently, an aggregate 10-year average conceals the timing and magnitude of Brexit's effect. A good counterfactual method must be able to show when divergence began, how it evolved, and whether it has persisted.

Fourth, the self-selection problem. The UK did not 'randomly' Brexit. The vote was correlated with pre-existing economic characteristics. A counterfactual that ignores this correlation will either overestimate or underestimate the causal Brexit effect.

The synthetic control method (SCM) best solves for these issues. So, what is a SCM?

The SCM constructs a 'Synthetic UK' — a weighted composite of economies that most closely matched the UK on GDP, investment, trade, labour market, and inflation metrics in the pre-referendum period. Post-2016 divergence between the UK and its synthetic twin is then our best available estimate of the Brexit causal effect — not perfect, but transparent, visually intuitive, and methodologically rigorous. The appendix presents more detail on the SCM, including its limitations.

How do we construct the UK synthetic control?

We measure the Brexit effect on UK GDP, employment and prices by comparing the UK to a counterfactual "doppelganger" built from a donor pool of comparable economies. Here, we rely on a donor pool from OECD economies.

For GDP, employment, and prices, we use the standard Synthetic Control Method (SCM): a weighted blend of donor economies chosen to track the UK closely before the 2016 referendum, with the post-2016 divergence as the estimated Brexit effect. The SCM's weights are constrained to be positive and to sum to one, so the counterfactual is always a plausible combination of real economies.

Figure 19: SCM Controls

SCM Controls	Control Variables
GDP	Gross Fixed Capital Formation, Exports, Trade Openness, Government Consumption, Net FDI, Years of Schooling, R&D Investment, Employment, Professional/Science Ratio to GDP
Employment	Consumption, Gross Fixed Capital Formation, Unit Labour Costs, Years of Schooling, Trade Openness, GDP, Professional/Science Ratio to GDP, Net FDI
Prices	Unit Labour Costs, Import Prices, Export Prices, Policy Rates, GDP, Trade Openness, Employment

Source : Deutsche Bank

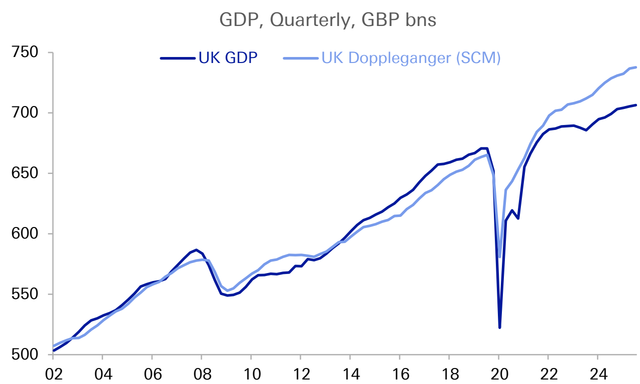
Impact of Brexit: results from a synthetic control

GDP: As expected, the UK economy underperformed relative to our 'UK doppelganger'. Our synthetic control doppelganger showed GDP tracking 4% stronger, relative to the UK's actual output (2025) (Figure 20). In pound terms, we estimate this to be equal to just around GBP 120bn – worth roughly GBP 1,700 per capita (in 2020 terms).

Our estimated output gap sits slightly better than other external estimates of a 6-8% GDP loss due to Brexit. However, note the breakdown in performance. The initial Brexit years, which encapsulated both the Brexit referendum shock and the period of constitutional crisis, saw UK GDP actually outperform its doppelganger – likely a function of early monetary stimulus, competitive exports (via the FX depreciation), higher net immigration, and some stockpiling.

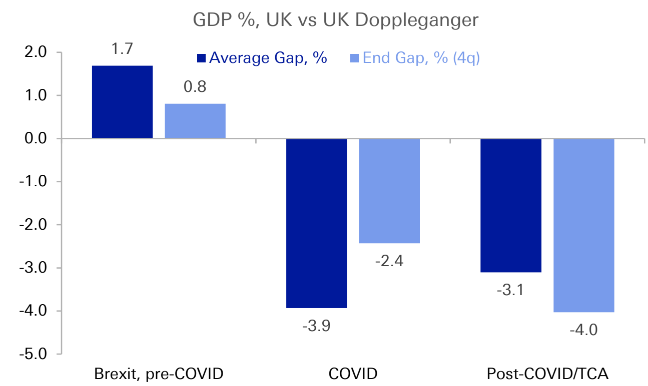
It wasn't until the pandemic began and the UK officially left the EU, that GDP started to materially underperform from its doppelganger. As of the most recent data point available, we estimate the GDP gap to have widened to 4.2% (Q4-25). It's worth noting, however, that on average since the Q2-2016, the GDP gap sits at a smaller -1.3% relative to its doppelganger – underscoring the wider variation in GDP trajectory over the last decade (Figure 21).

Figure 20: UK GDP sits around 4% lower relative to its doppelganger



Source : Deutsche Bank, Macrobond

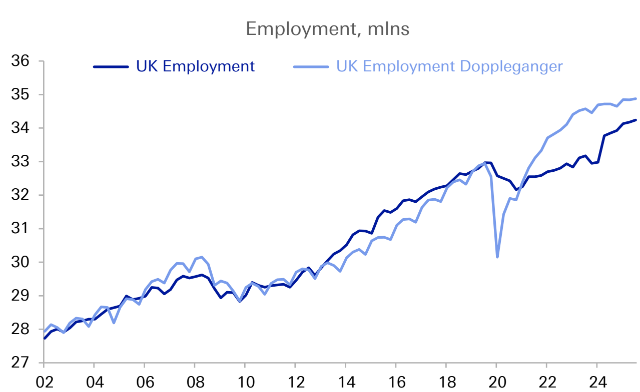
Figure 21: There is a wide variation in performance across the decade, with an output gap emerging only after 2019



Source : Deutsche Bank

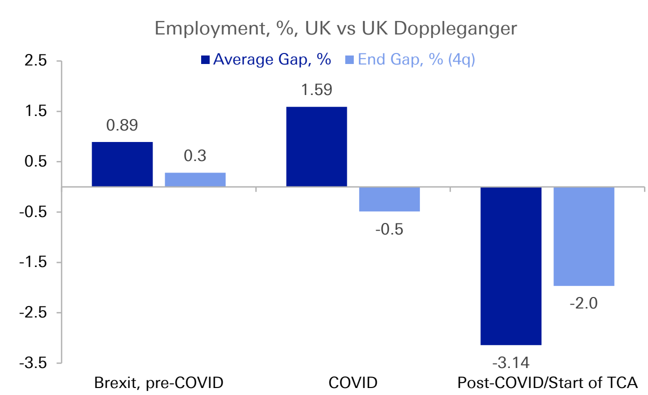
Employment: With GDP lower, employment is also likely to be lower. We estimate that the UK employment gap at the tail end of the sample (2025) sits closer to -2%. In numbers, that works out to just over 685k jobs (Figure 22).

Figure 22: We estimate the UK employment gap to be around 2%



Source : Deutsche Bank, Macrobond

Figure 23: The employment gap only really turned negative after the pandemic and the implementation of the TCA



Source : Deutsche Bank

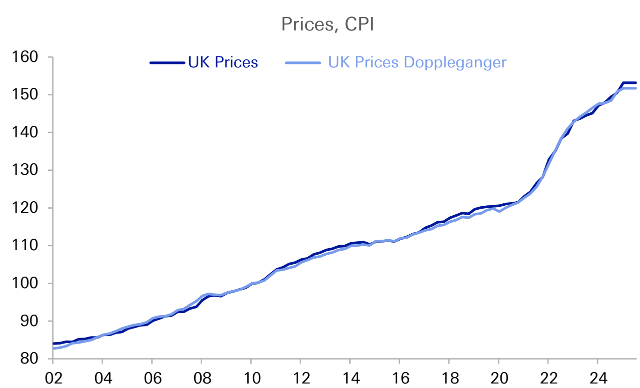
Again, we highlight the differences between the different phases of Brexit. By the end of the pre-pandemic period, employment levels were 0.3% above its estimated UK doppelganger. This, in our minds, chimes with the large inflows of workers seen just after referendum and in the run up to the pandemic where firms substituted away from capital and into cheaper labour, driving up employment. It wasn't until the pandemic period (2020-2021) where UK employment levels fell below the estimated synthetic (-0.5%). And in the TCA period (2022-2025), the gap between the UK and its doppelganger reached its widest levels, with both the average and end point showing a markedly bigger net fall (Figure 23).

Prices: Perhaps the most transparent impact of Brexit was inflation. On our SCM model, we estimate that UK prices (CPI) sits 0.7% above where it would have otherwise have been (i.e. in level terms) (Figure 24).

The end state, however, masks several phases of Brexit. The initial aftermath of the referendum saw sterling plummet, pushing CPI nearly 1% above its doppelganger. By the time the pandemic hit and the UK officially left the EU, prices were only marginally higher (0.4%) compared to the constructed UK synthetic.

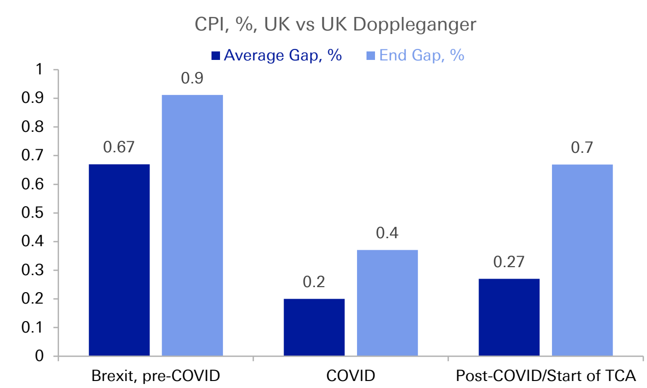
By the latest data point (2025), however, the gap between the UK and its doppelganger widened to around 0.7% (Figure 25). Overall, we estimate that over the last decade, the CPI level has tracked 0.5% higher, on average, due to Brexit forces with trade frictions, an initially weaker sterling, and structural changes to the UK labour market all likely playing some role in driving inflation higher, relative to the counterfactual.

Figure 24: CPI sits around 0.7% higher over the last year, compared to its doppelganger



Source : Deutsche Bank, Macrobond

Figure 25: CPI was stronger across all three time periods following the referendum



Source : Deutsche Bank

The Brexit Balance Sheet: what's worked and what hasn't

From setting out the UK scorecard on some of the effects of Brexit, we turn to the Brexit Balance Sheet, looking at what has worked and what hasn't. Indeed, while Brexit may not have been the success some claimed it would be, some things have worked better – while many others have not. We set them out in more detail below.

What's worked?

1/ Current account deficit has improved. In the trade scorecard we noted how the overall trade deficit had narrowed over the past decade. The income side has also improved, in part thanks to Brexit. Contributions to the EU budget ceased at the end of the transition period in 2020, but the UK's divorce payments - agreed as part of the Withdrawal Agreement - have also now tapered down to a rounding error from a balance of payments perspective. Indeed, net payments to the EU totaled nearly EUR 11bn in 2020 but by last year had fallen to under EUR 1bn. Only EUR 6bn of obligations for the next forty years remains outstanding. As such, the lesser explored secondary income component of the UK current account has materially improved from a decade ago - from averaging 1.3% of GDP in the 5 years before the Brexit vote to just over 0.5% of GDP in 2025 (Figure 27).

2/ Free trade across Irish border preserved. Figure 28 shows how trade between the Republic of Ireland and Northern Ireland jumped up immediately once the TCA came into effect in January 2021. With Northern Ireland de facto remaining part of the EU customs union with a border instead erected across the Irish Sea, goods trade on the island has thrived and is much higher than before the referendum.

3/ Regulatory Autonomy. The UK's freedom to set its own regulatory environment is the most significant theoretical advantage of Brexit. This has allowed the UK to move faster than its European counterparts in several areas, including:

- Financial Services: The Edinburgh Reforms (2022-23) introduced a more outcomes-based, less prescriptive prudential and conduct framework. The removal of the EU bonus cap for bankers (2023) was symbolically significant; more substantively, the UK has maintained its lead in derivatives trading volumes (LCH/LME), with the EU's attempts to force euro-denominated clearing to the continent only partially successful.
- Life Sciences: The MHRA's independence from the EMA has allowed faster clinical trial approvals and novel regulatory frameworks for advanced therapeutics (ATMP). The COVID vaccine programme was the most visible example; the MHRA's Innovative Licensing and Access Pathway (ILAP) has since attracted several early launches ahead of EU approval.
- Artificial Intelligence: The UK has deliberately positioned itself as a lighter-touch regulatory jurisdiction relative to the EU's AI Act. The 2023 AI Safety Summit at Bletchley Park and the establishment of the AI Security Institute have given the UK a soft-power position in global AI governance. This has allowed the UK to take a leading role in Europe when it comes to AI deployment and innovation.

- Green Finance: The UK's post-Brexit Sustainability Disclosure Standards diverge from EU SFDR in ways that some institutional investors consider more workable, potentially attracting more ESG capital.

4/ Independent Trade Policy. Post-Brexit, the UK has signed or rolled over approximately 70+ trade agreements covering economies representing around 67% of UK exports.

New agreements include the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP, formally joined May 2023) and updated deals with Australia and New Zealand (UKFTA 2023). The economic scale of these agreements is, to be direct, modest in aggregate — the CPTPP is estimated to add approximately 0.08% to UK GDP by the late 2030s.

Figure 26: The UK has been busy signing trade deals

Trade Deals				
Trade Agreement	Type	Status	GDP Impact (est)	Key Sectors
EU-UK TCA	Goods FTA / Services partial	In force Jan 2021	Net negative vs EU membership	All sectors
CPTPP	Regional trade agreement	In force May 2023	+0.08% GDP (long run)	Agri, autos, services
UK-Australia FTA	Bilateral FTA	In force May 2023	+0.08% GDP (long run)	Agriculture, finance
UK-New Zealand FTA	Bilateral FTA	In force Feb 2024	+0.03% GDP (long run)	Agri, digital
UK-India FTA	Bilateral FTA	Negotiations ongoing	+0.2% GDP (est.)	IT services, pharma
UK-US FTA	Bilateral FTA	Stalled / under review	Contested	Food, pharma, finance

Source : Deutsche Bank, UK Government

5/ Subsidy Flexibility. Outside EU state aid rules, the UK has been able to deploy targeted industrial subsidies at scale. The Freeport programme (10 English freeports operational by 2025) provides tax incentives and customs flexibility that replicate some single market benefits in a localised way. Net investment linked to freeport locations has been estimated at several billion over the programme period. Additionally, the UK's Semiconductor Strategy and advanced manufacturing support programmes benefit from freedom to exceed EU de-minimis thresholds.

What's not worked?

1/ UK trade is more unbalanced between goods and services than ever before. Increased frictions with the EU have decreased overall levels of trade in goods between the two blocs. From a deficit perspective, the trade balance in goods has widened, most notably in manufactured goods and food and drink. On the flip side, UK services trade continues to boom, especially telecoms and IT as well as "other business services" which captures things like law, management consultancy, and accountancy. Geographically, the UK's services surpluses with both the US and the EU have increased materially in recent years (Figure 29). At a high level, since leaving the EU, the UK has leaned further into its comparative advantage on services rather than rebalancing towards manufacturing and goods trade (Figure 30). Indeed, the UK is now far more unbalanced than countries of a similar wealth (Figure 31).

2/ Non-tariff barriers and trade friction. The TCA delivered zero tariffs and zero quotas – but raised substantial costs via non-tariff barriers (NTBs) on UK-EU trade. The TCA eliminated tariffs on goods meeting rules of origin requirements, but the reintroduction of customs declarations, regulatory border checks, and rules of origin compliance has imposed significant costs on UK-EU goods trade. Academic work from the LSE CEP, CEPR, and the UK Trade Policy Observatory saw a 15-25% reduction in UK goods trade intensity with the EU attributable to these NTBs, after controlling for the pandemic and other factors.

A recent paper by the Productivity Institute suggested that Brexit led to a drop in imports of 23.7% and exports by 18.6%, with business investment also suffering by 2.5% (as a result of the drop-off in trade intensity). A separate study showed that non-tariff barriers raised prices by 6% (implying a pass-through of 50-80%) with households losing nearly GBP 5.8bn in added welfare costs.

3/ Regional policy. EU regional policy was a major, long-running transfer to the UK's poorer regions: between 1975 and 2020 the UK received around GBP 66 billion from Brussels – historically supporting west Wales, Cornwall, the Scottish Highlands, Merseyside and south Yorkshire. Had the UK stayed in, external analysis suggests it would have been entitled to around EUR 13 billion of structural funds for 2021–27 — a 22% increase on the prior period, as worsening regional inequality pushed more areas into the "less developed" category. The replacement — the UK Shared Prosperity Fund — has been widely judged to fall short: the Welsh Government estimates it alone faces a shortfall of around GBP 772 million against what EU funds would have delivered over 2021–25.

To add to this, empirical work show that UK regions underperformed their counterparts elsewhere. Fetzer & Wang (2020) showed, via local synthetic controls (local doppelgangers) that in 2016–2019 many areas grew more slowly than their no-Brexit counterfactual. Interestingly, they also found that costs tended to be higher in areas that voted to leave the EU, had larger manufacturing sectors and more low-skilled workers — i.e. the places promised the most gains bore disproportionate early costs.

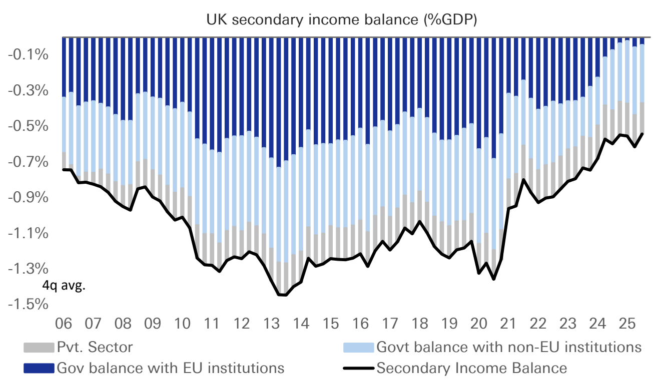
The grey zone: neither worked nor didn't work

1/ Some equity sectors have continued to outperform, some have continued to lag In the market's scorecard we highlighted how in aggregate both UK major and midcap equity indices have underperformed peers, most notably over the past few years. However, the sectoral breakdown presents a more nuanced picture, with a different story comparing UK sectors to their European peers and to their US equivalents (Figure 32). Interestingly, against European peers, some sectors that outperformed during the uncertainty period have largely continued this trend since 2020 (most notably materials), though others such as financials have lagged. Perhaps most striking of all, however, is the underperformance of technology shares (despite strong contributions to UK GDP growth), in both of our two Brexit periods and relative to both European and US equivalents.

2/ Defence and security. For most of the post-Brexit period defence barely registered on the EU–UK agenda, with NATO and bilateral frameworks (such as the 2010 Lancaster House treaty with France) serving as the default. The geopolitics of 2025 changed that abruptly: the May 2025 EU–UK summit produced the first-ever Security and Defence Partnership, establishing structured dialogue across maritime, space, cyber and hybrid-threat domains. In November 2025, talks on UK participation in the EU's EUR 150 billion "Security Action for Europe" (SAFE) defence-procurement programme broke down, exposing persistent post-Brexit friction even as third countries like Canada moved to associate. The UK's eventual role as a defence leader in Europe remains in question, and arguably has left the UK more exposed in a post-Brexit world.

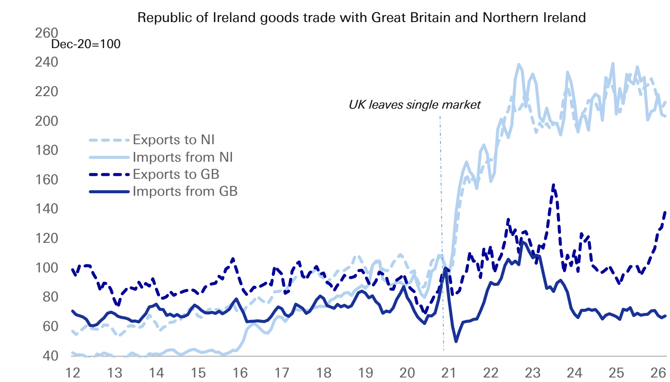
3/ Immigration. Ending free movement with the EU was meant to cut migration; instead, overall net migration rose to a peak of over 900,000 in 2023 before falling back to 431,000 in 2024. The CER–Portes counterfactual found that by 2024, Brexit had reduced EU-born workers by about 2.3% of the workforce, offset by a slightly larger rise in non-EU workers of around 2.95%, for a net increase of roughly 207,000 foreign-born workers — about 0.6% of the total workforce.

Figure 27: UK saves c. 0.6% of GDP on payments to EU with divorce payments having now tapered down



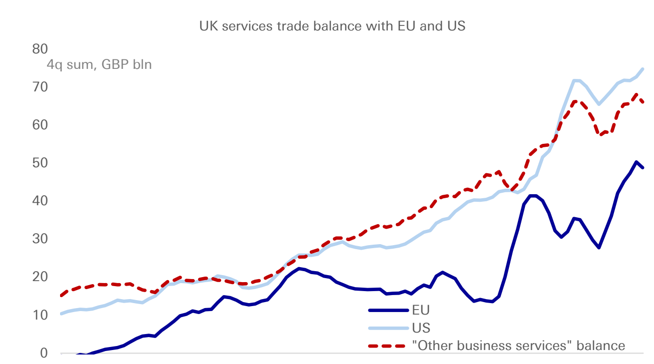
Source : Deutsche Bank, Haver Analytics

Figure 28: Brexit deals have successfully preserved free movement of goods across the Irish border



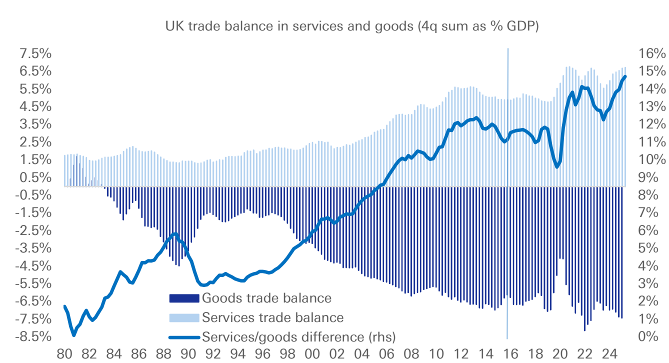
Source : Deutsche Bank, Haver Analytics

Figure 29: UK services surplus has grown with both the EU and the US....



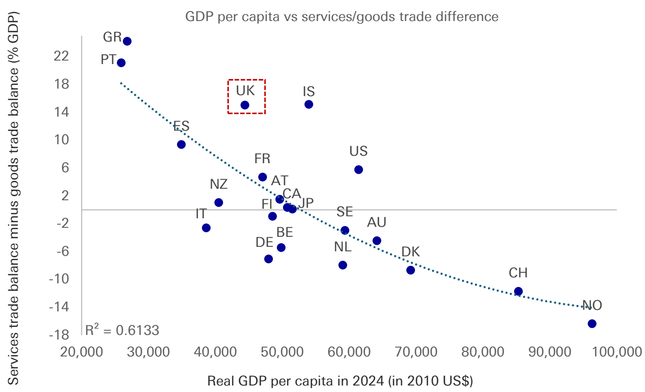
Source : Deutsche Bank, Haver Analytics

Figure 30: ..but overall UK trade more unbalanced than ever between service surplus and goods deficit



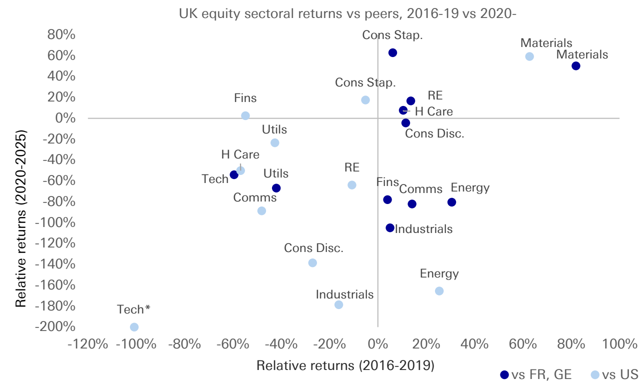
Source : Deutsche Bank, Haver Analytics

Figure 31: UK far more unbalanced between goods and services trade than nations of comparable wealth



Source : Deutsche Bank, OECD, Haver Analytics

Figure 32: Sectoral breakdown shows a mixed picture vs Europe and the US over the past decade



Source : Deutsche Bank, Bloomberg Finance LP
**tech vs US truncated for chart visibility*

The Look Ahead: What's Next?

The impact of Brexit – though uncertain in scale – remains clear: the UK has lost output, employment, and has seen a larger cost of living shock.

So, what next? In this section, we look at the potential paths ahead for the UK. First, we look at how the UK can build on the current TCA deal. Second, we ask the big medium-term question: is there a path back for the UK into the EU? And, if so, what form could this take? Finally, we end with some thoughts on what (and how) the UK should consider as it rethinks its relationship with its largest trading neighbour.

The near-term questions: how can the UK improve on the TCA?

While policy debates have centered around the bigger (perhaps politically challenging) questions, there's more the UK can do to improve on the current TCA.

Recent policy proposals have focused on reducing trade and non-trade barriers between the UK and EU. Progress on these could have a measurable impact on UK GDP. Such 'Brexit reset' proposals can be summarised below:

Figure 33:Brexit reset proposals

Improving the TCA		
Policy proposal	**Description**	**Main barrier reduced**
Sanitary and Phytosanitary agreement (SPS)	Simplifies border checks and provides for dynamic alignment of food safety and animal welfare standards with the EU.	Agricultural and food product trade barriers
Visa and transport barriers for touring artists	Introduces targeted visa waivers for performers and technical crews and reduces transport frictions for longer tours.	Cross-border mobility
Mutual recognition of professional qualifications	Supports recognition of regulated professional qualifications, including in sectors such as nursing and legal services.	Labour mobility
Youth Mobility Scheme	Allows UK and EU 18–30-year-olds to move for work, study or other purposes for up to four years, with proposals also covering domestic tuition-fee treatment.	Labour, education and mobility barriers
Emissions Trading Scheme linkage	Links the UK and EU carbon markets, reducing carbon-price-related frictions and potential CBAM-related costs for covered trade.	Trade frictions
Product safety and metrology	Aligns UK product-safety and metrology rules with EU requirements, reducing duplicate compliance checks and easing market access for traded goods.	Regulatory barriers

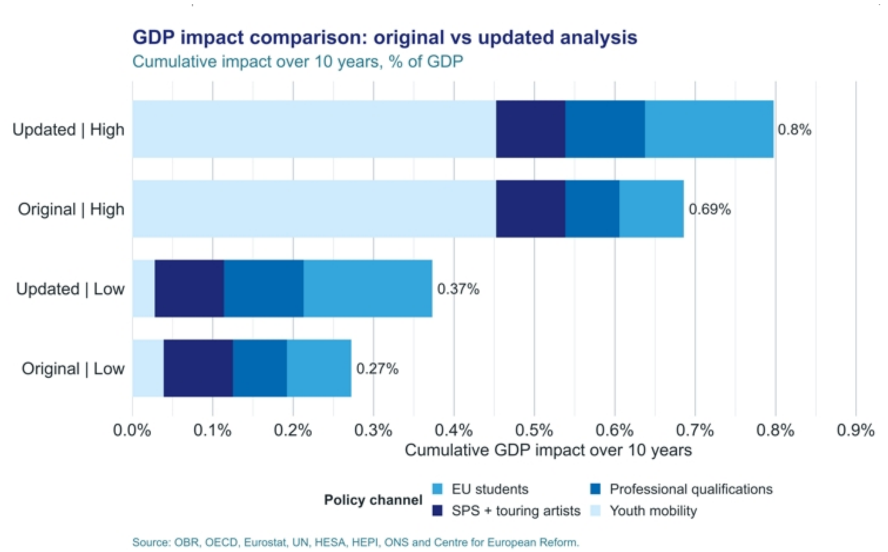
Source : Deutsche Bank

We quantify the GDP impact of the first four policy proposals in the table. Following Springford (2024), we update the underlying data and present a range of estimates for their effect on UK GDP. These calculations are approximate and draw on available OBR estimates and relevant academic literature. The estimates measure the cumulative impact on UK GDP over a 10-year horizon, relative to a counterfactual in which the policies are not implemented.

Overall, we estimate a cumulative GDP impact of +0.4–0.8% over 10 years. Our updated estimates are slightly higher at the top of the range, mainly because of the larger assumed effect from mutual recognition of professional qualifications. If the UK were treated as re-entering the relevant EU labour market, UK workers would account for around 15% of regulated professionals. Combined with higher assumed inflows of EU regulated professionals than outflows of UK regulated workers, this produces a somewhat stronger effect than in the original estimates. The low and high estimates vary primarily with assumptions about take-up of the Youth Mobility Scheme (see the appendix for more detail).

The chart below compares Springford's original estimates with our updated estimates:

Figure 34: Estimated impact of a Brexit reset



Source : Deutsche Bank, Springford (2024)

Because the Youth Mobility Scheme is an important source of uncertainty, we test two alternative assumptions: first, we assume a diminishing annual growth rate providing a more realistic projection of scheme take-up; and second, we vary the productivity assumption using OBR scenarios in which additional migration has a positive, negative or neutral effect on potential GDP. The effects of GDP are broadly negligible here, widening the overall impact to 0.33-0.84%.

Overall, the results suggest that the GDP effects are material, although the exercise does not capture wider general-equilibrium effects. The estimates should therefore be interpreted as a partial assessment of the potential impact of Brexit reset policies.

Finally, these estimates *exclude* the potential effects of UK-EU ETS linkage and product safety and metrology proposals. The table below summarises the main channels through which these policies could affect the UK economy:

Figure 35: UK-EU ETS and product safety changes can also boost GDP materially

Other policies to consider		
Policy proposal	**Channels**	**Impact on GDP**
UK-EU ETS Linkage	**1.** Lower UK carbon prices than otherwise, reflecting increased liquidity and access to the EU ETS market. **2.** Lower costs for UK exporters by removing the need to pay a carbon price differential.	**1.** Lower intermediate input costs due to lower carbon credit prices, with potential pass-through to consumer prices and GDP. **2.** Support for export competitiveness in affected sectors, raising export receipts and GDP growth.
Product safety and metrology	Reduces duplicate compliance and conformity-assessment costs for firms trading goods with the EU.	Lower compliance costs could improve exporter competitiveness and support GDP growth.

Source : Deutsche Bank

Including these additional proposals would likely raise the estimated GDP impact, as both ETS linkage and product safety and metrology could reduce costs for firms and support trade competitiveness.

Overall, the estimates point to a positive but partial view of the GDP impact from the Brexit reset proposals. However, they should provide some indication of the overall effects of adopting each of these proposals.

The big medium-term questions: is there a path back?

Beyond the very near term, the bigger question remains: is there a path towards closer alignment with the EU? In this section, we set out some of the thinking around the UK's options. Here, there are two: one, building on the TCA and pushing towards either a customs union or single market; or two, rejoining the EU.

Building on the TCA: customs union or single market?

The previous section's balance sheet pointed to a single asymmetry: Brexit's costs are concentrated at the core — via re-erected trade and regulatory friction with the UK's largest market — while its gains sit at the margin.

Any forward path therefore is based on how much of that core friction can be removed, and at what price in sovereignty. The two structural options on the table are a customs union and single-market membership (or a close emulation of it). They are not the same thing, they remove different frictions, and the empirical literature attaches very different magnitudes to each.

Customs Union: A customs union would align the UK's external tariff with the EU's and remove the customs border between them, which does two concrete things.

First, it eliminates rules-of-origin compliance. This is the requirement to prove what share of a product originated in the UK or EU to qualify for zero tariffs under the current TCA. Such compliance costs are estimated to add 2–8% to the cost of exporting goods, the burden falling heaviest on heavy industry and integrated manufacturing supply chains. The friction is real and quantified: in 2021, the TCA reduced the count of product-destination export relationships with EU countries per quarter by around 30%.

Second, a customs union would restore frictionless participation in European supply chains, where components cross the Channel multiple times before final assembly — automotive and chemicals being the textbook cases. Under a customs union those repeated crossings no longer trigger origin checks. A UK–EU mutual-recognition agreement on conformity assessment — a natural addition to a customs union — has been estimated to raise UK goods exports to the EU by 10% on average, and by up to 25% in chemicals and machinery.

What's the biggest limitation of a customs union? A customs union is a goods-only instrument. It does nothing for services. It also removes the UK's freedom to strike independent goods-trade deals, since the common external tariff would be set in Brussels. And it does not, by itself, remove SPS (food-safety) border checks, which require a separate veterinary agreement (something that would need to be agreed separately).

Single Market: The single market is a regulatory union: product and service regulations are harmonised or mutually recognised, so that non-tariff barriers between members fall to virtually zero. Re-entry would remove the bulk of the friction Brexit created, and crucially it would do so for services — the UK's structural strength.

Beyond the direct trade channel, single-market membership delivers a productivity dividend that a customs union cannot. Deeper market integration intensifies competition. OBR estimates suggest that the biggest cost from Brexit in the long run can be attributed to a productivity channel — a 4% long-run productivity loss. Re-entry would also restore the labour-supply flexibility, investment certainty and FDI attractiveness that the uncertainty channel of Brexit eroded.

What would a single market entry cost? The single market's price is sovereignty. Membership requires accepting the four freedoms — including freedom of movement — and submitting to EU regulation designed without a UK seat at the table. It also implies a budget contribution. These are precisely the red lines that successive governments have ruled out, which is why the single market is the larger economic prize but the smaller political probability.

To rejoin the EU or not to?

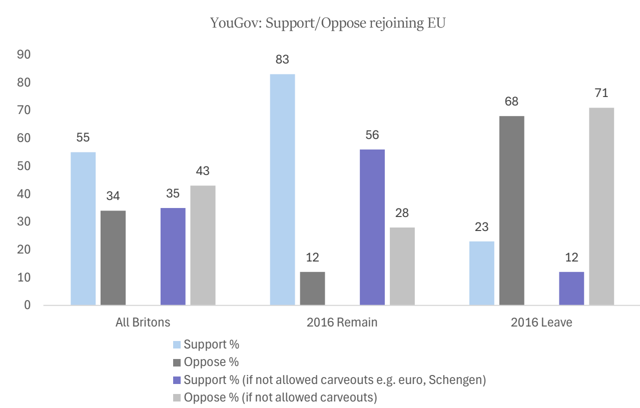
A key consideration when assessing the UK's longer-term relationship with the EU is public support, and what the UK public thinks about Brexit and its future.

This piece by polling expert John Curtice nicely summarises the state of play. When asked how they would vote in a hypothetical second referendum, polls on average this year have shown around 60% support for being in the EU, vs 40% for staying out. Brexit is seen by a majority of voters to have had a negative impact on both the economy and immigration. Interestingly, given the large scale of non-EU net migration into the EU over the past few years noted above, the view that it has had a negative impact on the view towards immigration is equally held by those who voted Remain as those who voted Leave in 2016.

To be sure, when given more context about the potential preconditions for rejoining the EU, the picture amongst voters becomes more split. For instance, if rejoining the EU had to be accompanied by the UK adopting the euro, polling suggests UK public support drops materially per YouGov, to 35% (Figure 36). On the economic side, UK fiscal picture would have to improve to meet the convergence criteria for the euro, both in terms of running a narrower fiscal deficit and lowering (or illustrating credible path towards lowering) debt/GDP (Figure 37).

Here, the wider continental context is important. It is not inconceivable that the next decade or two sees a variety of countries across northern and eastern Europe vote on either joining the EU (e.g. Norway, Iceland) or on adopting the euro (e.g. Sweden, Hungary, Czechia). By the time that the UK were to hold any hypothetical vote on closer ties, the bar for the EU to allow the UK to rejoin on its pre-Brexit terms would likely be even higher. Even now, the relationship with the UK does not feature near the top of the EU's priority list. We explore this and the wider view of Brexit from across the channel in the next section.

Figure 36: UK public supports rejoining EU, but only on similar pre-Brexit terms



Source : Deutsche Bank, YouGov

Figure 37: UK would likely need material fiscal consolidation if euro membership was made a condition to rejoin

Euro convergence criteria

	Rule	UK latest reading	EU reference rate or target
Inflation	Less than 1.5% above 3 lowest EMU member states	*3.0%*	*2.9%*
Long-term yields	No more than 2 ppts above three best EMU member states	*4.8%*	*3.4%*
Fiscal deficit	Not exceed 3% GDP	*4.2%*	*3.0%*
Debt/GDP	Gross debt to GDP below 60% or diminishing at satisfactory pace	*94.0%*	*60.0%*
Currency volatility	2 years in ERM II where EUR/GBP trades in +- 15% range of agreed central rate	*7.3%**	*+- 15% range from central rate*

Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP

What about the EU side?

Pragmatic rapprochement without reintegration. The UK’s relationship with the EU ten years after the 23 June 2016 referendum is best understood as a shift from membership to structured third-country cooperation. Institutionally, the relationship is now more structured than it was immediately after Brexit, with annual summitry and multiple negotiation tracks under the 2025 “reset” (see here). Substantively, however, it remains selective and transactional rather than integrated. The most likely near-term outlook from an EU perspective is incremental deepening in sectors where geopolitical pressure and economic interdependence are strongest.

The UK-EU reset has produced real, but limited results so far. The reset in UK-EU relations has produced real but limited results – primarily on defence, food trade and energy. Stumbling blocks on further progress are disagreements on financial contributions by the UK in return for access to EU defence loan programs and access to the joint electricity market.

- On defence, the first joint UK-EU summit after Brexit in May 2025 produced a Security and Defence Partnership. However, the fact that both sides could not find a compromise on the level of the UK's financial contribution for allowing British firms to participate in the EU's EUR 150bn SAFE defence loan program, is a setback in defence integration.
- On food trade, the UK and EU have agreed to work towards a Sanitary and Phytosanitary agreement on food and agricultural standards, which the UK hopes to be in play by 2027.
- And on energy, the EU Council green-lighted talks on electricity market integration in early 2026. However, the EU will likely insist that in return for access to the EU single market for electricity, the UK should contribute to EU cohesion payments, like Norway and Switzerland.

UK-EU relations do not feature prominently at the top of the EU agenda. The reality is while the UK continues to debate what its future with the EU looks like, the EU has other more pressing issues.

The EU Commission's key policy priorities are centred around managing trade relations with the US and China, supporting Ukraine and improving its competitiveness (see our recent note here). As such, improving UK-EU relations are of lower-order importance, although deeper defence, electricity and trade integration would be beneficial for both sides (e.g. by better reaping the potential of North Sea wind). However, there is little in UK-EU relations so far that offers tangible benefits for national leaders in EU capitals and thus, negotiations remain at the technical rather than strategic level.

Relationship remains constrained by UK red lines and EU caution. The political will exists on both sides to further deepen ties as global geopolitical developments strongly favour selective rapprochement. But the relationship remains constrained by UK red lines (not rejoining of the single market or the customs union, not reopening freedom of movement) and EU caution and fear of cherry-picking. Both sides can be described as transactional in their approach limiting the willingness to be creative in improving their relationship.[1]

Next critical milestones to watch. The TCA review and the upcoming 22 July 2026 summit are the next critical milestones. Legally, the 2026 review is anchored in Article 776 TCA, which provides for a review of implementation five years after entry into force and every five years thereafter. The review does not constitute a commitment to reopen the TCA and the EU prefers a "light-touch" assessment focused on implementation. EU council decisions on the youth experience scheme (see here), a common sanitary and phytosanitary area (see here) and emission trading (see here) show that the preferred EU-UK method in the reset phase is not to reopen the TCA, but instead, to build targeted new agreements alongside it. For areas where the EU thinks that the UK benefits more than the EU from new agreements, other package deals are likely to be asked for.

1 See also CER, EU-UK relations - Will 2026 be the year to reset the reset?, February 2026

A hypothetical path ahead

The path ahead for the UK remains riddled with uncertainty. Politically, the push for further integration with the EU remains slow and disparate. Economically, the case for integration remains strong.

Social perception of Brexit has also changed in the past decade. More people view Brexit as a mistake. And the latest polls suggest that the UK would command a majority to rejoin the EU – but only under the right circumstances.

The main question, however, is at what cost would further integration come? If the UK could retain some of the exemptions that it benefitted from prior to its departure from the EU alongside maintaining some of the trade, regulatory and immigration freedoms, the case to rejoin the EU would be strong.

The reality, however, is that any future deal with the EU won't have as many exemptions. Politically, as well, no major political party has pushed a mandate for wholesale changes to the UK's relationship with the EU. Moreover, as we have noted, it takes two (or really 28 countries) to tango. Fears around any rapprochement of UK-EU reintegration due to fragmented politics (and remaining EU scepticism) will likely hinder any progress on a better deal.

So, what should the UK do? A sensible approach could be for a two-staged approach. The first, given the political constraints, would be to build on the current TCA to close some of the gaps in non-tariff barriers left as part of the original Brexit deal. This would include things like implementing the agri-food deal, establishing a Youth Experience Scheme, discounting university prices for EU students, reducing rules of origin checks, and agreeing on mutual recognition of qualifications. Our own analysis suggests that this could boost output meaningfully, scraping back some of the lost output in GDP. Doing so could even give the UK enough of a growth uplift to partially offset its diminishing fiscal headroom.

The second-stage would involve broadening out the UK's integration with the EU on things like defence, AI, and financial services. Should there be political appetite to push further towards the EU via either joining the Single Market, customs union, or re-joining the EU, no move should be made without a clear public mandate in order to avoid the political paralysis that followed after the Brexit referendum. The UK should also look to prioritise carve outs on its existing trade deals (though we think this could be difficult for the EU to agree to).

To be sure, in the current global ecosystem, the UK should prioritise its ability to retain flexibility in sectors of the future such as AI. Its ever growing reliance on services relative to goods - far larger than nations of comparable wealth - means that the UK may still benefit from use of its increased sovereignty and flexibility, which Brexit in theory has allowed for.

Appendix

A. The Synthetic Control Method

A1. The counterfactual problem

The impact of Brexit cannot be simply measured by a before-and-after comparison. Why? Because the UK simultaneously experienced a pandemic, an energy crisis and a sharp monetary tightening. Isolating the Brexit-specific effect requires a credible counterfactual — an estimate of what UK GDP, employment and prices would have done had the 2016 vote gone the other way, holding the other shocks constant. The synthetic control family constructs that counterfactual from a donor pool of economies that did not undergo Brexit.

A2. Standard SCM (used for GDP)

The Synthetic Control Method (Abadie, Diamond & Hainmueller, 2010) builds a "synthetic UK" as a weighted average of donor economies. The weights are chosen so the synthetic UK matches the actual UK as closely as possible across the pre-referendum period, on both the outcome's own trajectory and a set of economic predictors. Two constraints are central: the weights are non-negative and sum to one. This makes the synthetic UK a convex combination of real economies — it can never lie outside the range of the donor pool, so the counterfactual is always economically plausible and the method cannot extrapolate to an implausible level.

The post-2016 gap between the actual UK and its synthetic counterpart is the estimated Brexit effect. Because the outcomes are modelled in logs, this gap reads directly as a percentage. Statistical credibility rests on small pre-treatment fit error (RMSPE), confirming the synthetic tracks the UK before treatment.

Why is the SCM the right tool? The UK exhibits a tight convex match on GDP, employment, and prices (low pre-RMSPE), for one. Additionally, GDP's principal post-treatment confound (the pandemic) enters largely as a level effect that the convex blend handles well. The SCM's constrained form — its core strength — is not a binding limitation here.

A3. SCM Weights

We include our estimated weights for our SCM (including factor-augmented) below for our key three variables: GDP, employment, prices.

Figure 38:SCM weights

SCM Weights			
	GDP	Employment	Prices
Australia	39.2%	4.0%	33.7%
Austria		0.6%	54.7%
Canada		11.3%	0.3%
Denmark	0.1%	0.5%	3.7%
France		1.1%	0.1%
Germany			0.1%
Greece		0.5%	0.1%
Hungary	17.9%	0.8%	5.9%
Ireland		1.2%	
Italy	15.9%	0.7%	0.3%
Japan		0.3%	
Netherlands		21.8%	
New Zealand		4.5%	0.1%
Norway		0.8%	0.2%
Portugal	4.3%	0.6%	0.1%
South Korea		0.5%	
Spain		2.3%	0.2%
Sweden		1.6%	0.3%
Switzerland	22.3%	2.5%	
United States		44.4%	

Source : Deutsche Bank

References: Abadie, Diamond & Hainmueller (2010), JASA; Abadie (2021), JEL; Xu (2017), Political Analysis; Bai (2009), Econometrica.

A6. Limitations of SCM

Our approach to estimating the impact of Brexit (thus far) has its own limitations. We set them out below.

Synthetic control. The synthetic control method, for all its transparency, rests on assumptions that aren't perfect.

The counterfactual is only as good as the donor pool: it assumes some weighted combination of economies can credibly reproduce the UK's pre-referendum trajectory, and that those donors were not themselves materially affected by Brexit or by idiosyncratic shocks of their own over the estimation window.

The method also assumes no anticipation — that the UK's pre-2016 path was not distorted by the prospect of the referendum — and that the relationship between the UK and its synthetic twin, calibrated before 2016, would have held absent the treatment. A close pre-treatment fit makes these assumptions more plausible but cannot prove them.

Inference is another constraint: the synthetic control has no conventional standard errors. Estimates are also sensitive to specification choices — the donor set, the predictor list, and the treatment date — and the further the analysis runs beyond the referendum, the more the accumulating overlay of unrelated shocks (a pandemic, an energy crisis, a monetary tightening cycle). This erodes the cleanliness of the counterfactual. We mitigate these concerns by ensuring a tight pre-treatment fit, but they cannot be eliminated.

Overall, use of a SCM does not deliver a definitive causal estimate; it merely provides a disciplined, transparent counterfactual under stated assumptions. We regard the convergence of our results with independent external estimates as more informative than any single point estimate. All figures should be read as central estimates surrounded by genuine, if unquantified, uncertainty.

B. Detailed workings on TCA adjustments

This appendix sets out the sensitivity analysis used to estimate the GDP impact of selected Brexit reset proposals. The approach updates Springford (2024) with more recent data and tests how the results vary under alternative assumptions. We refer to the original article for the full workings behind these calculations.

The quantified policies are the first four proposals listed in the main table.

- ETS linkage and product safety and metrology are treated qualitatively, given the broader range of assumptions and data required for a robust estimate.
- The recent data used to update the estimates are as follows:
- SPS: Updated UK agrifood exports and imports to and from the EU to 2025.
- EU students: Updated the number of EU-domiciled students in the UK to 2024/25. We assume that universities can accommodate the relevant increase in demand fully (given the drop in student visa applications).
- Mutual recognition of professional qualifications: Updated total intra-EU recognition decisions to 2024.
- Youth Mobility Scheme: Updated Eurostat data on young people in employment in the EU, HMRC data on EU young people in payrolled employment, and UN data on UK young people in the EU to 2024/25.

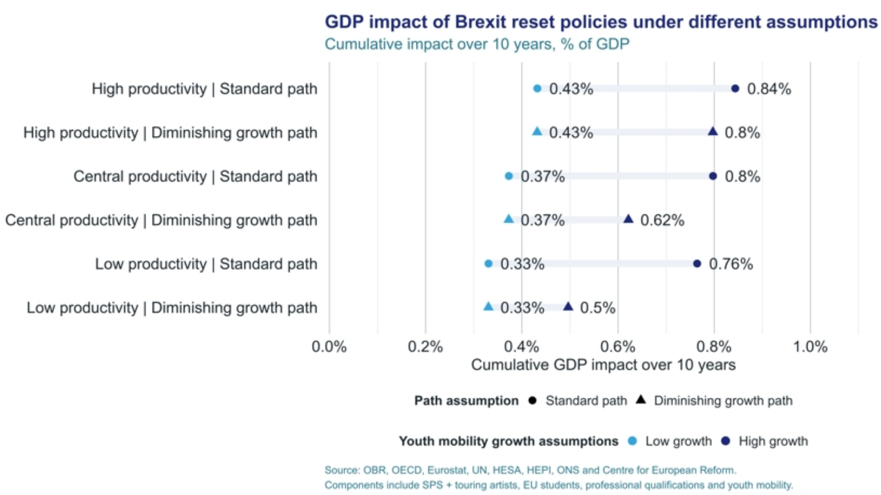
The scenario assumptions are as follows:

- Path assumptions: Instead of applying a constant average annual growth rate, we use a diminishing growth rate for projected take-up of the Youth Mobility Scheme. We model this using a simple AR(1) process that converges to a steady growth rate.
- Productivity assumptions: We apply OBR estimates for the impact of additional migration on potential GDP.
  - The central assumption assumes additional migrants have the same productivity per worker as the existing population, for example if capital per worker is unchanged or any dilution is offset by higher total factor productivity growth. Under this assumption, GDP is around 1.5% higher.

- The low assumption assumes lower overall productivity, for example due to lower capital per worker, resulting in a smaller GDP increase of around 1%.
- The high assumption assumes higher productivity, reflecting additional migrants having a much higher participation rate than the adult UK population. Under this assumption, GDP is around 2.2% higher.

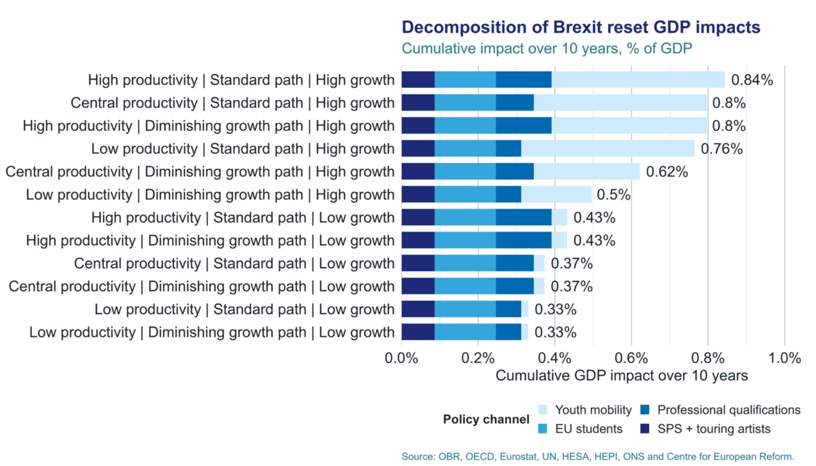
The figures below provide a more detailed overview of the policy proposals and their contributions to the estimates.

Figure 39: Brexit reset policies under different assumptions



Source : Deutsche Bank

Figure 40: Decomposition of Brexit reset GDP impacts



Source : Deutsche Bank

Appendix 1

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Twenty Miles That Shook the World

Analyzing the impact of the Iran conflict and charting a new normal in the Strait of Hormuz

June 2026

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With contributions from:

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Introduction and Table of Contents

Twenty miles, the width of the Strait of Hormuz (SoH) at its narrowest point, was enough to trigger the largest single disruption of global energy supply in recorded history and a far broader shock to the world economy. This report is being published following the US and Iran having signed a June Memorandum of Understanding (MoU) intended to reopen the Strait, broadening the key questions from the scale of disruption to the shape and speed of recovery.

When Iranian forces effectively closed the Strait in late February 2026, the immediate damage was visible in energy markets. Brent crude oil rose from roughly $61 per barrel at the start of the year to more than $120 at the peak, European natural-gas prices nearly doubled, and the IEA compared the scale of the disruption to both oil shocks of the 1970s happening at once.

To describe the episode simply as an energy crisis is to understate what happened. The same narrow passage also trapped fertilizer trade, disrupted petrochemical and industrial feedstocks, constrained helium supply critical to semiconductors and elsewhere, and repriced shipping, insurance, routing, and inventories. What emerged was not one shock but several layered on top of one another.

Report objectives: to gauge the crisis's broad impacts and assess how quickly normalization can proceed, once a lasting reopening agreement emerges. Furthermore, the report aims to identify changes to the global landscape that could prove permanent. With several key markets having been affected, we adopt a sectoral approach to addressing these issues that complements macro approaches.

Report Organization

This report provides a bottom-up roadmap from the physical disruption to its economic consequences. The macro analyses it complements include those by Deutsche Bank's macroeconomics research team, as well as analysis from the IMF, OECD, World Bank, and others. It does so by tracing the crisis impacts and likely recovery paths sector by sector and then drawing broader conclusions about inflation, market pricing, resilience, and the post-crisis global operating environment.

Following this introduction, the report is organized in three parts. This structure is important because the central lesson of the report is that not all sectors normalise at the same speed. Financial prices can move first, shipping lanes next, production later, and downstream consumer effects later still.

Part 1: State of Play and Path to Normalization. Here we first review the maritime status of the SoH, based on our dbDataInsights team's analysis of the cargo volume collapse, stranded vessels and recent crossings. Next, we discuss why the path to normalization will be protracted, not expeditious. The impediments are diplomatic, physical, and psychological. We analyse the risks around the MoU as evidenced by Iran's claiming the reclosure of the Strait just days after the "deal" was signed. We also review constraints on production and shipping that will add several months or more to normalization.

Part 2: Structural Implications. This section begins by assessing the macroeconomic effects of the crisis that can be gleaned from the sectoral analysis that follows. Here we compare our bottom-up findings with top-down macro estimates produced by Deutsche Bank's macro research team as well as by major international institutions. We then enumerate the top ten structural economic and geopolitical changes that we see as having been set in motion, reinforced, or exposed by the crisis. These include, among others, a persistent risk premium around the SoH, more investment in route diversification, a world economy with more redundancy and higher operating costs, a stronger shift towards security-driven energy-transition, more geopolitical fragmentation, higher defence spending, and a weaker commitment to regional nuclear non-proliferation.

Part 3: Sectoral Analysis. This part, which drives the rest of the report, examines the eight key sectors directly impacted by the SoH closure. They include: shipping and maritime logistics, crude oil, refined petroleum products, liquefied natural gas, fertilizers, petrochemicals, helium, and aluminum.

Each sector is presented using the same four-part structure, enabling comparison of transmission channels across markets:

(i) pre-war baseline, establishing why the sector mattered globally and how exposed it was;
(ii) war-induced disruption, identifying the specific impairments to trade, production, and pricing;
(iii) recovery considerations, physical normalization, price and supply effects; and
(iv) implications, including the impact on costs, pricing, trade, corporate behaviour and ultimately, consumers.

Key Conclusions

Conclusion 1: Normalization will be phased, uneven, and slower in the physical economy than in market prices. The report stresses a distinction between a diplomatic reopening and a full operational recovery. We expect diplomatic reopening to resume soon given strong economic incentives on both sides of the conflict. On the operational side, our dbDataInsights analysis shows the backlog of vessels trapped in the SoH as of mid-June would represent a week or more of pre-crisis daily exits. While an agreed reopening may allow for immediate vessel movement, various impediments persist. Transit activity will be constrained by factors like port congestion from inventory rebuilding, bunkering needs, complex insurance processing, and a scarcity of viable rerouting passageways. Many logistical pressures can be alleviated within several months once a sustainable reopening is agreed and assuming mine clearing has progressed. But major shipping companies may need more time to gain confidence that hostilities have subsided, and security and safety concerns have been addressed. Needed repairs to some facilities will take years before full normalization can occur, with expected supply diversification contributing to a new normal.

Conclusion 2: The inflation shock was temporary, but price level impacts will be more lasting. Oil, oil product and LNG prices have already reversed some or much of their spike as reopening has come into view, and we expect them to move gradually lower from recent levels, with Brent crude ending the year in the $75-80 per barrel range. This will still be noticeably above pre-war levels as inventories and reserves are rebuilt. That reversal will allow some substantial easing in headline inflation, but non-energy inflation will remain more elevated due to lagged pass-through from earlier energy increases, together with cost increases in other affected sectors. Consequently, these prices should remain materially higher over the longer term than they otherwise would have been. Relative to energy, shipping and petrochemicals added smaller effects through traded goods and manufacturing inputs. Fertilizers mattered mainly through lagged food-price effects, while helium and aluminum were more important as bottlenecks for selected industrial and technology chains than as major direct CPI drivers. Taken together, the sector work is consistent with a cumulative increase in consumer price level in the range of 0.5–1.5 percentage points globally this year under a constructive reopening scenario, with exposed importers in Europe and parts of Asia nearer the upper end and insulated economies nearer the lower end.

Conclusion 3: The crisis will have fundamental economic and geopolitical impacts lasting far beyond its immediate resolution. Elevated risks to transiting the SoH are here to stay. This will stimulate investment in diversification via alternative routes around the Strait as well as switching to other sources of supply like the US. It also ties energy security and energy sustainability more closely, likely stimulating greater investment in renewables. Firms and governments will place greater weight on redundancy, strategic inventories, supply-chain resilience, and defense of critical infrastructure. This means a global economy that is both more expensive and less frictionless. We will also be looking at a world that is likely trending more toward geopolitical fragmentation, and less toward nuclear nonproliferation, especially in the Gulf region.

The closure of the Strait of Hormuz…

In late February 2026, the closure of the SoH by Iranian forces sent immediate shockwaves through energy markets. The International Energy Agency (IEA) compared the scale of disruption to experiencing both oil shocks of the 1970s simultaneously. Framing this as an energy crisis, however, is misleading. Ultimately, the impact has been significant across several markets and commodities.

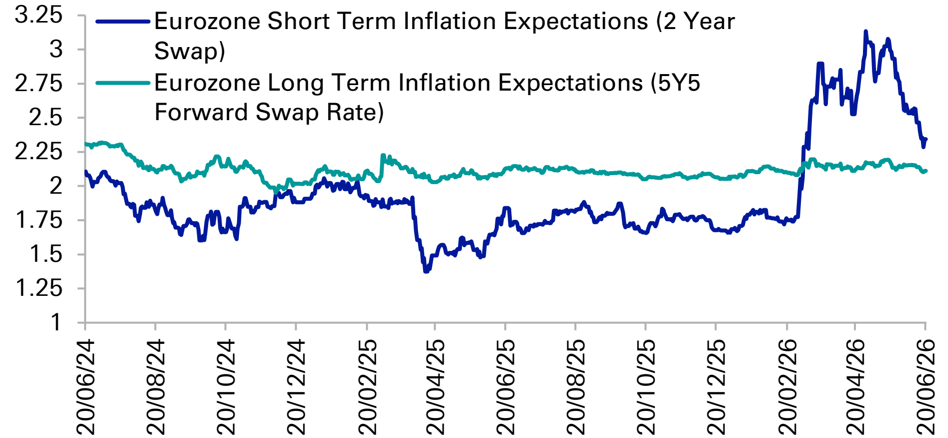
The subsequent US blockade, enacted in early April 2026, was enforced on vessels traveling to and from Iranian ports and significantly raised the risk of broader military conflict in the waterways. Beyond increased insurance premiums, shipping costs and transit times, the blockades significantly impacted supply chains and commodity prices leading to near-term inflationary pressures and goods shortages.

The fragile "reopening" of the Strait of Hormuz…

After more than three months of disruption, a Memorandum of Understanding (MoU) was signed by the US and Iran last week. As outlined in more detail below, the deal extended the ceasefire by 60 days for negotiations.

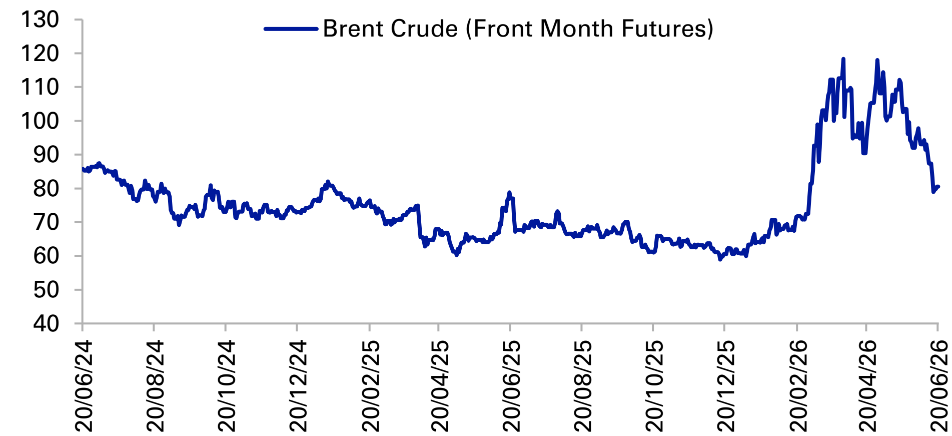
The MoU signing provided the conditions for a phased resumption of trade flows through the SoH. Consequently, Iran resumed limited safe transits while the US lifted its naval blockade, though US Naval vessels remain in the "general area", allowing traffic from Iran to be "in proportion" to transits restored by Iran. But on 20 June, days after signing, Iran announced reclosure of the Strait, citing Israeli strikes in Lebanon as a ceasefire breach, a claim the US disputed. This underscores the short notice reversibility of the reopening and the fragility of the MoU.

Eurozone inflation expectations
(%)



Source: Bloomberg Finance, LP, Deutsche Bank

Brent crude (front month futures)
(USD/bbl)



Source: Bloomberg Finance, LP, Deutsche Bank

Key Regional Ports in the Persian Gulf

Key regional ports in the Persian Gulf

Key regional ports inside the Gulf (in order of importance):

1. Jebel Ali (UAE):

— The region's dominant commercial hub by overall logistics importance, combining large scale container throughput with connectivity across the Gulf, South Asia, East Africa and Europe.

2. Khalifa Port (UAE):

— The second major Gulf gateway, with rising container and industrial cargo significance and strong integration with Abu Dhabi's manufacturing and logistics base.

3. King Abdulaziz Port (S. Arabia):

— The largest Saudi Arabian commercial port on the Gulf and a major inbound/outbound cargo node, important for containers, general cargo and industrial supply chains.

4. Bandar Abbas (Iran):

— Iran's principal commercial gateway and a key node for containerized, general and project cargo, though its international role was already constrained by sanctions.

5. Hamad Port (Qatar):

— Qatar's main commercial gateway. Strategically important for containerized imports and non-hydrocarbon trade, even though its tonnage profile differs from the larger UAE and Saudi hubs.

Regional alternatives outside the Gulf

1. Yanbu (Saudi Arabia):

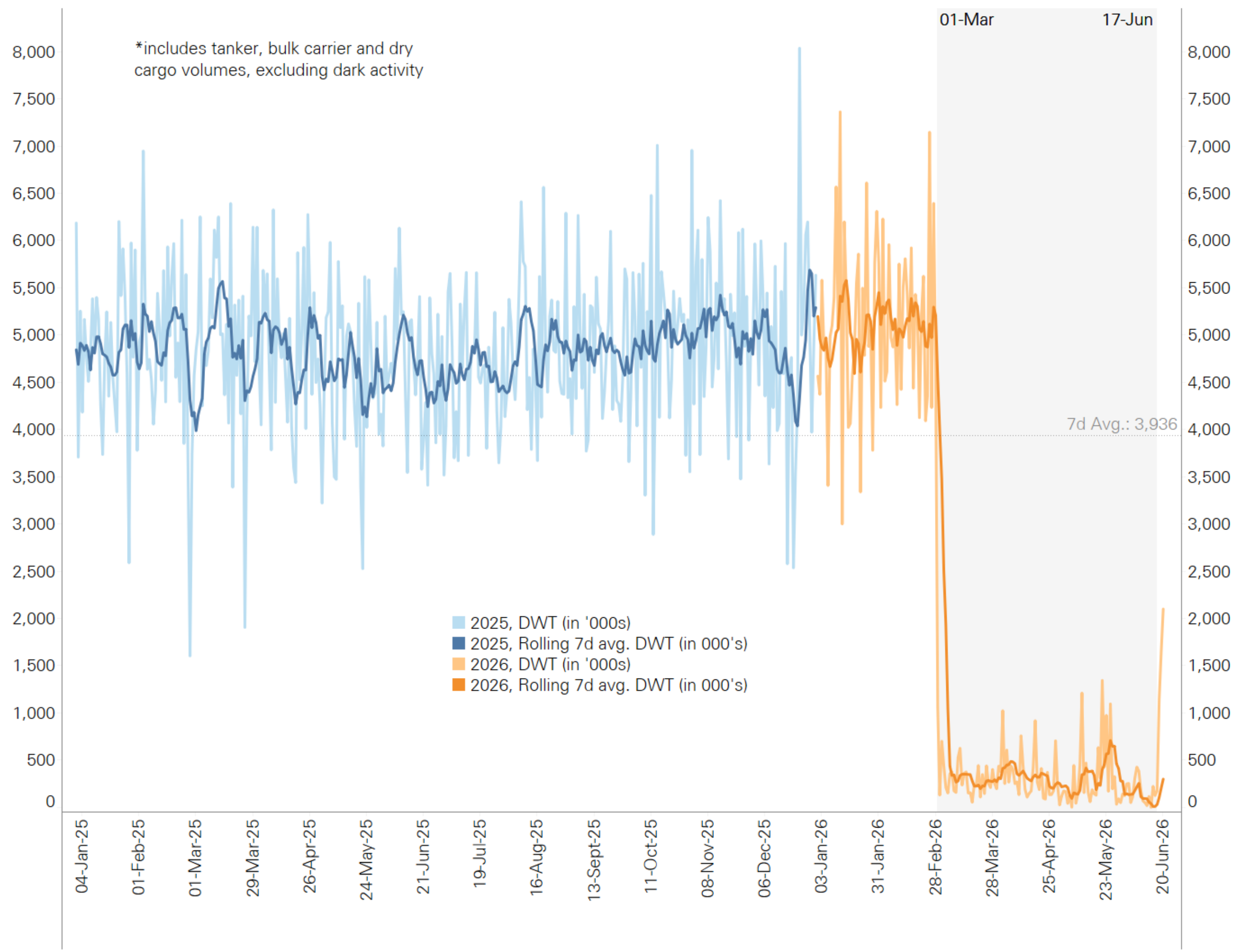
— A Red Sea outlet linked to westbound pipelines, providing limited bypass capacity, but ramping since March.

2. Fujairah (UAE):

— An east-coast hub on the Gulf of Oman, key for bunkering and storage, and the most immediate fallback, but faced attacks during the conflict.

Part 1: State of Play and Path to Normalization
Depth of SoH Commercial Cargo Volume Loss

Blue-water commercial cargo volumes exiting the Strait of Hormuz (from 2025)
(DWT in '000s)



Source: dbDataInsights

Notes on tracking vessels and volumes transiting the Strait of Hormuz.

Daily averages (1 March – 17 June):

— Prior to the crisis, daily exits from the SoH averaged c.70 "blue water" commercial vessels, with an estimated daily capacity of 5.1m DWT. Following the crisis and up to MoU signing, this plummeted to c.9 vessel exits per day, with estimated daily capacity of ~300K DWT (c.6% of pre-crisis levels).

— Our estimates discussed above include some dark activity as we were able to monitor several vessels exiting the strait with transponders off by cross-referencing departure logs from Persian Gulf ports with subsequent arrival records outside the SoH, or transits confirmed by the media. That said, if we use estimates for total dark activity since 1 March, it will point to daily transits in the low teens.

Dark activity:

— Dark activity (vessels concealing their identity or location) increased after the crisis began. Pre-crisis this tactic was generally only deployed by Iranian ships evading sanctions.

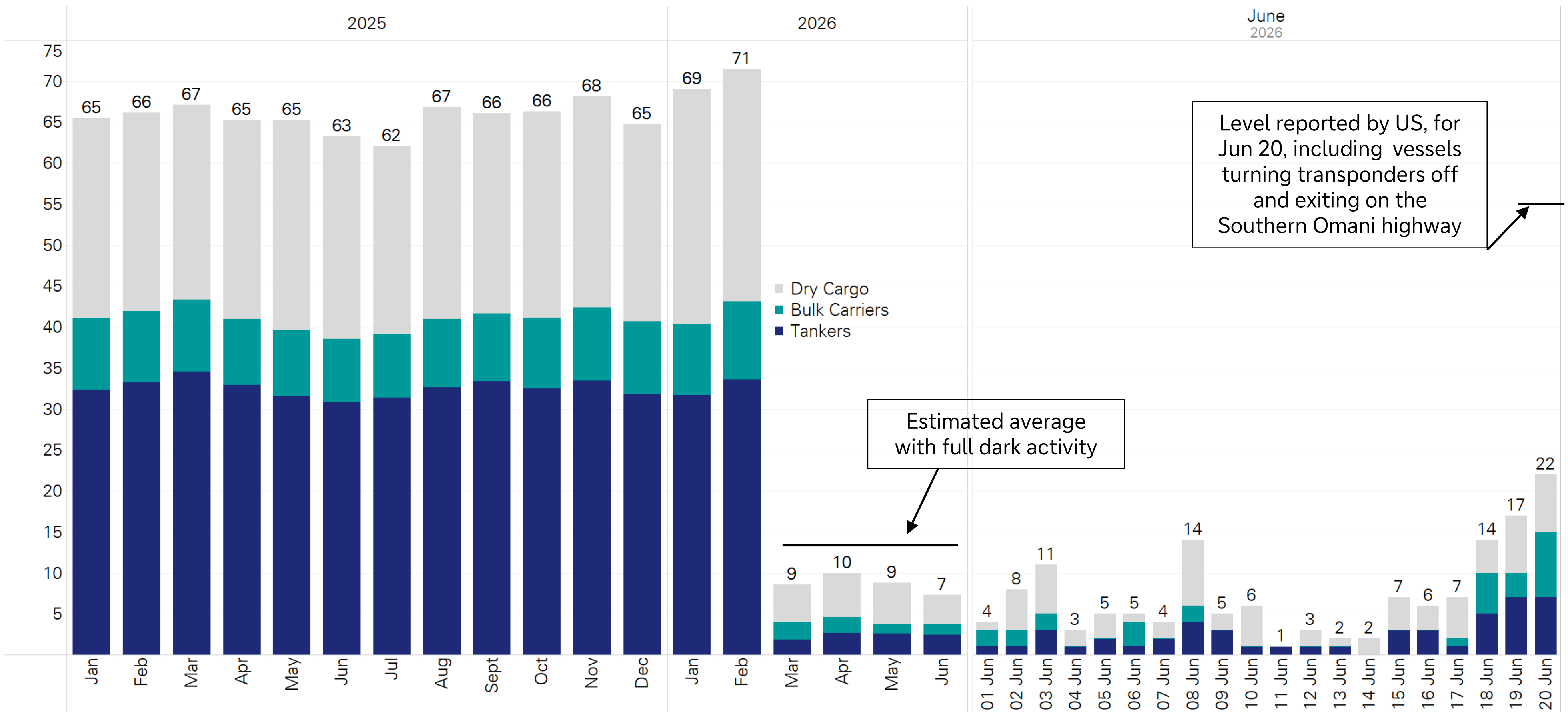
— Estimated pre-conflict dark activity was in the low-single digits for daily transit exits, (implied c.3-6 vessels).

— Kpler estimates that 40% of all transits through the SoH since 1 March occurred as "dark", with higher levels as the crisis continued (especially for tankers).

— Non-Iranian operators made up a sizeable proportion of the increase in dark activity as they sought to avoid conflict and go undetected.

Depth of SoH Commercial Cargo Volume Loss

Blue-water commercial cargo tankers exiting the Strait of Hormuz (from 2025)
(Vessel count)



Source: dbDataInsights

Note: Pre-crisis figures do not account for 'dark activity' (estimated 3-6 vessels per day with transponders off, mainly Iranian tankers). Following the crisis, we implemented a methodology to estimate partial dark activity by cross-referencing Persian Gulf port departures with confirmed arrival records outside the SoH, and media reports. Based on this, our recorded average daily vessel crossings from March to mid-June were nine. However, if we were to triangulate all estimated dark activity, average daily crossings during this period would likely have been in the low-teens. Dark activity appears to have picked up significantly since the MoU was signed as the U.S. Central Command said 55 commercial ships passed through the strait on 20 June.

Stranded Vessels Inside the Persian Gulf

Assessing stranded vessels…

— Our analysis at the signing of the MoU showed that among the roughly 2,600 vessels in the SoH, we considered 582 as "stranded" from a commercial cargo perspective, with an estimated capacity of 32.5 Deadweight Tonnage (DWT). This equates to 6-8 days of pre crisis traffic exiting the SoH. The type of vessel is important as the capacity of a common size tanker could be 6x that of an average cargo ship in DWT.

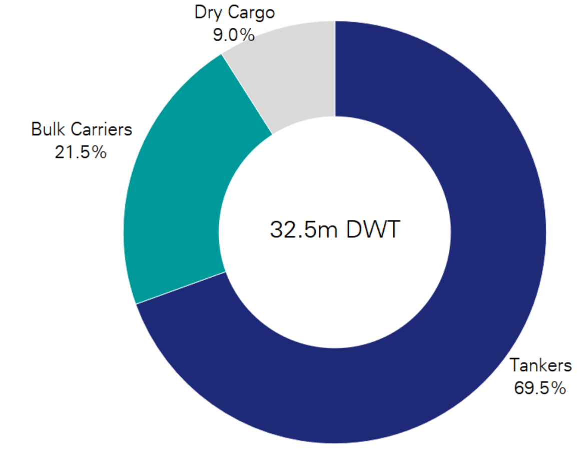
— Amongst the stranded vessels, we were able to identify the country of beneficial ownership of 471 ships (accounting for c.92% estimated DWT). Greece, China/HK and Japan had the most exposure. Looking at Persian Gulf countries, the UAE had the highest total at 6.6%.

Crossings have picked up since the MoU was signed but have subsided since Iran "reclosed" the SoH…

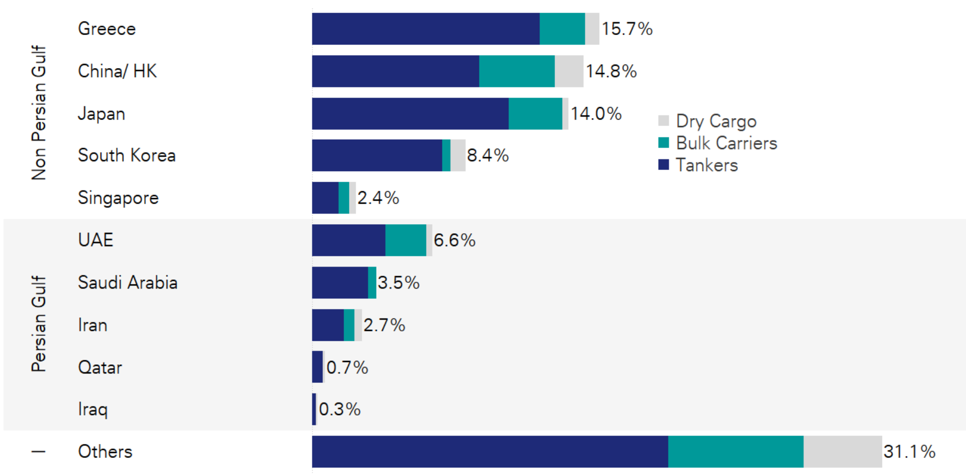
— Our data showed an increase in total vessel exits from the SoH since the MoU was signed, with more 20 vessels exits on Saturday, 20 June, the day Iran "reclosed" the Strait. In fact, U.S. Central Command said on the day that 55 commercial ships passed through the strait that day, implying many vessels turned off their transponders. There were reports of vessel crossings on the Sunday, but at slightly lower levels, though this appears to have picked up in both directions on Monday, 22 June, with further crossings into the SoH pointing to optimism.

— From here, transits may occur by way of the "Southern Highway" through Omani waters as the US advises and monitors this route. The Joint Maritime Information Center (JMIC) has also confirmed it is clear of mines.

Commercial cargo ships stranded in the Persian Gulf
(582 vessels as of 15 June, % of estimated DWT)



Stranded commercial cargo ships by country
Denominator = 471 ships with identifiable country affiliation)



Source: Bloomberg Finance, LP, Deutsche Bank

The MoU: Key Provisions and Risks

Key provisions now implemented by the signed MoU…

- **Cease-fire framework extended to all fronts, including Lebanon**. The MoU establishes an immediate cease-fire between the US and Iran and their respective allies, with the text explicitly covering Lebanon alongside reciprocal commitments to sovereignty, territorial integrity, and non-interference.
- **A 60-day interim window, not a final settlement**. The MoU is structured as a sequencing and implementation framework with a maximum 60 day negotiating period, leaving the war's permanent resolution, sanctions architecture, and nuclear and regional security questions to a later, final deal.
- **Phased military drawdown and Iranian Strait commitment**. US to remove naval blockade within 30 days and withdraw forces within 90 days. Iran commits to safe, toll-free Strait passage for 60 days while retaining future role in longer-term governance alongside Oman and Gulf states.
- **Phased sanctions relief**. Immediate removal of humanitarian designations, secondary/tertiary sanctions removed within 45 days; permanent architecture negotiated in final deal. Nuclear sanctions, explicitly deferred.
- **Nuclear enrichment remains to be addressed.** Iran has restated its previous commitment on this issue, but important details on down-blending, transfer, conversion or other neutralisation and especially verification will be the central technical and political test of the MoU.

Diplomacy risks…

- **Lebanon reescalation and Strait governance asymmetry**. Lebanon cease-fire inclusion is the deal's most fragile element – Iran has invoked Israeli strikes in Lebanon to justify Strait closure, showing third-party conduct can trigger unrelated disruption. US blockade removal is on fixed schedule in the MoU; Iran's Strait commitment has proved reversible with disputed cease-fire violations.
- **Sanctions relief political unsustainability**. US Congressional opposition to Iran sanctions easing and Trump Administration's unpredictability create risk of unilateral US rollback of sanctions relief mid-sequence. Iran views Strait closure as justified response to US violations; asymmetric sanctions relief reversal creates mutual recrimination and induces reescalation.
- **Most significantly for the regional order, the MoU points to future arrangements for Iran's enriched material rather than locking in irreversible neutralization now**. That risks setting a precedent that closing, or threatening to close, the Strait can extract sanctions relief and military disengagement before nuclear issues are resolved, weakening non-proliferation credibility and encouraging Saudi Arabia, the UAE or others to hedge toward their own enrichment capability.

Impediments to Reopening (Non-Diplomatic)

Physical and security frictions…

- **Mine clearance and routing constraints:** central SoH exit routes are restricted at present due to mines. This includes the International Maritime Organization (IMO)-recognized Traffic Separation Scheme that runs through both Iranian and Omani waters. These lanes must be cleared and certified. Optimistic estimates point to several weeks, with some reports suggesting six months.
- **Security guarantees:** shipmasters and owners may wait for proof of mine-free routes, accepted escorts and stable rules before committing crews and assets.
- **Port capacity and bunkering delays:** damaged or offline berths and long queues at regional ports will constrain bunkering and turnaround, especially at congested ports.
- **Vessel inspection/repairs:** ships idled may need hull, machinery, seaworthiness checks; repair-yard slots, spares, and class/flag approvals; this may add days/more.
- **Crew fatigue & rotation:** stranded crews need relief, shore leave, medical support and repatriation; expired contracts and depleted stores can delay departures.
- **Port and infrastructure repairs:** some locations could take significant time as detailed elsewhere in this report.
- **Residual controls:** inspections, toll or service-charge demands, escort requirements and monitoring could keep throughput below nominal capacity.
- **Increased congestion:** inventory buffering strategies will likely increase stockpiling, further contributing to port congestion and strain on logistics networks.

Commercial and sequencing impediments…

- **Early sequencing:** oil and LNG are likely prioritized over other cargoes, with expectations for early outflows to Asia.
- **Uneven first movers:** national companies controlling their own shipping (UAE, Kuwait, Iran) and holders of stored volumes are likely to move first, especially to Asia.
- **Flag/ownership asymmetry:** Greece-, China- and UAE-linked vessels that kept transiting may retain greater tolerance or preferential treatment.
- **War-risk insurance:** cover may be available, but case-by-case pricing is still cited around 1–4% of vessel value per transit versus below 0.1% pre-war; a $200m tanker can add $2m–$8m, pointing to concern over the outsized impact vessel damage (e.g., oil spills, pollution, etc.).
- **Higher transit costs:** insurance, escorts, fuel, congestion, inventories and supply-chain reorganization likely keep costs above pre-crisis levels.
- **Congestion**: efforts to rebuild inventories and supply-chain reorganization will add to initial crowding in both directions.
- **European exposure:** Europe remains more exposed to Qatari LNG and refined products, especially jet fuel, sustaining supply-chain and inflation pressure.
- **Sanctions and compliance lags:** even with sanctions lifted, banks and insurers may take months to change internal compliance restraints and resume dealings. That said, press reports have suggested that pre-crisis, Iran has been able to evade sanctions to some degree.

Phasing of SoH Reopening

Several transport and materials exposed companies have pointed to 2027 for normalization assuming a mid-year opening. A constructive reopening should be viewed in phases, with the early part of Phase 1 having occurred the last few days as several vessels, including large tankers, have transited out of the SoH, while others have begun to position themselves to enter from the Strait of Oman.

— **Phase 1: Safety and bottlenecks (~1-2 months):** Crew and vessel safety concerns will likely keep several major shippers from immediately resuming operations in the region. From a logistical perspective, mine-clearing, lane-certification (especially central routes), and high war-risk insurance approvals will cause delays. Bunkering availability and congestion from vessel repositioning/ backlogs will add to bottlenecks.

Sheikh Nawaf, CEO of Kuwait Petroleum Corp, stated on 18 June that Kuwait aims to "exceed 2 mb/d within a week, and resume pre-war production within weeks, pending the availability of international commercial shipping to reach Kuwaiti ports." Shippers willing to take on these risks and logistics should be able to exit via the Southern Omani route, monitored by the US and partners. What is unclear, at present, is the ability to transit along the northern Iranian route.

In recent days, tankers eager to leave the region for Asia appear to be turning transponders off to avoid detection. Despite this activity, vessel placement remains a critical issue. As Saudi Aramco's CEO noted in May, the "biggest issue is the tanker fleet," with vessels in "the wrong places" requiring time to "reassemble."

— **Phase 2: Improved logistics with caution (~2-4 months):** Mine-clearing estimates suggest central pathways through the Strait may not be fully cleared at this stage (some estimates point to 6 months for clearance), but overall logistics should improve with some backlog cleared. The war-risk premium will likely still be in effect, with major shipping companies still slow to return. The precedent here being the Red Sea/Gulf of Aden situation, where sporadic attacks caused dramatic resets for years, suggests a high level of caution will continue.

— **Phase 3: Partial recovery, continued war-risk premium (~4-6 months):** A decent level of flow could be expected, assuming hostilities have subsided enough to establish confidence, and mines have mostly been cleared. Key questions involve the level of regional friction and whether all major shippers will fully return.

— **Can full normalization occur in 2027 – the 4th Phase?**

Full operational normalization, involving infrastructure repairs, is not expected until 2027 and beyond. Critical facilities, such as Qatar's Ras Laffan LNG complex, face multi-year recovery timelines, with some specialized repairs estimated to take up to five years.

We expect a structural shift, rather than a temporary workaround, as exposed countries and companies strategically invest in diversified supply chains and alternative routes to reduce risk. This should drive increased demand for LNG and oil from the US, Russia, and non-OPEC countries, in addition to efforts to expand pipeline capacities like Yanbu (Saudi Arabia) and Fujairah (UAE), and overland transport through Syria (from Iraq).

What the sector work suggests…

- **Sector pattern:** Energy dominates the consumer-price effect. Crude is the largest channel, with refined products next through diesel, jet fuel, freight and transport. Shipping and petrochemicals add smaller second-round effects through traded goods, packaging and manufactured inputs, while helium and aluminium matter more as bottlenecks than as broad CPI drivers. Fertilisers mainly affect prices through delayed food channels, since the shock works through farm-input costs, planting decisions and then retail food prices with a lag.
- **Sector ranges:** Crude appears to have added roughly 0.3–0.8pp to headline inflation in major importers, with refined products adding another 0.1–0.3pp, especially in Europe and Asia. Shipping and petrochemicals likely added only modest increments. Fertiliser plus diesel and freight likely raised food commodity prices by around 4–5%, but only about 0.2–0.3% overall consumer food prices so far because retail food prices adjust with a lag.
- **Rough macro view:** A rough takeaway is a cumulative inflation effect of about 0.5–1.5pp globally under a constructive reopening scenario. That should be read as an order-of-magnitude range, not a precise forecast, with Europe and parts of Asia toward the upper end and the US and China nearer the lower end. The broad message is a meaningful but uneven inflation shock, dominated by energy and amplified by transport and food channels.

Why aggregation is difficult and how it compares with macro estimates…

- **Double counting:** Adding sector estimates is inherently imprecise. Crude feeds into refined products and petrochemicals; diesel and shipping both affect food; and some sectors affect availability more than prices. Timing matters because pass-through depends on inventories, margins, regulation, subsidies and repricing speed.
- **A cross-check**. Sector-by-sector summation is better read as a bound than a forecast. OECD input-output analysis helps in principle, but the framework is not granular enough to cleanly separate overlapping effects across the Hormuz-exposed sectors. The results are therefore best viewed as a cross-check on macro estimates rather than stand-alone projections.
- **Deutsche Bank World Outlook:** DB's early-June World Outlook is broadly consistent with the sectoral range. Under a mid-year reopening baseline, DB estimated the disruption would add about 1.4pp to 2026 inflation in the euro area, 0.9pp in the US, 0.7pp in Japan and India, and 0.3pp in China. A more protracted disruption would push those forecasts materially higher.
- **IMF/OECD/World Bank:** These estimates point in the same direction. The OECD and World Bank revised 2026 global inflation up by nearly 1.5pp relative to pre-war projections, while the IMF revised it up by somewhat less. This is best read as validation rather than exact one-for-one comparability, given differences in aggregates, timing and baseline assumptions.

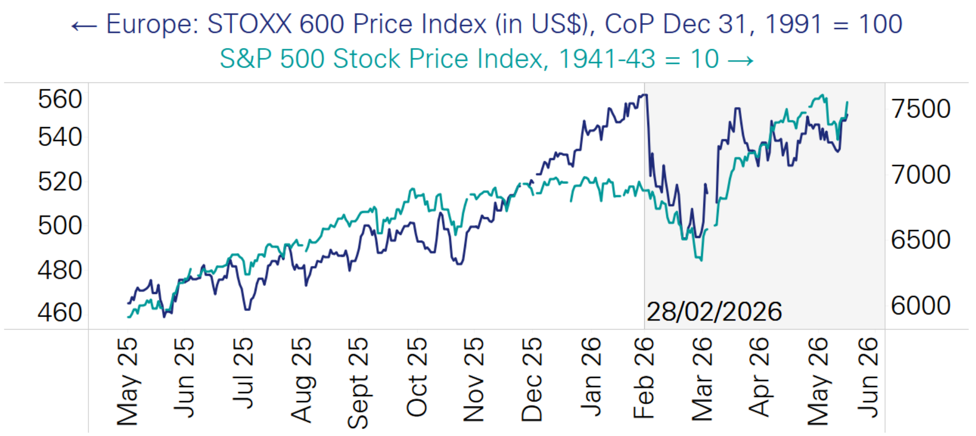
Beyond commodity and consumer prices, the crisis also generated a broader risk-off episode in financial markets. Equities sold off, bond yields rose as investors repriced inflation and put off expectations of policy easing. Currencies in energy-import-dependent economies came under pressure.

The macro damage therefore ran not only through physical shortages and higher consumer prices, but also through tighter financial conditions for a time and weaker confidence. These channels amplified the growth drag in exposed regions, even if headline equity indices later recovered more quickly than the underlying supply shock.

Central-bank policy implications…

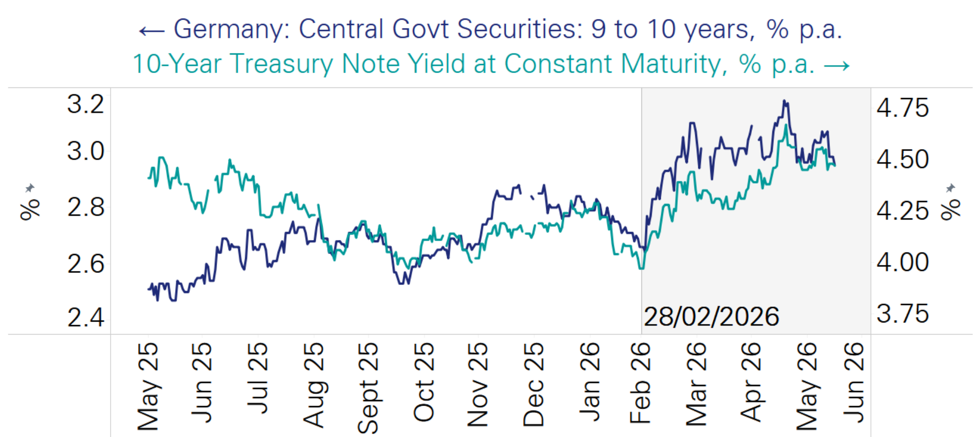
- The inflation shock materially changed the anticipated paths of monetary policy. DB's macro view of the ECB shifted from a path that would hold policy rates unchanged well into 2027 to the expectation of two rate hikes before the end of summer. The first of those hikes occurred in June as the surge in imported energy prices lifted inflation and risked de-anchoring expectations.
- The US faced much less immediate pressure from energy imports, now being a significant net exporter. But the jump in global energy prices and other anticipated price pressures caused the outlook for Fed policy to move in a similar direction. Longstanding market expectations of Fed rate cuts later this year shifted to no change in policy and eventually to a possible rate increase. DB's US economics team sees Fed policy rates unchanged over the year ahead.

Equity markets



Source: STOXX Limited, Standard & Poor's/ Haver Analytics

Bond markets



Source: Deutsche Bundesbank, Federal Reserve Board/ Haver Analytics

Deutsche Bank Research Institute

Top 10 Macro and Geopolitical Implications

War risk cost: The post-crisis economy is unlikely to return fully to the old equilibrium. Instead, the most probable outcome is a higher-friction global economy characterized by increased war-risk premiums, greater inventory buffering, more diversified supply chains, higher hedging costs, and a renewed corporate and policy emphasis on resilience over efficiency. This shift also includes significant geopolitical changes, with some spanning both economic and geopolitical realms.

Economic implications…

- **A lasting risk premium in the Strait of Hormuz.** This will be the most immediate permanent change on transit through the SoH. The conflict has shown that Iran can assert control over the waterway with an impact not fully understood pre-crisis. Lessons learned during this period will leave firms and governments reluctant to assume that this chokepoint can be treated as reliably secure, even in more "normal" times.
- **War-risk cost structure: more inventory, redundancy and an upward shift in costs.** The post-conflict economy is unlikely to return fully to the old efficiency-first equilibrium. Companies and governments are instead likely to hold more inventory, maintain more redundancy, spend more on insurance, hedging and transport, and accept more duplication of routes and suppliers. The main macro consequence is not necessarily permanently higher inflation in the long run, but a more persistent upward shift in costs and price levels and a world economy that is structurally more expensive to operate.
- **The resilience equation has changed.** Substantial reserves had already been built for crude oil and gas going into this crisis, but SoH related disruption proved governments and companies will need to broaden this basket to meet a resilience standard. This will include refined fuels, petrochemicals, fertilizers, food, metals, critical minerals and other industrial and materials inputs whose interruption can cascade rapidly through supply chains and consumer prices.
- **Energy transition policy is likely to become (even more) security-driven.** The link between energy transition and security strengthened after the Russia-Ukraine war and is further solidified by the US-Iran conflict. Governments will prioritize investments in renewables, power grids, storage, nuclear generation, and electrified transport not just for climate reasons but to reduce reliance on imported fuels and vulnerable shipping routes. Concurrently, many states may continue to secure fossil fuel access in the near term, making the transition more strategic and less purely decarbonization-driven.
- **Regional supply diversification and circumvention.** Major importers in Asia and Europe are likely to seek more diversified energy and commodity supplies, including both Gulf product that circumvents the SoH and more non-Gulf oil, gas and refined products where infrastructure permits. Over time, that could redirect trade flows toward the US and elsewhere, support new long-term contracts, and encourage fresh investment not only in domestic storage and power resilience but also in pipelines, terminals, rail links and other bypass routes that reduce dependence on a single chokepoint.

Economic and geopolitical implications...

— **The new reality of low-cost coercion.** Cheap drones, missiles and disruption tactics have shown that relatively low-cost capabilities can threaten shipping, ports, refineries, power systems and major militaries. The growing ascendency of drones was already evident in the Russia-Ukraine war and has been further confirmed in the Iran conflict, where relatively inexpensive systems and proxy-enabled attacks have been able to impose outsized disruption risk. That lowers the threshold for coercion, raises the cost of protecting infrastructure and commerce, and gives middle-tier powers and proxy networks more leverage over the global economy than traditional force rankings alone would imply.

— **Regional tension and realignment of Gulf states, highlighted by the UAE's departure from OPEC**. Gulf states will increasingly prioritize strategic autonomy, export-route redundancy, and diversified security relationships. The United Arab Emirates' exit from OPEC, effective 1 May 2026, exemplifies this, driven by a desire for full control over its oil production and to meet national economic goals. More broadly, this could mean more investment in westbound pipelines, Red Sea and Mediterranean outlets where possible, a broader mix of security partnerships, and greater willingness to hedge between Washington, Beijing and other external powers.

Geopolitical implications...

— **The conflict reinforces a global shift towards a fragmented world order with competing security and trade blocs.** States are likely to make more economic and diplomatic choices through a geopolitical lens, including sourcing, sanctions alignment, technology partnerships and strategic commercial relationships. For example, some countries may deepen ties with Western security and financial systems while others expand arrangements with China, Russia or more non-aligned networks to preserve room for manoeuvre under sanctions and supply stress.

— **The conflict will leave a durable legacy of higher spending on missile defence.** The SoH crisis highlighted the need for maritime security, port and pipeline protection, air defence and grid resilience. An important part of that is likely to be investment in lower-cost drone interceptor technologies, including systems better suited to counter large volumes of relatively inexpensive UAVs without relying only on costly high-end interceptors. This implies a more costly global infrastructure model in which critical assets are designed and financed with disruption risk more explicitly in mind.

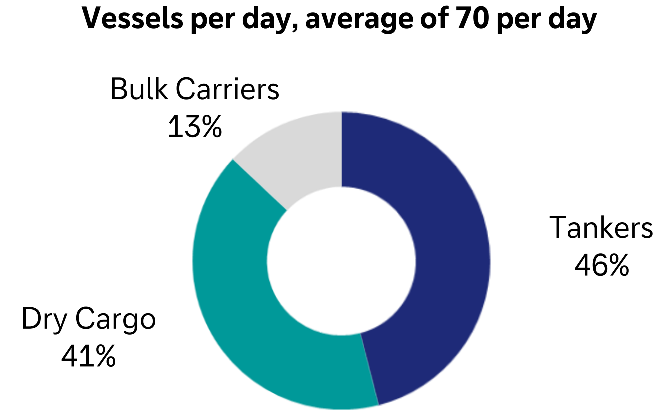
— **Weaker confidence in the regional non-proliferation order will be of the most consequential long-run risks.** Even if few states openly move to acquire nuclear weapons, some of Iran's neighbours may place more value on nuclear latency, missile capability, civilian fuel-cycle options and stronger deterrence relationships. The result could be a gradual erosion of confidence in the non-proliferation treaty (NPT) framework and a more proliferation-prone regional security environment.

Part 3: Sectoral implications

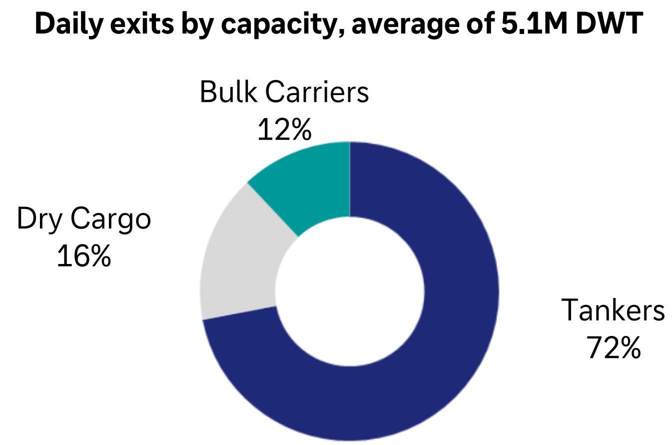
Before the war, the SoH was more than just a critical shipping route for oil and gas markets. It was a high-value, multi-cargo potential chokepoint. The baseline matters because even partial disruption quickly feeds through into freight, insurance, inventories and a variety of important regional supply chains. Consequently, disrupted trade activity carried outsized economic importance relative to its share of total shipping.

- **Traffic density:** Approximately 140 vessel transits per day during Jan-Feb 2026 on a broad cargo-and-tanker definition. This is equally split between vessels entering/exiting the Gulf.
- **Traffic composition:** Nearly half of the economically significant transits were dominated by tanker class vessels, which made up c.72% total volume in Deadweight Tonnage (DWT) terms. Crude oil tankers accounted for c.76% of this volume, followed by chemicals/oil products (16%) and LNG (8%). Per day, ~29 dry cargo/passenger ships exited the SoH followed by nine bulk carriers. The corridor also carried c.2.8% of global container trade and c.2.4% of global dry-bulk trade.
- **Energy concentration:** about 20mb/d of crude and products (3/4 of it crude), or c.25% of global seaborne oil trade, moved through the strait in normal conditions. In total, just over 70% of exports transited to Asia, while c.4% were sent to Europe.
- **LNG focus:** Qatar- and UAE-linked sailings accounted for close to one-fifth of global LNG trade. In total, ~90% of exports transited to Asia, with ~10% sent to Europe.

Daily vessel exits from the SoH Pre-War
(% of total vessels)



Daily vessel exits from the SoH pre-war
(% of daily capacity, DWT)



Source: IHS, dbDataInsights

At peak disruption, 85-95% of normal Hormuz trade flows were effectively blocked. The near total cutoff of commercial shipping through the SoH resulted from three compounding mechanisms:

1. **Iranian interdiction and insurance collapse.** Iranian authorities imposed restrictions on vessel transit, including toll demands and selective interference with commercial shipping. Attacks on, and physical risk to, hulls and crews prompted war-risk underwriters and Protection & Indemnity (P&I) clubs to either withdraw insurance coverage entirely or reprice it at cost-prohibitive levels. The effect was a soft closure as even vessels with navigational access could not move commercially insurable cargo, functionally freezing the market before any formal blockade was imposed. Iranian cargo was getting through at this point, as well as vessels that were paying tolls to Iran.
2. **US blockade of Iranian ports.** The enforcement of this blockade, together with interdiction and redirection risk for vessels attempting to serve them, converted the soft closure into a near-hard one on 13 April, 2026. Commercial operators now faced a broader enforcement regime rather than just narrower risk around Iranian-flagged or Iranian-affiliated shipping. Some vessels were able to slip through, but these reflected exceptions rather than a functioning market.
3. **Physical degradation of infrastructure at key Gulf facilities.** The largest shipping-relevant damage was concentrated at Bandar Abbas and Shahid Bahonar on the Iranian side. Elevated risk conditions were also prevalent at Fujairah in the UAE.

Factors behind the initial increase in shipping rates following the closure of the Strait of Hormuz…

— Several tankers were trapped inside the strait and unable to leave, while some were waiting outside the Strait in case of a quick re-opening, with both contributing to the removal of effective capacity from the market.

— Several ships that had just loaded were already en route to buyers, and not yet available for re-routing, which also removed effective capacity from the market,

— Ships were largely out of position to service emerging trade routes resulting from cargoes and volumes available at other ports, which also limited effective available capacity in the market.

— Buyers of crude and refined oil products were scrambling to secure cargoes and volumes, resulting in several unusual long-haul (less efficient) trading patterns, tying up additional effective capacity. This explains why freight and shipping rates went up despite effective volumes of crude oil and other energy commodities being down.

Why rates retreated from peak levels as the closure of the Strait persisted…

— Some tankers were able to escape or leave the region to re-position to other trade routes.

— More volumes being diverted from the SoH via pipeline and truck before being exported at alternative ports.

— More countries started drawing down crude inventories rather than buying additional seaborne cargoes.

In the near-term, we expect shipping rates to move higher initially and then begin to normalize over time as trade routes move back to efficiency. Shipping rates tend to spike during periods of rapid change or disruption. We would expect rates to react positively, at least initially, in a sustained re-opening scenario. This is largely due to similar dynamics which characterise the initial closure including:

- Ships being out of position to immediately service volumes from inside the SoH and needing to re-position.
- Likelihood of confusion, congestion, and slow loading and transiting of the SoH during the initial re-opening.
- Several countries needing to restock inventory levels that were drawn down over the past several months.

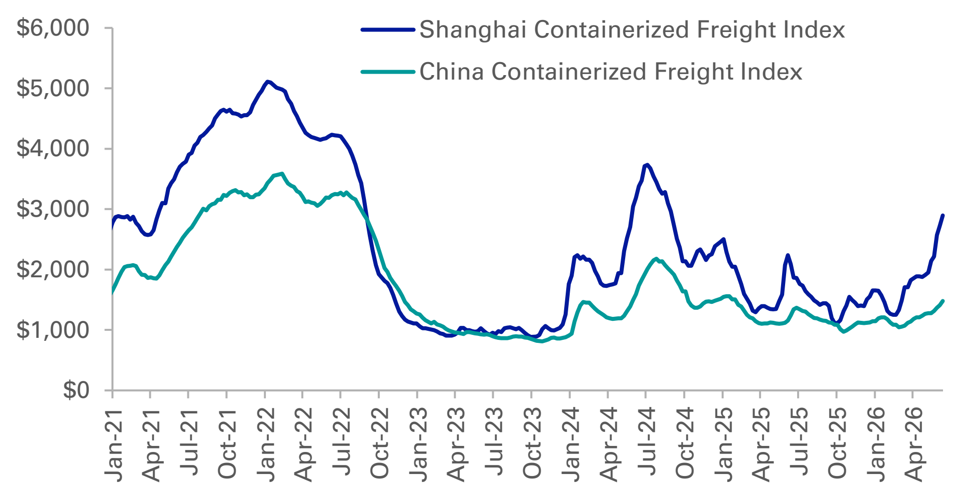
Impact of inventory and congestion will have an impact near-term but normalization in the SoH will not occur for 6-12 months, with longer term implications.

- Addressing damage to ports in the region will take some time and, in some cases, (e.g. LNG) several years.
- Longer term, owners and operators expect global trade to evolve towards supply diversification, regionalization and de-globalization. This may not necessarily imply less trade, but rather lower logistical efficiencies, more strategic storage requirements, and higher redundancies in supply chains. This combination should be fundamentally supportive for shipping tonne-mile demand and rates. When voyages are longer, loading/discharge is slower and less efficient, and vessels reroute or avoid unsafe or dangerous regions, the result is the same: less effective available supply

For container ships, the greater regional exposure is in the Red Sea and Suez Canal, which is important.

- There is a question of whether increased stability in the SoH could have a significant knock-on implication for the Red Sea. Given the Houthis' alignment with Iran, de-escalation in the SoH might lead to a reduction in aggression. This, in turn, could instil greater confidence among shipping companies to resume operations through the Red Sea-Suez Canal route, potentially reducing the rerouting of vessels around the Cape of Good Hope that has occurred in recent years.

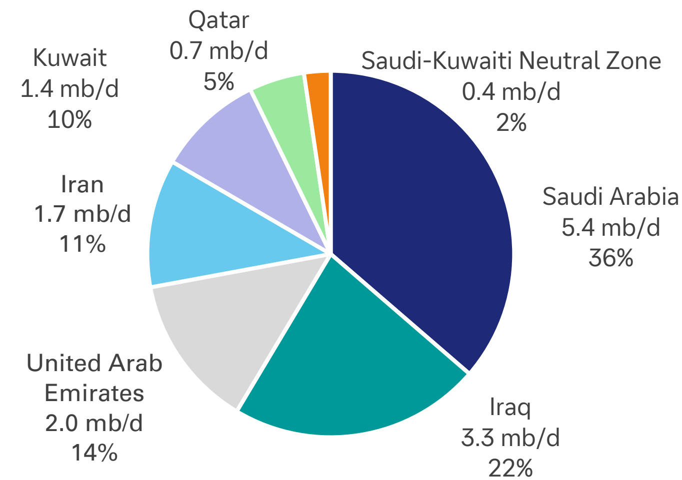
Container freight export rates
($/teu)



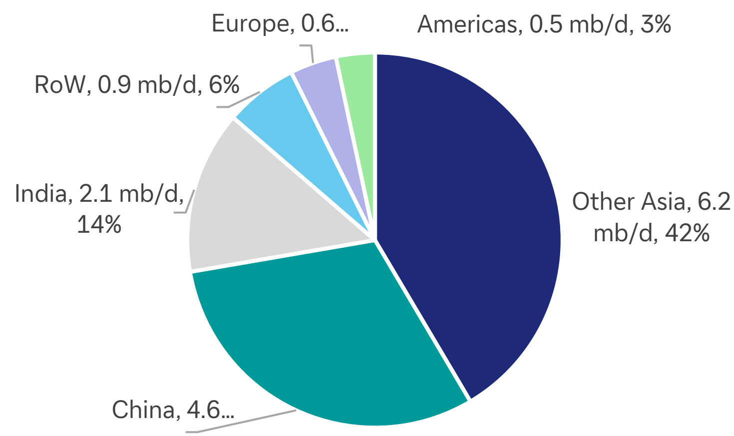
Source: Clarksons Research Services Limited, Shanghai Shipping Exchange, Deutsche Bank

- **Global relevance:** Gulf producers most exposed to the SoH (Saudi Arabia, Iraq, the UAE, Iran, Kuwait, Qatar, and Bahrain) accounted for a large share of globally traded crude. The key point is not just their production size, but export concentration: much of the world's marginal seaborne supply and most spare capacity sat behind a single chokepoint. Overall, the exports from the region accounted for ~22% of global oil demand in 2025.
- **Geographic exposure:** Asia is the main destination for Gulf crude, while Europe and the US were more exposed through price, refining and product-channel effects. For the US in particular, DB's macro framing is important: the issue is less headline energy independence than refinery-quality mismatch and global price pass-through.
- **Normal flow pattern:** in 2025, roughly 15-16 mb/d of crude transited the SoH. More than 80% was destined for Asia, with China and India alone taking roughly 45% of crude flows. Another 2.0-2.5 mb/d was exported through Saudi Arabia's East-West system from Yanbu on the Red Sea and the UAE's Habshan-Fujairah pipeline exiting on the Gulf of Oman. These were normally operating well below full capacity. Another ~5 mb/d of oil products was exported via the Strait.
- **Why crude mattered first:** unlike many industrial commodities, crude is globally fungible and forward-priced, so even a localized transit shock immediately reprices benchmarks, cracks spreads, freight and inflation expectations. That is why crude became the fastest macro transmission channel from the conflict into broader markets.

Crude oil exports via the SoH in 2025
(exports – 15 mb/d)



Crude oil exports via the SoH by destination
(% in 2025)



Source: IEA

War-Induced Disruption

The crude shock was not a single event; it moved through shipping, storage, infrastructure and then production.

1. Shipping paralysis and shut-ins

— As tanker liftings collapsed and onshore/floating storage filled, Gulf producers were forced into emergency shut-ins and operational curtailments.

— Not all shut-ins are cleanly reversible. Prolonged interruptions can reduce near-term output efficiency, delay well restarts and leave export systems restarting faster than upstream production.

2. Direct war damage

— Attacks or shutdowns affected KSA facilities around Ras Tanura/Abqaiq and the East-West system and caused repeated disruption at Fujairah/Habshan in the UAE, while there were wider strikes across Gulf refining, terminal and support infrastructure.

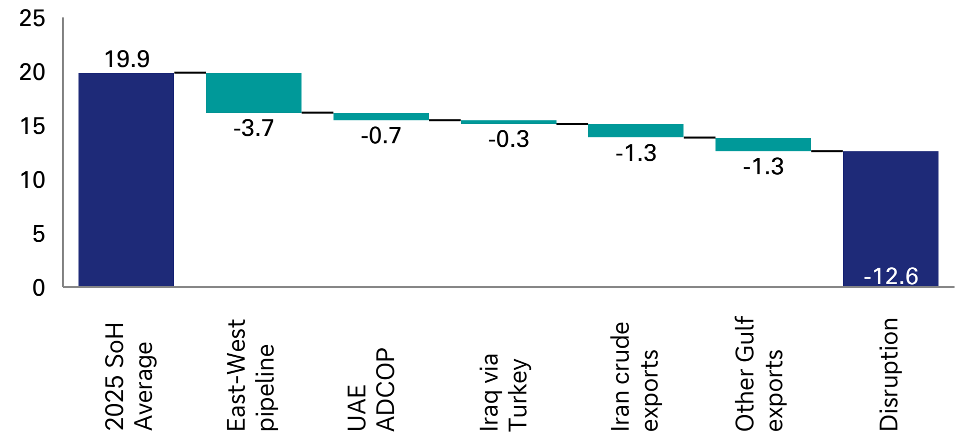
— Iraq avoided the worst physical damage but cut southern field output sharply as exports could not clear. Thus, the shock increasingly became physical as well as logistical.

3. Overall supply loss:

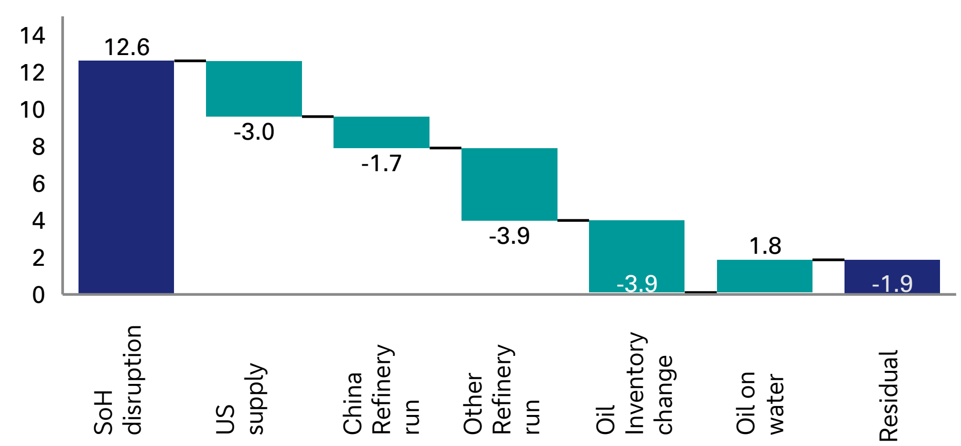
— In April, the estimated supply loss from the SoH region was 12.6 mb/d, compared to a 2025 baseline of 19.9. Approximately 4.4 mb/d of this loss was offset by rerouting crude through Saudi Arabia's East-West pipeline to Yanbu port and the UAE's ADCOP pipeline connecting to Fujairah port.

— The supply shock was cushioned by a sharp draw on inventories, especially in the US and lower refinery runs from China and rest of the world.

Physical disruption of oil supply in April
(millions of barrels per day)



Reconciling Gulf disruption & global inventory change (Apr)
(millions of barrels per day)



Source: IEA, Reuters, Bloomberg Finance LP, Kpler, Deutsche Bank

IEA stock release...

— The IEA's March coordinated action made roughly 400mn barrels of emergency stocks available to the market, with the US carrying a large share of the burden via a planned 172mn-barrel Strategic Petroleum Reserve (SPR) release over roughly 120 days.

US cushion thinned...

— The US SPR stood at roughly 415mn barrels at the start of the release programme, had already fallen to the mid- to high-300s by late May, and would fall to roughly 243mn barrels if the full planned release were complete, its lowest level since the early 1980s.

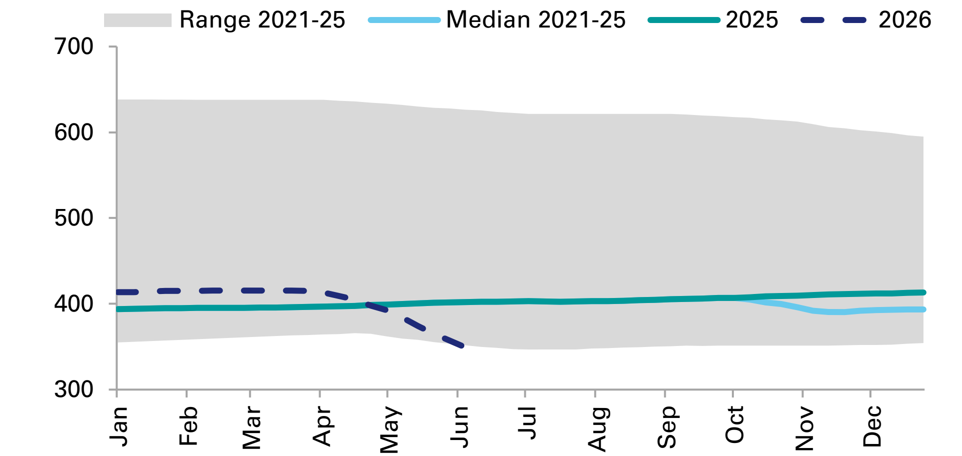
— Physical drawdown had also been running slower than awards suggested by late May, reflecting cavern and discharge constraints rather than a change in policy intent.

Broader system still had cover...

— Other major holders, especially Japan, Korea and parts of OECD Europe, retain substantial cover and remain above the IEA's 90-days-of-net-imports framework.

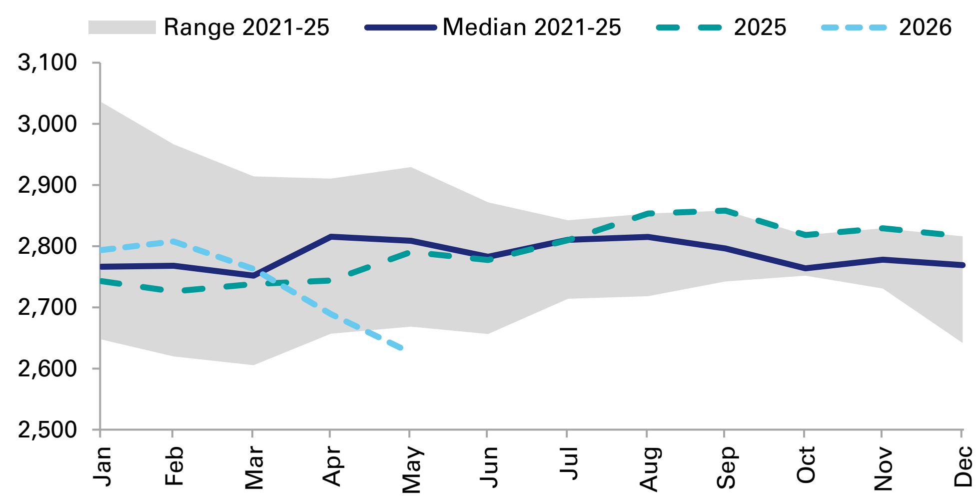
— At present, the system is not close to exhaustion in an aggregate sense, but the marginal buffer is clearly thinner than it was at the start of the crisis, meaning that if more meaningful transit totals were delayed significantly further, the market would shift increasingly from a stock-release story to a rebuilding-cost story.

US Strategic Petroleum Reserve (SPR) crude stocks
(2021 – 2026 YTD, millions of barrels)



Source: Bloomberg Finance L.P., Deutsche Bank

OECD commercial liquid inventories (5-year range)
(millions of barrels)



Source: Bloomberg Finance LP, Deutsche Bank

War-Induced Disruption

Workarounds helped, but only partly…

- Ultimately, the ramp of Saudi Arabia's East-West pipeline to Yanbu and the UAE's Habshan-Fujairah ADCOP pipeline approached 5 mb/d by May.
- These moves materially reduced effective oil export shortfall, especially for those two countries, but did not solve the problem for Iraq, Kuwait, Qatar, Bahrain or most Iranian exports. Thus, bypass routes cushioned the hit; they did not remove it.

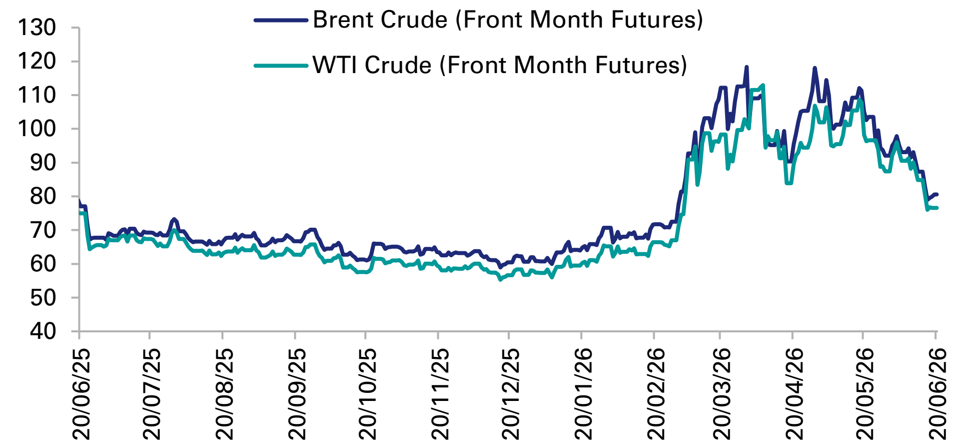
Higher prices drove non-Gulf incremental supply…

- US crude production and exports appear to have reached record levels during the crisis, and production in Latin America surged, which helped cushion the seaborne shortfall at the margin.
- However, the market's central constraint remained unchanged: spare capacity existed largely inside the Gulf, which meant even nominal OPEC spare barrels were not fully available to the seaborne market while the SoH remained impaired.

Higher prices supported demand destruction…

- The IEA's May assessment projects global oil demand to contract by about 420 kb/d year-on-year in 2026, with the sharpest weakness in 2Q26, when demand is expected to be down roughly 2.45 mb/d y-o-y.
- Petrochemicals and aviation appear to be the sectors most directly affected so far, but the broader macro point is that higher prices, weaker confidence and policy-driven conservation were already suppressing fuel use before full physical normalization had begun.

Brent & WTI crude oil front month futures
(USD / bbl)



Source: Bloomberg Finance LP

Physical recovery is multifaceted…

- Crude normalization depends on three separate clocks: i) repair of damaged upstream/export infrastructure, ii) restoration of loading and shipping through the SoH, and iii) the willingness of producers to unwind emergency shut-ins. Those clocks do not move at the same speed.
- Exports can rebound faster than production because some barrels were already produced or partially staged and simply waiting for a loading window.
- As ports clear and storage constraints ease, producers with usable infrastructure should be able to raise output,
- KSA spare capacity is the key swing factor. In principle it offers upside; in practice the market only benefits fully once both export routes and market belief are restored
- Full crude normalization requires a lower war-risk premia, steady tanker scheduling, repaired facilities where relevant, and confidence that the SoH will remain functionally open. That is why prices may normalize well before volumes and logistics fully do.

Key factors in the "new normal"…

- The need to restock inventory will be the most obvious one, with US refiners who have participated in the SPR exchange having to return 150-160 kb/d.
- There may be an inclination to restore depleted commercial inventories, while industry may aim to boost stockpile levels and establish new stockpile locations.
- The other new normal is that the UAE is no longer a member of OPEC but has pledged to act responsibly.

Price path and normalisation…

- Deutsche Bank's broad macro framing as presented in the World Outlook (WO) was consistent with a substantial Brent shock versus pre-war levels and a gradual path to normalization.
- In the WO baseline, Brent stayed elevated in 2Q before falling toward the mid-$80s in 4Q and ~$80 in 2027 as Hormuz reopens and the market moves back into surplus.
- Following the news last week of the MoU agreement and the partial reopening of the SoH, prices had dropped substantially faster than expected. This development likely reflected the combination of i) unwinding of the war/disruption premium, ii) signs Hormuz shipping and exports exceeded reported numbers recently and now could normalize sooner than assumed previously, and iii) some signs of easing demand pressure in China and elsewhere.
- However, news over the weekend that Iran has once again closed the Strait due to clashes in Lebanon has somewhat disrupted this narrative.
- Should traffic continue, a new sustainable path could see crude end the year in the $75-80 per barrel range (Brent). This will still be noticeably above pre-war levels as inventories and reserves are rebuilt. We caveat that this estimate comes with considerable uncertainty attached, with these concerns only amplified by the further closing of the SoH.

Input-cost transmission…

- On a broad level, DB's macro view is that the crude shock was large enough to lift headline inflation materially.
- Input cost pressures for exposed sectors should dissipate meaningfully within one to three months of a sustained SoH reopening as crude prices and freight normalize, but full downstream pass-through can take time and will unwind more slowly wherever inventories were replaced at elevated prices.

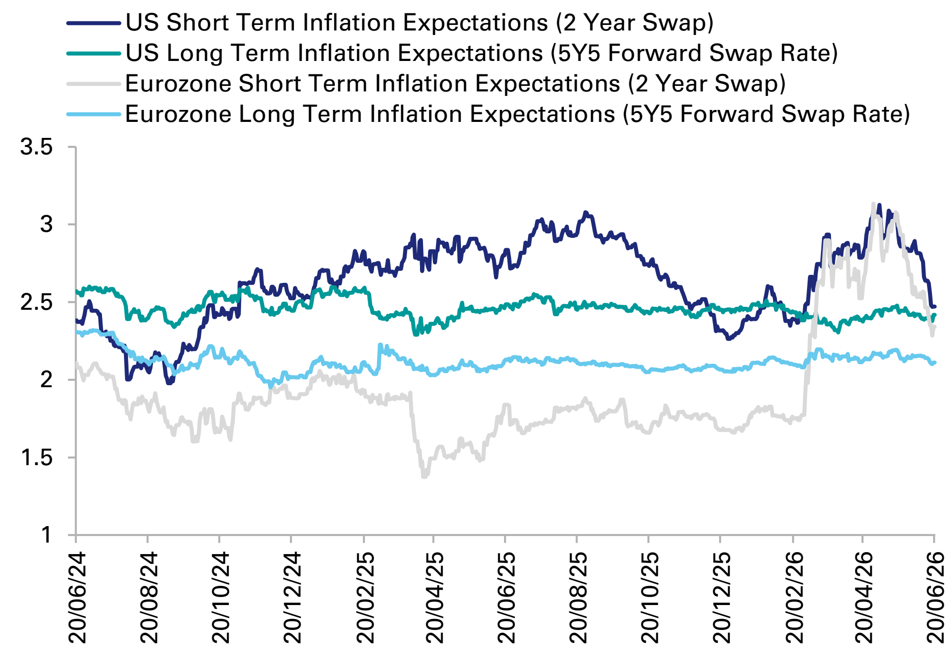
Consumer-price implications…

- In the scenario where oil moves gradually lower during a more sustained reopening of the SoH (with Brent crude ending the year in the $75-80 per barrel range), we expect some substantial easing in headline inflation.
- However, non-energy inflation will remain more elevated due to lagged pass through from earlier energy increases coupled with cost increases in other affected sectors.
- Consumer price impacts will be more pronounced in exposed oil importers, notably Europe and parts of Asia.
- Pass-through also varies materially across economies depending on fuel taxes, subsidies, regulation and retail fuel structures. As of early June, our view was that roughly half of that crude-driven impulse to consumer prices could unwind within 2-3 quarters after credible reopening if oil prices progressed into the mid-low 80s.
- A more sustained reopening of the SoH would point to a lower path for oil prices and a somewhat quicker unwind.

Bottom Line…

- Crude prices can normalize faster than physical flows, especially if geopolitical risk premia compress before inventories and trade routes fully normalize, but the inflation and earnings effects do not disappear as quickly because inventory replacement, refining margins and consumer fuel prices adjust with lags.
- That is why DB and external estimates both point to a sizable, but partially reversible, headline inflation impulse even under a constructive reopening scenario.

Inflation expectations across the US and Eurozone
(%)

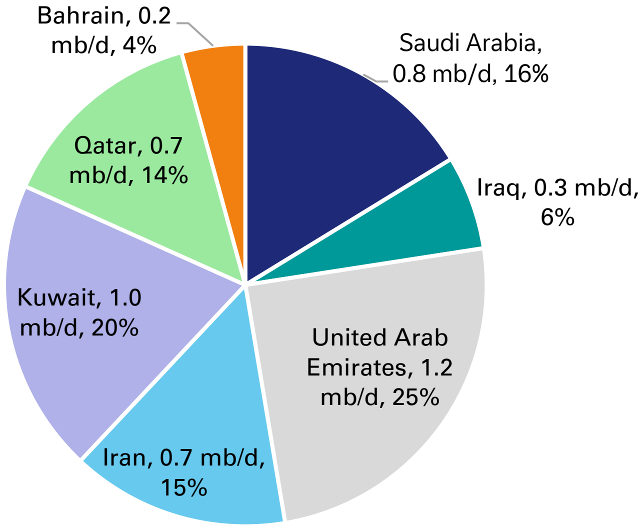


Source: Bloomberg Finance L.P., Deutsche Bank Research

Refined Petroleum Products
Pre-War Baseline

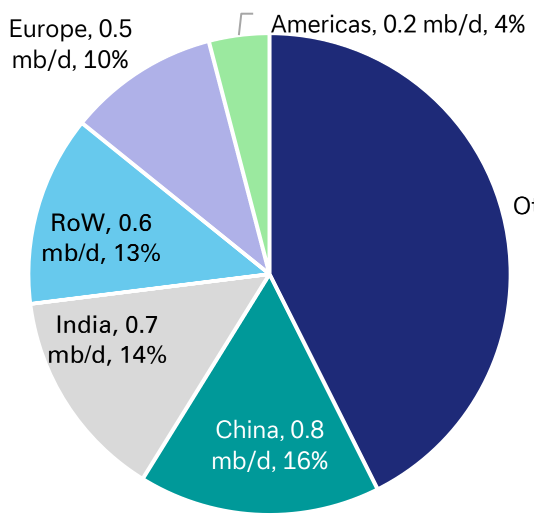
- **Global relevance:** before the war, the Gulf was not just a crude-exporting region; it was also a major supplier of refined products to global markets, especially middle distillates. In 2025, Gulf producers exported roughly 4.9 mb/d of oil products via the SoH, making the region materially important for jet fuel and diesel balances even where direct crude dependence was lower. Oil products include gasoline (petrol), diesel, jet fuel, kerosene, fuel oils, LPG, naphtha and asphalt.
- **Product mix and regional exposure:** the most important transmission channel was middle distillates, not gasoline. Europe had meaningful direct dependence on Gulf jet fuel and diesel, while Asia was exposed both directly through product imports and indirectly through crude feedstock losses to regional refineries. By contrast, gasoline markets were relatively better cushioned because they are broader, more Atlantic Basin-linked and somewhat easier to rebalance.
- **Why products mattered differently:** refined product markets are less fungible than crude because the shortage is product-specific. Refiners cannot easily re-optimise output toward jet fuel or diesel at scale, which is why cracks and regional physical premia moved much more violently than benchmark crude during the disruption.

Oil products exports via the SoH in 2025
(exports - 5 mb/d)



Source: Bloomberg Finance L.P., Deutsche Bank Research

Destination Share of oil products exports via the SoH
(2025)



Source: Bloomberg Finance L.P., Deutsche Bank Research

War-Induced Disruption

The refined-product shock came through three channels at once: (i) direct refinery outages in the Gulf, (ii) the loss of export routes for products that were still being made, (iii) the feedstock squeeze imposed on refineries outside the region that relied on Gulf crude. This combination made products tighter than crude in several regional markets.

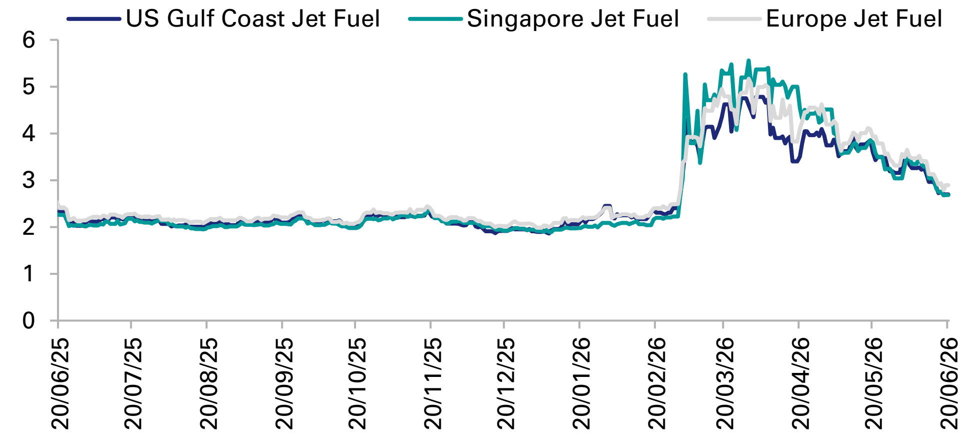
Market effects from export collapse and physical damage

- Pre-war Gulf exports amounted to roughly 3.3 mb/d of refined products, but by March/April much of that flow had collapsed alongside Hormuz shipping.
- The IEA estimates more than 3 mb/d of refining capacity in the region was shut down because of attacks and a lack of viable export outlets, while refinery runs outside the Gulf were also constrained by feedstock shortages.
- Direct physical damage mattered most in Kuwait and Bahrain, where repeated strikes hit refining assets at Mina al-Ahmadi, Mina Abdullah and Sitra/Bapco.
- Saudi Arabia and the UAE saw important but mixed disruption through hits to Ras Tanura, SAMREF and the wider Fujairah/Ruwais logistics-refining chain.

Jet fuel surged on capacity concerns, driving prices higher, though ultimately only Asia struggled to get supply.

- Per a report by Boston Warwick, the closing of the SoH disrupted ~25% of European jet fuel supplies. Unlike crude oil, jet fuel cannot easily be substituted, leading to a sharp decline in jet fuel stocks during March and April.
- While there were clearly scheduling decisions impacted in Europe, issues with fuel availability was more pronounced in Asia leading to flight cancellations.
- By mid-April, there were fears European inventories could fall to critical levels. Airlines responded with measures such as the cutting of marginal capacity and introduction of fuel surcharges, with these policies spurring sufficient demand reduction to prevent widespread shortages.
- By early June, the market had stabilised due to measures including: i) The emergence of US Gulf Coast refiners as a key swing supplier, with exports redirected to Europe, ii) European refiners increased output of jet fuel, iii). Maintained capacity discipline by airlines.
- The current dynamic is stable, but vulnerabilities persist. The present equilibrium is largely due to ongoing intervention rather than a return to pre-crisis conditions. Larger hub airports have proved more resilient; regional and leisure-focused airports remain more exposed.

Jet fuel pricing
(USD/Gallon)



Source: Bloomberg Finance L.P., Deutsche Bank Research

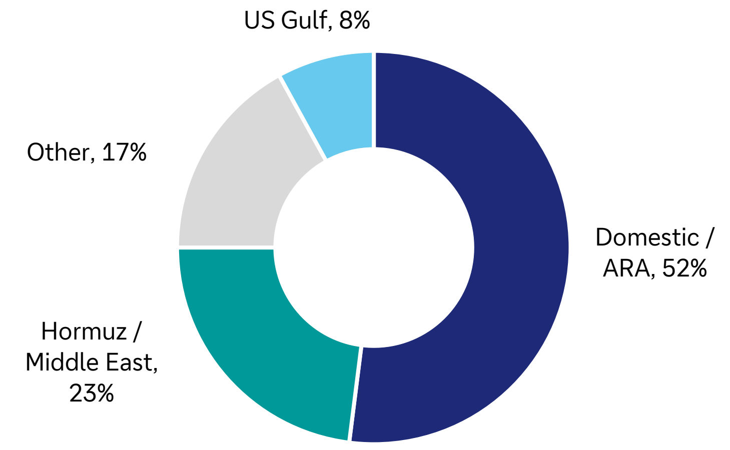
Recovery Timelines

- Product flows can restart faster than refinery output fully normalizes if undamaged barrels are already in storage or partly staged for export.
- In practice, the earliest improvement should come from the resumption of insured product tanker loadings rather than from a full recovery in Gulf refinery runs.
- As crude flows recover through Hormuz and bypass systems, refineries in Asia and elsewhere should be able to raise runs and partially rebuild jet fuel and diesel availability.
- Middle-distillate normalization is likely to lag crude because product balances need both feedstock and refinery operating stability.
- Full normalization in products requires repaired refinery and terminal infrastructure where damaged, lower freight and war-risk costs, and time for regional inventory buffers to rebuild.

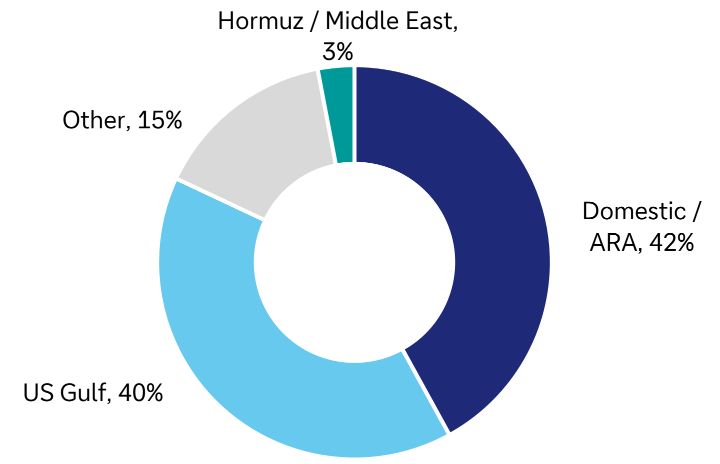
The pre-crisis European jet fuel model is unlikely to return

- While a sustained reopening of the SoH would undoubtedly be positive for European jet fuel pricing, supply recovery from the region will be gradual.
- A report by Boston Warwick in June, opined a (e.g., the US) return to previous supply patterns is unlikely. New supply from other regions, operational changes, and a reassessment of risk have encouraged a more diversified approach to sourcing fuel.

European jet fuel supply shares pre-crisis (early 2026)



Current European jet fuel supply shares (mid-June 2026)



Source: Boston Warwick

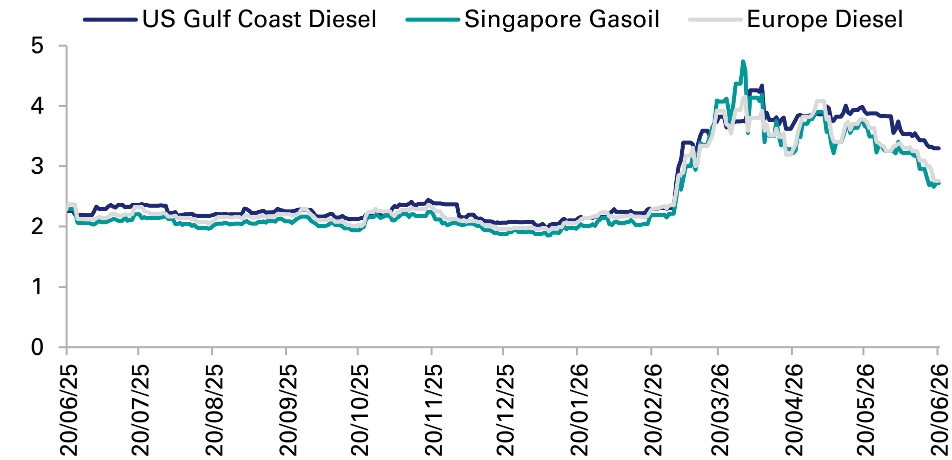
Price path and dissipation

- DB analysis and market reporting suggest the refined-products shock was sharper than the crude shock, especially in jet fuel and diesel. While exposed prices have dropped from peak levels, they likely continue to have a high degree of sensitivity to oil prices/supply.
- Peak pricing distortion could continue to unwind with a credible reopening (which we are already seeing), but middle-distillate tightness may persist into late 2026 if refinery repairs and inventory rebuilding lag.

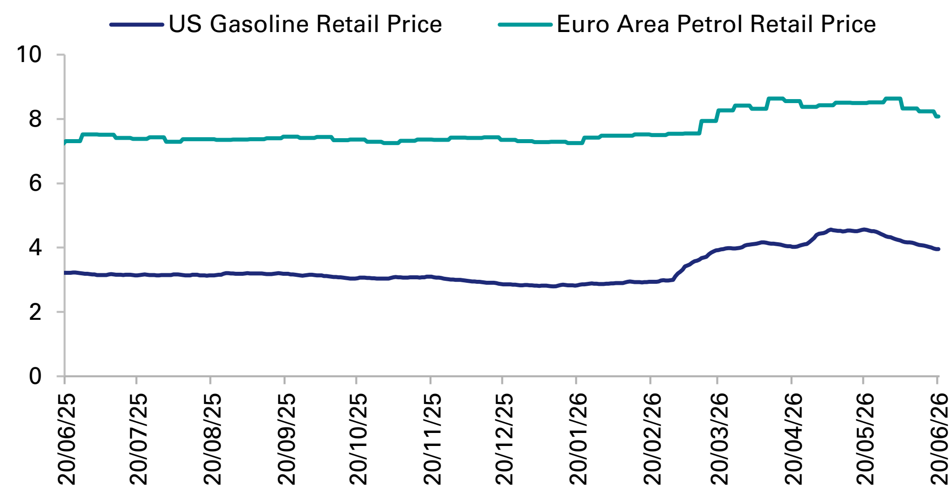
After more than doubling during the crisis, jet fuel prices have receded in recent weeks; post crisis implications point to a new reality for European jet fuel.

- The shift to a system that places greater emphasis on domestic European refining and imports from the US offers greater resilience at the expense of higher structural costs (also noted by Boston Warwick). Relative to Middle-East supplies, US Gulf jet fuel possesses a material freight premium due to the longer voyage needed to import it. Increasing cost curve.
- Boosting domestic European production has necessitated configuration changes and higher crude input costs. Consequently, these alternative sources come at a premium, meaning even after a full SoH reopening, European costs will likely be higher than pre-crisis.

Diesel
(USD / Gallon)



Gasoline
(USD / Gallon)



Source: Bloomberg Finance L.P., Deutsche Bank Research

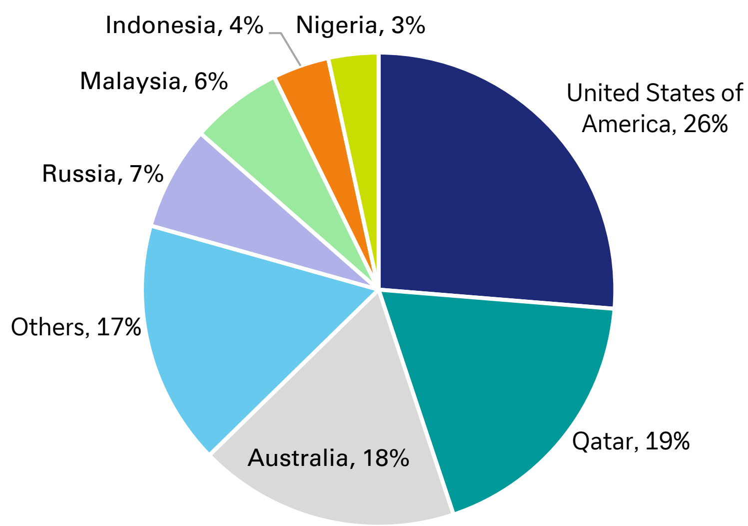
Liquefied Natural Gas (LNG)
Pre-War Baseline

— **Global relevance:** before the war, the LNG market was already tighter than many investors appreciated. Qatar and the UAE together supplied roughly 110–112bcm of LNG through Hormuz in 2025—close to one-fifth of global LNG trade—with Qatar overwhelmingly dominant via Ras Laffan. For transparency, that is broadly equivalent to around 80-plus mtpa from Qatar plus a much smaller UAE contribution when converted into gas-equivalent volumes. The market implication was straightforward: a single regional chokepoint sat in front of one of the world's largest baseload gas supply systems.

— **Market structure and exposure:** ~90% of Hormuz-linked LNG exports were normally destined for Asia, making Northeast and South Asia the most directly exposed regions, while Europe was more exposed through price competition for Atlantic cargoes than through direct Gulf dependence. LNG dynamics also differ from oil: once liquefaction stops, storage fills quickly, cargoes cannot be rerouted easily, and the chain from upstream gas to regasification becomes much less flexible.

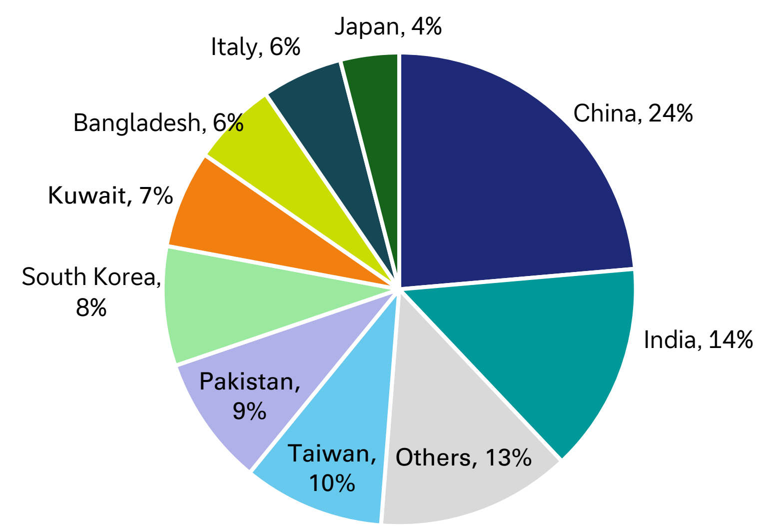
— **Why LNG mattered disproportionately:** LNG supply is concentrated, contract-heavy and physically rigid. That means a disruption in Qatar does not just lift benchmark prices; it reshapes regional netbacks, forces buyers into fuel-switching and demand curtailment, and quickly turns Asia into the market's balancing point.

Global LNG exports by exporting country (2025)
(Total 442mn metric tonnes)



Source: Bloomberg Finance L.P., Deutsche Bank Research

Qatar LNG exports by importing country (2025 delivered)
(Total 75mn metric tonnes)



Source: Bloomberg Finance L.P., Deutsche Bank Research

War-Induced Disruption

The LNG shock came through both shipping and infrastructure at once. The effective closure of Hormuz halted LNG tanker traffic, while attacks on Ras Laffan shut liquefaction trains and resulted in force majeure on long-term contracts. Unlike crude, this was not a market where floating storage and rerouting could absorb the shock for long.

Scale and location of damage

— The key structural hit was at Ras Laffan in Qatar, where reported strikes damaged two of fourteen LNG trains, taking roughly 12.8 mtpa, about 17% of Qatar's LNG export capacity, offline for what QatarEnergy and market analysts describe as a multi-year repair cycle.

— UAE LNG exports from Das Island were also constrained by shipping and storage limits, but the market's central physical loss was Qatari. Force majeure and cargo disruption were especially relevant for buyers in China, South Korea, Italy and Belgium, highlighting how concentrated the downstream exposure was among large contract importers.

Outside-region response

— The US, Australia and other exporters could tighten utilization and redirect cargoes, but they could not replace the lost Gulf volumes quickly at scale. In practice, the adjustment came more through demand destruction and fuel-switching than through a clean supply substitution. This was especially in South and Southeast Asia.

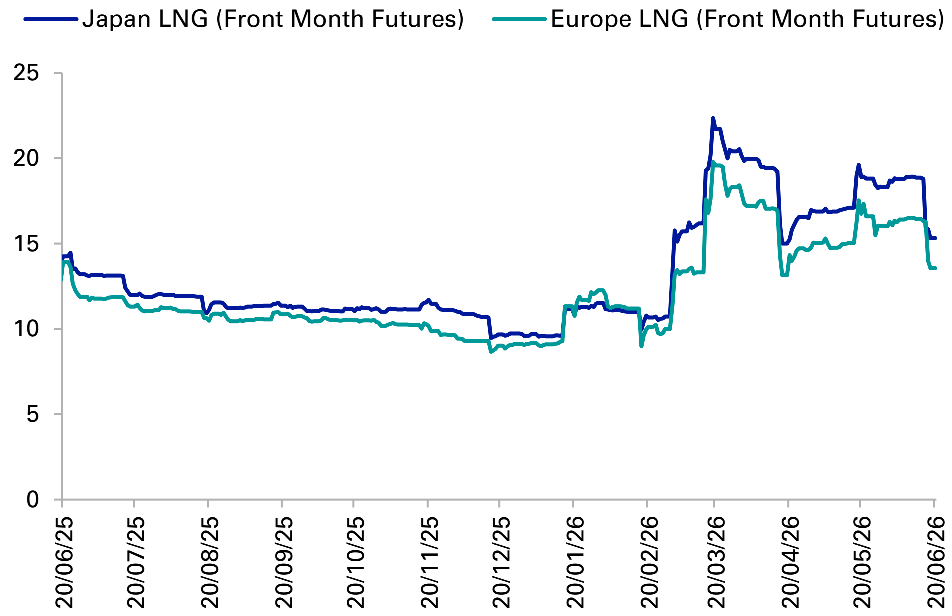
Production and exports drop

— The IEA estimates the disruption removed close to 20% of global LNG tradeable supply from the market at peak and also drove observed global LNG production down about 8% year on year in March.

Regional adjustment

— Asia bore the sharpest physical stress, Europe retained more optionality through Atlantic sourcing, and both regions responded with price-led demand adjustment.

LNG pricing
(USD / MMBtu)



Source: Bloomberg Finance L.P, Deutsche Bank Research

Deutsche Bank Research Institute

Restart is slower than oil

— LNG recovery depends on more synchronized systems than crude, including feed gas, liquefaction trains, refrigeration equipment, storage tanks, loading arms, specialized carriers and downstream regasification schedules.

— Consequently, reopening Hormuz is necessary but not sufficient; LNG can resume only as shipping and plant reliability recover together.

Phased recovery path

— Unaffected Ras Laffan trains could restart in phases once secure tanker transit resumes, but full system normalization is likely to lag into late summer even in a constructive scenario.

— More importantly, the damaged trains represent a structural loss: market reporting indicates roughly 12.8mtpa may remain offline for three to five years

— The damage to these trains implies that post-reopening flow normalization and medium-term supply normalization are not the same thing.

Bottom line

— LNG shipping can restart faster than the market fully normalizes, but the combination of physical damage at Ras Laffan and LNG's inherently rigid logistics means gas balances are likely to stay tighter for longer than the crude market.

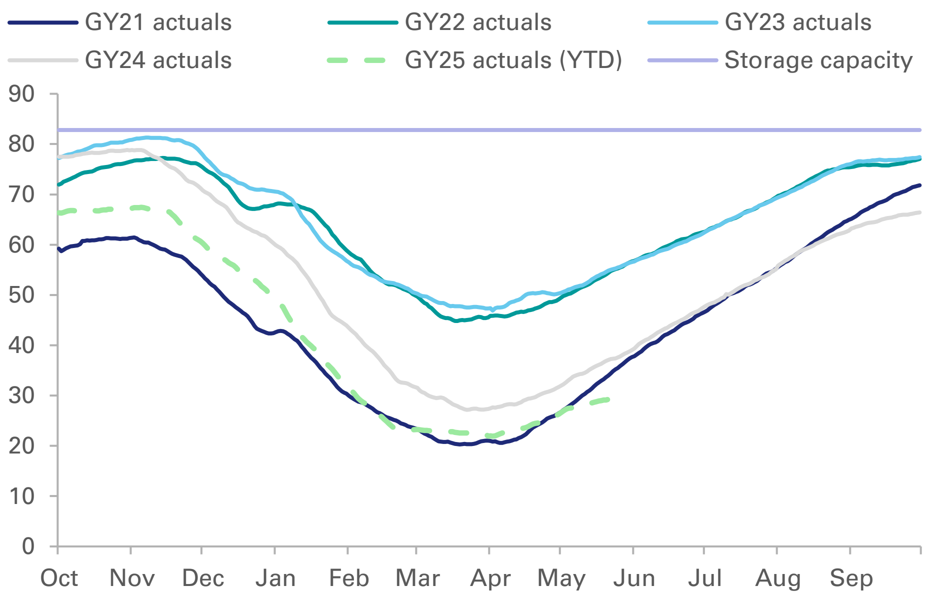
Price path and persistence

— LNG likely delivered the most persistent energy-price shock after crude and products.

— With roughly 19–20% of global LNG trade disrupted at peak and some Qatari capacity structurally impaired, spot LNG prices likely overshot well beyond the immediate physical loss.

— Even after reopening, prices should normalize only gradually because LNG shipping can resume faster than full liquefaction and contract performance normalizes.

European underground gas storage inventories
(billion cubic meters)



Source: Bloomberg Finance L.P., Deutsche Bank Research
Note: GY represents gas years. Europe definition as per BNEF Europe Perimeter.
GY25 actuals through May 24th.

Pre-War Baseline

The importance of the Middle-East to the global fertiliser market

- The Gulf is a major production and export hub for fertilizer markets, making up ~1/4 to 1/3 of globally traded volumes across the main nutrient and feedstock chains.
- The Strait of Hormuz matters not only for fertilizer shipments themselves, but also for globally relevant inputs that sit upstream of fertilizer production including for farm/food costs.

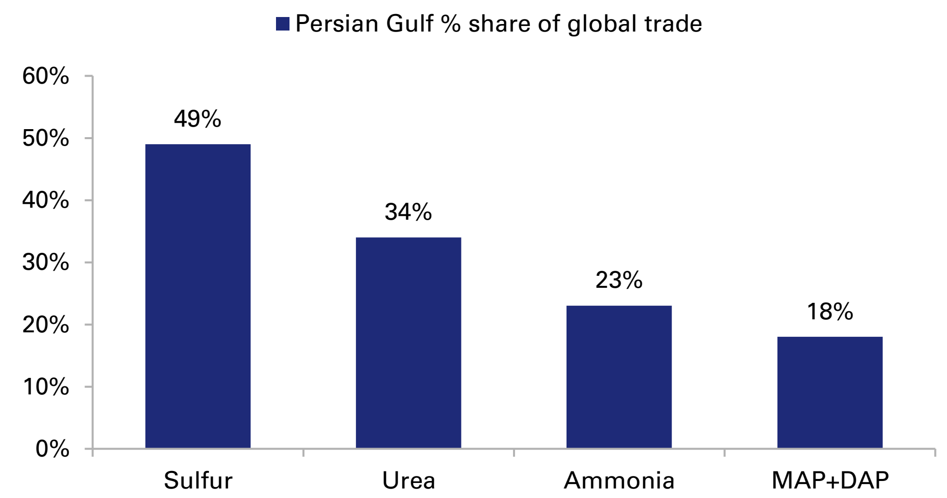
Product-specific exposure

- **Sulfur:** nearly half of global sulfur trade is Hormuz-linked, making it an important component because it is a critical feedstock for phosphate production globally (phosphate is one of the three key fertiliser nutrients).
- **Urea:** roughly 34% of global urea trade is linked to the Gulf (it is made from natural gas which is cheap in the Gulf). Urea is the most important nitrogen fertiliser.
- **Ammonia:** around 23% of global ammonia trade is Hormuz-linked. Ammonia is important as a fertilizer product and as an intermediate for downstream nitrogen markets.
- **Phosphates (MAP/DAP):** around 18% of global phosphate trade is linked to the Gulf. This is smaller than nitrogen and sulfur in trade-share terms but still meaningful.

Geographic exposure

- Saudi Arabia, Qatar, and Iran are the three most important suppliers, though other Gulf countries also feature.
- By destination, South Asia, Brazil, parts of Africa and segments of Latin America are the most directly exposed import regions as they rely heavily on seaborne fertilizer and feedstock flows linked to the Gulf.
- The US is more exposed through fertilizer pricing than through direct ammonia dependence.
- Europe is more exposed through second-round input-cost and food-price effects rather than through direct fertilizer dependence.

Persian Gulf region share of global fertilizer trade (2024)



Source: International Fertilizer Association

Supply shock mechanics

— The fertilizer shock came through shipping, feedstocks (gas) and production simultaneously.

— Commercial paralysis in Hormuz trapped exportable urea, ammonia, sulfur and phosphates behind the strait, while higher gas prices and some plant outages raised production costs across nitrogen markets.

Scale and location of disruption

— The biggest immediate hit was to tradability rather than outright destruction of fertilizer production facilities.

— Finished fertilizer exports from all Gulf suppliers were severely disrupted, while there were supply losses in Qatar, Iran, and Bahrain.

— Blocked Gulf shipments of sulfur reduced feedstock availability for fertilizer producers in places like Morocco, China and elsewhere.

— Reported missile and drone strikes around Ras Laffan and broader Gulf industrial sites mattered, but shipping logistics were still the dominant constraint.

Inventories and buffers

— Given that there is no true fertiliser equivalent of a strategic petroleum reserve (long term storage can be problematic), inventory matters more and policy backstops matter less.

— Timing can be a key macro amplifier in the sector (crop-calendar constraint). The market reaction has so far been less extreme than in 2021–22, in part because the Northern Hemisphere spring planting had already passed its most acute procurement phase before the disruption peaked, moderating the immediate hit,

— However, inventories are uneven, sulfur stocks are less forgiving, and the cushion will thin quickly if the conflict persists into future application windows.

Likely agronomic effect

— The main risk is not that crops fail immediately, but that farmers economize on application rates, switch to less input-intensive crops, or accept lower yield potential.

— The most severe effects may have been avoided so far, but this is a particularly relevant consideration for corn, wheat, and rice where nitrogen and phosphate intensity is high and substitution is limited.

Regional next watchpoints

— The next deadlines that matter are southern hemisphere pre-plant purchasing, African import tenders and India's fertilizer procurement for the monsoon cycle.

— If Hormuz-related friction persists into 2026Q3, the shock becomes less about spot fertilizer prices and more about 2026/27 crop supply and food inflation.

Recovery Considerations

Production recovery is uneven

- Where fertilizer plants are physically intact, production can recover relatively quickly once gas, sulfur and export logistics normalize.
- The larger issue is not plant rebuild time but restoring economically viable trade flows. Nitrogen output can in principle rebound faster than phosphate balances normalize, because sulfur bottlenecks can continue to constrain downstream phosphate production even after shipping improves (there is a longer chain that needs to restart).

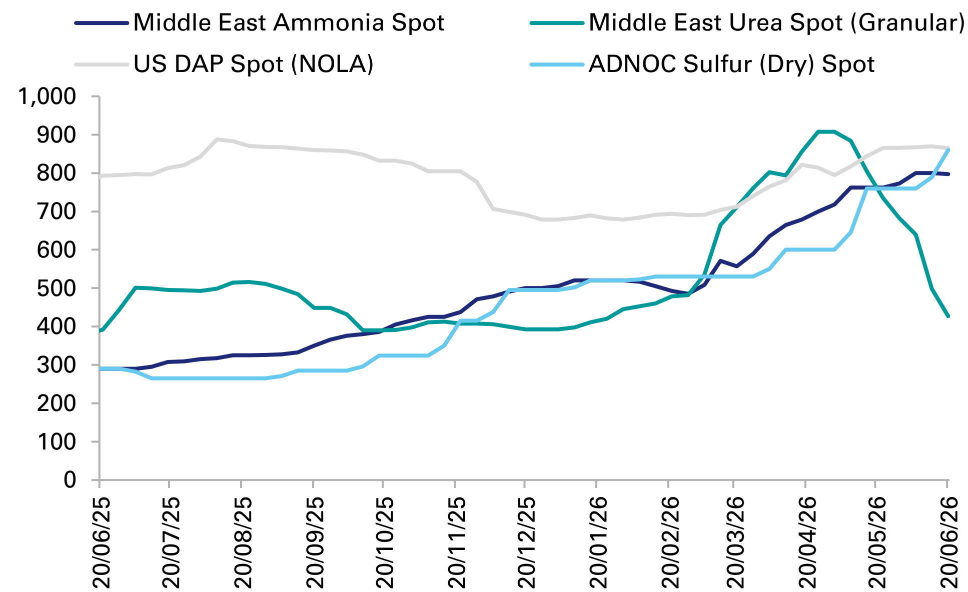
Recovery timeline

- Fertilizer shipping can restart faster than agricultural impact would normalize.
- Initial vessel movement could resume within days of a credible reopening, but price normalization is likely to lag (e.g. in DAP) while insurers, charterers and buyers rebuild confidence.
- More importantly, agronomic normalization lags physical normalization: theoretically missed applications or altered crop choices in one season cannot be fully reversed just because product flows recover later.
- All of that said, it is worth noting that the timing of the season and reasonable nitrogen availability has kept urea prices at a lower level, meaning as it stands urea "normalisation" upon an end to the conflict would be fairly limited.

Why fertilizers transmit differently

- Unlike oil, fertilizer demand is constrained by crop calendars rather than there being continuous industrial consumption.
- Farmers can defer or reduce application, but this comes at the cost of lower yields, weaker crop quality or shifts in crop mix.
- Timing is key, as even if physical fertilizer supply later normalizes, a missed planting or top-dressing window cannot always be recovered within the same season.

Fertilizer spot prices
(USD / MT)



Source: Bloomberg Finance L.P., Deutsche Bank Research

Price path and agricultural timing...

— So far the fertilizer shock has been less explosive than 2021–22 in headline price terms, while timing and availability has also been supportive.

— A reasonable peak range is +20–90% for urea/ammonia and +15–40% for phosphates and sulfur-linked chains versus pre-war norms, with the largest risks shifting into the next planting and top-dressing cycles rather than immediate crop failure.

— The key macro point for the sector is crop-calendar timing matters more than spot prices alone: even if trade flows normalize within weeks of reopening, missed application windows and altered crop choices could still feed through into yields, farm profitability and food inflation with a lag.

Earnings and sector effects...

— Fertilizer producers with non-Gulf production and sulfur access have been relative winners, while farmers, import-dependent distributors, and some food processors have faced margin pressure.

— The main macro loser is downstream agriculture, where the earnings hit arrives through lower farm profitability, lower acreage, or lower yields.

Combined fertilizer and diesel effect...

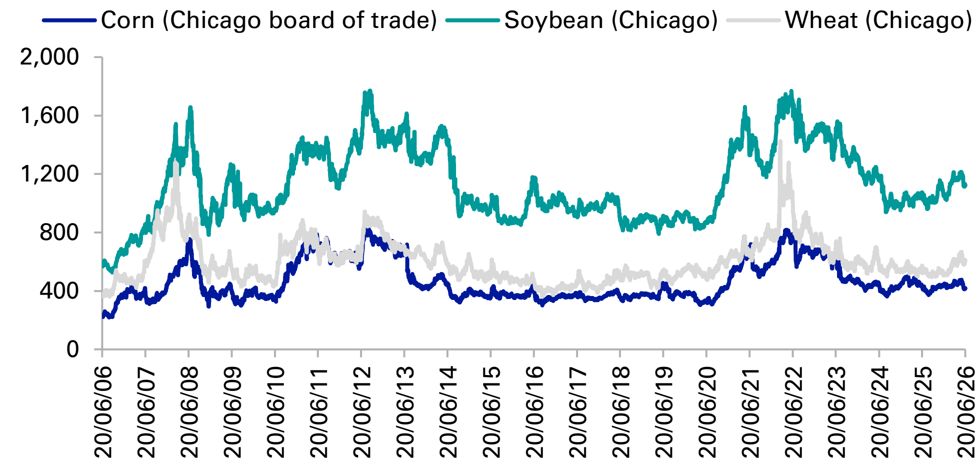
— The consumer impact is broader than fertilizer alone, because higher diesel prices also raise the cost of running farm machinery, irrigation and harvesting equipment, and of transporting grain, produce and food products through storage, distribution and retail chains.

— Physical fertilizer trade can normalize within weeks of reopening, but food-price dissipation usually takes one to three crop cycles because farm-input, distribution and retail-price adjustments all occur with lags.

The key question is the durability of a peace deal...

— The longer supply is restricted the more there is a risk that fertiliser prices increase more significantly than they already have. In particular, there is upside risk to nitrogen prices since price spikes have really been reduced so far due to fortunate timing. Of course, if fertilisers spike even more from here there is additional risk to crop output and hence crop prices and food inflation.

Key crop prices
(USd/bu)

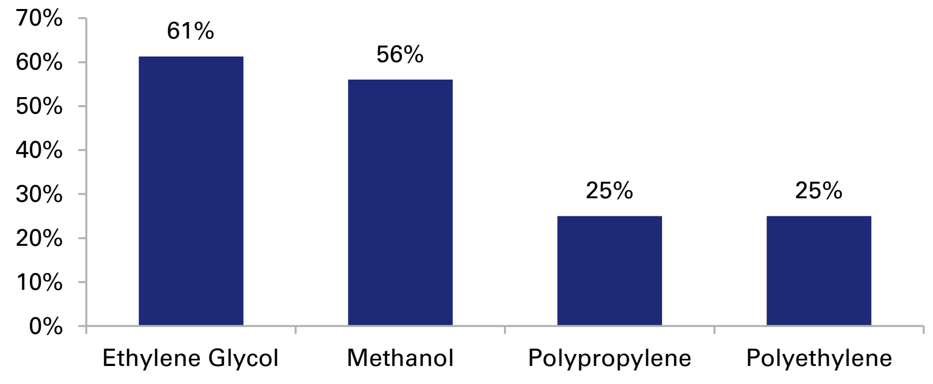


Source: Bloomberg Finance L.P., Deutsche Bank Research

Pre-War Baseline

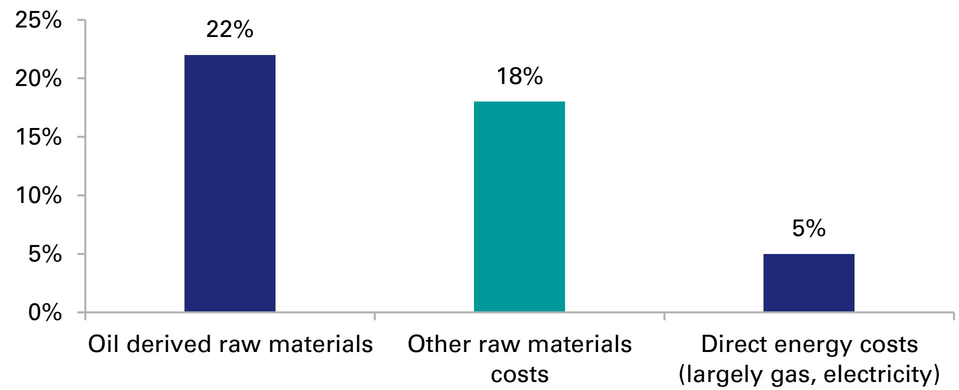
- **Global relevance:** The Gulf is one of the world's most important export hubs for petrochemicals and polymers. Its access to low-cost hydrocarbons (oil and gas) underpins its position as a globally competitive producer of upstream chemicals. These upstream chemicals are used as feedstocks for downstream production worldwide, with Asia in particular acting as a major import market. The most important products include methanol, polyethylene, ethylene glycol and polypropylene, while the region is also central to the supply of naphtha, LPG and other feedstocks used by chemical crackers and other derivative plants across Asia and Europe.
- **Middle East supply:** Importantly, 24% of global naphtha supply and 29% of global seaborne trade in LPG transits via the SoH, respectively. Saudi Arabia is the dominant regional chemicals player, led by Jubail-based production, with meaningful additional exposure in Qatar, the UAE, Kuwait and Iran. By destination, China and the rest of Asia are the most directly exposed to the conflict in the Middle East because they depend on Gulf feedstocks and polymers both for domestic manufacturing and for export-oriented conversion industries, while Europe is more exposed through feedstock cost (higher gas and oil prices) and margin pressure.
- **Why petrochemicals matter macroeconomically:** petrochemicals are used in a wide range of downstream manufactured goods and can cause disruption to several markets include packaging, consumer electronics, auto parts, appliances, textiles, construction materials, industrial intermediates and other parts of the economy. For investors, the key point is that this is not just a chemicals-sector shock; it is a broad manufactured-input shock with downstream margin implications.

Persian Gulf share of global petrochemical trade
(% share of global trade)



Source: S&P International Trade Publications

Chemical company exposure to key inputs
(% of sales)



Source: Deutsche Bank research

War-Induced Disruption

The petrochemical shock came through both exports and feedstocks. Hormuz restrictions cut off shipments of methanol, polyethylene, ethylene glycol and polypropylene, while the same conflict also tightened naphtha and LPG availability for crackers in Asia. The combination of these factors matters because it hits both the product itself and the feedstock base needed to replace it elsewhere.

Scale and location of disruption…

— The biggest initial impact was a tradability shock, but there has also been some physical disruption through strikes (e.g. Iran, Asaluyeh).

Inventories and substitutability…

— There is no strategic reserve system for petrochemicals (there are too many of them), with inventories varying sharply by chemical chain.

— Most chemical markets entered 2026 in cyclical oversupply, which may have reduced the shock, but that cushion has been thin in certain areas; for example in feedstocks such as naphtha and in products such as methanol where seaborne trade is concentrated in the Middle East. The result is a classic sequence: spot disruption first, then contract repricing. In theory this is likely to be followed by downstream margin pressure.

Geographic exposure…

— Asia is the most directly exposed region with Korea, Japan and India relying heavily on Gulf-linked feedstocks and polymers for manufacturing chains.

— China has been a large importer of chemical feedstocks from the Middle East, but it has more flexibility in sourcing (e.g. it can switch to coal-based chemicals in areas, and it can use its strategic reserves as support for some time).

— European chemicals is not that reliant on the Middle East in terms of direct imports but overall, higher gas and oil prices mean more expensive feedstocks, which companies try to pass into pricing.

— The US can offset some of the shortages, for example, in polyethylene and ethylene glycol, but it does not have sufficient production to restore global balance quickly.

Recovery timeline…

— Facility recovery is uneven. Where outages reflected shipping paralysis or precautionary shutdowns, restart can be relatively fast once feedstocks, labor and export routes normalize.

— Recovery will be slower for integrated hubs that suffered direct damage with crackers, derivative units and utilities often needing to restart in sequence.

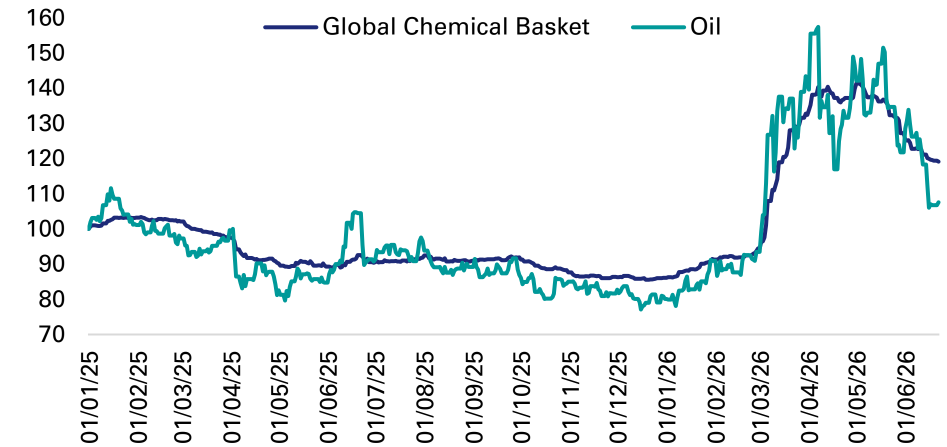
— Chemical companies tend to estimate a 6-9 month time-frame for supply chains to normalize upon the end of the conflict (consultants lower at 4-5 months).

Shipping and market normalization…

— Chemical and polymer cargoes can begin moving again relatively quickly after a credible reopening, but market normalization is likely to lag because feedstock costs, insurance premia and contract renegotiation all adjust more slowly than vessel traffic.

— In practical terms, methanol and some polymers may see trade flows recover before margins normalize, while integrated downstream chains could remain impacted into late 2026 even in a constructive reopening scenario.

Chemical pricing
(1st January 2025 = 100)



Source: Bloomberg Finance L.P., Deutsche Bank Research

Price path and manufacturing sensitivity

— Petrochemical price reactions to the conflict were mixed with more pronounced reactions in areas like methanol, ethylene glycol and select polymers, while price reactions were less dramatic in oversupplied chains. That said, overall prices broadly tracked oil…

Winners and losers…

— The winners are US-based chemical players who could take advantage of higher chemical prices while feedstock prices in the US remained low compared to other regions. This has particularly been the case for commodity focused chemical producers.

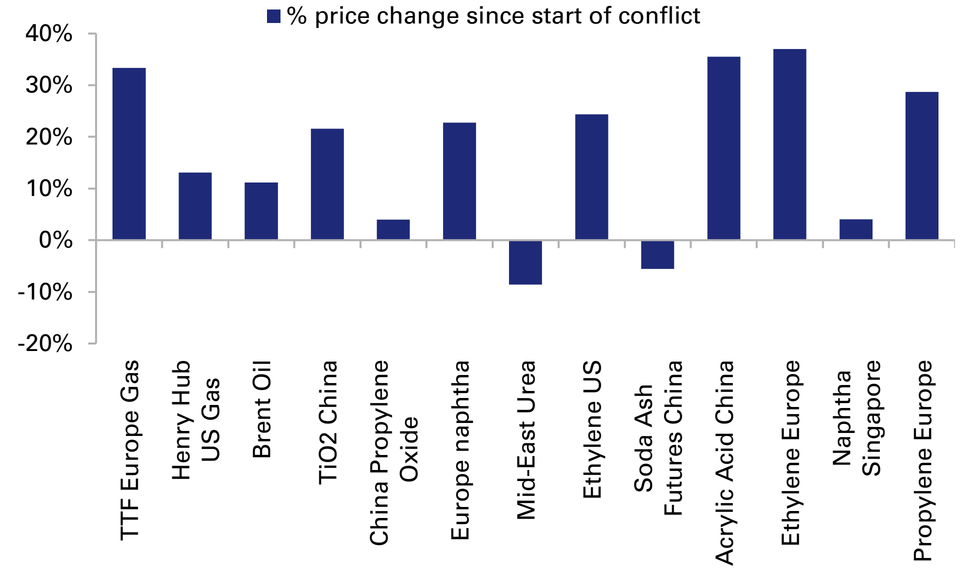
— For European players, some chains have been pressured by higher costs, but at the same time demand has remained at a stable level, perhaps supported by market share gains at the expense of those unable to provide supply.

— In Asia, this is where some companies are feeling the effects of feedstock shortages and market share losses, although if companies can find feedstocks (e.g. by switching to coal or importing from the US) profitability can also be strong.

— While downstream packagers ultimately pass through price increases to customers, many will choose to hold bigger inventories on supply concerns resulting in short term cashflow impacts.

— The main investment implication is that for companies who have seen exceptional profits linked to the conflict there is likely to be normalization when the conflict ends.

Key chemicals-related price changes since start of conflict



Source: Bloomberg Finance L.P., Deutsche Bank Research

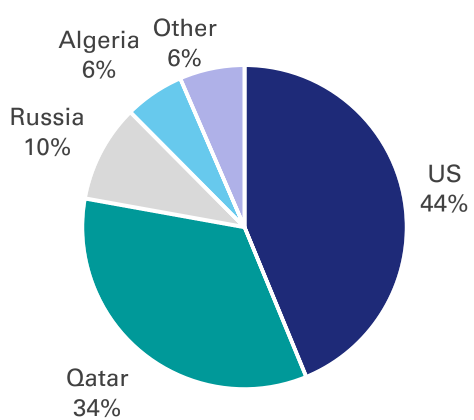
Pre-War Baseline

— **Global relevance:** Helium is a small market in value (~$4.5B pre-conflict) but strategically important due to some unique properties (low density, low boiling point, small molecule size, excellent heat-conducting capabilities) which make it irreplaceable in certain high-tech and medical applications including semiconductors, magnetic resonance imaging (MRI), scientific equipment, aerospace systems, welding and fiber-optic processes.

— **Difficult and expensive:** The extreme requirements needed to refine helium mean only a handful of countries have significant reserves and production capabilities. Thus, global supply is primarily traded.

— **The natural gas connection:** Helium is intrinsically linked to LNG as natural gas contains a small percentage of helium. After extraction, the two gases are separated via various processes including cryogenic liquefaction.

2025 Helium production by country
(Total 185mn cubic metres)



Source: Reuters

— **Disrupted supply from Qatar:** Pre-conflict, Qatar was the world's second largest helium producer (behind the US) accounting for more than 30% of global supply. This came from QatarEnergy's Ras Laffan industrial complex which houses three helium plants. These plants extract helium from LNG production processing trains operating in Qatar.

According to QatarEnergy, 2 of 14 LNG trains at Ras Laffan have been damaged representing 17% of Qatar's LNG capacity. This reduction in LNG capacity is expected to reduce QatarEnergy's helium output by 14%.

Estimates are that it will take 3-5 years to repair QatarEnergy's damaged LNG trains and restore Qatar helium production to pre-conflict levels.

— **Exposure to Qatari disruption:** The majority of Qatar's helium production is exported to Asia, primarily Taiwan, South Korea and China, for use in semiconductor fabrication. A portion of Qatar's helium production is exported to Europe for use in medical imaging, research and industrial-gas markets.

— **Russian restrictions:** Russia, the world's third largest helium producer (10% of global supply), has introduced export controls on helium until the end of 2027. Russia's largest helium production site is at Gazprom's Amur Gas Processing Plant, located near the border of China, which is the largest buyer of Russian helium.

War-Induced Disruption

Global inventories:

— Helium users generally operate with much thinner inventories than do those in oil or gas markets. This is because the high cost of liquefaction (helium liquefies at -269°C) combined with the small size of helium molecules means helium has a high boil-off (or leakage rate). This makes storing large volumes of helium in liquid form uneconomic.

— **A just-in-time product:** There is no strategic reserve system for helium that is comparable to that for key energy markets.

— **A market disrupted:** With Qatari production halted, the helium market that was long pre-conflict has become extremely tight-to-short post conflict.

— **Supplier storage:** Large underground helium storage fields exist in the US, Germany and Russia. Amongst the leading industrial gas companies, Air Liquide, Air Products and Linde act as distributors for a large portion of the market.

Their underground storage caverns contained, on average, 6-12 months of helium supply based on their annual demand heading into the conflict.

The supply shortfall from Qatar is being made up in the near-term by the following four factors:

— **i) Excess supply:** Heading into the conflict, the helium market was long (~6.5 bcf of supply vs ~6.0 bcf of demand). This oversupply was due to new Russian capacity which came on-stream in '24 – with more Russian capacity scheduled to start-up in mid-'26 and '27-'28;

— **ii) Some mobile inventory:** Most of the helium industry's inventory is contained in ISO containers shipped around the world. However, the leakage rate from standard helium transport containers is 0.2-0.3% per day. So, in a 30-day voyage at a 0.3% daily boil-off, 9% of the helium is lost before delivery.

— **iii) Existing short-term inventories:** Key industrial suppliers can cover immediate demand

— **iv) Potential to limit non-industrial applications:** It is estimated that lifting applications consume 5-10% of global helium demand. This includes balloons for weather forecasting, atmospheric testing (essential), blimps, toys and party and gift balloons (non-essential).

Recovery Considerations and Implications

Pricing and availability...

— Helium is the clearest example in this report where availability matters more than price. That said, spot helium prices have more than doubled since the start of the Middle East conflict

— **Impact on semiconductors:** Industry reports suggest some users entered the conflict with several months of inventory. Coupled with improved recycling, they have been able to maintain production up to now.

Major Asian producers, especially in South Korea where exposure to Qatari helium is highest, are reported to have held reserves sufficient to last at least through mid-year. The risk is therefore less an immediate pricing shock than a later-year production bottleneck if damaged Qatari LNG and helium-linked facilities remain impaired even after the Strait of Hormuz reopens.

— **Secondary shock to AI semis supply:** Semiconductor prices have risen significantly in recent months, but the available evidence suggests this has been driven primarily by strong demand linked to the rapid expansion of AI-related activity rather than by helium constraints.

Of note, our dbDataInsights webscraping effort to track the semi-conductor product category shows price growth in AI exposed products to be materially higher than non exposed products. This could escalate, should SoH supply be restricted and further supply shocks ensue.

— **Other industries:** Many scientific, medical and industrial users have all faced tighter allocations, sharply higher prices (2x), surcharges and force majeure within weeks of the conflict starting.

— Aside from semis, the most exposed chains to helium are MRI/medical imaging and scientific instrumentation. The issue here is production continuity rather than headline cost share. Thus, the helium shock is best understood as a bottleneck risk to selected semiconductor and high-performance computing supply chains, not a large macro inflation shock in its own right.

Recovery is constrained by both plant repair and export logistics:

— Helium shipments can only normalize once LNG processing at Ras Laffan is restored, liquid-helium production lines are functioning, and specialized export (ISO) containers can move again through a secure SoH corridor.

— Yet, even after reopening, damaged helium capacity and container logistics could take quarters rather than weeks to normalize.

— Even in a constructive reopening scenario, some trapped inventories could move relatively quickly, but a full return to pre-war supply appears much slower because of damage to Qatar's LNG capacity as a result of Iranian missile and drone strikes.

Aluminium
Pre-War Baseline

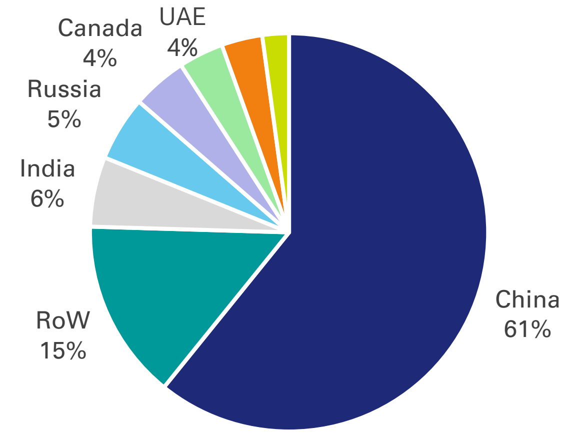
— **Global relevance:** The Gulf region is a globally significant hub for primary aluminium, producing c.8% of worldwide output. However, its importance is amplified by its outsized role in the export market, accounting for a much larger share of traded supply into key importing regions, notably c.23% of ex-China output, thus SoH exposure is far more significant than production share alone suggests.

— **Producer concentration and exposure:** the key producers are Emirates Global Aluminium in the UAE, Aluminium Bahrain (Alba) and Qatalum in Qatar, alongside capacity in Saudi Arabia and Iran. Europe, Japan, South Korea and the US are all meaningfully exposed through metal imports, while downstream users span autos, aerospace, construction, packaging and electrical infrastructure.

— **Why aluminium matters differently:** Aluminium differs from many other metals due to its energy-intensive production process, which demands a continuous power supply and uninterrupted smelter operation. This makes disruptions particularly difficult to absorb. Furthermore, the region's heavy reliance on alumina imports via the SoH creates significant vulnerability to logistical issues. Consequently, even initially moderate production losses can lead to extended recovery timelines.

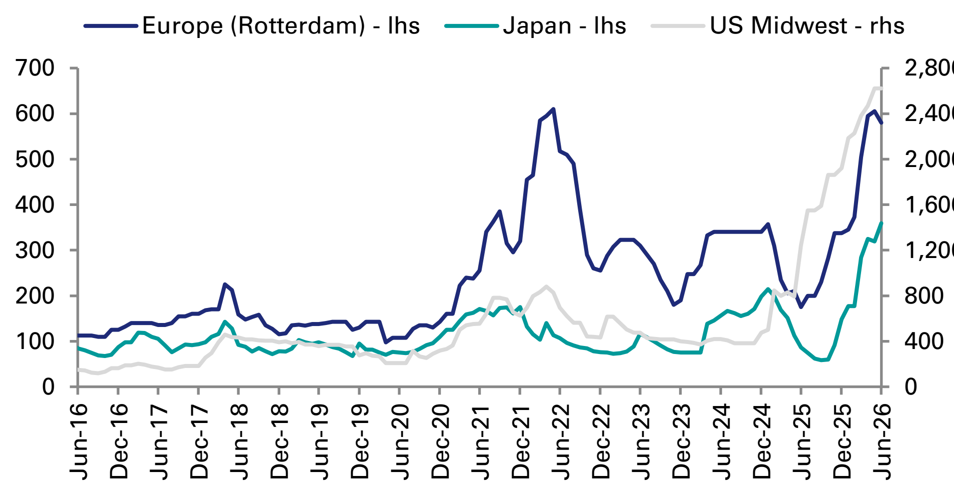
— **Exposure:** The most exposed importing regions are Europe, Japan and the US as Gulf metal represents a much larger share of their import mix. Impacted industries include autos construction, metal packaging, electronics, and aerospace. Many of these end markets have contractual price pass throughs.

2025 Aluminum production by country
(Total 74MT)



Source: USGS, International Aluminium Institute

Aluminum: Regional physical premiums (month-end/latest)
(USD/t)



Source: Bloomberg Finance, LP

War-Induced Disruption, Recovery Timeline and Implications

Supply shock mechanics

The aluminium shock began as a logistics and raw-material problem but increasingly became a production problem. The SoH closure stranded finished metal and, more importantly, constrained inbound shipments of alumina and other inputs. This forced producers to manage inventories, reduce operating rates and seek costly rerouting options.

Scale and location of disruption

— April production in the region fell c.35% year on year as smelters in Bahrain, the UAE and Qatar cut output or suffered direct disruption.

— Alba is believed to be operating at c.30% of capacity due to alumina shortages and direct attacks from Iran and is seen as the most vulnerable to alumina shortages. Qatalum is reportedly operating at c.60% of capacity due to falling alumina stockpiles. EGA has been the most severely impacted, with the major Al Taweelah smelter expected to be offline for at least 12 months due to a missile attack on the power plant, while its other smelter, Jebel Ali, is believed to have pre-emptively reduced production due to alumina shortages.

Workarounds and market status

— Producers are re-routing metal exports and raw material imports through Oman and, in some cases, across Saudi land routes, but those channels are slower and costlier.

— Aluminium prices likely rose materially less than oil or LNG in percentage terms, but with more persistent industrial consequences.

Recovery timeline is mixed, but pointing to 2027

— Smelters that have pro-actively managed down production (Qatalum, Jebel Ali, Alba), should be able to return to normal production levels within 6-9 months once the flow of raw materials normalises.

— The repair work to the Al Taweelah smelter is expected to take at least 12 months, meaning production from the region is unlikely to return to pre-war levels until H2 2027.

Price path and downstream sensitivity

— The key point is that the market impact is larger than the Gulf's production share alone would imply because the region accounts for a much bigger share of traded supply into exposed importing markets.

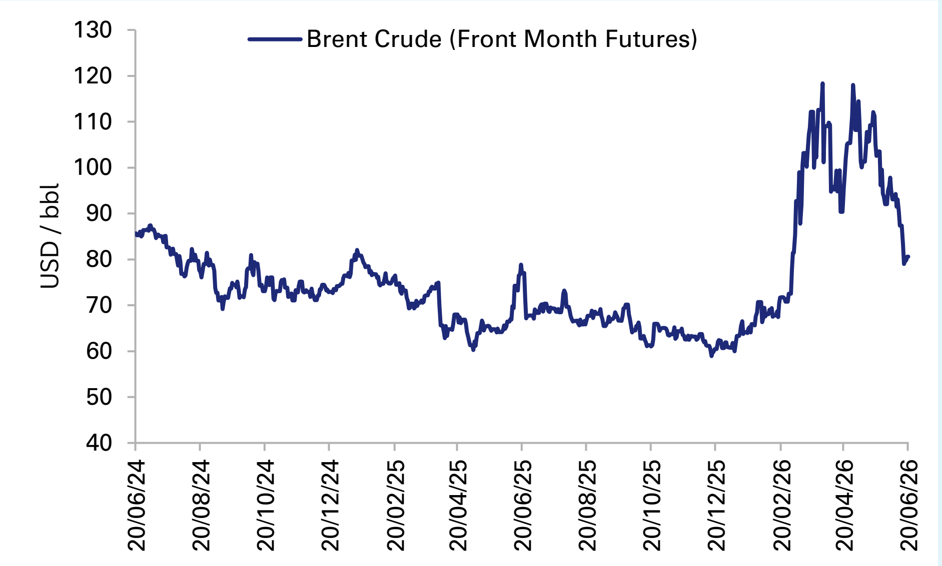
Earnings and sector effects

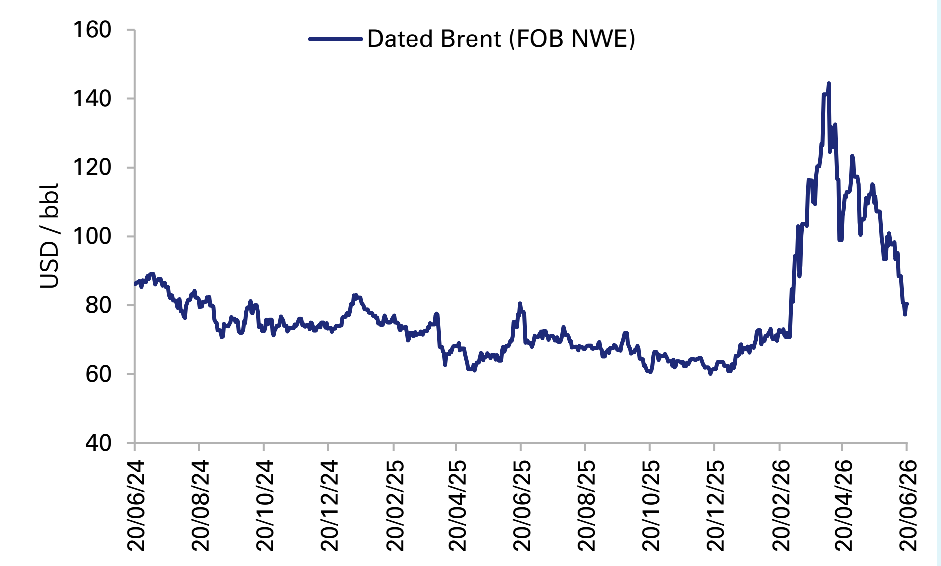
— Non-Gulf aluminium producers and scrap/recycling-linked suppliers were relative winners, while autos, packaging, construction products and import-dependent fabricators faced margin pressure. The key investment distinction is that aluminium users with pricing power can pass through the shock, but lower-margin fabricators and transport-equipment suppliers are more exposed.

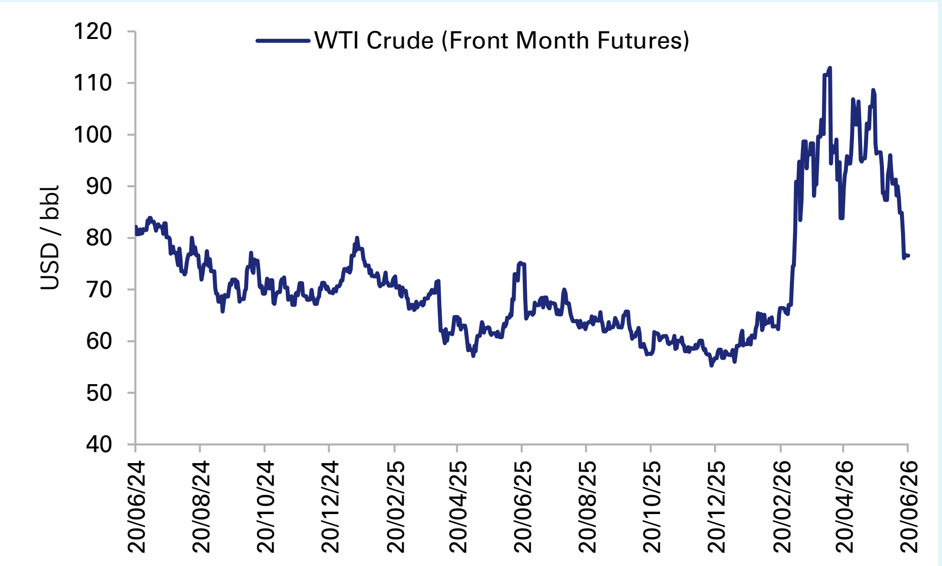
Appendices

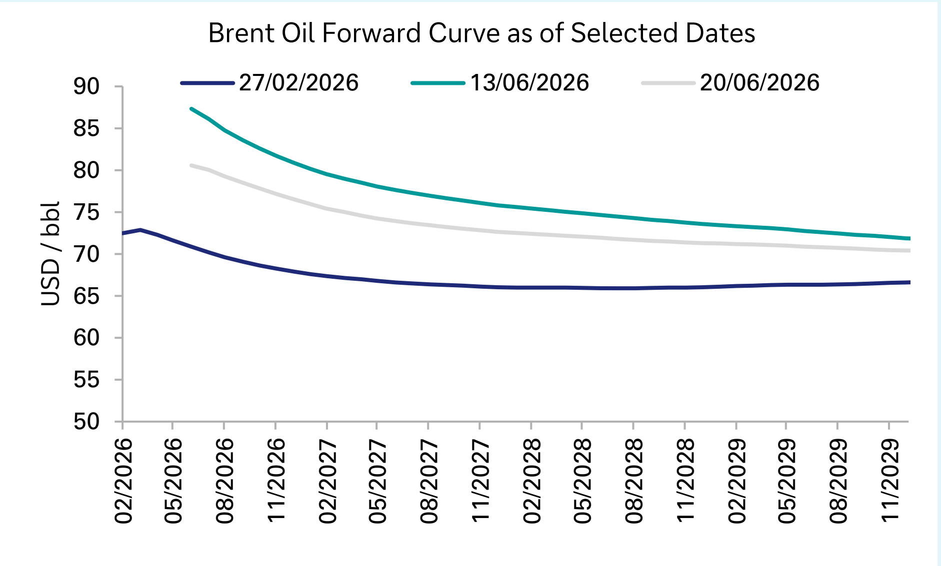
Key insights and considerations across exposed sectors

Oil



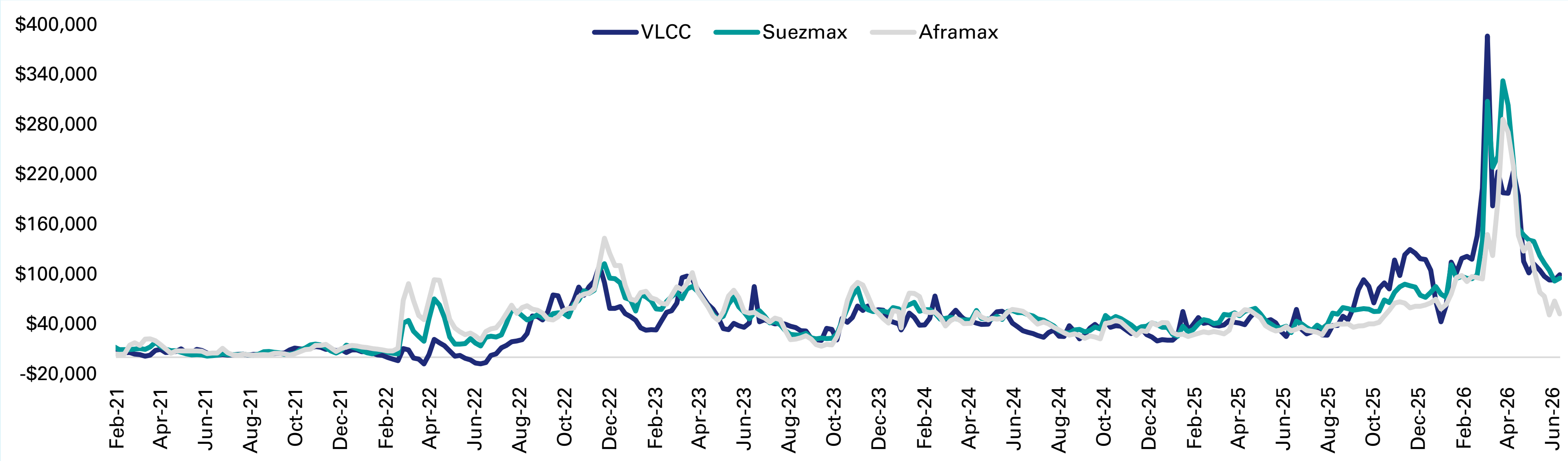


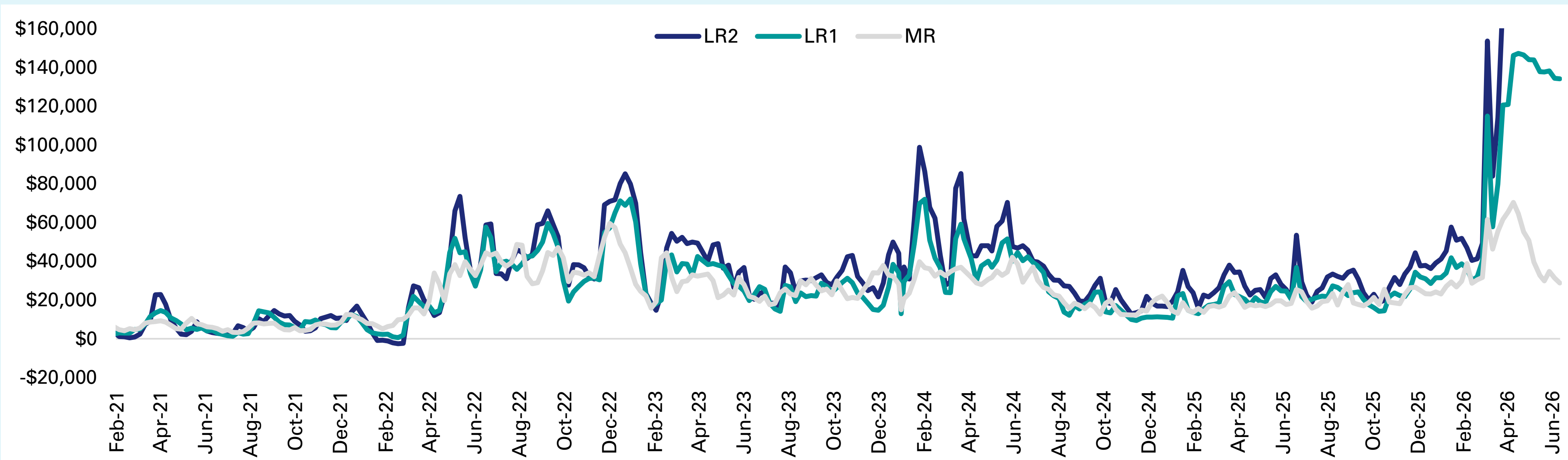




Source: Bloomberg Finance L.P., Deutsche Bank Research

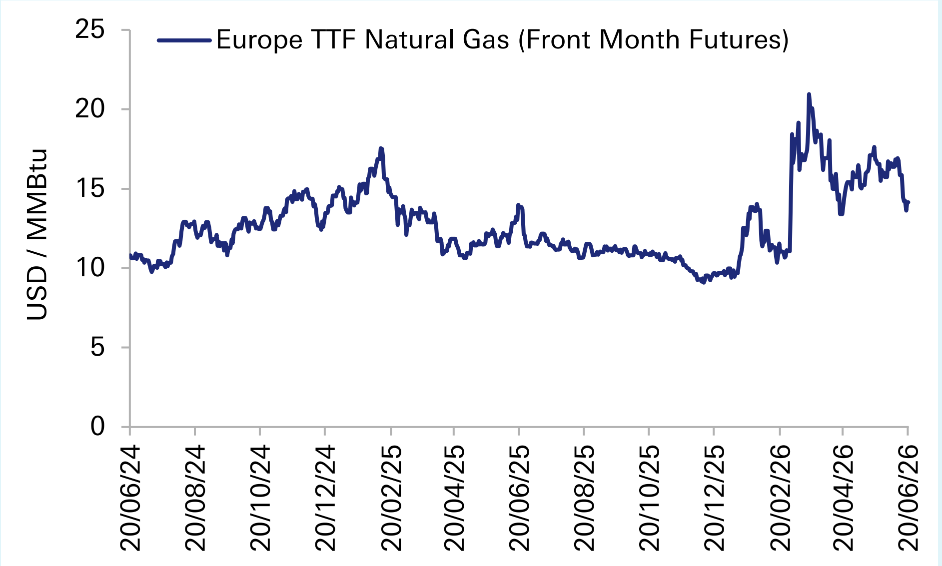
Shipping Rates

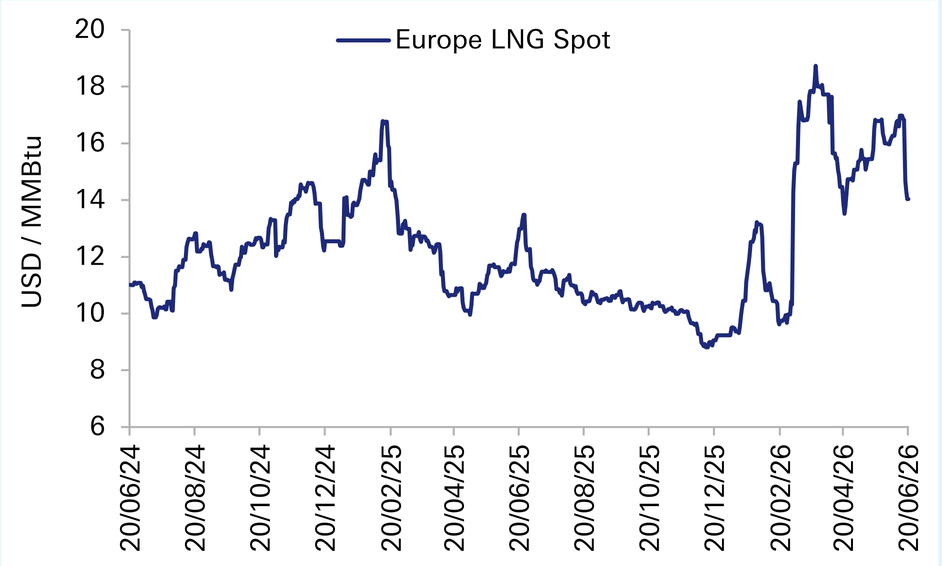


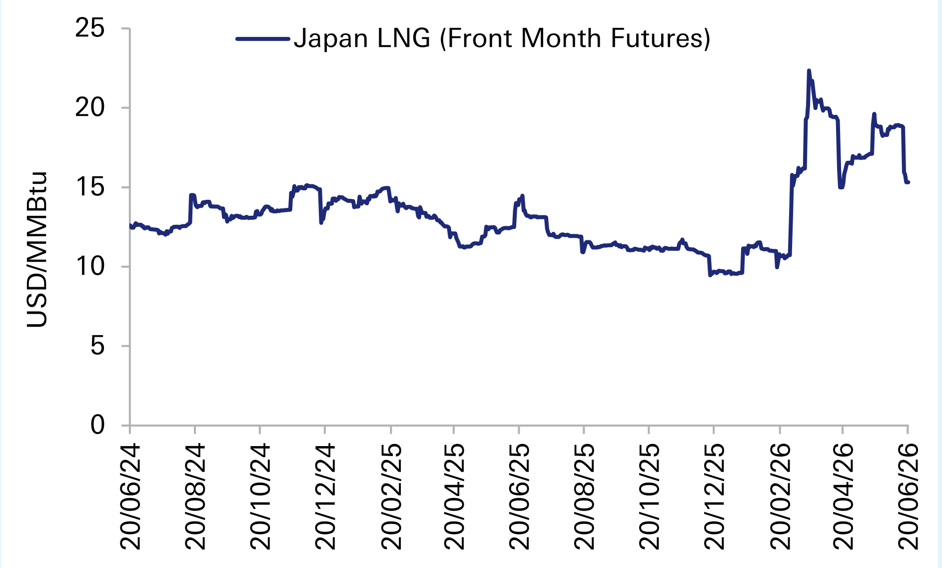


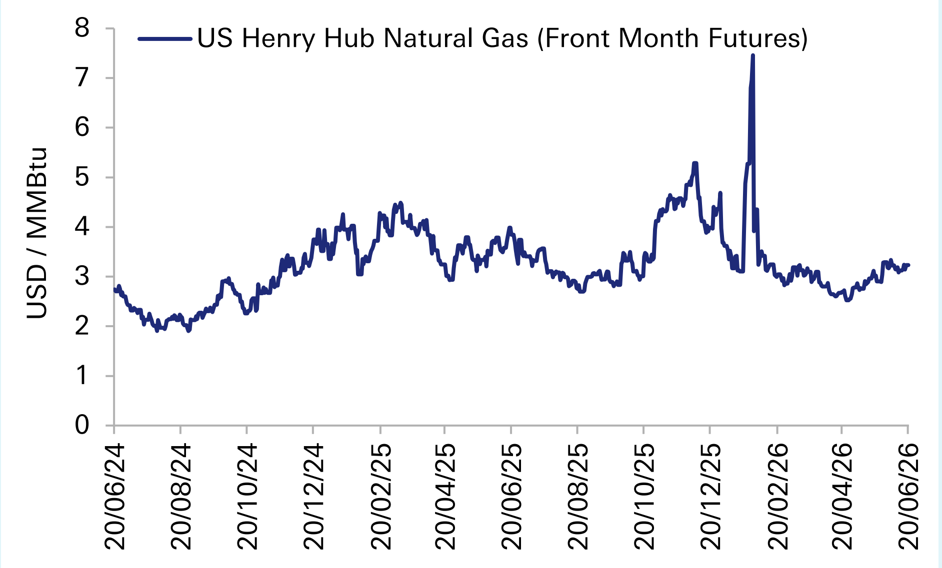
Source: Clarksons Research Services Limited, Deutsche Bank Research

Appendix A – Key Exposures and Economic Indicators
Natural Gas and LNG





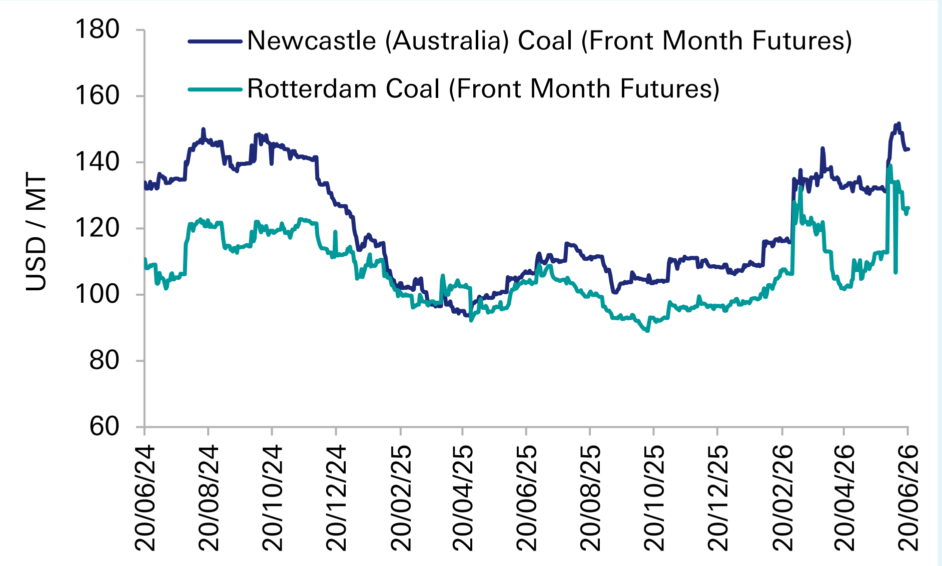


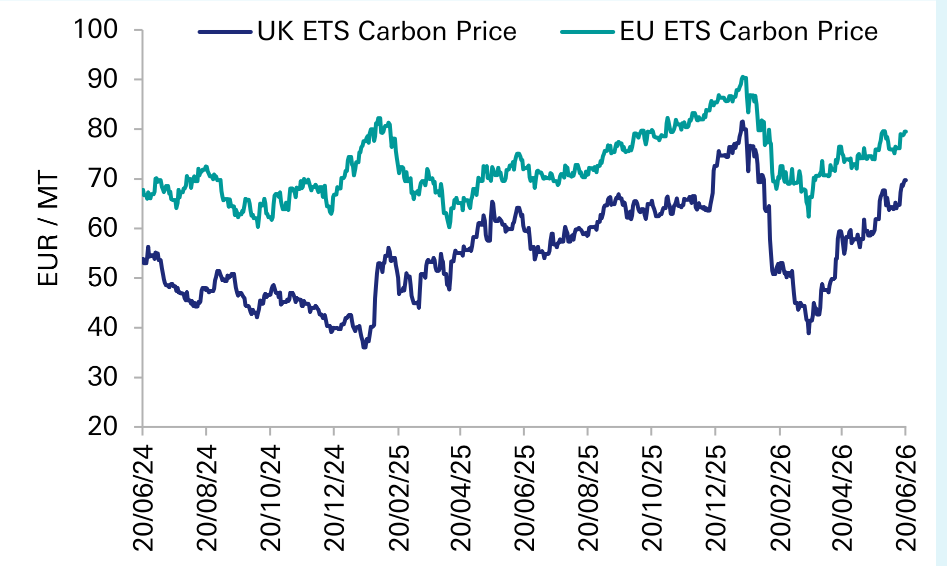


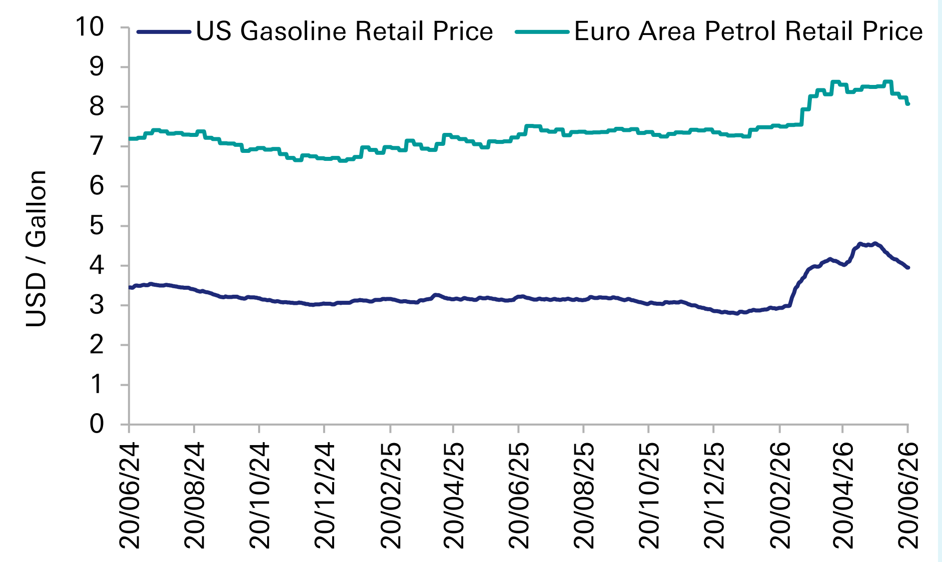
Source: Bloomberg Finance L.P., Deutsche Bank Research

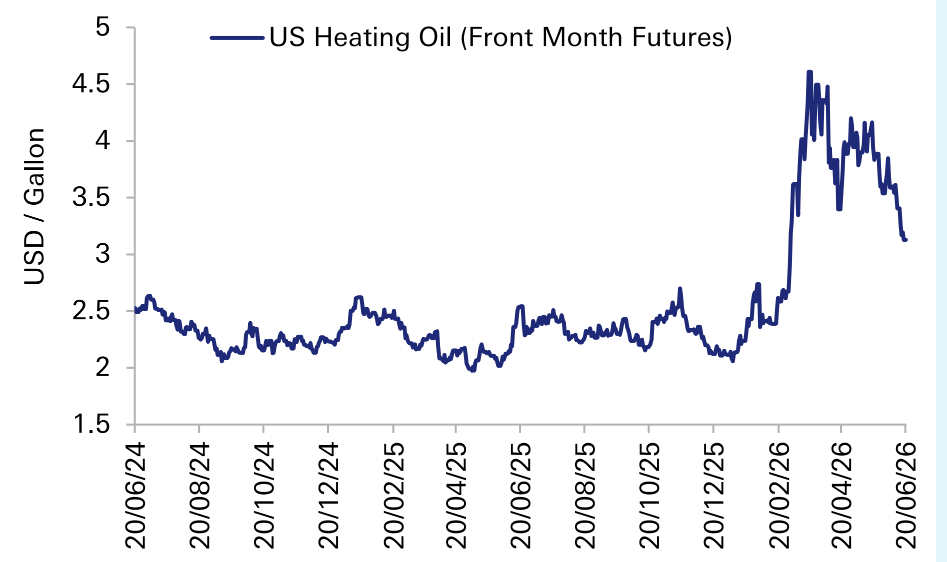
Appendix A – Key Exposures and Economic Indicators

Coal, Carbon, Gasoline / Petrol and Heating Oil



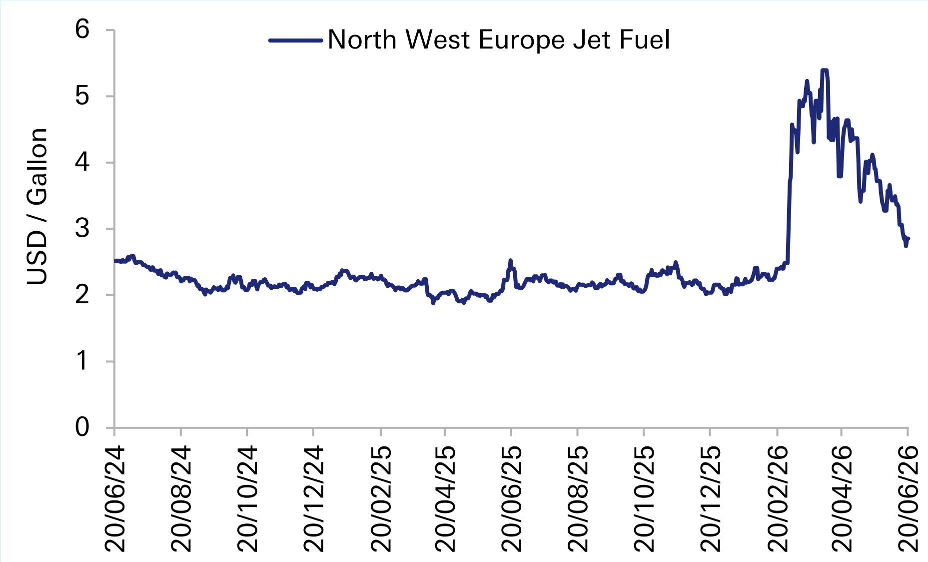


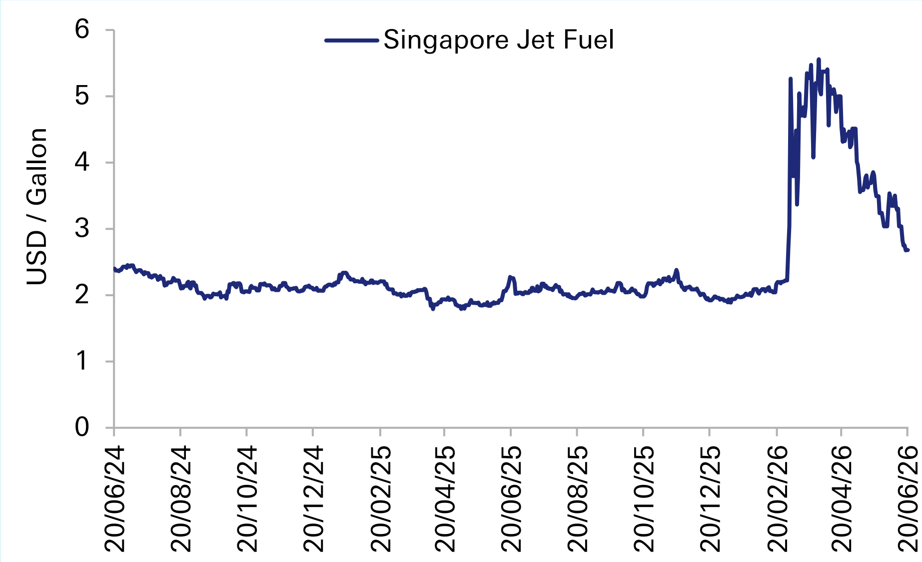


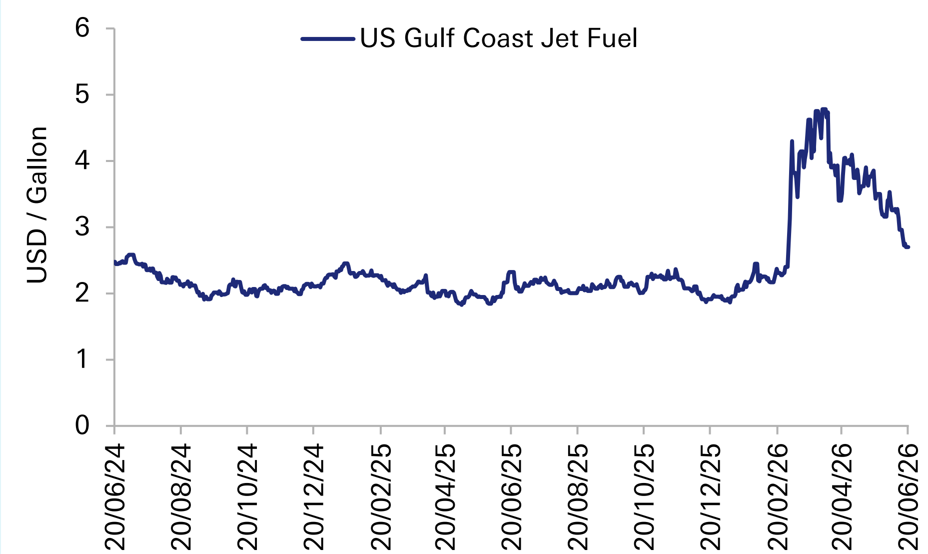


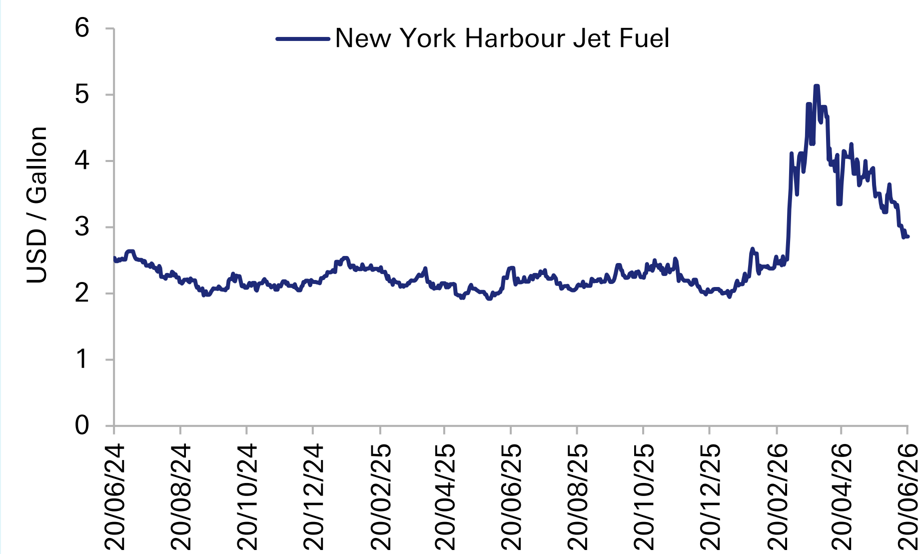
Source: Bloomberg Finance L.P., Deutsche Bank Research

Appendix A – Key Exposures and Economic Indicators
Jet Fuel



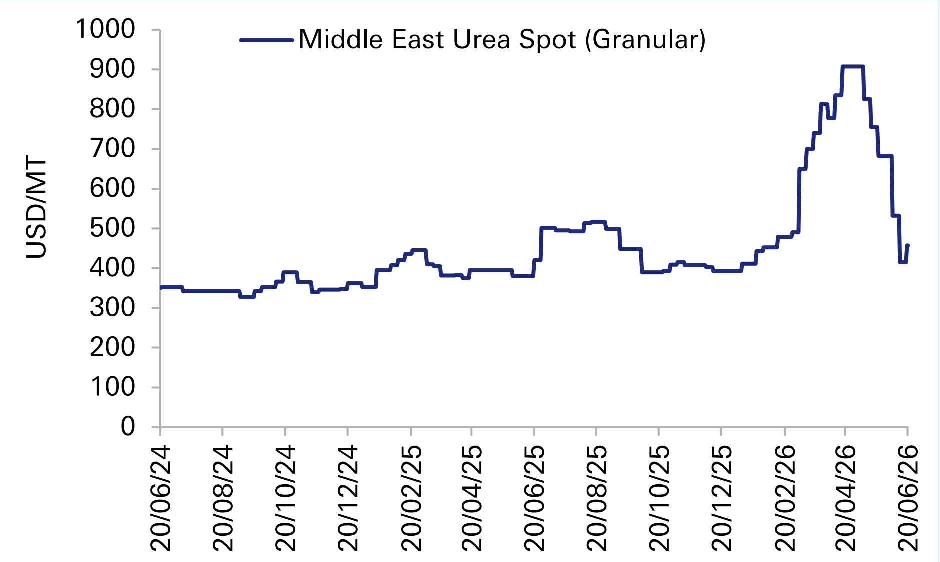


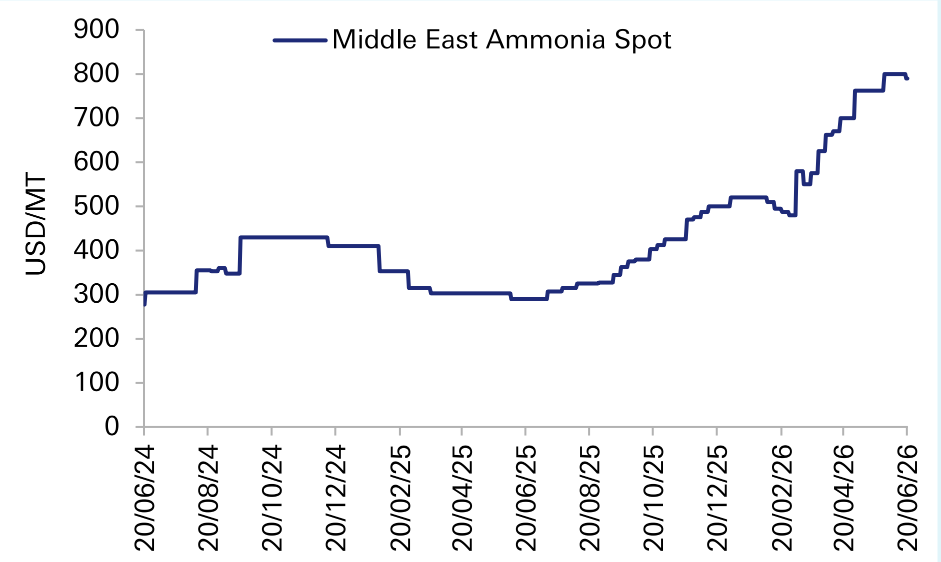


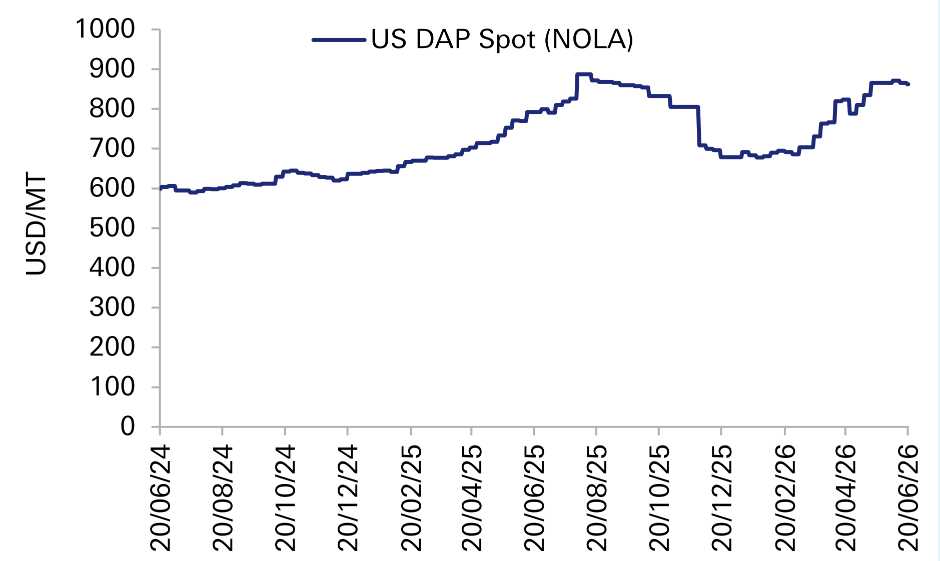


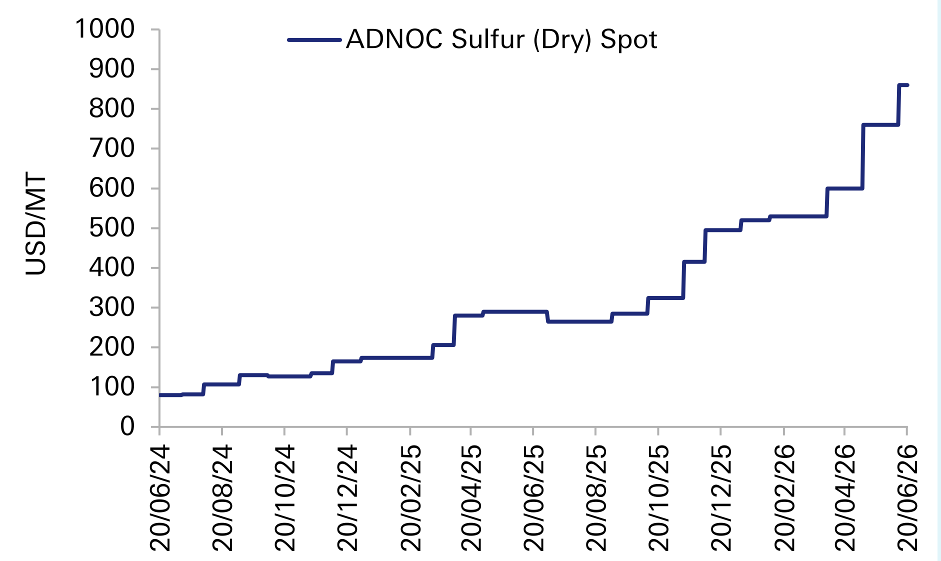
Source: Bloomberg Finance L.P., Deutsche Bank Research

Fertiliser



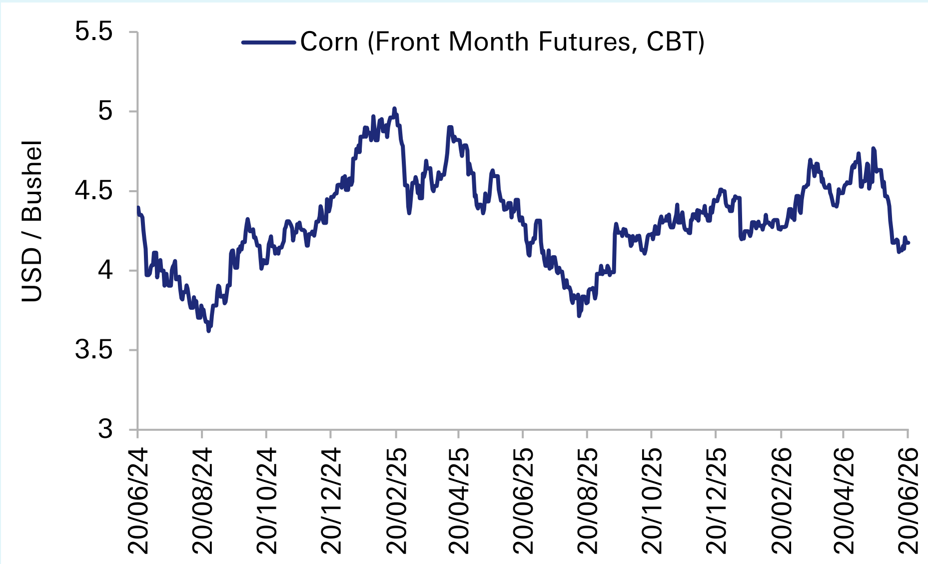


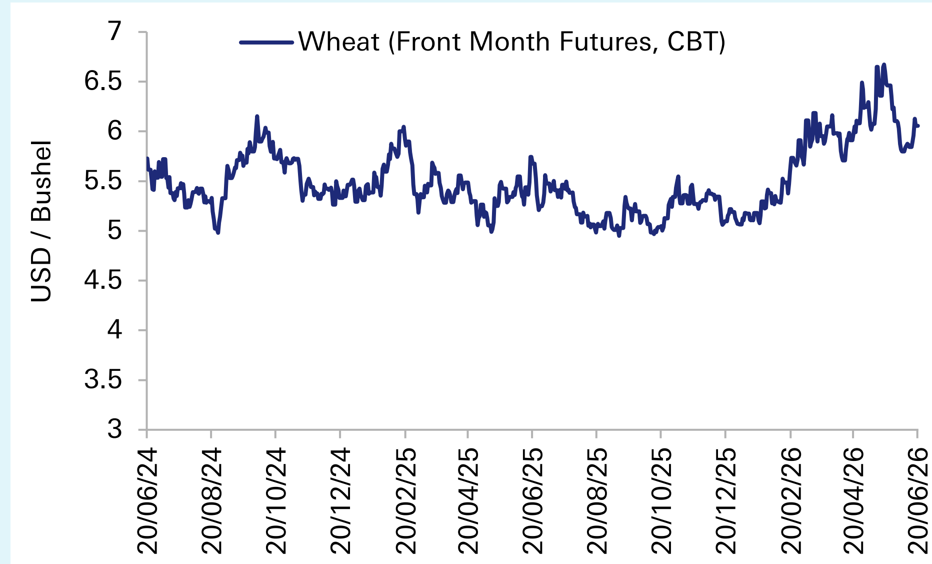


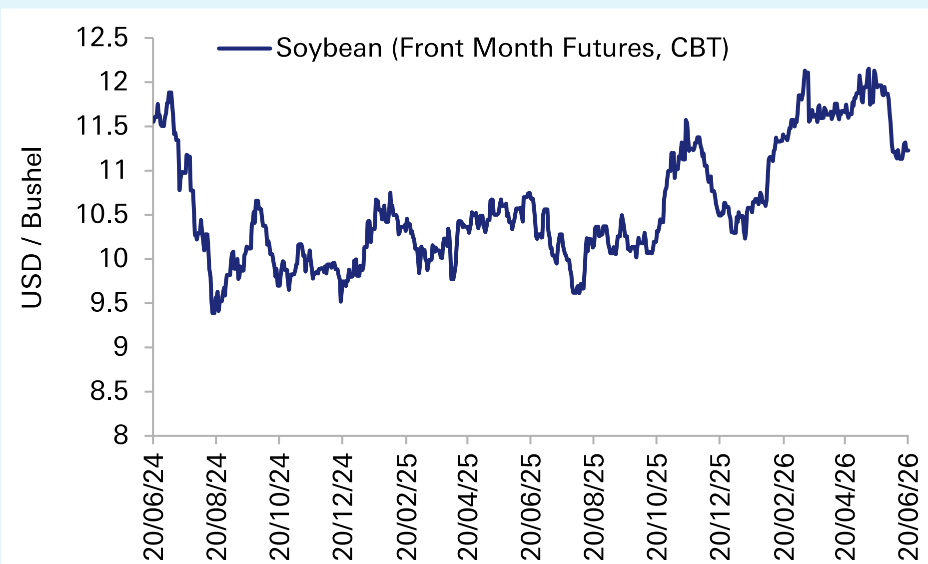


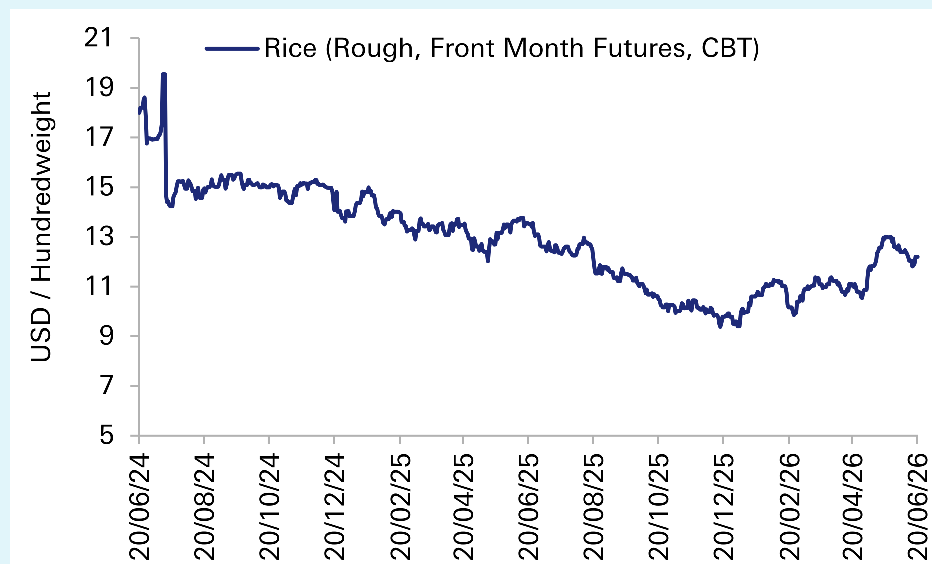
Source: Bloomberg Finance L.P., Deutsche Bank Research

Appendix A – Key Exposures and Economic Indicators
Agriculture



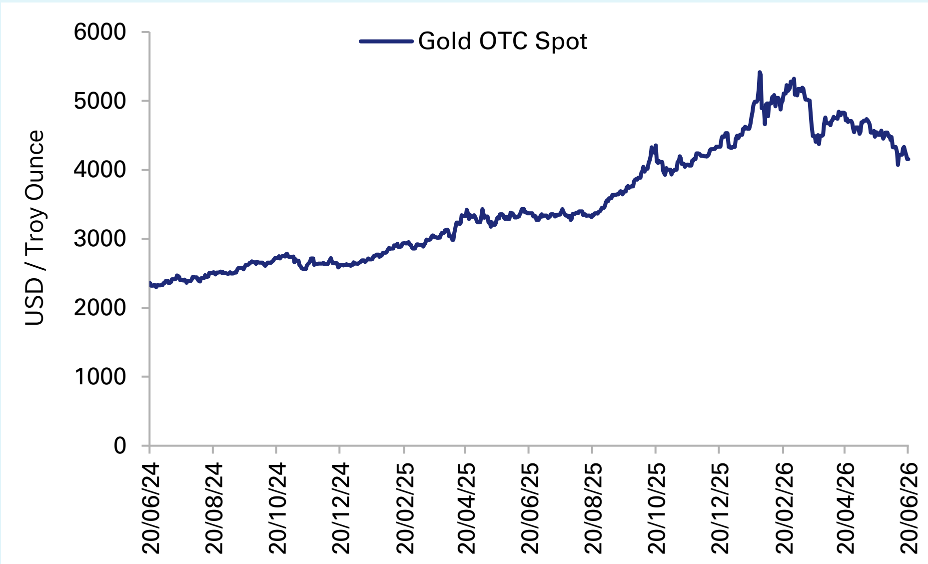


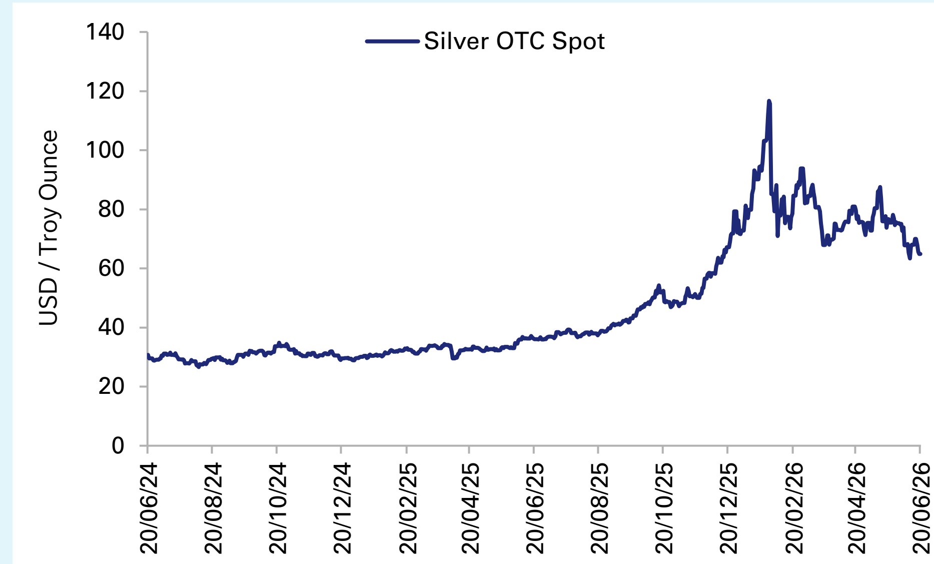


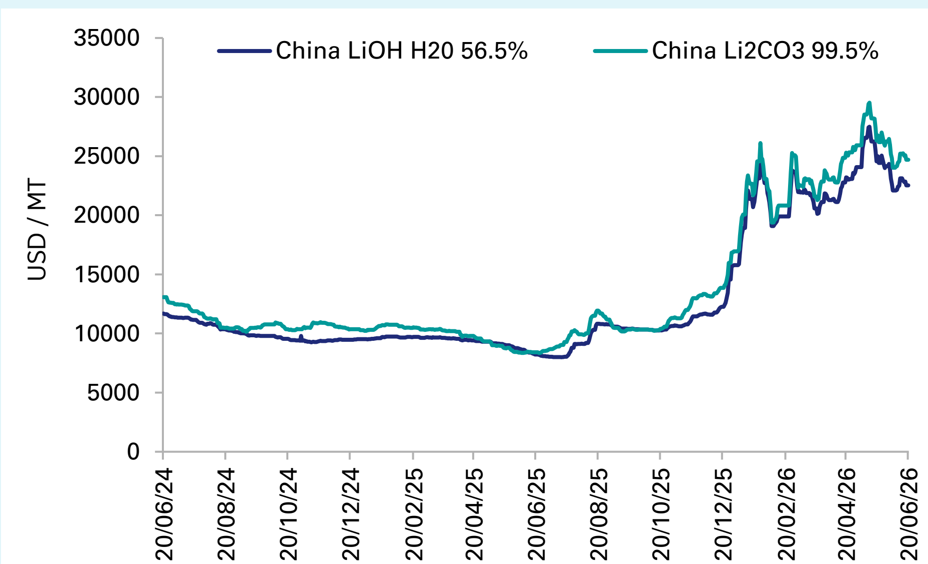


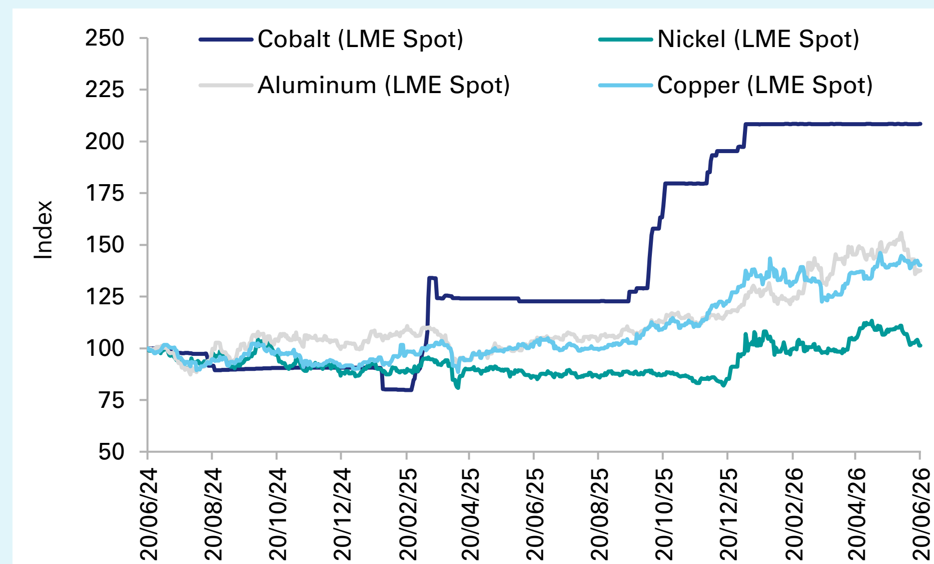
Source: Bloomberg Finance L.P., Deutsche Bank Research

Appendix A – Key Exposures and Economic Indicators
Metals



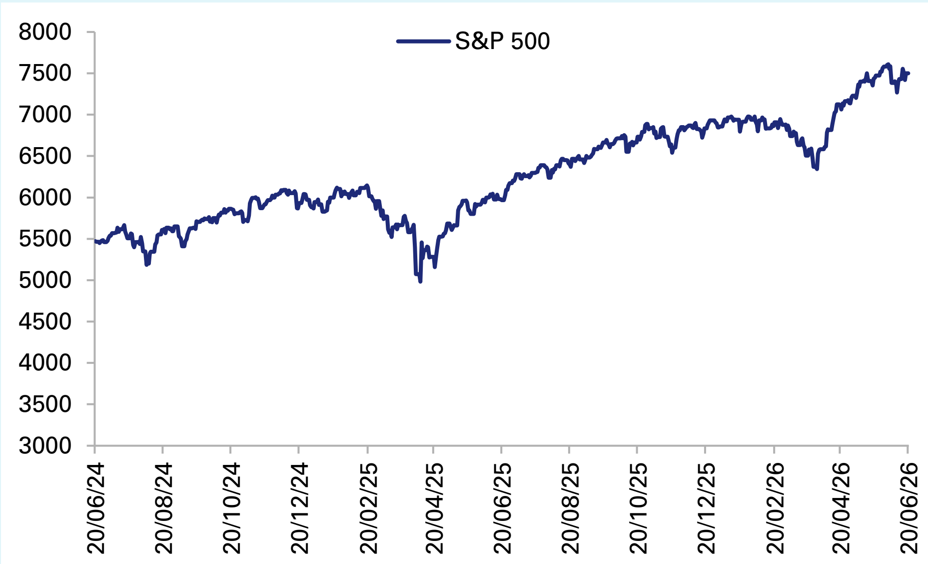


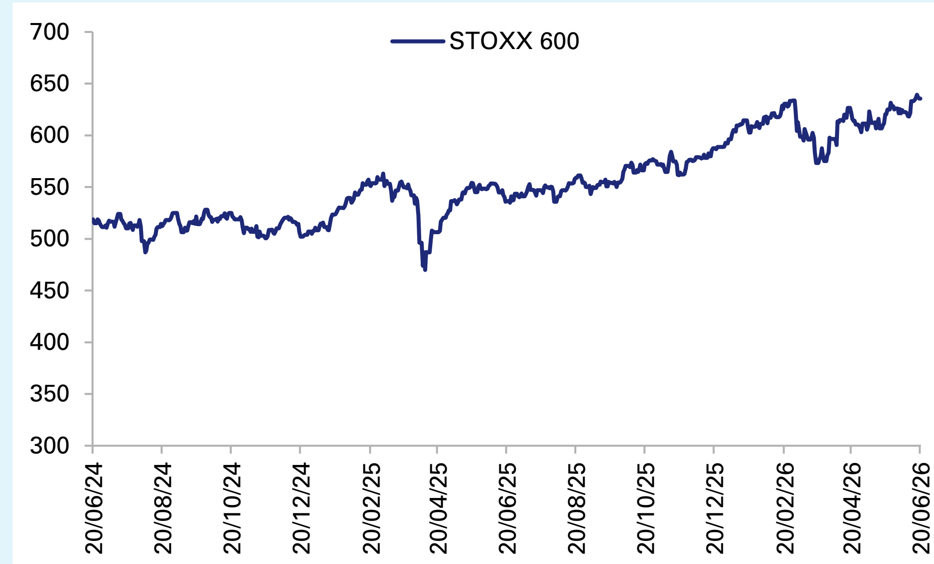


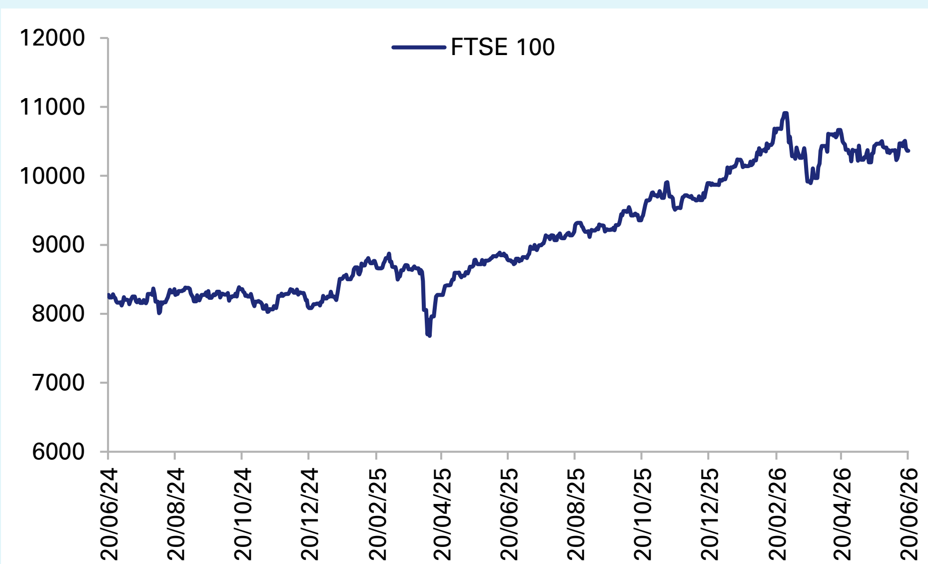


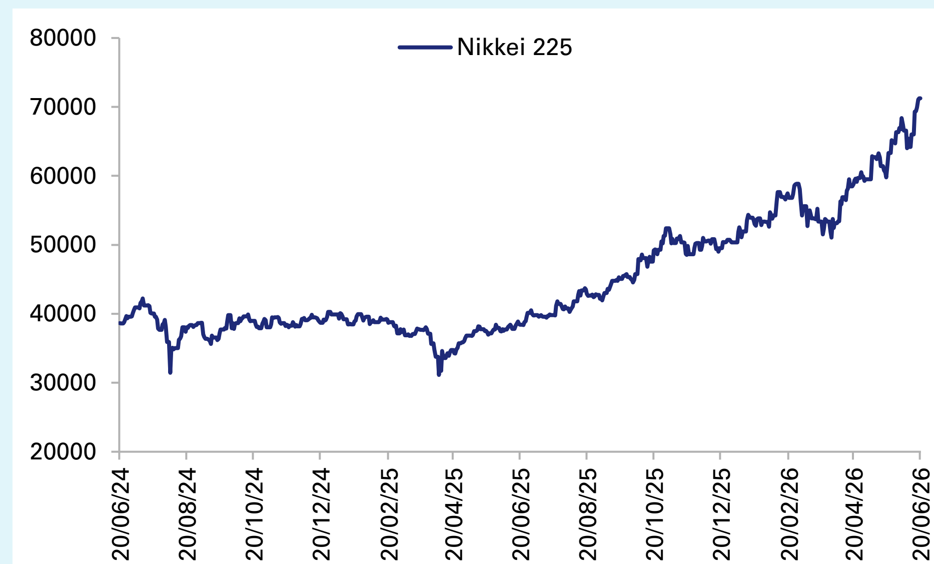
Source: Bloomberg Finance L.P., Deutsche Bank Research

Equities



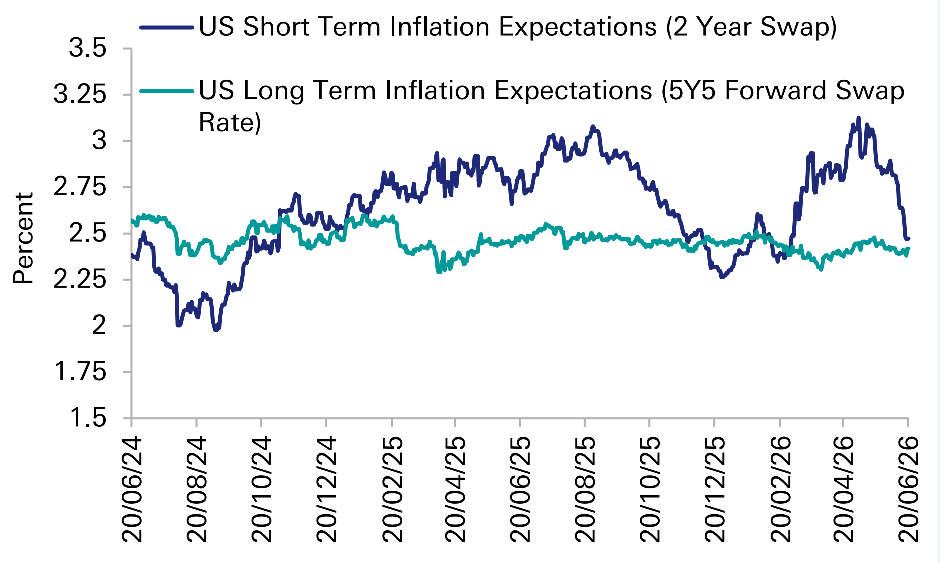


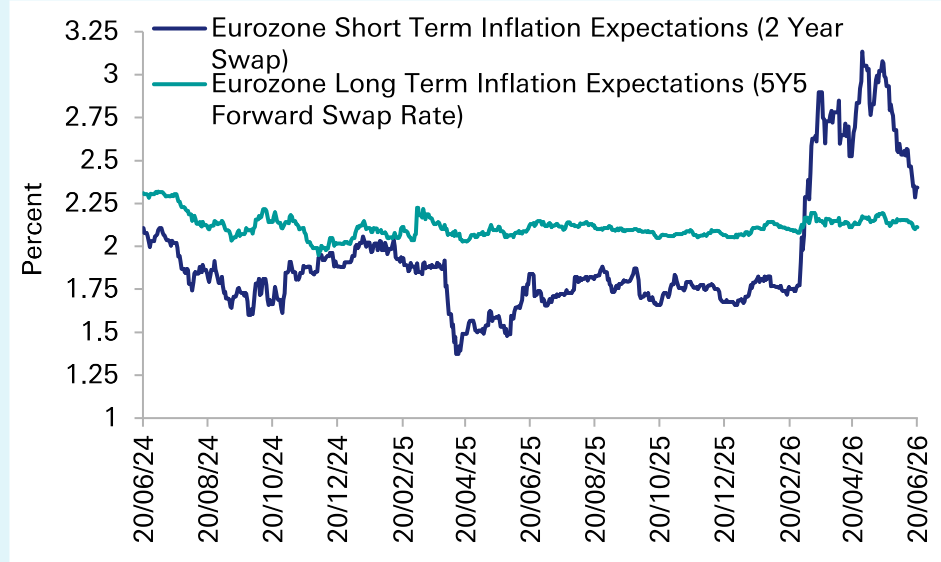


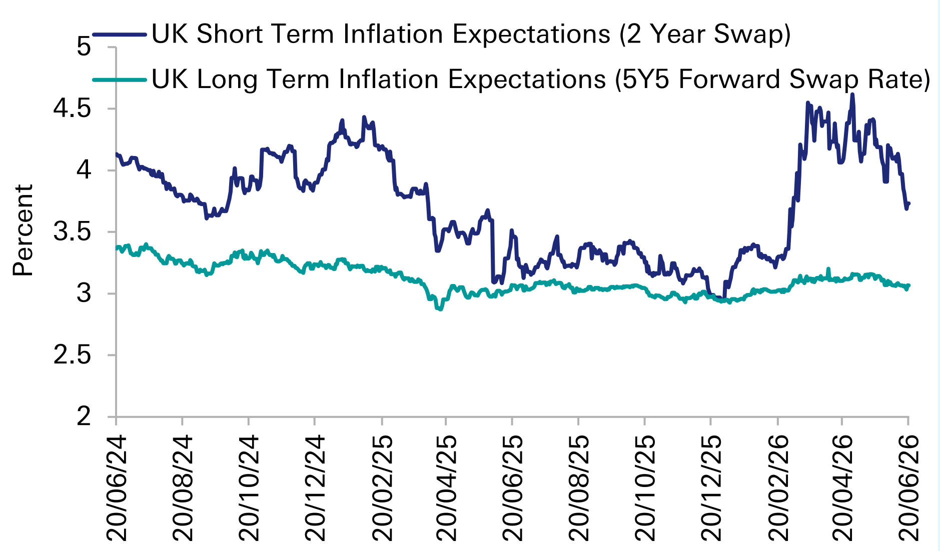


Source: Bloomberg Finance L.P., Deutsche Bank Research

Appendix A – Key Exposures and Economic Indicators
Inflation Rate Expectations

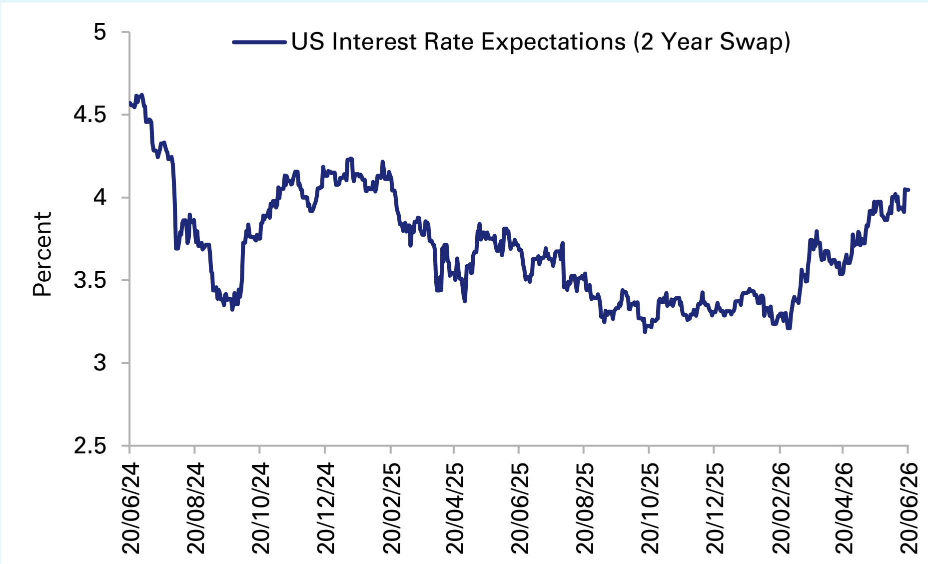


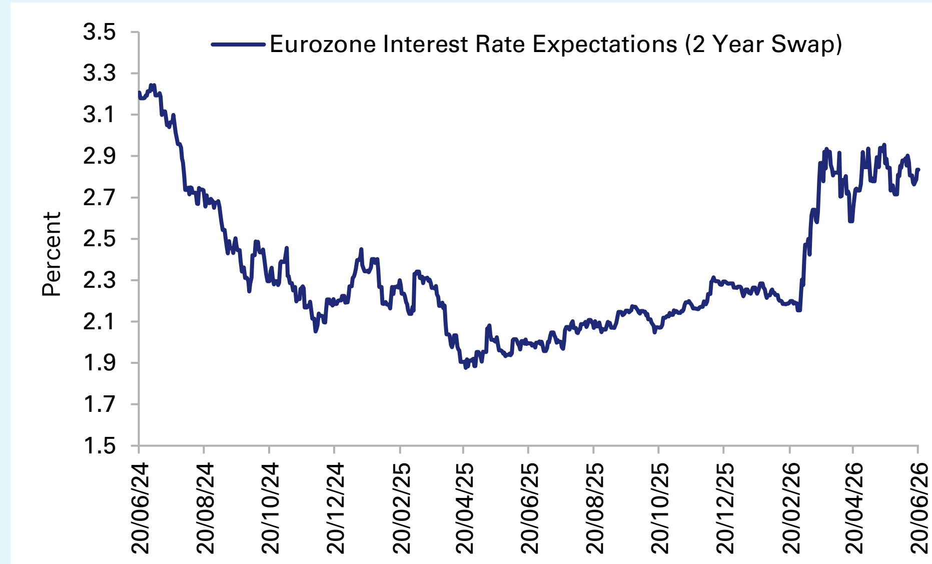


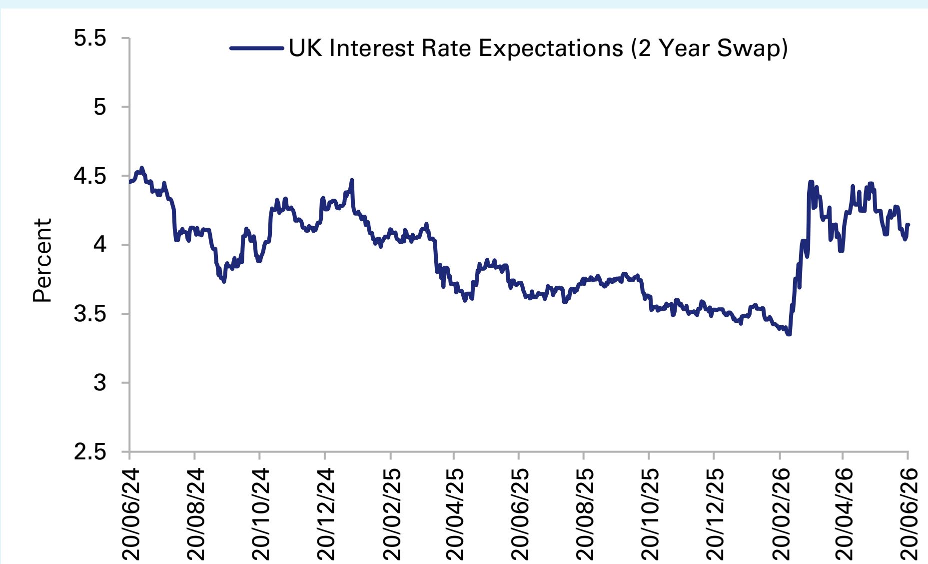


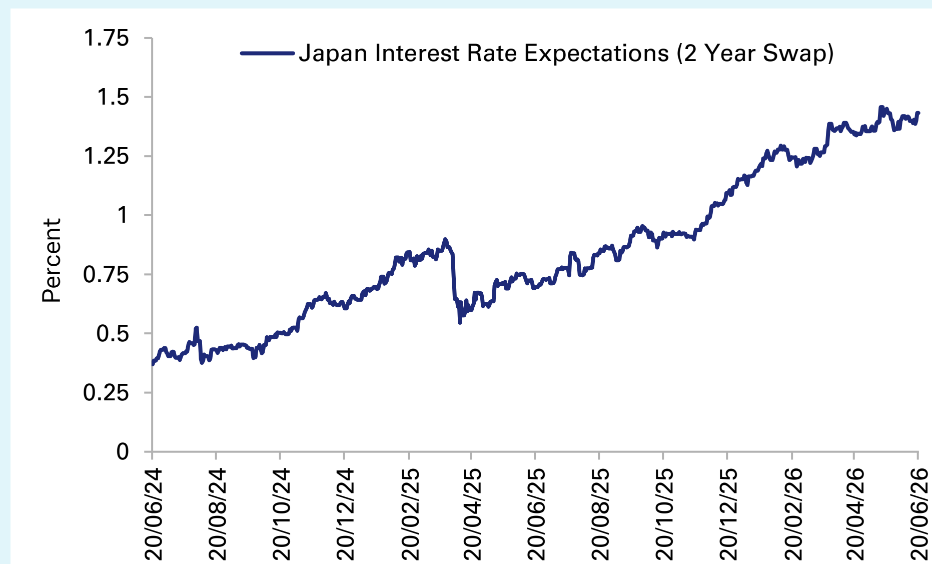
Source: Bloomberg Finance L.P., Deutsche Bank Research

Appendix A – Key Exposures and Economic Indicators
Interest Rate Expectations









Source: Bloomberg Finance L.P., Deutsche Bank Research

Appendix B: Consolidated source notes by section

Deutsche Bank research was used as the primary analytical framework, supplemented by the external sources listed below.

Report Section	Sources
Macro / financial transmission and post-war geopolitical frictions	Deutsche Bank research, including World Outlook (1 June 2026); IMF, OECD, World Bank; Reuters, Bloomberg and market reporting on equities, bonds and FX; reporting and public-source material on post-war Strait governance, tolling and maritime control; LNG repair-timeline reporting; Oxford Institute for Energy Studies / Oxford Energy scenario modelling on LNG and European gas-market spillovers; Dallas Fed scenario analysis on oil-price pass-through to US inflation; Kiel Institute work on energy bottlenecks and food-security spillovers; Asia Global Institute regional analysis on Europe/Asia/US exposure by fuel type; UNCTAD material on trade and development effects of Hormuz disruption.
Shipping and maritime logistics	Sources: Deutsche Bank research; IEA; Reuters, Bloomberg, Lloyd's List; UNCTAD; Clarksons; IMO/maritime authority reporting; maritime security and mine-clearance reporting; container-fleet exposure / vessel-tracking reporting; OECD/BEA.
Crude oil market	Sources: Deutsche Bank research; IEA; World Bank; OECD; Reuters, Bloomberg, EIA; DOE/SPR materials; crude production and export market reporting; Dallas Fed scenario analysis on the pass-through from oil-price shocks to US inflation; Asia Global Institute regional analysis on Hormuz crude dependence, Asian stockpiles and bypass-pipeline limits.
Refined petroleum products	Sources: Deutsche Bank research; IEA; Reuters, Bloomberg, CNBC, API, S&P Global, Financial Times; refining-margin and middle-distillate market reporting; airline and airspace reporting; Asia Global Institute regional analysis on Europe's jet-fuel and distillate dependence versus lower direct crude dependence. Boston Warwick.
LNG	Sources: Deutsche Bank research; IEA; Reuters, Bloomberg, S&P Global; LNG buyer-country and force-majeure reporting; Oxford Institute for Energy Studies / Oxford Energy scenario modelling on LNG export-capacity loss, European storage effects, and regional demand destruction under short and prolonged Hormuz-closure scenarios; Asia Global Institute regional analysis on Asian and European LNG dependence.
Fertilizers	Sources: Deutsche Bank research; IFA, IFASTAT, World Bank, UNCTAD; Reuters, Bloomberg, CNBC; Kiel Institute work on Hormuz-related energy and food-security risks; World Economic Forum summary material on fertilizer, sulfur and downstream agricultural exposure; Asia Global Institute regional analysis on fertilizer-export exposure and import-dependent regions.
Petrochemicals	Sources: Deutsche Bank research; ICIS, Wood Mackenzie; Reuters, Bloomberg.
Helium	Sources: Deutsche Bank research; USGS; Reuters, Bloomberg, CNBC, C&EN, Centre for Materials and Resilience; LNG/helium linkage and Pearl GTL reporting; World Economic Forum summary material on semiconductor and healthcare exposure to helium shortages.
Aluminum	Deutsche Bank research; International Aluminium Institute; Reuters, Bloomberg, CNBC, S&P Global; OECD/BEA; World Economic Forum summary material on Gulf aluminium's role in traded supply and exchange-stock withdrawals.

Appendix

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Deutsche Bank
Research Institute

AI's tightest bottleneck: Memory chips

June 18, 2026

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Figure 1: Key Terminology

DRAM	**Dynamic Random-Access Memory:** One of two main memory technologies (50-60% of annual industry revenue). DRAM is fast, volatile working memory that serves as a temporary system memory. It offers low latency for frequent reads/writes, but is volatile and must be refreshed constantly, which adds to power use. It primarily sells into data centre, client PC, graphics, industrial and automotive end-markets, and is highly concentrated, dominated by three players: SK Hynix (36%), Samsung (33%), and Micron (24%).
NAND	**NOT-AND:** One of two main memory technologies (30-50% of annual industry revenue). NAND is non-volatile flash storage used to keep data when power is off. Because of this it is useful for applications in the data centre, client PC, consumer and automotive end markets.
HBM	**High Bandwidth Memory:** A specialized form of DRAM, built by stacking individual DRAM dies vertically to create the ultra -fast, high-bandwidth memory used directly in AI accelerators and high-performance computing. Key applications of HBM include AI/Machine Learning, Data Centres, Graphics Processing and High-Performance Computing.

Source: Deutsche Bank Research

In 2026, we can no longer ignore the memory market's parabolic rise. The largest memory chipmakers, SK Hynix, Micron and Samsung Electronics are each now valued above $1trn in market cap – a feat considering they represent over 90% of the DRAM market. What is happening in the memory market to drive these geometric valuations? Behind the ascent, the answer is more worrying than current prices would seem to suggest.

The reason is that the global race for AI is driving a supply-displacement shock into the memory market. And as the industry shifts from standard large language model (LLM) response to autonomous agents that observe, reason, plan and act (Agentic AI), this problem will only get more acute.

While memory has historically suffered a boom-and-bust cycle, the exponential demand from AI suggests the current cycle is more structural than cyclical. In this case, chipmakers' ability to deliver cannot keep pace with the surge in hyperscale demand for memory used in AI. And given it takes years for chipmakers to build fabs for memory chip production, we should expect a multi-year supply shock that constrains the rate of progress in AI. More worryingly for corporations and consumers, the vast amount of memory being swallowed to produce HBMs means less is being allotted to produce other types such as standard DRAM, which is used in the everyday applications the global economy depends on.

As a result, the shortage facing the memory industry has massive repercussions for the individual consumer. Just this month, nine US trade associations representing the automotive, consumer electronics, medical devices, telecoms and retail industries sent a letter to US Treasury Secretary Scott Bessent and Commerce Secretary Howard Lutnick warning of the consequences the AI-fuelled rush for memory will have on the US economy.[1]

Lastly, the memory shortage creates a new dimension in US-China tech competition. For China, the repercussions are double-edged. On the one hand, the shortage constrains Chinese chipmakers further in the broader logic-chip and AI build-out. China's AI self-

[1] https://cryptobriefing.com/us-memory-chip-shortage-ai-boom/

sufficiency is constrained by its access to both high-end compute and memory. Without them, its performance in AI processing and inference remains limited. On the other hand, the supply imbalance of memory chips needed to power traditional devices creates opportunities for additional suppliers, including mature-node (older, less advanced process node) Chinese memory makers like ChangXin Memory Technologies and Yangtze Memory Technologies Corporation. As Western non-AI corporations are squeezed out of the memory market, the possibility they could turn to Chinese memory manufacturers instead is growing.

1. The squeeze: What's actually happening in memory?

(i) The demand side: AI's bottomless appetite

The problem facing the memory industry is that there is not enough supply to meet the exponential demand for AI inference.

The crisis partially lies in high-bandwidth memory (HBM), a "memory stack" that layers individual DRAM dies into the ultra-fast, 3D memory needed to process AI models. For example, an AI chip, such as an Nvidia GPU, can only compute on data that is loaded onto it; memory does the holding and feeding of that data into the models. Memory has two dimensions: capacity (how much data fits) and bandwidth (how fast it moves). Without memory, chips cannot compute or train AI models – and without it, companies cannot run a trained model to perform an agentic task (inference).

In addition, the fast-evolving AI landscape is beginning to require more expansive memory consumption across other areas of the memory stack beyond. In particular, the shift from human-driven generative AI to inference and agent-driven AI is driving another leg up for DRAM bit demand. Agentic AI systems, unlike traditional AI models, can store and recall past experiences, learn from interactions, and maintain context, leading to more coherent outputs. This requires various types of memory to function effectively, spanning across other memory devices (DDR5, LPDDR, NAND).

Memory also dictates the pace of AI advancement, because companies cannot scale on compute alone. This is the "memory wall" problem: beyond a point, peak computational performance cannot be reached without expanding memory bandwidth. In other words, increasing computing power without increasing memory bandwidth eventually limits the benefits of the compute itself.

Thus, the accelerating need for memory is – alongside compute – a defining bottleneck for AI progress[2]. But memory has arguably become a tighter near-term constraint, given its manufacturing and supply limits: HBM bandwidth for inference-heavy workloads, HBM/packaging capacity, and capacity for traditional DRAM for long-context queries as the determining factor for how fast compute can be deployed. Micron's CEO has called memory the backbone of AI, noting: "as AI models grow larger and inference requirements continue to rise – AI moving from training to inference, and from data centers to the edge – the demand for storage will only increase. It requires greater capacity, higher performance, and lower power consumption.[3]" As a result, hyperscalers (the largest cloud operators) like Meta, Amazon and Microsoft need vast quantities of HBM both to hold the model and to keep the compute from sitting "idle." In recent years, they have driven the memory market and last year total revenues increased by 35% y/y to a record $223bn[4].

Our equity analysts forecast HBM demand to grow at a ~40% CAGR through 2030, versus a ~21% CAGR in demand for standard DRAM (the chips used for everyday consumer

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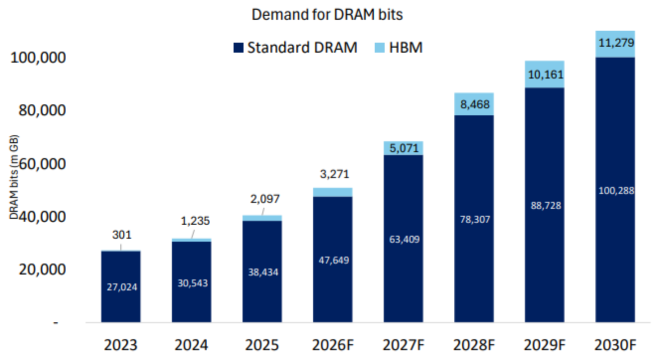
[2] In the AI tech stack, compute is made of GPUs, TPUs, and other AI accelerators, while memory refers to where the data sits. That can include the model's weights, the intermediate activations produced during a forward pass, the KV cache used during inference, etc. Memory has two relevant dimensions: capacity (how much data it can hold) and bandwidth (how fast data can move between memory and the compute units) For those interested in reading more of our work on the compute bottleneck, please reach out.
[33] https://news.futunn.com/en/post/74269829/micron-ceo-memory-is-the-overlooked-bottleneck-in-ai-supply?level=1&data_ticket=1781437411630811
[4] Global Automotive: DRAMa ahead? - Deutsche Bank Research

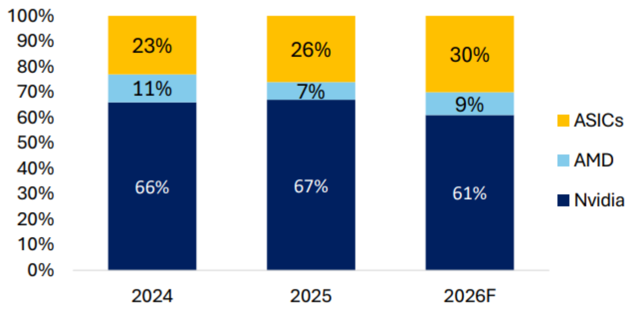
computing and standard cloud databases).[5] This is because hyperscalers will pay premiums over automotive, consumer-electronics, and industrial buyers, and will sign multi-year lock-ins to secure supply.

Figure 2: Standard DRAM demand grows at +21% CAGR through 2030, vs HBM at +40%



Source: Deutsche Bank equity analyst research.

Figure 3: HBM demand breakdown by major customers



Source: Deutsche Bank Equity Team. For more, see here and here.

(ii) The supply side: Why fabs can't follow

The inability of supply to meet outsized demand is compounded by the lead-time needed to construct more fabs. Memory fabs (fabrication plants) are among the most expensive and complex facilities to build, with a 2-3 years construction timeline. Most announced projects will not contribute meaningfully to HBM capacity until 2027 at the earliest.

The complications of HBM production exacerbate the memory shortage crisis. HBM is made from DRAM, but producing one additional bit of HBM requires around 3x more silicon.[6] That means every wafer directed to HBM consumes multiple wafers' worth of standard DRAM/NAND capacity needed for end-markets like automotives and PCs – creating shortage risk across not just the AI segment but all memory types. As this ratio moves from 3x to 4x silicon needed with HBM4/HBM4e, the crowding-out effect will worsen. For example, facilities and tools that would have served conventional memory are being redirected to HBM; "clean room space" – the ultra-sterile environments where wafers are processed – is at its limits, forcing manufacturers to choose what to build on finite factory floors. Anecdotally, Micron has said it can fulfil only 50% to 2/3 of demand from several key customers, describing the demand-supply gap as the largest it has ever seen.[7] Qualcomm has also said that the size of the 2026 handset market will be defined by DRAM availability rather than consumer demand.

Our Deutsche Bank equity analysts' DRAM demand-supply model projects continually tight supply demand conditions in the coming years (beyond 2028). Growth in WSPM is expected to still lag accelerating demand for DRAM wafers over the coming years, even despite new fab announcements and project expansions by top DRAM suppliers.[8]

DRAM now represents ~70% of the memory market – up from its historical 50-60% range, as the HBM boom lifts DRAM's share of revenue.

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[5] Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist
[6] Otherwise known as the HBM trade ratio.
[7] https://www.techradar.com/pro/we-are-only-able-to-supply-for-our-key-customers-in-the-midterm-about-50-percent-to-two-thirds-of-their-requirements-micron-ceo-forecasts-production-spend-increase-to-meet-the-insane-demand-for-memory-but-the-ram-crisis-will-only-get-worse
[8] WSPM stands for Wafer Starts Per Month, which measures the number of raw silicon wafers that enter a fab's manufacturing pipeline every month. That is the university industry standard metric to measure and express the max production capacity of a fab.

2. The spillover: Why it matters beyond AI

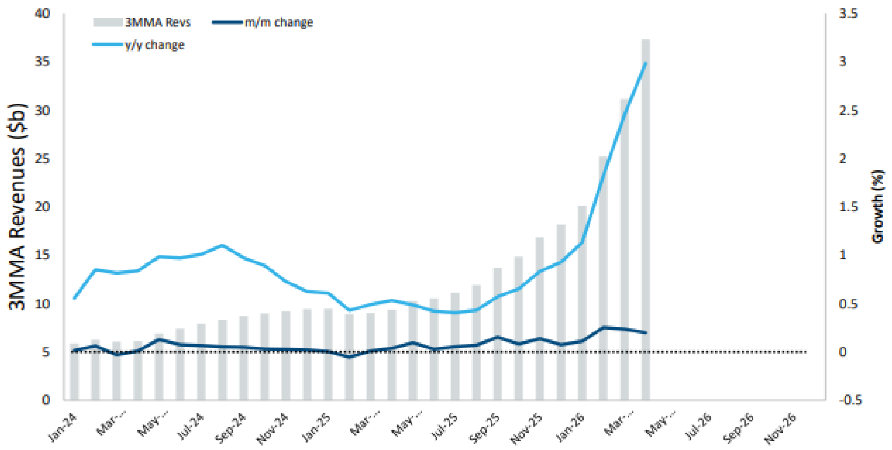
(i) Prices, mix shift, and the HBM-vs.-DRAM squeeze

The overwhelming demand for AI compute and inference means US hyperscalers will pay premiums to lock in future supply. Examples include SK Hynix's supply agreement with OpenAI for the Stargate Project, Samsung reported long-term deals with Google and Microsoft, and Micron's first five-year strategic agreement.

To meet surging AI demand, chip suppliers are reallocating their capacity towards more HBM production and pulling out of older segments. The result is an industry cycle shift, in which standard DRAM and NAND are also pulled into data centres and tighten supply, hitting non-HBM buyers in energy, consumer electronics and appliances. TrendForce predicts standard DRAM contract prices will rise 58-63% q/q in Q2 2026, with NAND flash contract prices up 70-75% q/q.[9]

We are seeing this today not just in leading-edge pricing: in Q1 2026, there was an uptick in mature-node pricing for the first time in two years, as increases filter into mature chip categories.[10] Price rises are set to spread to server CPU, storage, substrates and passives, while several of the largest mature foundries – Vanguard, Powerchip and Nexchip – reported they would raise prices by up to 10% or more, driven by a 5% increase in costs.[11]

Figure 4: 3MMA DRAM revenues continued to surge in April, growing to $37.3bn (+299% y/y)



Source: Deutsche Bank equity analyst research, SIA. Notes: 3MMA = 3 month moving average.

(ii) Who can cope, and who gets rationed?

The shortage is affecting all direct users of memory. But hyperscalers and AI cloud providers remain the best positioned overall to weather it, because they can price AI compute services to reflect the underlying input costs.

With AI absorbing all the silicon, broader corporates and consumers are being squeezed and rationed. Lenovo, Dell Technologies, Apple and HP have sounded alarms on rising memory costs. The effect is global: Taiwanese manufacturers Acer and ASUS, and Chinese smartphone company Xiaomi have also issued warnings.

(iii) The most exposed sectors, bypass-through power

Of the key memory end-markets – data processing, communications, consumer and industrial – those most exposed to the AI squeeze are PCs (within data processing), consumer electronics (within consumer), and automotives (within industrials).

[9] https://www.trendforce.com/presscenter/news/20260331-12995.html
[10] Semis skimmed: Pricing and lead time tracker: Mar 2026. Deutsche Bank Equity Research.
[11] Semis skimmed: Pricing and lead time tracker: May 2026 - Deutsche Bank Equity Research

Starting with automotives: suppliers that sell complete technology systems to manufacturers such as Ford and Toyota have among the highest exposure to memory chips. These suppliers embed DRAM across the vehicles to help drivers use cockpit infotainment and navigation, advanced driver-assistance systems, and more. In 2026, companies such as Aptiv, Aumovio and Ford have signalled low DRAM supply despite efforts to secure inventory and contracts. With inventories set to drain through 2026, our equity analysts see potential impact on production volumes from 2027.[12]

Still, these suppliers may be best positioned among non-AI buyers, given their recovery mechanisms established after the 2021 chip shortages and these companies' ability to pass costs to the equipment manufacturers like Ford and Toyota, who have more limited consumer pricing power. Our DB models estimate the inflated cost of DRAM raises the price of an average vehicle by $150-300 for luxury high-end models, and $400-600 for vehicles with higher-level autonomy.[13] These carmakers therefore can either absorb the cost and compress margins, pass it to consumers, or strip out some DRAM-intensive features such as L2+ autopilot capabilities or GenAI chatbots.

Consumer-electronics manufacturers and PCs are also highly exposed to the memory crisis, via margin compression and softer unit volume. Compared with automotives, this sector faces a greater challenge because it has the weakest pass-through capacity among the sectors. Price sensitive categories –smartphones, traditional phones, PCs, smart TVs, gaming consoles and tablets – all face rising memory costs per unit. Total consumer revenues also fell -10% q/q in Q1 this year, with most weakness from AMD's and Nvidia's consumer businesses. Our equity team expects total consumer end-market revenues down -15% y/y in 2026, recovering +9% y/y in 2027.[14]

The consequences of memory's supply disruption amongst consumer electronics are being echoed publicly across large corporations. In Apple's Q1 earnings call, its CEO warned that higher memory costs are just the beginning and that the company was facing supply constraints.[15] While Apple has not detailed its options, it has reportedly quietly reduced the maximum memory capacity offered on several of its computers.[16] It is not just Apple: Dell and Microsoft are also paring back their configurations to manage input costs, rather than pass them onto the individual buyer. or Microsoft, even as the company advances in AI with its Copilot+– quietly cut its new surface for Business laptops to as little as 8GB of RAM (down from 16GB previously) on entry-level configurations to mitigate the DDR5 costs.[17]

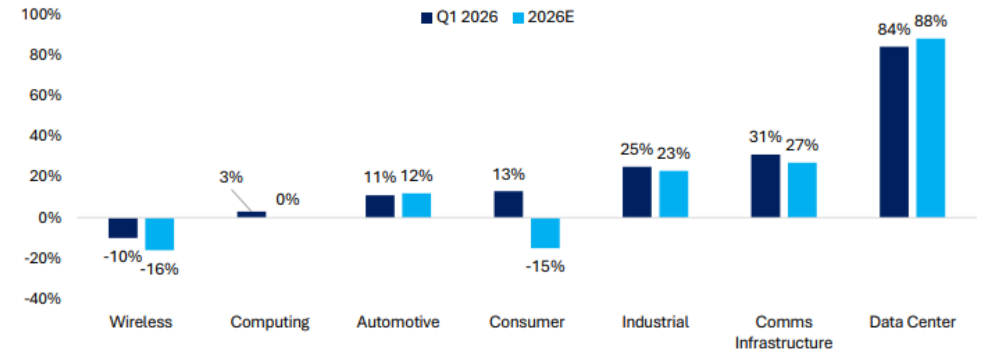
Figure 5: Who is exposed?

Data Processing	Communications	Consumer	Industrials
PCs Servers	Wireless Wireline	Consumer electronics	Manufacturing Automotive Aerospace & Defence MedTech

Source: Deutsche Bank Research

[12] Global Automotive: DRAMa ahead? - Deutsche Bank Research
[13] Global Automotive: DRAMa ahead? - Deutsche Bank Research
[14] US Semiconductors: Quarterly End-Market Monitor - Deutsche Bank Research
[15] Apple CEO warns of memory crunch. 'We'll look at a range of options'
[16] Apple quietly axes 128GB Mac Studio amid supply constraints and local AI frenzy — highest memory capacity reduced to 96GB, two months after discontinuation of 512GB model | Tom's Hardware
[17] https://www.tomshardware.com/laptops/8gb-of-ram-is-back-on-laptops-companies-are-lowering-memory-offerings-to-make-affordable-notebooks-during-component-crisis

Figure 6: End-market growth, Q1 vs. 2026 estimates



Source: Deutsche Bank equity analyst research

(iv) The macro spillover: the chipflation tax

With companies in consumer electronics and appliances for example forced to bear rising costs, the risk of a "memory inflation tax" on consumers is growing.

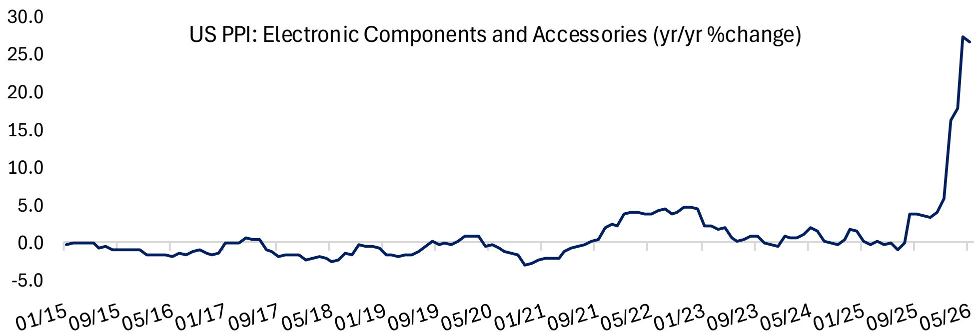
This is happening because end-product manufacturers – across PCs and smartphones for example – face severe Bill of Materials (BOM) cost pressure from the reallocation of silicon towards AI infrastructure. When memory prices surge, products with memory-heavy BOMs meet sharp production-cost increases, forcing companies to redesign or pass the "tax" down to consumers. We detailed the automotive and smartphone trade-offs above; if firms pass costs through, the average person takes a direct price impact. PC vendors Lenovo, Dell and ASUS have already warned of potential 15-20% price hikes and contract resets, with Lenovo and Dell starting increases this July.[18]

Consumers may also face financing and long-term stress: higher vehicle prices, for example, could push buyers into longer loan terms and higher total interest payments over the life of the asset.

In short, the memory-supply crisis has the potential to inflict acute second-order macroeconomic effects. The AI squeeze is crowding out the economy's ability to serve the day-to-day needs of ordinary people. Qualcomm and AMD have both flagged memory constraints actively hitting consumer-device demand.

We are already seeing it in US purchasing price inflation (PPI): the electronic components and accessories – which captures semiconductor and memory-devices prices – rose +26.9% y/y in May, up from +5.9% y/y rise in January. That, in turn, is pushing up upstream input costs between the businesses that trade these materials.

Figure 7: The sharp rise in PPI for electronics is driven by surging memory-driven chip costs



Source: Deutsche Bank Research, Haver.

[18] https://www.trendforce.com/news/2026/06/10/news-lenovo-reportedly-set-for-july-price-hikes-across-product-portfolio-as-memory-costs-pressure-pc-market/

3. Fault lines: The geopolitics of memory

(i) Winners and losers, country by country

South Korea is heavily exposed to the energy-intensive global AI capex boom, as its two tech giants (SK Hynix and Samsung) benefit from specializing in memory for AI data centres. The two companies account for around 69% of global DRAM production. However, that success heavily dictates Korea's economy, and when memory falls it drags Korean markets down with it. For example, after semiconductors melted down on June 8, the tech-heavy Kospi fell -8.29%, its ninth worst day in more than 12,000 trading sessions since Bloomberg data began in 1980. And earlier, in late March, Google's publication of a research algorithm called TurboQuant – which reduces the memory needed to run AI inference – also sent SK Hynix and Samsung shares lower. In short, the winners of the AI trade can quickly become the biggest losers if the gamble for global AI buildout fails.

China's semiconductor exports have also risen, as firms like CXMT (which specializes in DRAM), and Yangtze Memory Technologies (which specializes in NAND) benefit from industry's shift towards HBMs, which has left supply for standard DRAMs tighter. Exports surged by 72.6% y/y (in dollar terms) between Jan-Feb 2026, with memory the largest and fastest-growing category: customs data show memory exports soared 81.6% y/y (in dollar value) in Dec 2025.[19]

At the same time, Chinese companies are also hurt by the memory squeeze, as waiting lists for advanced memory and compute lengthen. Vendors (CXMT, JHICC) are expanding aggressively but are constrained to mature-node DRAM by US export controls. For example, CXMT's Shanghai fab expansion will add up to 200k wafers per month by 2030 but only begins high-volume manufacturing in 2027. Chinese producers will relieve some pressure in standard memory segments but are less exposed to HBM and automotive-grade DRAM segments. US export controls have shaped domestic fabrication capacity, so Chinese supply cannot substitute for the premium segments most affected. Ultimately, Chinese memory companies are currently focused on standard memory applications and are at an earlier stage in HBM development.

The United States is home to Micron (24% of global market share) and is the largest region by memory revenue. Outside AI buyers, there is real potential for US corporates and government to lean on China's mature-node advantage as a partial answer to the AI memory squeeze. For instance, in February, the Pentagon quietly removed China's CXMT and YMTC from its restricted-companies list, which makes US sourcing Chinese memory and NAND more plausible – though it is too early to judge whether adoption follows.[20] Several tech companies, including Apple and major PC OEMs like HP and Dell, are reportedly exploring or qualifying partnerships with Chinese memory makers to diversify supply chains and combat rising memory costs.

The geopolitical stakes are serious. Every dollar into CXMT and YMTC also benefits companies widely viewed as national-security risks – a hard case to make when the current US administration's priority has been to decouple from China (as evident from the MATCH Act). Whether US tech companies actually embark on these partnerships is uncertain. As noted earlier, China is confined to the lagging edge, so it would be difficult to justify a company like Apple shipping more phones built on a less advanced node than prior versions.

Congress has introduced the US MATCH Act, which aims to restrict countries from buying semiconductor manufacturing equipment technology from the US and its trade partners that they cannot build themselves. Importantly, MATCH designates restrictions on CXMT and YMTC across exports, servicing, and technical support. It has received broad bipartisan support but blowback from US chipmakers, who warn of further memory price

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[19] Gavekal, China Customs Administration
[20] https://wccftech.com/cxmt-ymtc-removed-from-pentagon-list-opening-door-for-chinese-dram-adoption/?utm_source=substack&utm_medium=email

rises, as the Act could cap global production capacity and shift pricing power back to a dominant Western-aligned oligopoly. Micron alone has publicly backed MATCH – unsurprisingly given it is in the company's interest to maintain price advantage.[21] If MATCH were to be implemented, it would have severe consequences for China's ability to industrialize.[22]

4. Breaking the bottleneck: What could actually fix it

(i) Building more fabs (and why it's slow)

To meet ongoing demand, the big 3 players are raising capex to help break the memory bottleneck. Our latest DB estimates place total DRAM wafers per month estimate to increase by +1,475k WSPM over the next five years, due to a variety of new fab announcements/expansions/accelerations.[23] Micron remains at the heart of a transformative moment in the memory industry, with a $200bn commitment to expand its memory capacity in the US, and will benefit from sustained tightness in demand. As we expressed earlier, however, fab construction is a multi-year process and will come online only after the peak of the shortage in the industry occurs.

Samsung, SK Hynix and Micron are trying to pull the timeline forward by buying partially finished or existing facilities. One example is Micron's $1.8bn acquisition of a used fab from Powerchip Semiconductor Manufacturing Corporation (PSMC) in Taiwan earlier this year, which lets Micron skip the roughly 2-year timeline needed to build a facility from scratch.[24] We expect other memory companies to follow in Micron's lead. Samsung is also reportedly negotiating long-term deals with Google and Microsoft, while Micron has disclosed its first five-year strategic customer agreement.

Whether China can be a credible player is still hard to gauge, as many new Chinese fabs remain under construction. But as we said before, these would primarily address mature-node memory demand, and the US export controls have shaped China's ability to fabricate at more advanced nodes.

(ii) Beyond fabs: Algorithms, SRAM and yield

Google's unveiling in March of an algorithm that requires less memory to serve LLMs ("TurboQuant") caused a sharp decline in the share price of Samsung (-6%), Micron (-7%) and SK Hynix (-7%) on the day.

However, the subsequent recovery in the memory market underscores that experiments like TurboQuant have yet to convince investors or AI accelerators that they a breakthrough is at hand that would turn the memory boom to bust. The key reason is scope: TurboQuant compress the inference-time KV cache – not model weights or training memory. While both focus on reducing memory bottlenecks, KV caching speeds up generative inference, whereas training memory optimization allows models to process massive datasets during the learning phase. Therefore, TurboQuant's potential impact was only limited to a part of total AI memory demand. More fundamentally, because cheaper inference historically expands total usage – the Jevons Paradox – such efficiency gains may ultimately lift, not cut, aggregate memory demand.

Still, TurboQuant is likely the first of several attempts to reduce reliance on HBM. Other strategic efforts include improved KV-cache eviction (such as Nvidia's inference context memory storage system) and greater reliance on SRAM.

There will likely be more attempts to upgrade node conversion (upgrading the manufacturing process to print smaller circuits) and yield improvements (the share of chips on a wafer that actually work). Both aim to lift the capacity of existing memory

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[21] https://finance.yahoo.com/economy/policy/articles/micron-backs-match-act-policy-180733909.html
[22] https://www.kim.senate.gov/press_release/senators-kim-and-ricketts-introduce-match-act-to-level-the-global-playing-field-for-u-s-tech/
[23] Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist - Deutsche Bank Research
[24] https://investors.micron.com/news-releases/news-release-details/micron-signs-letter-intent-purchase-tongluo-site-begin-0

production rather than building new fabs. The memory industry will nonetheless continue to face headwinds in bringing incremental capacity online.

Conclusion: From commodity to macro variable

The production of memory chips is becoming a zero-sum game. For every wafer devoted to HBM stacks for AI servers, others are unavailable for smartphones, PCs, or vehicles. And because the shortage will reverberate across key economic systems well beyond AI – hurting the average consumer via memory-cost inflation – memory chips have transitioned from a pure commodity to a distinctly macroeconomic variable.

The companies best positioned to survive the structural dynamics of the memory industry are, of course, the hyperscalers and AI cloud providers. And Micron stand out as a clear winner of the current shortage. For traditional PCs and smartphone manufacturers, memory is critical to device function, but unlike in AI, adds little value at higher performance or capacity; their objective is to minimize cost per bit while keeping systems efficient. As a result, the insatiable needs of the AI buildout are consuming capacity that would have gone to consumer-end markets. Consumer electronics, automotives and automotive-system suppliers will face headwinds large enough that sustained DRAM inflation will have a measurable inflationary impact in 2027, hurting average customers.

Looking ahead, the question worth considering is what the macroeconomic consequences would be if the AI gamble fails. Where would all the HBM and silicon capacity go? The sharp swings triggered by events like TurboQuant show that if AI demand moderates even modestly, the market could swing from severe shortage to meaningful oversupply relatively quickly.

References:

Thematic Research: Chips: The Biggest AI Bottleneck

Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist

Monthly SIA Data Digest: April 2026: Slight m/m dip, but better than expected vs. seasonal norms

Micron: A trip down memory lane; Initiate with Buy rating

US Semiconductors: Quarterly End-Market Monitor

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Appendix 1

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Deutsche Bank
Research Institute

Sovereign Ratings vs. Reality: India's Case for a Higher Rating

June 18, 2026

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Chief Economist

A cross-country comparison based on latest IMF data reveals important insights into India's fiscal position and sovereign rating relative to global peers. Advanced economies such as Japan, Italy, the United States, France and the United Kingdom continue to carry substantially higher public debt burdens—ranging from around 100% to over 200% of GDP—yet maintain significantly stronger sovereign ratings. In contrast, India's public debt ratio, is lower relative to these economies. This divergence suggests that sovereign ratings are influenced by a broader set of factors beyond debt levels alone, and that India's rating appears conservative when viewed purely through the lens of debt metrics.

India's macroeconomic fundamentals remain robust, characterized by sustained growth outperformance, anchored inflation, positive real interest rates, and a sound financial system. **Debt sustainability analysis** indicates that public debt is likely to follow a declining trajectory, supported by a favorable growth-interest rate differential.

India's sovereign credit profile reflects a combination of improving fiscal dynamics, strong growth prospects, and a resilient debt structure. While short-term fiscal deficits remain relatively high, medium-term consolidation, stable debt dynamics, and structural strengths suggest a stronger underlying credit position than implied by current ratings. **On balance, there appears to be a credible case for a reassessment of India's sovereign rating toward the upper end of the BBB range, particularly if fiscal consolidation continues broadly as envisaged.**

Sovereign Ratings vs. Reality: India's Case for a Higher Rating

Introduction

The pandemic in 2020 had a severe impact on the deficit and debt dynamics of almost all countries in the world, though in varying degrees. India was also severely impacted, with the general government deficit and debt rising sharply in the immediate aftermath of the pandemic, which was unavoidable. Since then, thanks to the strong resolve of the authorities, significant improvement has been achieved on the fiscal front, culminating first in a ratings outlook upgrade to positive from stable by S&P Global Ratings on 29 May 2024 and then a subsequent sovereign ratings upgrade to BBB from BBB- on 14th August 2025 after a gap of 18 long years.

Fiscal consolidation has gathered significant momentum in the last few years, thanks to robust growth, fiscal marksmanship and particular focus on improving the quality of fiscal spending. Indeed, both the interim budget in Feb 2024 and the post-elections union budget in July 2024 exceeded expectations as regards to the fiscal deficit target, which is praiseworthy. The FY26 Union Budget also saw the central government announcing a further moderation in the fiscal deficit target to 4.4% of GDP, vs. a revised estimate outturn of 4.84% of GDP in FY25, which was finally met, despite concerns related to higher tariffs imposed by USA in 2025, a lower-than-expected nominal GDP growth profile and revenue uncertainty due to GST rates rationalisation.

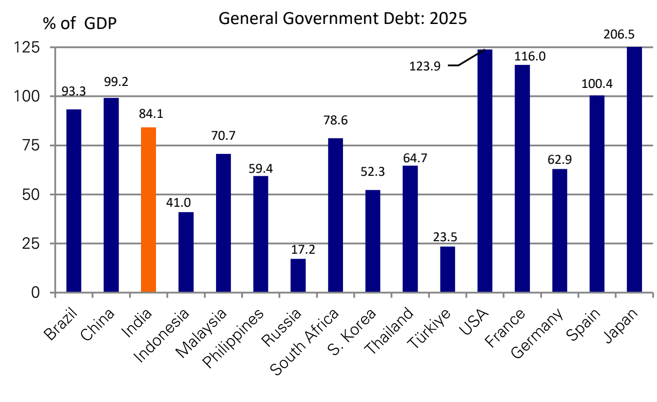
Amidst ongoing concerns related to tariff uncertainty in the beginning of this year and in the backdrop of heightened geopolitical tensions, the government presented the Union Budget for FY27 on 1st February 2026. Striking a delicate balance between the dual objectives of supporting growth and employment, while remaining on the path of fiscal consolidation, the budget set a fiscal-deficit target of 4.3% of GDP for FY27, with the authorities reemphasizing the medium-term fiscal goal of reducing the centre's debt/GDP ratio to the pre-pandemic level of 50% by FY31, while setting the FY27 target at 55.6% of GDP.

Indian sovereign ratings relative to rest of the world

In the following pages, we present a series of charts and tables using latest IMF data for major advanced and emerging economies (including India) covering six key dimensions:

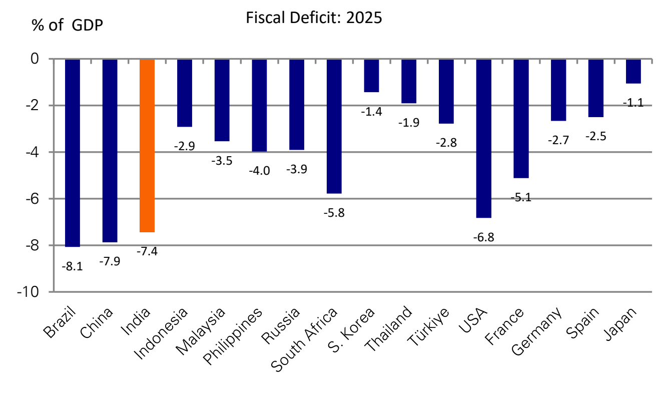
1. General government debt as a % of GDP;
2. General government budget deficit as a % of GDP;
3. Expected change in debt and deficit between 2025 and 2031 based on IMF's forecasts;
4. Ranking of each of these economies by their size (2025 data; nominal GDP in USD trn terms);
5. Sovereign credit ratings assigned by S&P, Moody's and Fitch;
6. We also provide a table which show the IMF's interpretation of credit rating scale of major credit rating agencies.

Figure 1: General Government Debt: 2025



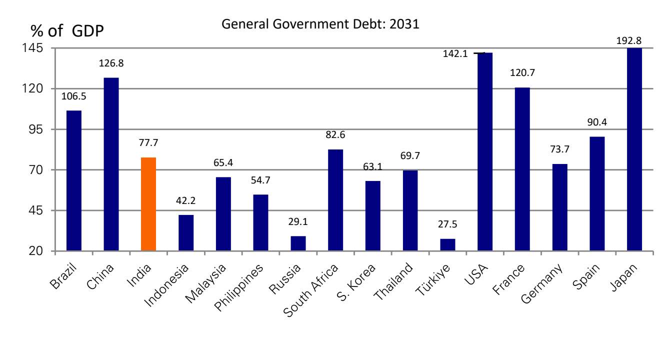
Source : IMF: World Economic Outlook Database, April 2026

Figure 2: General Government Deficit: 2025



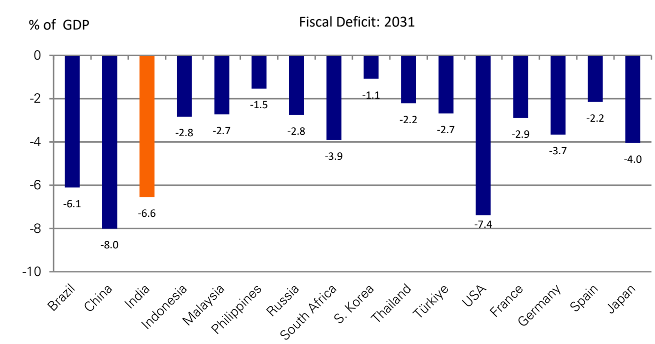
Source : IMF: World Economic Outlook Database, April 2026

Figure 3: General Government Debt: 2031 F



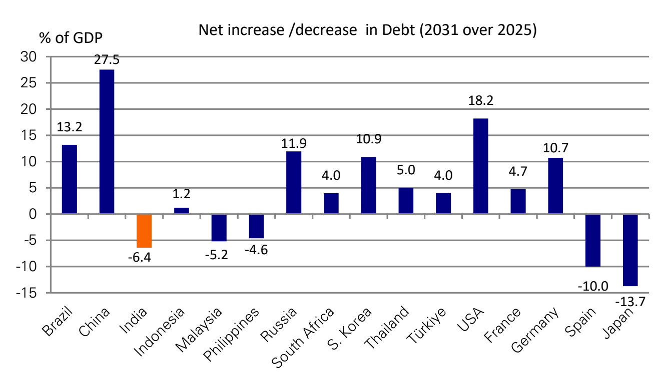
Source : IMF: World Economic Outlook Database, April 2026

Figure 4: General Government Deficit: 2031 F



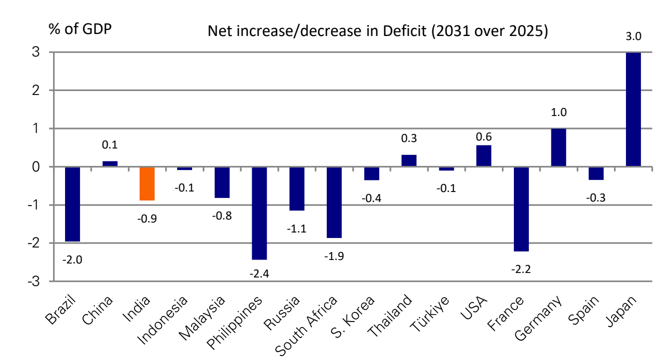
Source : IMF: World Economic Outlook Database, April 2026

Figure 5: Net change in Debt: 2031 over 2025



Source : IMF: World Economic Outlook Database, April 2026

Figure 6: Net change in Deficit: 2031 over 2025



Source : IMF: World Economic Outlook Database, April 2026

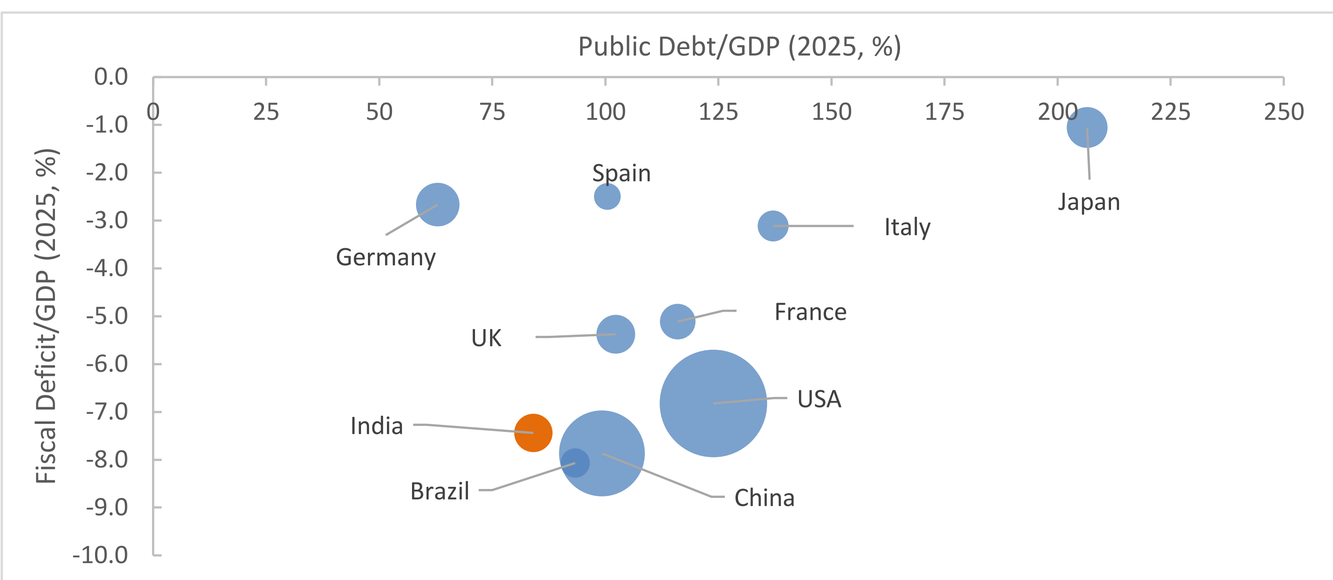
India's sovereign credit ratings, debt and deficit profile and economic size relative to other economies - a summary

Figure 7: Debt, deficit and creditworthiness - advanced and emerging economies

	Country	Nominal GDP USD trn, 2025	Rank By size	Public debt/GDP 2025	Fiscal deficit/GDP 2025	Credit Rating		
						S&P	Moody's	Fitch
1	USA	30.8	1	123.9	-6.8	AA+	Aa1	AA+
2	China	19.6	2	99.2	-7.9	A+	A1	A
3	Germany	5.0	3	62.9	-2.7	AAA	Aaa	AAA
4	Japan	4.4	4	206.5	-1.1	A+	A1	A
5	UK	4.0	5	102.3	-5.4	AA	Aa3	AA-
6	India	3.9	6	84.1	-7.4	BBB	Baa3	BBB-
7	France	3.4	7	116.0	-5.1	A+	Aa3	A+
8	Italy	2.6	8	137.1	-3.1	BBB+	Baa2	BBB+
9	Brazil	2.3	10	93.3	-8.1	BB	Ba1	BB
10	Spain	1.9	11	100.4	-2.5	A+	A3	A
11	S. Korea	1.9	12	52.3	-5.8	AA	Aa2	AA-
12	Turkiye	1.6	15	23.5	-2.8	BB-	B1	BB-
13	Indonesia	1.4	16	41.0	-2.9	BBB	Baa2	BBB
14	Thailand	0.6	29	64.7	-1.9	BBB+	Baa1	BBB+
15	Philippines	0.5	33	59.4	-4.0	BBB+	Baa2	BBB
16	Malaysia	0.5	34	70.7	-3.5	A-	A3	BBB+
17	South Africa	0.4	40	78.6	-5.8	BB	Ba2	BB-

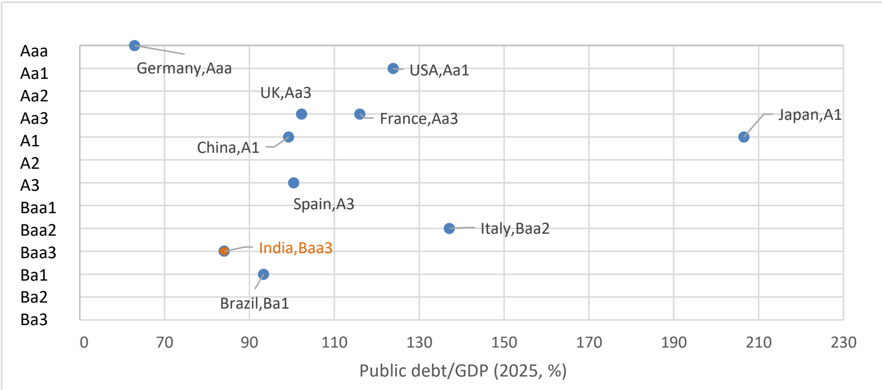
Source : IMF: World Economic Outlook Database, April 2026, S&P, Moody's and Fitch

Figure 8: Debt, deficit and GDP size across major economies (2025)



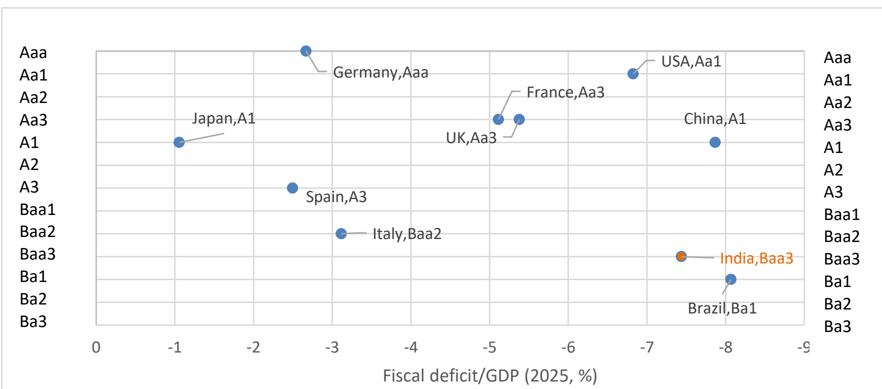
Source : IMF, S&P, Moody's and Fitch. Note: Bubble size = Nominal GDP

Figure 9: Debt/GDP vs. Moody's credit ratings across major economies (2025)



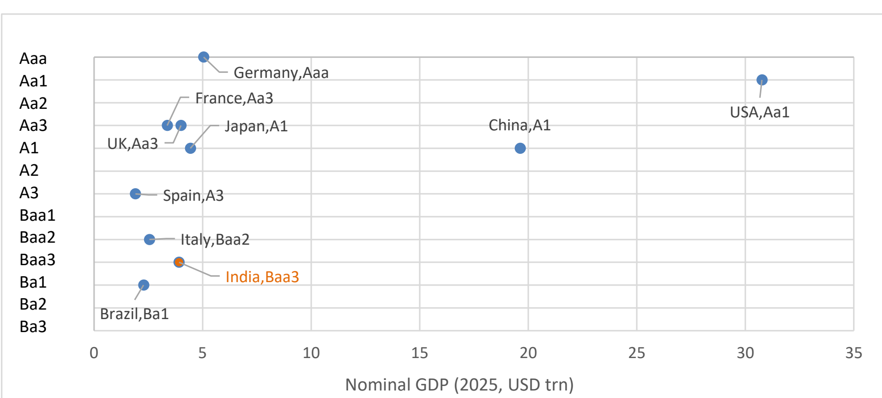
Source : IMF, Moody's

Figure 10: Deficit/GDP vs. Moody's credit ratings across major economies (2025)



Source : IMF, Moody's

Figure 11: Nominal GDP vs. Moody's credit ratings across major economies (2025)



Source : IMF, Moody's

Figure 12: IMF's interpretation of credit rating scale of some major credit rating agencies

Interpretation	Fitch and S&P	Moody's
Highest quality	AAA	Aaa
High quality	AA+	Aa1
	AA	Aa2
	AA–	Aa3
Strong payment capacity	A+	A1
	A	A2
	A–	A3
Adequate payment capacity	BBB+	Baa1
	BBB	Baa2
	BBB–	Baa3
Likely to fulfill obligations, on going uncertainty	BB+	Ba1
	BB	Ba2
	BB–	Ba3
High-risk obligations	B+	B1
	B	B2
	B–	B3
Vulnerable to default	CCC+	Caa1
	CCC	Caa2
	CCC–	Caa3
Near or in bankruptcy or default	CC	Ca
	C	C
	D	D

Source : Economic Survey 2020-21 Volume 1, IMF: 2010: The Uses and Abuses of Sovereign Credit Ratings

Observations

*** Do India's current sovereign ratings adequately reflect its economic fundamentals?** India appears fiscally stronger than several countries that enjoy significantly higher sovereign ratings**,** particularly when debt levels, economic growth prospects and future fiscal trends are considered together. While India still runs a relatively large fiscal deficit , its debt burden is moderate and is expected to improve over time. This raises a legitimate question as to whether India's current sovereign ratings fully reflect its economic fundamentals and growth potential.

*** Public debt/GDP and sovereign ratings:** The 2025 IMF data show a wide heterogeneity among countries. The highest debt/GDP ratios are observed in the case of Japan (206%), Italy (137%), USA (124%), France (116%), and UK (103%). These countries have accumulated substantial public debt over many years due to aging population dynamics, welfare commitments, financial crises and pandemic-related spending. Despite these elevated debt levels, they continue to enjoy substantially higher sovereign ratings compared to India. As per the IMF's data, India's public debt/GDP and general government budget deficit stood at 84% and 7.4% respectively. Only Germany and some emerging economies have materially lower debt levels compared to India.

*** General government deficit/GDP and sovereign ratings:** India's general government budget deficit remains elevated at around 7.4% of GDP, which is on the higher spectrum of the economies considered in our analysis. However, the IMF's projections till 2031 factor in a meaningful consolidation of India's deficit (from 7.4% of GDP in 2025 to 6.6% of GDP by 2031) and public debt (from 84.1% of GDP in 2025 to 77.7% of GDP by 2031) by the end of this decade. The improvement suggests i) continued economic growth (10.8% average nominal GDP growth between 2026-2031); ii) rising tax revenues; and iii) better fiscal and expenditure discipline. **Indeed, India's debt trajectory appears more favorable than that of several major economies, where debt ratios are projected to increase further in the coming years**.

*** Size of the economy and sovereign ratings:** As far as size of the economy is concerned, **India is currently the 6th-largest economy, comparable to the UK and larger than France, and slated to become the third largest before the end of this decade, as per our and IMF forecasts.** Yet, India's sovereign ratings currently remain at the lowest investment grade (Moody's and Fitch). Therefore, on the basis of the size of the economy, India appears underrated, in our view.

*** Funding of the public debt and sovereign ratings:** In this respect, it is instructive, in our view, to reflect on the IMF's interpretation of credit rating scale of major credit rating agencies. As per Fitch/S&P scale, A range (A+, A, A–) signifies "Strong payment capacity", while BBB range (BBB+, BBB, BBB–) reflects "Adequate payment capacity" and BB range and below highlight increasing uncertainty and risk, as per the IMF's interpretation. India currently sits at BBB as per S&P, while ranked at the lowest level of investment grade by Fitch (BBB-) and Moody's (Baa3). That places India at the lowest rung of "adequate payment capacity".

In our view, **there is little risk associated with the funding of the public debt** as bulk of India's debt is in domestic currency. Indeed, foreign currency denominated debt is small - less than 5% of GDP - and even within that, more than half of it is owed to multilateral institutions. A country, like India, with mostly domestic currency debt, does not face currency mismatch risk (unlike EM crisis cases), can rely on its domestic financial system to absorb debt, has more monetary flexibility (central bank support) and is less exposed to external funding crises. **These are typically characteristics associated with higher creditworthiness (often A-category countries).**

India, in our view, shows traits consistent with "strong payment capacity", being a large, diversified domestic economy (#6 globally; on its way to become #3 before the end of this decade); holding predominantly local-currency debt; deep and growing domestic financial market; limited external vulnerability compared to many EM peers. These characteristics align more closely with "A category" descriptors than the BBB range in our view. While S&P upgraded India's sovereign ratings one notch higher to BBB, from BBB- in August 2025, both Fitch (BBB-) and Moody's (Baa3) have kept India's sovereign ratings at the lowest level of investment grade.

Even if the rating agencies do not provide India with an A-category sovereign rating, reflecting "Strong payment capacity", due to lower-per-capita income relative to peers, **there could be merit to at least consider upgrading India to the upper rung of the "Adequate payment capacity" (BBB+ for S&P and Fitch) and Baa1 for Moody's, which will require one more rating upgrade by S&P and a double-notch ratings upgrade by Fitch and Moody's, in our view.**

Figure 13: India's sovereign credit rating (1998 -2024)

Date	S&P	Moody's	Fitch
June 1998		Ba2*	
October 1998	BB*		
March 2000			BB+*
November 2001			BB*
February 2003		Ba1*	
January 2004			BB+*
January 2004		Baa3	
February 2005	BB+*		
August 2006			BBB-
January 2007	BBB-		
November 2017		Baa2	
June 2020		Baa3	

*Speculative Grade; Green highlights ratings upgrade; Red highlights ratings downgrade, Black indicates first rating

Source : Economic Survey 2020-21 Volume 1, Compiled from S&P Global, Fitch and Moody's

A debt sustainability exercise for India

Next, we undertake a debt sustainability exercise for India, to ascertain whether the forecasted debt consolidation path has changed materially on account of the current oil-price shock. In our view, this is useful for providing a reference point of the progress that is likely under the baseline scenario. Then, using this reference point, we calculate the upside and downside risks, based on various scenarios. This assumes particular importance, as Government of India's medium-term fiscal roadmap is now aimed towards achieving a sustained debt consolidation path by the end of this decade, but without pre-committing explicitly to any fiscal deficit target. **As per the plan, the centre aims to reduce its debt/GDP ratio to the pre-pandemic level of 50% by FY31, from about 55.6% currently**.

Fiscal snapshot at the general government level

The two tables below summarize the fiscal performance at the general government level pre- and post-Covid. General government budget and primary deficits were on a path of consolidation until FY19, while interest payments remained broadly sticky at around 4.8% of GDP. The Covid-19 pandemic impacted the fiscal and debt position severely, with **general government deficit rising to 13.1% of GDP by FY21, from 5.8% of GDP in FY19.** This is despite the central government being extremely prudent with its targeted expenditure support for the absolutely vulnerable and middle-class section of the population. Interest payments as a % of GDP increased to 5.4% of GDP by FY21, from 4.8% of GDP in FY19, while **general government debt increased to 88% of GDP in FY21, from about 70% in FY19.**

Figure 14: General Government fiscal snapshot

% of GDP	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27F
Revenue	20.7	19.8	21.6	22.2	22.5	22.5	22.6	22.7
Expenditure	27.9	32.9	32.0	31.2	30.9	30.0	29.9	30.0
Interest payment	4.8	5.4	5.3	5.2	5.3	5.2	5.3	5.2
General government deficit	**7.2**	**13.1**	**10.3**	**9.0**	**8.4**	**7.5**	**7.3**	**7.3**
Primary deficit	2.4	7.7	5.1	3.7	3.2	2.3	2.2	2.3

Source: Ministry of Finance, Deutsche Bank forecasts

Figure 15: Liability position of the centre and the states

% of GDP	FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
I. Central govt debt	47.1	47.4	45.5	48.9	49.4	52.3	61.5	58.8	57.9	58.1	57.2	57.2
II. State govt debt	21.0	22.5	23.8	25.1	25.3	26.7	31.0	29.1	28.0	27.5	27.5	27.5
III. Central loans to state	1.2	1.1	1.0	3.7	3.2	2.8	3.5	3.6	3.4	3.4	3.3	3.3
IV. Investment in treasury bills by states	1.0	1.2	1.3	1.2	1.0	0.9	1.2	1.3	1.0	1.1	1.1	1.1
General govt debt = I + II - III - IV	**65.9**	**67.6**	**67.1**	**69.0**	**70.4**	**75.2**	**87.8**	**83.0**	**81.3**	**81.1**	**80.3**	**80.3**

Source:Ministry of Finance, Deutsche Bank forecasts

States' debt/GDP was also affected severely, rising from 21% of GDP in FY19 (excluding central loans to states and investment in treasury bills by states) to 26% of GDP by FY21. States' combined fiscal deficit also increased to 4.1% of GDP in FY21, from 2.5% of GDP in FY19. Meanwhile, the **centre's** fiscal deficit increased to 9.2% of GDP in FY21, from 3.4% of GDP in FY19 and debt/GDP worsened to 61.5% of GDP in FY21, from 49.4% of GDP in FY19.

Figure 16: Centre and state fiscal snapshot

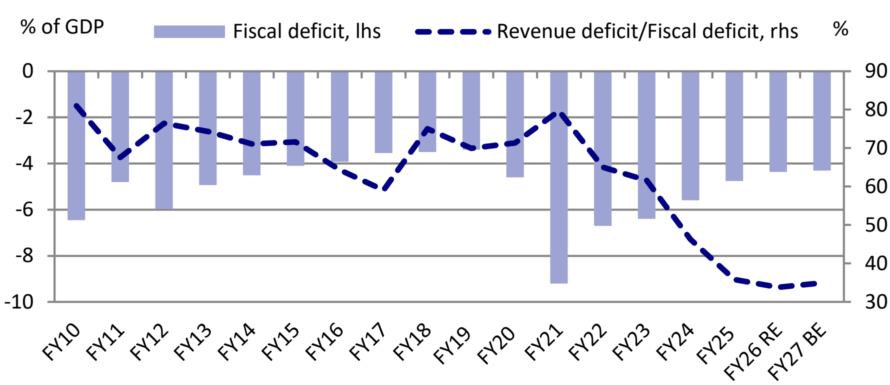
% of GDP	FY22	FY23	FY24	FY25	FY26	FY27F	FY28F
Central government balance	**-6.7**	**-6.4**	**-5.7**	**-4.9**	**-4.4**	**-4.3**	**-4.2**
Government revenue	9.4	9.0	9.6	9.7	9.9	9.4	9.8
Government expenditure	16.2	15.4	15.3	14.6	14.3	13.8	14.0
Central primary balance	-3.3	-3.0	-2.0	-1.4	-0.8	-0.6	-0.5
Consolidated deficit	**-9.5**	**-9.1**	**-8.5**	**-7.7**	**-7.2**	**-7.1**	**-7.0**
Memo							
State fiscal deficit	-2.8	-2.7	-2.8	-2.8	-2.8	-2.8	-2.8
Public sector debt/GDP	**83.0**	**81.5**	**81.1**	**80.3**	**80.8**	**79.8**	**79.5**

Source : Ministry of Finance, Deutsche Bank forecasts

Centre's quality of fiscal spending has improved significantly, but has peaked out

An indicator to track in this regard is the **revenue deficit/fiscal deficit ratio** (lower the ratio, the better). Since FY17 (which recorded a low of 59.1%), this ratio moved up to a peak of nearly 80% in FY21, but it has improved once again and fallen below 40%. Revenue deficit/fiscal deficit moderated to 33.8%, from 35.8% in FY25. Beyond this, we see little scope for this ratio to improve significantly, given that the centre's capex spending allocation is reaching a peak, while it may be difficult to reduce revenue expenditure further as most of these expenditures are by definition "committed expenditure". Indeed, the authorities are targeting a revenue deficit/ fiscal deficit ratio of 34.9% in FY27, higher than 33.8% in FY26.

Figure 17: Centre's quality of fiscal consolidation has improved significantly



Source : Various budget documents, RBI, MOF and Deutsche Bank

Short-term fiscal projections

Since the peak deterioration of the fiscal and debt dynamic in FY21, owing to the adverse impact of the Covid-19 pandemic, significant progress has been made on the fiscal front, both at central and state government level.

- **Centre's fiscal deficit** has come down to 4.4% of GDP in FY26 and the government has set a target of 4.3% of GDP for FY27. **Despite the oil-price shock, we think the central government has enough buffers to be able to meet the FY27 fiscal deficit target of 4.3% of GDP.** The below table shows our latest projections for FY27 over the FY26 outturn.

Figure 18: Latest FY27 fiscal projections: Deutsche Bank vs. Government estimates

	Key items	2024-25 Actual	2025-26 RE	2026-27 DB Estimate	2026-27 Budget Estimate	2024-25 Actual	2025-26 RE	2026-27 DB Forecast	2026-27 Budget Estimate	FY26 RE over FY25 Actual	FY27 DB est over FY26 RE	FY27 BE over FY26 RE
S. no		INR bn	INR bn	INR bn	INR bn	% of GDP	% of GDP	% of GDP	% of GDP	% yoy	% yoy	% yoy
1	Revenue Receipts (2+3)	30366	33423	35280	35332	9.55	9.67	9.20	9.0	10.1	5.6	5.7
2	Tax Revenue (Net to Centre)	25000	26747	27840	28669	7.86	7.74	7.26	7.29	7.0	4.1	7.2
3	Non Tax Revenue	5366	6677	7439	6662	1.69	1.93	1.94	1.7	24.4	11.4	-0.2
4	Recovery of Loans	246	302	307	384	0.08	0.09	0.08	0.1	22.6	1.6	27.2
5	Other Receipts (disinvestments)	172	338	499	800	0.05	0.10	0.13	0.2	96.7	47.3	136.4
6	**Total Receipts (1+4+5)**	**30784**	**34064**	**36085**	**36515**	**9.68**	**9.86**	**9.41**	**9.29**	10.7	5.9	7.2
7	**Total Expenditure (8+10)**	**46529**	**49648**	**52728**	**53473**	**14.63**	**14.37**	**13.75**	**13.61**	6.7	6.2	7.7
8	On Revenue Account	36009	38691	41300	41255	11.32	11.20	10.77	10.5	7.4	6.7	6.6
9	of which: Interest Payments	11156	12743	14189	14040	3.51	3.69	3.70	3.6	14.2	11.3	10.2
10	On Capital Account	10520	10958	11428	12218	3.31	3.17	2.98	3.1	4.2	4.3	11.5
11	Revenue Deficit (8-1)	5643	5268	6021	5923	1.77	1.52	1.57	1.51			
12	**Fiscal Deficit (7-6)**	**15744**	**15585**	**16643**	**16958**	**4.95**	**4.51**	**4.34**	**4.31**			
13	Primary Deficit (12 - 9)	4589	2842	2454	2918	1.44	0.82	0.64	0.74			
	Memo item											
14	Nominal GDP, %yoy	318073	345472	383473	393004					8.6	11.0	13.8

Source : Various budget documents and Deutsche Bank

- **States' combined fiscal deficit** is likely to stay steady at 2.8% of GDP, as per our conservative baseline assumption. Prior to Covid, states' fiscal deficit was at 2.4% of GDP and 2.6% of GDP in FY19 and FY20, respectively.
- **General government deficit** is likely to moderate to 7.1% of GDP in FY27, from 8.5% of GDP in FY24.
- **General government debt** is likely to moderate to 79.8% of GDP in FY27, due to higher-than-expected nominal GDP growth, from 80.3% of GDP in FY26.
- **Central government deficit with and without EBR (Extra Budgetary Resources):** Prior to the Covid-19 pandemic, the central government accounted certain Extra Budgetary Resources as off-balance sheet items, which ideally should have been calculated as on-balance sheet expenditure, resulting in a higher-than-reported fiscal deficit. For example, in FY19, when the reported fiscal deficit was 3.4% of GDP, the EBR in that fiscal year was 0.9% of GDP; therefore, incorporating the EBR would have resulted in fiscal deficit rising to 4.3% of GDP. In FY17 and FY18, the reported fiscal deficit was 3.5% of GDP, but adding 0.5% of GDP of EBR for both those years, the augmented fiscal deficit of the central government stood at 4.0% of GDP. Post FY22, the central government has taken all expenditure on balance sheet and the practice of providing off-balance sheet EBR has stopped, in a bid to promote fiscal transparency. We assume that, in the coming years, the centre's fiscal deficit will not fall below 4.0% of GDP, which would be similar to the pre-pandemic low of 4.0% of GDP (after adding 0.50% of GDP worth EBR to the headline fiscal deficit of 3.5% of GDP). Assuming the states' fiscal deficit stabilizes at 2.8% of GDP, this will likely result in general government deficit stabilizing at the current 6.8% of GDP within the forecast horizon.

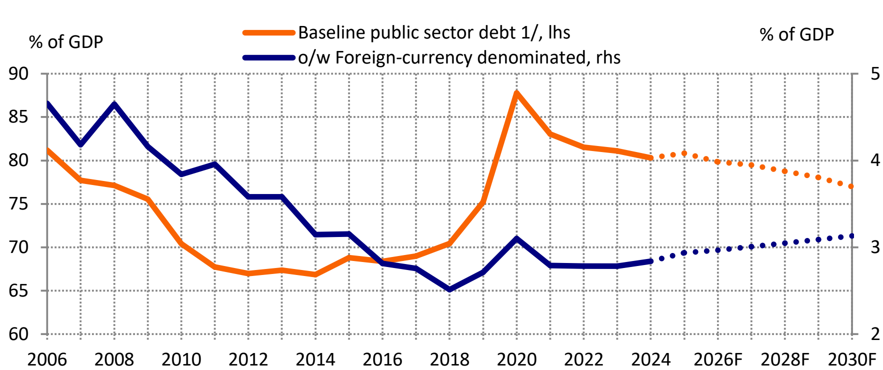
In the rest of this report, we use our own economic forecasts to present a debt sustainability analysis of the general government to ascertain the medium-term debt/GDP trajectory under various scenarios.

Baseline assumptions and projection

- Real GDP grows around 6.5% average in the rest of this decade (till FY31); nominal GDP growth averages about 10.5-11% during the forecasting period;
- With a flexible inflation targeting framework firmly in place, CPI inflation averages in the 4.0-4.5% range for the rest of this decade; we have assumed GDP deflator (mix of CPI and WPI) to average 4.5% in the years ahead;
- Real interest rate on public debt works out to 2.5% on an average (derived from 7.0% nominal rate and 4.5% GDP deflator-based inflation);
- We assume general government deficit (centre + state) to trough at 6.8% of GDP through the forecasting period, with the centre's fiscal deficit stabilizing around 4.0% of GDP, and 2.8% of GDP in case of states. There is scope for general government deficit to be lowered further beyond this, but we have assumed for now that, given the need for supporting growth through fiscal impulse and accommodating 8th Pay Commission recommended higher wages in the coming years, the combined deficit may not fall significantly below the 7.0% of GDP mark (pre-Covid 10-year average was 6.9% of GDP).
- The primary deficit eases in the period ahead, averaging close to 2.2% of GDP; primary spending rises by 5-6% in real terms, while revenues rise 10.5-11% on an average in nominal terms, in line with the projected nominal GDP growth for the rest of this decade.
- The rupee maintains a depreciation bias in the medium term; but given India's relatively small public external debt burden (2.8-3.0% of GDP), the exchange rate assumption matters little in the debt sustainability exercise.

Under this baseline scenario, public debt is projected to decline to about 75.1% of GDP by 2030 (FY31), from about 80.3% in FY26. This outcome is primarily driven by the assumption that the real growth-real interest rate differential stabilizes at 4-4.5% during the forecasting period. **India's debt burden does not look onerous under this scenario, but the interest cost of servicing the debt remains high during the forecasting period, accounting for 16-16.5% of total consolidated spending (or about a quarter of central government spending).** Given that India needs to ramp up total public spending on key sectors such as health, education and defense, the importance of reducing debt and interest costs is all too apparent.

Figure 19: Baseline Debt/GDP ratio - on a steady downward trajectory



Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

Problems associated with long-term projections - a cautionary tale

A word of caution - medium-term forecasts are subject to various risks which are difficult to anticipate and therefore the actual outturn could be appreciably different from the projected path. The Covid-19 pandemic is a testimony to this fact. A few years prior to the Covid-19 pandemic in 2020, the central government had formed an expert committee in 2016 to recommend a desirable fiscal and debt consolidation path for India over the medium term.

The FRBM committee recommended the central government continues on the path of fiscal consolidation, with the medium-term goal of bringing the centre's fiscal deficit down to 2.5% of GDP by FY23 (from 3.5% of GDP in FY17), which would result in the centre's debt/GDP coming down to 40% of GDP (from about 49% in FY18). The committee also recommended a desirable fiscal path for the state governments, so that cumulatively the debt/GDP of states can remain constant at around 20% of GDP throughout the forecasting period. This would require state fiscal deficit to reduce from 3.0% of GDP in FY17 - by about 1% of GDP - to 2.0% by FY23. If this was achieved, then India's overall public debt/GDP ratio would have moderated to about 60% of GDP by FY23, a 10% drop from FY18 levels (general government deficit was close to 70% of GDP in FY18).

But then Covid-19 happened in 2020, and the seismic growth shock pushed general government debt/GDP close to 90% in FY21, a 20% increase from FY18 levels. By FY23, public debt moderated but it was still above 81.5% of GDP. India's public debt/GDP is expected to moderate to 75.1% by 2030 (FY30/31), but that will still be higher than the 70% debt/GDP that was recorded in FY18. Or, in other words, a seismic shock like the Covid-19 pandemic set India's debt/ GDP trajectory back by a decade. This is true not just for India, but for most countries around the world.

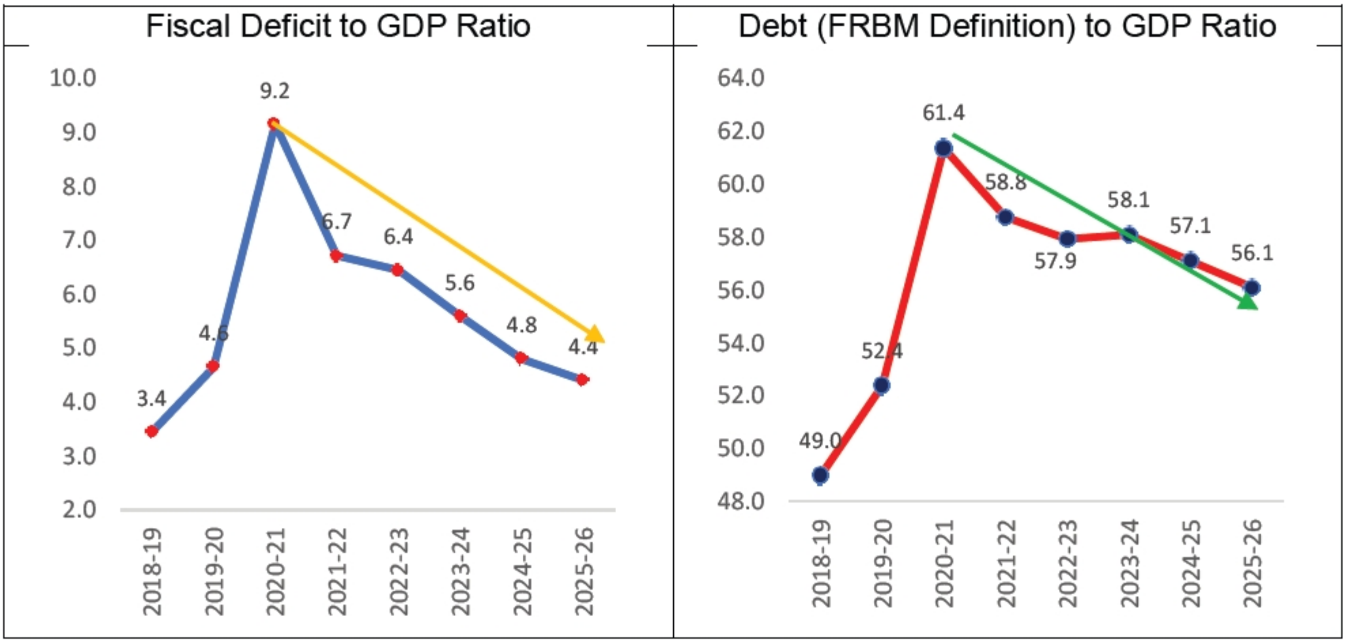
Figure 20: FRBM committee recommended fiscal target

	FRBM committee recommended target			DB estimate
% of GDP	Central govt.	Sate govt. fiscal	General govt.	General govt.
FY17	3.5	3.0	6.5	6.5
FY18	3.0	2.8	5.8	6.2
FY19	3.0	2.7	5.7	6.0
FY20	3.0	2.5	5.5	6.0
FY21	2.8	2.3	5.1	6.0
FY22	2.6	2.2	4.8	6.0
FY23	2.5	2.0	4.5	6.0

Source: FRBM Review Committee Report, Deutsche Bank

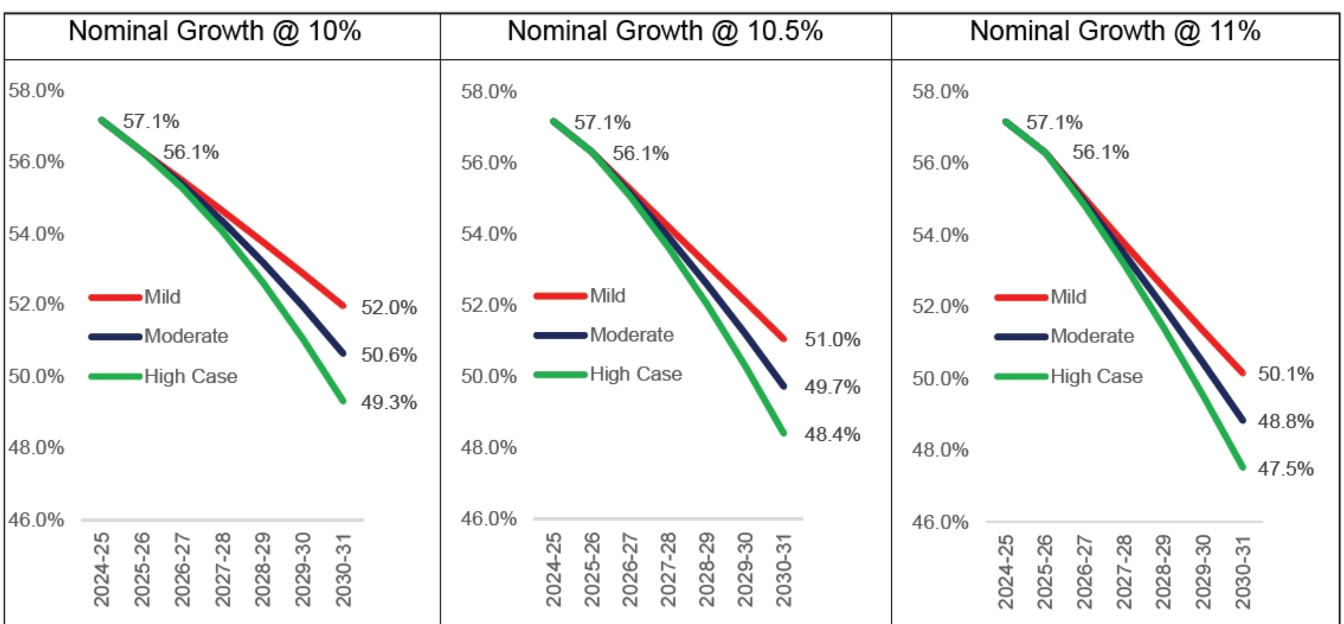
Therefore, it is important to recognize that the baseline scenario of debt/GDP projections can easily be derailed, if there is a large growth shock within the forecasting horizon. **The baseline debt/GDP projection does not incorporate any domestic or external shock. But, as has been illustrated vividly through the last decade, India's fiscal regime is vulnerable to various shocks.** It has also been seen that, even after an extended high-growth period, fiscal consolidation proves to be transitory when growth shocks materialize, or an external shock (global tariff shock, for example) weakens the fiscal position indirectly by impacting growth and revenue adversely.

Figure 21: Central government fiscal deficit and debt/GDP



Source : Ministry of Finance - Government of India

Figure 22: Central govt. debt/GDP projection by Ministry of Finance, Government of India



Note: Mild/Moderate/High case refer to degrees of fiscal consolidaton.

Source : Ministry of Finance

Keeping this in mind, in the rest of this note, **we introduce some shocks to our baseline scenario and see how the projections change** if indeed some of the key macro variables associated with debt dynamic were to take a turn for the worse in the coming years.

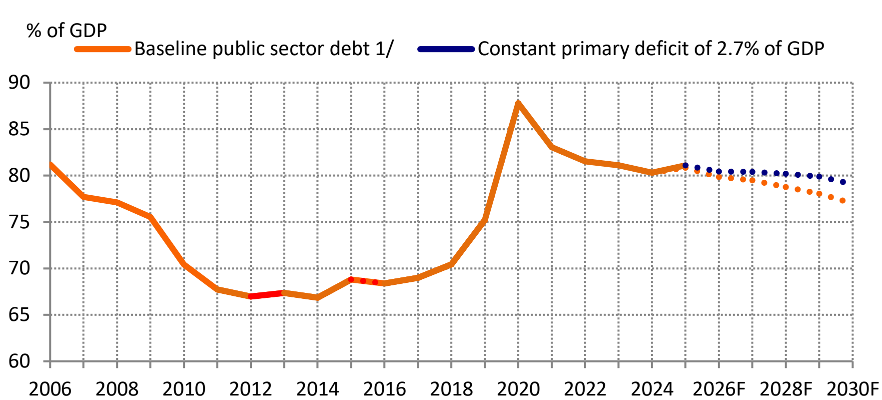
Stress Tests

Shock 1: No change in fiscal effort - primary deficit stays around 2.7% of GDP

If the primary deficit stays at around 2.7% of GDP (higher than our baseline assumption of about 2.2% of GDP), and the growth/interest rate nexus discussed above remains in place, India's debt path would still be on downward path, but the **debt/GDP ratio would fall to around 77.3% of GDP by 2030 (FY30/31).** A flatter path of debt adjustment should be a source of worry as fragility to shocks would rise.

State fiscal finances pose a big risk to the overall fiscal consolidation agenda. While on an aggregate basis, states' combined fiscal deficit has moderated once again below 3% of GDP (after rising to 4.1% of GDP in FY21 due to the impact of Covid-19), there is a large degree of heterogeneity and vulnerability regarding the fiscal health of certain states, which warrant close monitoring. **The states that would likely remain the most vulnerable to debt sustainability risks are Punjab, West Bengal, Bihar, Rajasthan and Kerala, given their weak fiscal and debt metrics even prior to the Covid-19 crisis, as per our analysis.**

Figure 23: Debt dynamic favorable to achieve decline in debt path even with modest fiscal effort

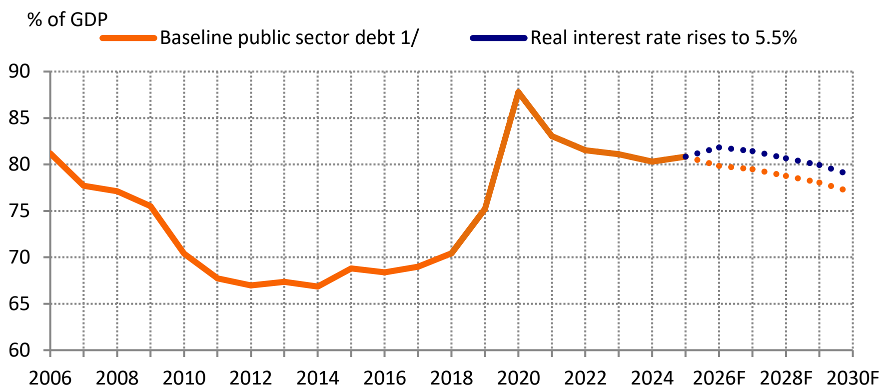


Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

Shock 2: Interest rate shock - real interest rate rises to about 5.5% on an average

If CPI inflation increases above the 6% upper threshold inflation target range in the years ahead, RBI would be compelled to hike the policy rate, which may increase the real interest rate to mid single digit, when measured in terms of WPI and GDP deflator. Under the interest rate shock scenario, real interest rate paid on public debt is raised by one standard deviation of the past average; **under this scenario, the debt trajectory flattens and debt/GDP comes down only to 76.9% of GDP by FY31 (2030).**

Figure 24: A rise in real interest rate could cause adverse debt dynamic

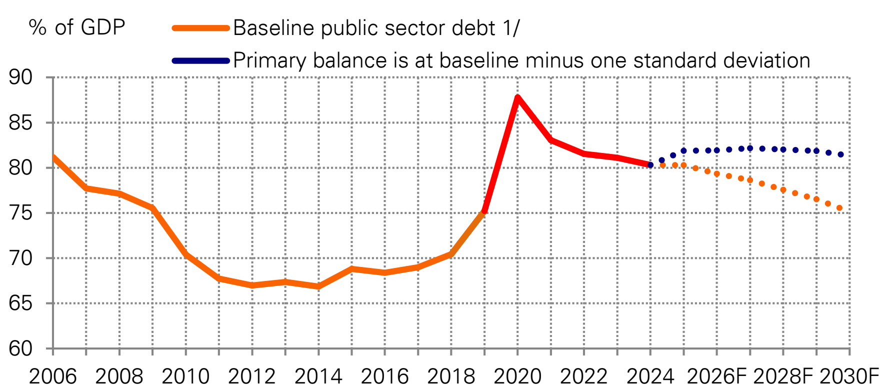


Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

Shock 3: Fiscal shock - primary deficit rises to 3.5% of GDP within the forecasting period instead of coming down to 2.2% of GDP

If the primary deficit rises to around 3.5% of GDP (higher than our baseline assumption of about 2.2% of GDP), and the growth/interest rate nexus discussed above remains in place, India's debt path would flatten considerably, with the **debt/ GDP ratio likely to remain unchanged at about 81.3% of GDP by 2030** (FY31).

Figure 25: A fiscal shock will affect the debt dynamic

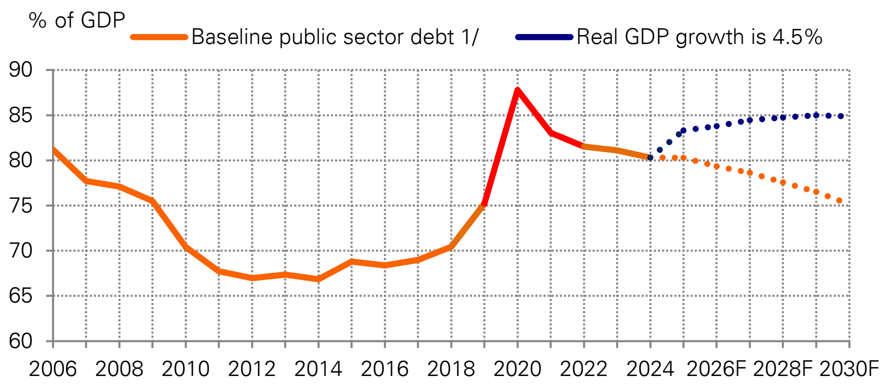


Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

Shock 4: Growth shock - real GDP growth moderates to 4.5%

Here we lower real GDP growth by 200bps, which pegs economic growth to around 4.5%.The shock impacts the debt path severely, **pushing up the debt ratio to 84.9% by FY31 (2030),** 10% higher than our baseline estimate of debt/GDP falling to 74.8% by the end of this decade. Indeed, the biggest risk to debt sustainability arises from a growth shock, as was witnessed during the time of Covid-19 pandemic. Of course, under such a scenario, interest rates would ease, real rates would fall, but the overall impact of lower growth and presumably higher fiscal deficit would push the general government debt/deficit significantly higher.

Figure 26: Debt sustainability hinges critically on sustained, strong growth



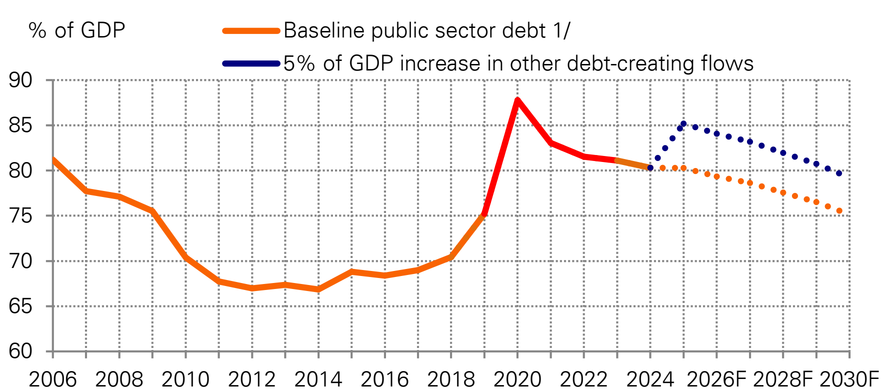
Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

Shock 5: Contingent liability shock of 5% of GDP

In this scenario, we assume that the government's contingent liabilities would rise suddenly. To account for such a risk, we raise the government's liabilities by 5% of GDP in FY25/26. **This first raises the debt/GDP to 85.2% in FY26, thereafter moderating to 79.2% by the end of FY30/31 (2030).**

Earlier such risks arose mainly from banking sector NPAs which had deteriorated significantly in the last decade. However, the public sector bank balance sheets have been cleaned up and the NPAs have fallen below 3%, a record low. Now the risk of contingent liabilities arises mostly from state finances. As per RBI data, overall state guarantees amounted to about 3.7% of GDP by FY21. **The guarantees are likely to have moderated to 0.8% of GDP, as per FY23 budget estimate figures are concerned.** If these guarantees (contingent liabilities) are accounted as on balance sheet expenditure at any point of time, then both general-government deficit and debt would go up by a commensurate amount, ceteris paribus.

Figure 27: A one-off increase in liabilities would delay debt reduction



Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

While the centre has stopped the practice of providing off-balance sheet resources (Extra Budgetary resources or EBR) and now everything is accounted as on-balance sheet expenditure, a similar accounting treatment by states at a later point of time could increase deficit and debt, a non-trivial risk, in our view. Recall that in past years, even before the Covid-19 crisis, a decision to accommodate the interest burden of the UDAY scheme as an on-balance sheet expenditure item along with the costs of farm loan waivers given by certain states deteriorated the health of the states' fiscal balances materially, a risk which exists even today, given the guarantees that exist in the form of contingent liabilities.

Figure 28: 'Freebies' announced by the states in 2022-23

	% of GSDP	% of Revenue Receipts	% of Own Tax Revenue
Andhra Pradesh	2.1	14.1	30.3
Bihar	0.1	0.6	2.7
Haryana	0.1	0.6	0.9
Jharkhand	1.7	8.0	26.7
Kerala	0.0	0.0	0.1
Madhya Pradesh	1.6	10.8	28.8
Punjab*	2.7	17.8	45.4
Rajasthan	0.6	3.9	8.6
West Bengal	1.1	9.5	23.8

*Source : RBI and Deutsche Bank. Note: * Dhasmana, I. (2022). "Not all states are so financially weak that they can't announce freebies". Business Standard. April 2022.*

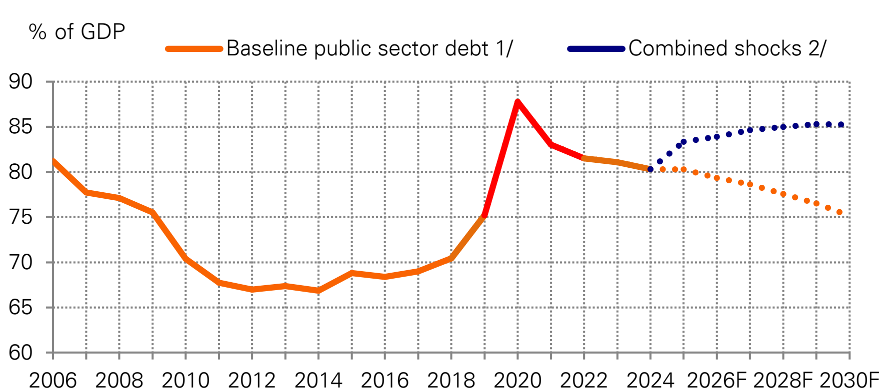
Figure 29: Guarantees issued by State Governments (% of GSDP)

	Bihar	Kerala	Punjab	Rajasthan	West Bengal	Andhra Pradesh	Uttar Pradesh	Haryana
2017-18	1.1	2.5	4.5	7.5	0.9	4.6	6.3	2.2
2018-19	1.0	3.4	0.9	7.6	0.6	6.2	6.9	2.6
2019-20	0.9	3.2	4.1	8.1	0.5	8.1	6.7	2.7
2020-21	2.6	4.6	5.3	8.2	0.6	11.0	7.9	2.3
2021-22 (RE)	3.6	-	5.7	-	1.1	9.8	-	-

Source : RBI and Deutsche Bank. Note: RE refers to revised estimates.

Shock 6: Combined shock - What if shocks were multiple in nature? There could be occasions when a growth shock eventually leads to a fiscal shock which then translates into an interest rate shock. What happens to the debt profile in case of such a combined shock scenario? With this in mind, we construct a scenario incorporating shocks half the size of the ones discussed above but in conjunction with one another. We find that the impact on debt is immediate and unambiguously adverse (the **debt/GDP ratio rises to 85.2% by FY31**). While the probability of such a manifestation is low, it is important to be cognizant of such eventualities.

Figure 30: Debt path can turn higher if hit by multiple shocks



Source : MOF, Deutsche Bank. Note: 1/ Combined Central and Sate level debt

India's debt servicing ability remains healthy

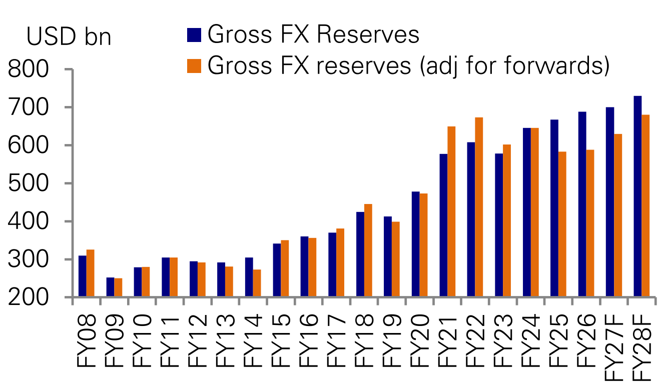
As our exercise shows, public debt/GDP is likely to be on a sustained downtrend due to positive growth-interest differential and there is little risk associated with the funding of the public debt as bulk of the debt is in domestic currency. Foreign currency denominated debt is small - less than 5% of GDP - and even within that more than half of it is owed to multilateral institutions.

India's macro outlook - both in the short and medium term - remains bright, with likelihood of sustained strong growth outperformance amidst anchored inflation, positive real rates, continued fiscal consolidation, healthy external and domestic financial sector and expected macro dividend from ongoing supply-side reforms. As our exercise shows, public debt/GDP seems likely to be on a sustained downtrend due to positive growth-interest differential and there is little risk associated with the funding of the public debt as bulk of the debt is in domestic currency.

Our analysis reveals that, despite GST rates rationalisation of 2025 and an oil price shock this year, **it is possible for India to lower its debt/GDP to 75.1% of GDP by FY31,** provided nominal GDP growth averages around 10.5-11% during the forecasting period. In our view, barring some unforeseen shocks, a 10.5-11% nominal GDP growth assumption is realistic based on an average 6.5% real GDP growth and 4-4.5% inflation (in terms of GDP deflator) over the forecasting period.

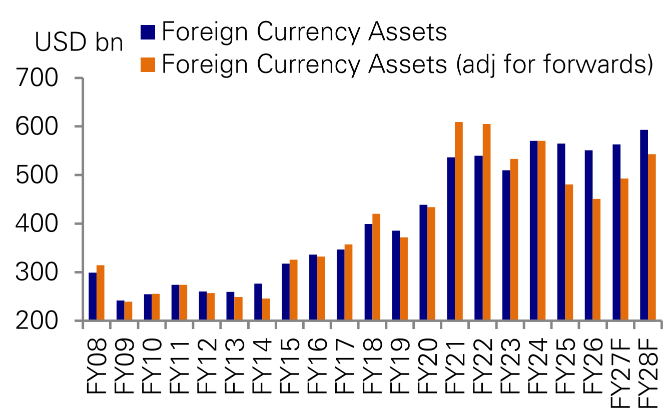
Our forecast is similar to IMF's estimate, which sees India's public sector debt/GDP moderating to 75.4% by 2031 (see, "Staff Report for the 2025 Article IV Consultation for India", Nov 2025), but is slightly higher compared to RBI's forecast of 73.4% over the same forecasting horizon (see, "The Shape of Growth Compatible Fiscal Consolidation", Feb 2024).

Figure 31: Gross Foreign Exchange Reserves



Source : CEIC, RBI and Deutsche Bank. Note: FY27,28 F refer to DB forecasts.

Figure 32: Foreign Currency Assets



Source : CEIC, RBI and Deutsche Bank. Note: FY27,28 F refer to DB forecasts.

Figure 33: India's key external debt (% unless indicated otherwise)

End-March	External Debt (US$ billion)	Ratio of External Debt to GDP	Debt Service Ratio	Ratio of Foreign Exchange Reserves to Total Debt	Ratio of Concessional Debt to Total Debt	Ratio of Short-Term Debt to Foreign Exchange Reserves	Ratio of Short-Term Debt (original maturity) to Total Debt
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*	*(8)*
1991	83.8	28.3	35.3	7.0	45.9	146.5	10.2
1996	93.7	26.6	26.2	23.1	44.7	23.2	5.4
2001	101.3	22.1	16.6	41.7	35.4	8.6	3.6
2006	139.1	17.1	10.1#	109.0	28.4	12.9	14.0
2007	172.4	17.7	4.7	115.6	23.0	14.1	16.3
2008	224.4	18.3	4.8	138.0	19.7	14.8	20.4
2009	224.5	20.7	4.4	112.2	18.7	17.2	19.3
2010	260.9	18.5	5.8	106.9	16.8	18.8	20.1
2011	317.9	18.6	4.4	95.9	14.9	21.3	20.4
2012	360.8	21.1	6.0	81.6	13.3	26.6	21.7
2013	409.4	22.4	5.9	71.3	11.1	33.1	23.6
2014	446.2	23.9	5.9	68.2	10.4	30.1	20.5
2015	474.7	23.8	7.6	72.0	8.8	25.0	18.0
2016	484.8	23.4	8.8	74.3	9.0	23.2	17.2
2017	471.0	19.8	8.3	78.5	9.4	23.8	18.7
2018	529.3	20.1	7.5	80.2	9.1	24.1	19.3
2019	543.1	19.9	6.4	76.0	8.7	26.3	20.0
2020	558.3	20.9	6.5	85.6	8.8	22.4	19.1
2021	573.4	21.1	8.2	100.6	9.0	17.5	17.6
2022	618.8	19.9	5.2	98.1	8.3	20.0	19.7
2023	623.9	19.1	5.3	92.7	8.2	22.2	20.6
2024 PR	668.9	19.2	6.7	96.6	7.4	19.7	19.1
2025 PR	736.4	19.8	6.6	90.8	6.9	20.1	18.3
end-June 2025 PR	746.8	19.7	6.6	93.5	7.1	19.3	18.0
end-September 2025 PR	747.2	20.1	6.0	93.7	7.0	19.7	18.5
end-December 2025 P	765.5	20.4	5.8	89.8	6.7	21.9	19.7

Source : RBI. Note: PR: Partially Revised; P: Provisional.#: Works out to 6.3 per cent with the exclusion of India Millennium Deposits (IMDs) repayments of US$ 7.1billion and pre-payment of external debt of US$ 23.5 million.Note 1: Figures are rounded to one decimal place.Note 2: Ratio of external debt to GDP from end March 2024 (P) onwards is based on the new GDP series (Base Year: 2022/23)

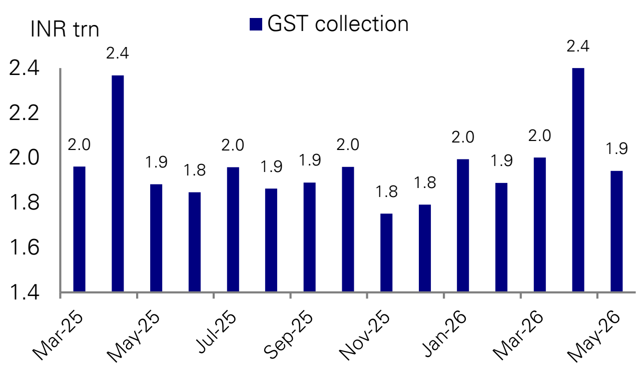
Appendix A: Tax collection trend

Figure 34: Monthly tax collection trend including Apr'26 (% yoy)

Revenue receipts, %yoy	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	FY25	FY26
Gross Tax Revenue	**8.4**	**13.8**	**-2.8**	**32.4**	**10.1**	**-19.2**	**2.1**	**-1.9**	**9.6**	**6.0**
Corporate Tax	0.2	76.4	61.2	22.7	180.8	-96.2	7.7	17.4	8.3	11.4
Income Tax	33.7	26.9	5.8	-9.2	12.6	-45.2	-4.3	6.8	17.0	0.0
Central Goods and Service Tax	8.9	8.1	0.7	6.9	10.6	5.9	10.5	37.7	10.8	6.4
Union Territories Goods and Service Tax	56.5	61.6	253.5	72.1	31.3	90.1	-51.6	268.1	7.2	32.5
Goods and Service Tax Compensation Cess	-5.5	-39.7	-70.3	-55.1	-57.5	-63.5	-100.2	-101.5	6.5	-33.4
Customs	29.0	13.4	-36.3	229.7	19.7	50.9	-19.7	25.0	0.1	13.4
Union Excise Duties	10.3	6.9	17.5	14.6	10.9	23.4	26.5	-1246.2	-1.7	13.9
Service Tax	-145.2	12300.0	-322.2	-51.7	-110.3	-15850.0	1050.0	115.4	-103.5	11726.7
Other Taxes	-33.0	-53.2	-33.0	4.1	11.2	39.4	19.8	66.6	5.8	-14.5
Less: Transfer to state	14.1	14.0	14.1	14.0	-41.3	19.0	19.0	7.4	13.9	8.1
Net Tax Revenue	**7.1**	**12.9**	**-13.6**	**36.5**	**156.1**	**-54.5**	**-1.3**	**-5.9**	**7.5**	**4.9**
Memo item										
Direct tax	11.0	41.7	19.3	10.7	33.1	-53.8	2.1	8.6	12.9	5.2
Indirect tax	1.5	-5.0	-19.4	96.7	-3.1	6.9	2.2	-11.6	5.4	7.1

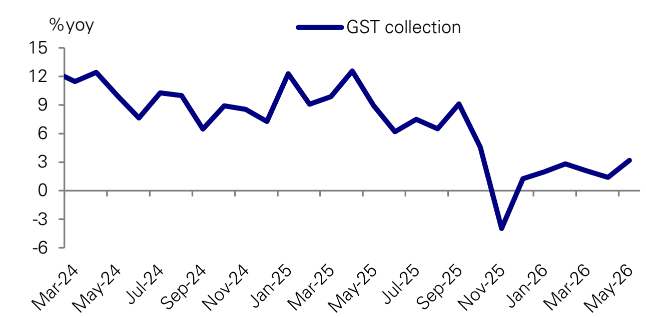
Source : Controller General of Accounts and Deutsche Bank

Figure 35: Monthly GST collection - INR trn



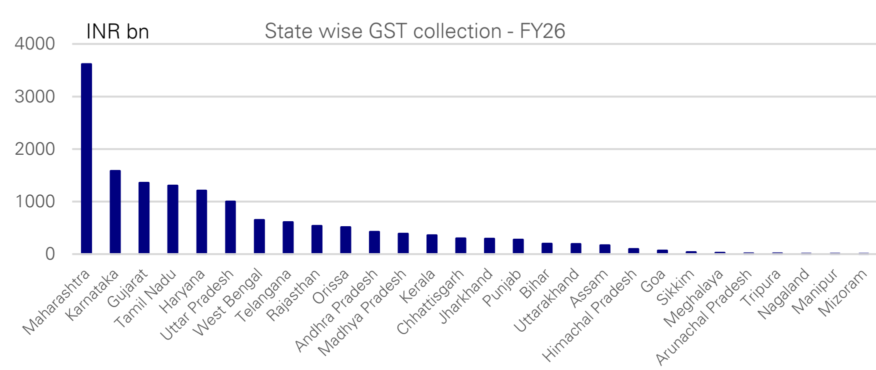
Source : CEIC and Deutsche Bank

Figure 36: Monthly GST collection - %yoy



Source : CEIC and Deutsche Bank

Figure 37: State-wise GST collection - FY26 (INR bn)



Source : CEIC and Deutsche Bank

Appendix B: A snapshot of state fiscal finances

Figure 38: Ranking of 17 key states based on FY26 BE

States	Ranking based on FY26 budget estimates	10 year SDL yield (%)						Fiscal deficit	Own tax revenue	Debt	Interest payment to revenue receipts
		2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	% of GSDP	% of GSDP	% of GSDP	%
Maharashtra	1	7.0	6.8	7.9	-	-	-	2.8	7.8	19	11.5
Jharkhand	2	6.8	6.9	-	-	-	-	2.0	6.3	25	5.1
Telangana	3	7.0	-	-	-	-	-	3.0	8.1	28	8.4
Uttar Pradesh	4	6.9	7.0	-	-	-	7.5	3.0	9.6	29	9.7
Orissa	5	-	-	-	-	-	-	3.2	6.2	20	2.8
Chhattisgarh	6	-	-	-	-	-	-	3.8	8.5	30	6.7
Gujarat	7	6.7	7.0	7.8	-	-	7.4	2.0	5.3	18	12.4
Karnataka	8	6.6	7.0	-	-	-	7.4	2.9	6.8	27	15.6
Haryana	9	7.4	7.0	7.8	7.4	-	-	2.7	6.8	30	20.5
Tamil Nadu	10	6.7	7.0	7.8	-	7.4	7.4	3.0	6.2	29	20.8
Bihar	11	-	7.2	-	-	-	-	3.0	5.4	37	8.8
Madhya Pradesh	12	6.9	6.9	7.9	7.4	7.4	-	4.7	6.4	31	9.8
Rajasthan	13	6.7	7.0	7.8	7.4	-	-	4.3	7.2	36	13.6
Kerala	14	7.2	7.1	-	-	-	-	3.2	6.4	36	20.9
Punjab	15	6.8	7.0	7.9	-	-	-	3.8	7.1	46	22.4
Andhra Pradesh	16	6.5	-	7.7	7.4	-	-	4.4	6.0	35	16.1
West Bengal	17	6.7	7.1	7.8	-	-	-	3.6	5.5	49	18.3
Avg. of 17 states		**6.8**	**7.0**	**7.8**	**7.4**	**7.4**	**7.4**	**3.4**	**6.8**	**32.0**	**13.8**

Source : RBI, Various state budget documents and Deutsche Bank.

Figure 39: Fiscal ranking of states over time

States	Rank based on 2004-08 avg. fiscal data	Rank based on 2008-10 avg. fiscal data	Rank based on FY13 fiscal data	Rank based on FY14 fiscal data	Rank based on FY15 fiscal data	Rank based on FY16 fiscal data	Rank based on FY18 fiscal data	Rank based on FY19 fiscal data	Rank based on FY20 fiscal data	Rank based on FY21 fiscal data	Rank based on FY22 fiscal data	Rank based on FY23 fiscal data	Rank based on FY24 fiscal data	Rank based on fiscal data b/w 2004 to 2016	States
Maharashtra	6	6	1	2	4	4	1	1	1	2	2	-	1	1	Karnataka
Chhattisgarh	4	1	4	10	2	1	2	3	9	3	1	1	9	2	Tamil Nadu
Orissa	8	4	2	7	9	6	3	4	10	1	3	-	2	3	Chhattisgarh
Karnataka	1	3	5	1	1	2	4	5	2	7	7	2	4	4	Maharashtra
Uttar Pradesh	14	16	10	12	14	15	5	2	3	4	11	4	7	5	Orissa
Goa	11	13	12	13	10	12	6	8	-	-	-	-	-	6	Andhra Pradesh
Gujarat	9	10	7	9	6	8	7	6	5	6	5	3	5	7	Madhya Pradesh
Tamil Nadu	3	2	3	3	3	3	8	7	6	12	12	-	12	8	Gujarat
Madhya Pradesh	10	7	8	5	5	7	9	10	7	9	10	-	10	9	Haryana
Haryana	2	14	11	8	11	13	10	9	15	10	9	7	3	10	Jharkhand
Rajasthan	13	12	9	14	15	17	11	15	14	16	14	-	13	11	Punjab
Kerala	7	8	16	16	13	9	12	12	8	13	15	6	11	12	Kerala
Bihar	15	15	15	15	17	10	13	13	12	11	13	11	17	13	Goa
Punjab	12	11	14	11	8	11	14	17	16	17	17	10	15	14	Rajasthan
Andhra Pradesh	5	5	6	4	12	5	15	14	13	14	8	8	16	15	Uttar Pradesh
West Bengal	16	17	17	17	16	16	16	16	17	15	16	9	14	16	Bihar
Jharkhand	17	9	13	6	7	14	17	11	11	8	4	5	8	17	West Bengal

Source : Deutsche Bank

Figure 40: Fiscal ranking of states based on FY24 actual data & comparison with FY25 RE and FY26 BE (best to worst)

States	Ranking based on FY24 actual fiscal data	Ranking based on FY25 revised estimates	Ranking based on FY26 budget estimates
Maharashtra	1	1	1
Orissa	2	4	5
Jharkhand	3	2	2
Gujarat	4	5	7
Karnataka	5	8	8
Uttar Pradesh	6	7	4
Chhattisgarh	7	6	6
Telangana	8	3	3
Madhya Pradesh	9	11	12
Haryana	10	9	9
Kerala	11	13	14
Tamil Nadu	12	10	10
Bihar	13	14	11
Rajasthan	14	12	13
Andhra Pradesh	15	15	16
West Bengal	16	17	17
Punjab	17	16	15

Source : Various state budget documents and Deutsche Bank. Note: RE refers to revised estimates and BE refers to budget estimates.

Figure 41: Fiscal snapshot of states

States	Ranking based on FY24 actual fiscal data	2023-24 Actuals				2024-25 RE				2025-26 BE			
		Fiscal deficit (% of GSDP)	Own tax revenue (% of GSDP)	Debt (% of GSDP)	Interest payment to revenue receipts (%)	Fiscal deficit (% of GSDP)	Own tax revenue (% of GSDP)	Debt (% of GSDP)	Interest payment to revenue receipts (%)	Fiscal deficit (% of GSDP)	Own tax revenue (% of GSDP)	Debt (% of GSDP)	Interest payment to revenue receipts (%)
Maharashtra	1	2.2	7.5	18.5	10.6	2.9	8.1	18.6	10.2	2.8	7.8	19.0	11.5
Orissa	2	1.7	6.3	19.7	2.9	3.1	6.3	20.6	2.6	3.2	6.2	20.3	2.8
Jharkhand	3	1.4	6.1	27.2	7.8	2.3	6.7	25.5	6.5	2.0	6.3	25.4	5.1
Gujarat	4	1.0	5.5	18.3	12.2	1.9	5.5	17.8	12.4	2.0	5.3	17.9	12.4
Karnataka	5	2.6	6.4	25.3	13.2	2.9	6.3	25.4	14.2	2.9	6.8	26.5	15.6
Uttar Pradesh	6	3.2	7.5	30.9	10.1	3.4	7.6	30.0	9.8	3.0	9.6	28.7	9.7
Chhattisgarh	7	5.3	7.6	27.3	6.6	5.3	8.1	29.2	7.4	3.8	8.5	29.8	6.7
Telangana	8	3.4	7.6	27.0	14.4	2.9	8.0	27.3	8.8	3.0	8.1	27.5	8.4
Madhya Pradesh	9	3.3	6.7	30.1	9.9	4.2	6.4	30.8	10.3	4.7	6.4	31.3	9.8
Haryana	10	2.9	6.7	31.3	21.3	2.7	6.8	30.2	21.9	2.7	6.8	29.9	20.5
Kerala	11	3.0	6.5	36.3	21.7	3.5	6.4	36.2	22.4	3.2	6.4	35.5	20.9
Tamil Nadu	12	3.3	6.1	31.0	20.2	3.3	6.2	27.3	20.5	3.0	6.2	29.2	20.8
Bihar	13	4.2	5.7	38.7	9.1	9.2	5.6	38.0	8.4	3.0	5.4	36.8	8.8
Rajasthan	14	4.3	6.2	37.4	16.8	4.1	7.1	37.0	14.9	4.3	7.2	35.9	13.6
Andhra Pradesh	15	4.4	6.0	34.6	17.0	4.6	6.0	35.5	17.6	4.4	6.0	35.4	16.1
West Bengal	16	3.3	5.4	39.7	21.3	4.0	5.5	40.0	20.3	3.6	5.5	48.9	18.3
Punjab	17	4.4	6.3	46.5	25.3	4.5	7.2	46.9	23.1	3.8	7.1	46.4	22.4
Avg. of 17 states		**3.2**	**6.5**	**30.6**	**14.1**	**3.8**	**6.7**	**30.4**	**13.6**	**3.3**	**6.8**	**30.8**	**13.1**
All States and UTs		2.9	6.5	28.1	12.9	3.5	6.8	28.4	12.3	3.3	7.1	29.2	12.2

Source : Various state budget documents and Deutsche Bank. Note: RE refers to revised estimates and BE refers to budget estimates.

Annexure: Credit ratings methodology

Figure 42: Moody's credit ratings methodology

Sovereign Bond Ratings Sector Scorecard Overview

Factor	Sub-factor	Sub-factor Weighting	Metric/Sub-sub-factor	Metric / Sub-sub-Factor Weighting
Factor: Economic Strength	Growth Dynamics	35%	Average Real GDP Growth $_{t-4 \text{ to } t+5}$	25%
			Volatility in Real GDP Growth $_{t-9 \text{ to } t}$	10%
	Scale of the Economy	30%	Nominal GDP (US$ bn) $_{t}$	30%
	National Income	35%	GDP per Capita (PPP, Int. USD) $_{t}$	35%
	Adjustment to Factor Score	0 - 9 notches	Other	
Factor: Institutions and Governance Strength	Quality of Institutions	40%	Quality of Legislative and Executive Institutions	20%
			Strength of Civil Society and the Judiciary	20%
	Policy Effectiveness	60%	Fiscal Policy Effectiveness	30%
			Monetary and Macroeconomic Policy Effectiveness	30%
	Adjustment to Factor Score	0 - 3 notches	Government Default History and Track Record of Arrears	
		0 - 3 notches	Other	
Factor: Fiscal Strength	Debt Burden	50%[1]	General Government Debt / GDP $_{t}$	25%
			General Government Debt / Revenue $_{t}$	25%
	Debt Affordability	50%[1]	General Government Interest Payments / Revenue $_{t}$	25%
			General Government Interest Payments / GDP $_{t}$	25%
	Adjustments to Factor Score	0 - 6 notches	Debt Trend $_{t-4 \text{ to } t+1}$	
			General Government Foreign Currency Debt / General Government Debt $_{t}$	
			Other Non-Financial Public Sector Debt / GDP $_{t}$	
			Public Sector Financial Assets and Sovereign Wealth Funds / General Government Debt $_{t}$	
		0 - 3 notches	Other	
Factor: Susceptibility to Event Risk	Political Risk	Minimum Function[2]	Domestic Political and Geopolitical Risk	
	Government Liquidity Risk	Minimum Function[2]	Ease of Access to Funding	
			0 - 2 scoring categories	Adjustment to Sub-factor Score High Refinancing Risk
	Banking Sector Risk	Minimum Function[2]	Risk of Banking Sector Credit Event (BSCE)	
			Total Domestic Bank Assets / GDP $_{t}$	
			0 - 2 scoring categories	Adjustment to Sub-factor Score
	External Vulnerability Risk	Minimum Function[2]	External Vulnerability Risk	
			0 - 2 scoring categories	Adjustment to Sub-factor Score
	Adjustment to Factor Score	0-2 scores		

Source : Economic Survey 2020-21, Volume 1

Figure 43: Fitch's credit ratings methodology

Analytical pillar	Structural features	Macroeconomic performance, policies & prospects	Public finances	External finances
SRM weights (%)	53.7	10.9	18.0	17.4

SRM variables	Measure	Impact	Weight (%)	Coefficient
Governance indicators	Latest	Positive	20.4	0.075
GDP per capita	Latest	Positive	12.3	0.040
Share in world GDP	Latest	Positive correlation with size	13.2	0.607
Years since default or restructuring event	Latest	Negative	6.4	-2.481
Broad money supply (% of GDP)	Latest	Positive	1.4	0.185
Overall weight in SRM			**53.7**	

SRM variables (%)	Measure	Impact	Weight (%)	Coefficient
Real GDP growth volatility	Latest	Negative	4.9	-0.767
Consumer price inflation	3-year centred average	Negative	3.1	-0.056
Real GDP growth	3-year centred average	Positive	2.9	0.093
Overall weight in SRM			**10.9**	

SRM variables	Measure	Impact	Weight (%)	Coefficient
Gross general govt debt/GDP	3-year centred avg	Negative	8.0	-0.021
General govt interest (% of revs)	3-year centred avg	Negative	4.7	-0.046
General govt fiscal bal./GDP	3-year centred avg	Directional	3.0	0.055
FC govt debt/gross govt debt (%)	3-year centred avg	Negative	2.4	-0.006
Overall weight in SRM			**18.0**	

SRM variables	Measure	Impact	Weight (%)	Coefficient
Reserve currency flexibility	Latest	Positive	7.8	0.551
Sovereign net foreign assets (% of GDP)	3-year centred avg.	Positive	6.7	0.011
Commodity dependence	Latest	Negative	0.8	-0.003
Foreign exchange reserves (months of CXP)[a]	Latest	Positive	1.3	0.027
External interest service (% of CXR)	3-year centred avg.	Negative	0.7	-0.012
Current account balance + foreign direct investment (% of GDP)	3-year centred avg.	Directional	0.2	0.002
Overall weight in SRM			**17.4**	

Source : Economic Survey 2020-21, Volume 1

Appendix 1

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Research Institute

AI twist in the tale for Anthropic's Fable and Mythos

June 16, 2026

Authors

Adrian Cox
Research Analyst

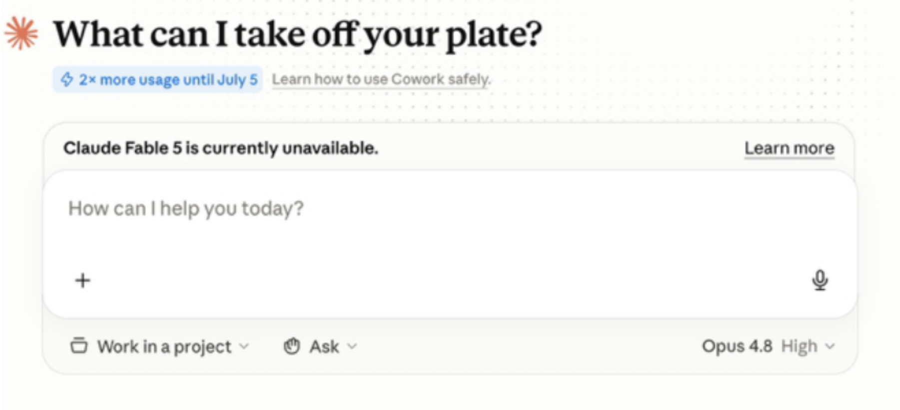
The story so far...

If you want to understand the implications of the abrupt switching off of Anthropic's Claude Fable 5 and Mythos 5 AI models globally due to US national security orders, just consider the likely reaction in the corridors of OpenAI, Washington and Brussels.

The US Commerce Department issued an export control directive late on Friday to suspend all access by foreign nationals to Anthropic's most advanced models, prompting Anthropic to disable access to all customers to ensure compliance.

The US said it had become aware of a method of bypassing, or "jailbreaking", Fable 5, the public version of Mythos 5, according to Anthropic. Mythos has only been rolled out to a few partners because of fears that it could be misused to exploit cybersecurity vulnerabilities. The startup disputed the concerns, which were reportedly raised by its partner Amazon with administration officials, saying: "We believe this is a misunderstanding." Further details remain hazy.

Figure 1: Anthropic's Claude Fable 5 is "currently unavailable"



Source: Anthropic's Claude desktop app, accessed June 14, 2026

Anthropic is in a race for funding in what could be a historic year for technology IPOs. After it announced on June 1 that it had confidentially filed for an IPO, OpenAI followed suit last week. Both are expected to take place later this year at valuations of more than $1 trillion apiece.

What could this latest twist in the tale mean for a company growing at breakneck pace; its now smaller rival, whose ChatGPT chatbot is still the byword for AI outside Silicon Valley; a US government that sees an existential threat from China; and European governments already painfully aware of the continent's lagging position in the AI race?

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1. Short-term pain, but potentially long-term gain, for Anthropic

- What greater compliment than to be officially labelled too powerful to release? Nuanced questions of guard rails are likely to be overlooked in the face of marketing that money can't buy. It's like the Adidas running shoes that are so good they're banned in elite competitions. Even if Anthropic has to keep its most advanced models in a cage in the short term, in the longer term its reputation will likely have received a significant boost.

- The timing may be ideal, with a series of mega tech IPOs making huge valuations seem attainable, but also potentially forcing investors to pick a favourite or two amid concerns about how much the market can absorb.

- This is not the first time Anthropic has had a run-in with the US administration. In February, the Department of Defense designated the startup as a supply-chain risk, effectively blacklisting it from working with the US military and contractors, after Anthropic asked that its technology not be used for mass surveillance of Americans and lethal autonomous warfare. President Donald Trump posted that every federal agency should stop working with Anthropic. The company has taken the Pentagon to federal court. But if there were to be a change in the political winds, that could put Anthropic back on the government's books with its reputation stronger than ever.

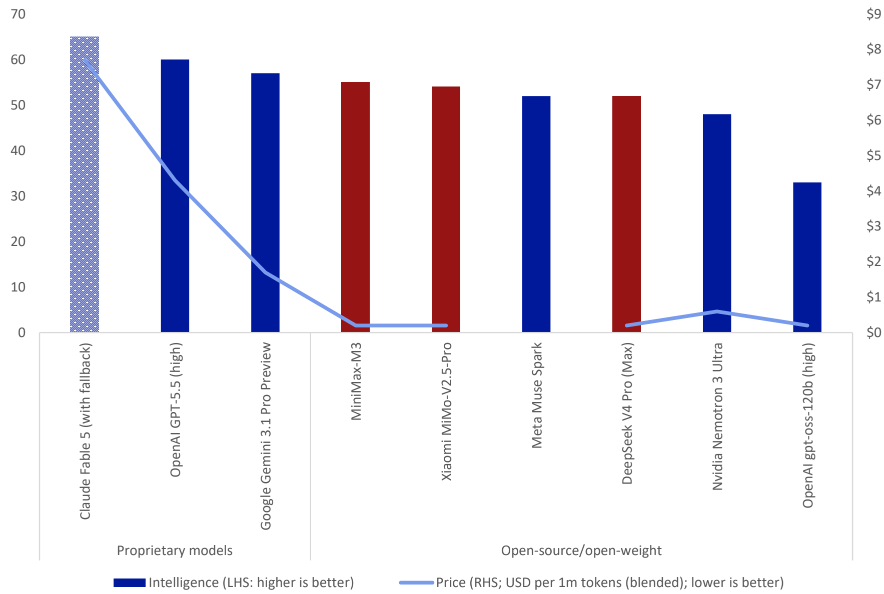
2. Rising pressure on OpenAI's technology and business model

- OpenAI is under pressure as a potential IPO approaches. Anthropic has recently overtaken it in revenues ($47bn annual recurring revenue at the end of May versus OpenAI's $24bn at the end of March) and valuation ($965bn post-money at the end of May in versus OpenAI's $852bn in March). While models produced by the two companies have now been neck-and-neck for many months, this has been the first time that a model has been banned for being too smart.

- While OpenAI is still the leading consumer app by a country mile, reportedly having crossed the billion monthly users mark in May, only 50 million were paying for it at the end of February. Consequently, the company has sought to refocus around businesses, an area that Anthropic already dominates. OpenAI had nine million enterprise subscribers in February.

- The company is considering drastic price cuts ahead of similar cuts it expects at Anthropic, the Wall Street Journal reported last week, amid stiffer competition from open-source Chinese models that may be good enough for most applications at a fraction of the cost. When it was available, Claude Fable 5 cost $7.70 per million tokens and GPT-5.5 cost $4.30, according to a blended score from Artificial Analysis. Compare that with around 20 cents for a number of open-source models.

Figure 2: Slightly more intelligent proprietary models are significantly more expensive than open-source alternatives (chart includes selected models)



Source : Artificial Analysis Intelligence Index. "Fallback" mechanism routes safety-flagged messages to Opus 4.8 model. Meta Muse Spark price not available. US models in blue; Chinese models in red

3. Double-edged sword for US aim to dominate AI

- The move is the first time the US government has used export controls to stop access to a commercial AI model that is already widely used by the public. There is a risk that it backfires not only for a country that considers itself to be in an existential race for "global AI dominance" but also for US companies that rely on a global market to justify AI investments.
- The move echoes the now well-established policy of putting export controls on hardware, notably Nvidia's most advanced AI chips. Indeed, an unattributed report from Semafor suggested that the decision to ban foreign use of Mythos had been driven partly by suspicions that a China-linked group had accessed it.
- Five years ago, Chinese companies relied on US chips and developers built their software on its industry-standard CUDA platform. However, since the imposition of stiff export controls, starting under the Biden administration, Chinese companies led by Huawei have invested heavily in creating their own chips and AI ecosystem, eroding US leadership. While they are not yet at the cutting edge, they are gaining fast and may get a foothold in the rest of the world with unaligned countries that prefer cheaper alternatives to US technology.

4. Intensified calls for sovereign AI amid rising trade tensions

- The move casts an unforgiving light on more profound regulatory and strategic issues about sovereign AI: who controls the AI models, data and infrastructure. The US has half of the world's data centres and last year clocked up more than 20 times as much private investment in AI companies as China, the No. 2 country, with European companies far in the distance.
- The speed of the ban is likely to reinforce concern that access to all US technology could be suddenly switched off without warning. It may further propel efforts to diversify by companies such as financial institutions that depend on US technology for systemically critical infrastructure.

- The ban "further underlines Europe's need for technological sovereignty," the European Commission said over the weekend. The problem is that although Europe is home to a number of key AI-related engineering companies, it only has one notable AI model maker, France's Mistral, and relies on non-EU countries for over 80 percent of key digital products, services, infrastructure and intellectual property. The Tech Sovereignty package announced on June 3 is an attempt to accelerate initiatives to build domestic capacity and autonomy across the stack.
- The move also plays into a wider theme of technological independence that has gained momentum with this year's conflict in the Middle East and attacks on data centres highlighting the vulnerabilities of supply chains and dependence on other countries for data and compute.

Related relevant content

Assessing AI in finance with Evident Co-CEO Alexandra Mousavizadeh (June 9, 2026)

ChatIPO: Quick take on OpenAI's record-breaking IPO plans (May 21, 2026)

Will Claude's AI finance agents change your life? (May 6, 2026)

AI, Iran and the Gulf 101 (April 13, 2026)

AI Q&A: Why have market vibes suddenly shifted on AI? (Feb 16, 2026)

Three AI themes for 2026: the honeymoon is over (Jan 20, 2026)

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Is the ceasefire a reset? Iran, the US and the Middle East they can't restore

June 15, 2026

The US and Iran have announced a framework agreement to end hostilities, setting the stage for a formal signing in Switzerland and sending markets rallying with oil tumbling. This landmark deal, a potential "win" for the Trump administration, Iran's resilience, and Israel's strategic gains, redefines the Middle East landscape. However, the path ahead is complex: watch for implementation disagreements, Gulf states' security demands, and Iran's continued leverage over the Strait of Hormuz. We're also seeing accelerated UAE-Saudi tensions and a multipolarizing Gulf as China and Russia position themselves, marking a critical juncture for regional security and global energy markets.

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Context

On 14 June, the US and Iran announced a framework agreement to end hostilities, with a formal signing ceremony to take place on 19 June in Switzerland. Under the tentative agreement, the US would gradually lift its naval blockade on Iranian ports and relax sanctions/release frozen Iranian assets, in exchange for progress towards a technical agreement on Iran's nuclear programme. The agreement is meant to reopen the Strait of Hormuz and address the stickier questions of freedom of navigation through the Strait, Iran's nuclear programme, reconstruction, and sanctions relief among other issues.

Everyone "Wins"?

We have suggested since early March that President Trump would seek an off-ramp (see Thematic Research: The Persian Chessboard: Decoding Iran's Strategy Beyond a US "Victory") given the greater incremental gains for the US versus diminishing returns.

Despite only a framework agreement and the tremendous costs of military actions and the Iranian responses, the Trump administration can likely present a "mission accomplished" based on the extensive damage to Iran's military infrastructure, nuclear sites and air defences, a route to reopening the Strait of Hormuz, and a pathway to some nuclear agreement.

For the Iranians, Tehran's strategy of weaponizing regional and economic instability to make the conflict prohibitively expensive for its adversaries was largely successful (though much will depend on concessions to be made during the forthcoming negotiations). Nevertheless, the regime has demonstrated its resilience. For Israel, it was able to significantly harm Iran's military capabilities, address Iran's nuclear programme, and degrade Hezbollah's capabilities in Lebanon. This may provide President Netanyahu with a political boost in advance of competitive Israeli elections later this year.

What should we be watching next?

While markets are celebrating with significant rallies and oil tumbling to a three-month low, the reversion to the old Middle East status quo may take time to restore, if indeed that is possible.

What could derail the deal? There are many moving parts in negotiations and we should watch closely for disagreements over implementation (timelines, verification, etc.), Gulf states' requirements for security guarantees, and domestic political opposition from key parties. However, we assess that both the US and Iran are keen to press ahead with the framework agreement even if this doesn't lead to a full-scale peace agreement. A critical factor to watch will be the potential for Israeli aspirations in Lebanon to upend the deal. We suspect that there is significant US pressure on President Netanyahu to cease operations and accept the accrued gains to prevent a derailing of the deal.

Strait of Hormuz: Shipping isn't likely to be restored immediately given the challenges of de-mining, insurance premia remaining high, lack of trust until we see sustained stability over time, infrastructure repair to ports, terminals and regional energy infrastructure, and security guarantees. While Iran will probably be keen to show goodwill, the Strait of Hormuz is its most important bargaining chip in negotiations – one that it will likely seek to protect. Whether any agreement on the Strait of Hormuz entails a “toll”, Iran's strategic leverage over the Strait could translate into long-term, persistent, albeit possibly intermittent, threats to navigation.

Why Iran may have won more than it lost: Despite significant military losses, Iran has demonstrated resilience through its military actions and a willingness to close the Strait of Hormuz. Crucially, the mere fact that the US came to the table validates Iran’s long-standing insistence that it cannot simply be bombed into submission. Iran’s ballistic missile programme remains and its proxy relationships with groups across the region, while weakened, have not been dismantled.

The regime that enters these negotiations is a harder one – the strikes killed many moderate figures – leaving a leadership cadre that appears more hawkish and defiant. In addition, surviving a confrontation with the US and Israel without surrendering also strengthens the regime’s hand domestically. The deal’s promises of potential sanctions relief and access to $24 billion in frozen assets provides a much-needed economic lifeline to an Iranian government under severe financial pressure. For the broader Middle East, this sets an uncomfortable precedent for engagement with Iran in the longer term.

The New Middle East?

UAE-Saudi tensions: The conflict has exposed and accelerated fault lines between the UAE and Saudi Arabia that have simmered for years. The UAE’s decision to withdraw from OPEC in May 2026, reportedly taken without prior consultation with Riyadh, is the most visible sign. This deepening rupture is driven not just by energy policy but also by broader strategic rivalries and competing visions for the region’s future. We will be watching what that rivalry means for oil markets, regional security arrangements and Washington’s ability to manage its Gulf alliances as a coherent bloc once the Iran crisis settles.

Gulf-US Relations: The Gulf states have responded cautiously but positively to the framework agreement, welcoming any steps that could reduce regional tension and improve stability. That said, we judge that they are keen to ensure that any lasting agreement entails enforceable security assurances and regional participation in future arrangements. In the aftermath of this crisis, Gulf ties with the US may be more complex, and we could see efforts toward greater economic and security diversification. Moreover, any outcome that leaves the Gulf states to deal with the consequences of a conflict that has already caused significant regional disruption would risk deepening existing fissures between the Gulf and the US and undermine confidence in long-term US commitments to regional security.

Multipolarizing the Gulf? Both China and Russia have welcomed the framework agreement. While Beijing has focussed on restoring stability in the Gulf and safeguarding international trade routes, Moscow has underscored respect for Iran’s sovereignty and the importance of a negotiated political settlement. Both seem keen to ensure that any final settlement does not reinforce long-term US dominance in the Gulf. Indeed, both are benefiting from positioning themselves as defenders of international law and the post-WWII order, and may be able to make inroads with regional Gulf allies in the aftermath of the crisis, potentially at the US’ expense.

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Capital allocation without Big Tech

June 15, 2026

The sharp sell-offs that have punctuated the AI-related stock market appreciation over the last few years have all come at times when investors have contemplated whether the AI business model is at a tipping point. This year's related IPOs have certainly entered into that realm.

The recent wobbles have also come as investors consider the financing methods of some the largest hyperscalers which have signalled they are ready to turn to equity issuance to invest in AI capacity.

Apart from testing the pecking order theory of financing, these developments also have implications for aggregate capital allocation of the S&P 500 given the size of tech firms. So, in this short note we disaggregate buybacks, debt issuance and capex of the S&P 500 to quantify the impact of tech firms.

Amongst our key takeaways is that while tech dominance in the stock market is growing, other S&P 500 corporates offer solid fundamentals which means the bulk of companies in the index do not look overpriced. That should keep the potential fallout from any further AI disenchantment more contained than many currently fear.

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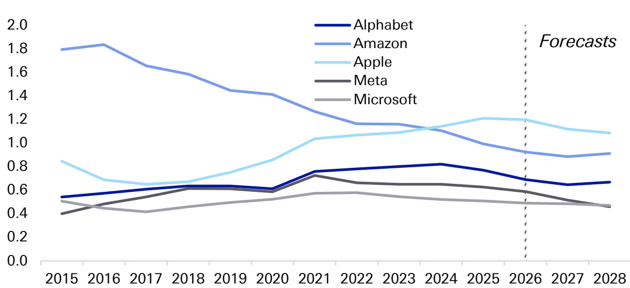
Tech dominance is back but its ranks are changing

Figure 1: TMT companies market cap as % of S&P 500 total

Source : Bloomberg Finance LP, Deutsche Bank.

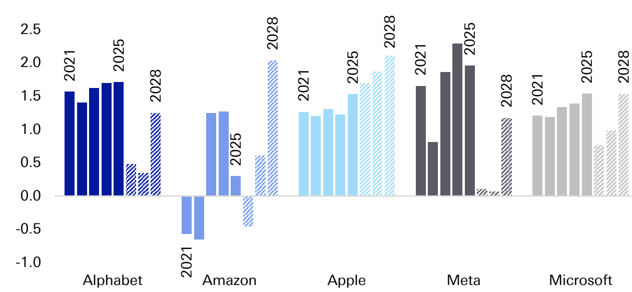
The run-up in the share of TMT companies' market cap relative to the S&P 500 total comes just as the biggest tech players that have propelled index fundamentals in recent years are expected to experience declines in free cash flow. That comes on top of slowing sales growth for some which will weigh on their asset efficiency, particularly given their capex plans.

Figure 2: Asset turnover of selected tech companies



Source : Bloomberg Finance LP, Deutsche Bank.

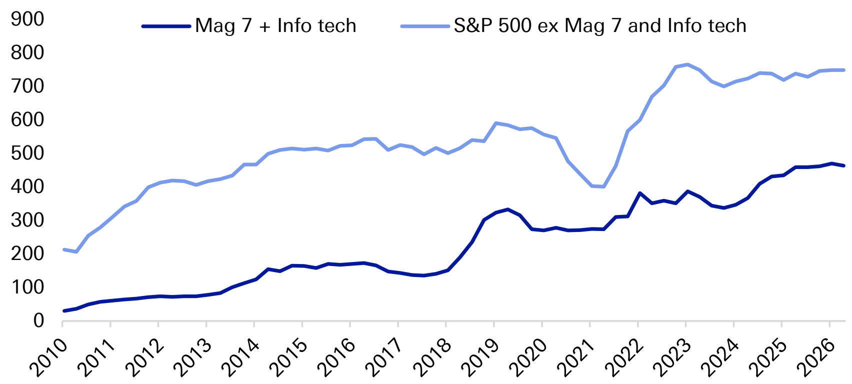
Figure 3: Free cash flow of selected tech companies, indexed to 2020 levels *



*Source : Bloomberg Finance LP, Deutsche Bank. * Note: shaded areas are forecasts.*

The cash drain from AI capex is a concern for investors in part because tech companies have offered strong payouts in recent years amidst record cash generation, including during covid when the aggregate for rest of the index declined.

Figure 4: Total payouts ($bn, LTM) *



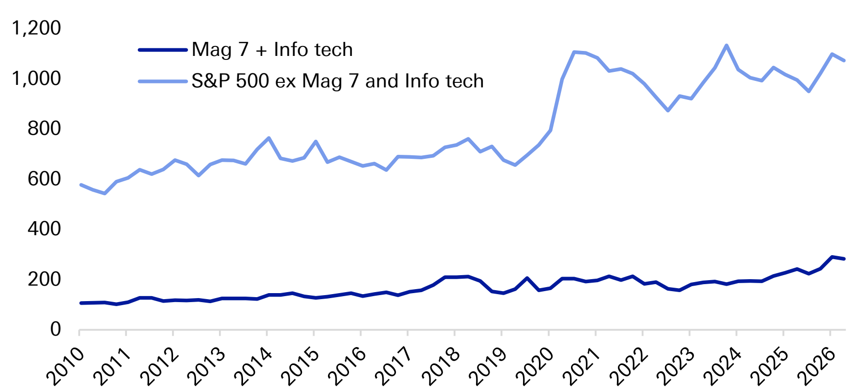
*Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.*

When we now look at large non-tech American corporates, there are strong signs of resilience. This leads us to expect them to increase their investing and payout activities in the coming few years. That would be a pleasant contrast to the first half of this decade which was beset by various shocks and uncertainty amidst a changing geopolitical and economic backdrop.

Capital allocation trends of non-tech corporates look solid

Compared with the trend in payouts, the relative size of aggregate tech companies' actual cash pile is much smaller. While this reflects the differences in cash needs of business models and profitability, aggregate cash coffers are up by c 8% p.a. since 2019, outpacing inflation.

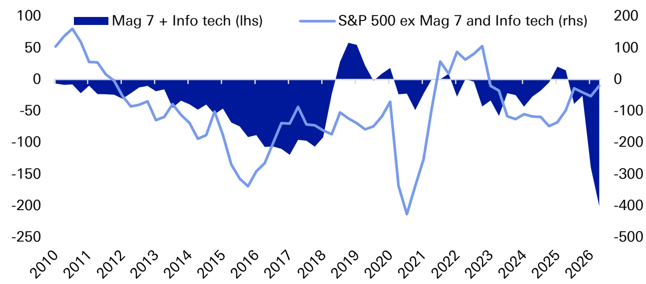
Figure 5: Cash & cash equivalents balances ($bn) *



*Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.*

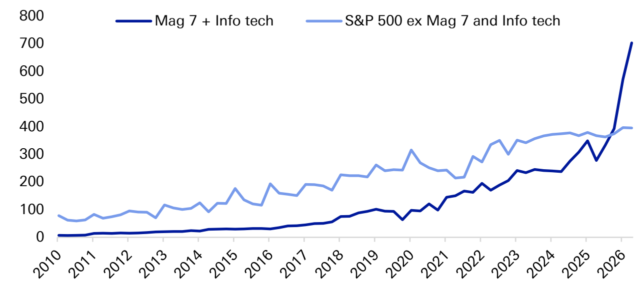
Part of the reason for the increase in cash in non-tech companies is a turn towards relatively conservative capital allocation. Unlike tech companies, the rest of the S&P 500 corporates have been in cash-preservation mode, which is reflected in almost no new net borrowing in the last 12 months as well as a relatively slower increase in capex (+876% for tech and +62% for non-tech since 2019).

Figure 6: Cash spent on net debt (negative - more borrowing) *



*Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.*

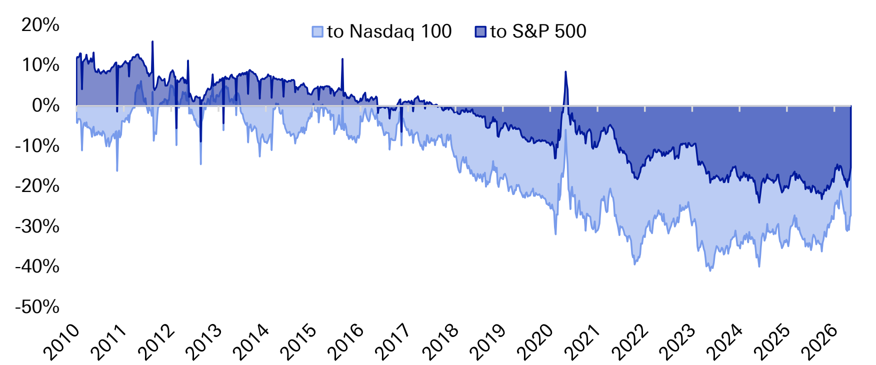
Figure 7: Reported capex, LTM $bn *



*Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.*

We believe this cash preservation is likely to change. Big M&A deals are already one of the signs that companies are looking to decrease their cash piles. This may help make non-tech corporates more attractive to investors and acquirors and, as the chart below shows, the equal-weighted S&P 500 continues to trade at c.15% discount to the market-cap index, and at 27% to the Nasdaq 100 (on a forward P/E basis). And while the non-tech firms are expected to have only 13.2% EPS growth in 2026 (23.7% for tech), that is ahead of Europe (11.2%) and Japan (10.5%).

Figure 8: Equal-weighted S&P 500, premium (+) / discount (-) to...



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